



DEPARTMENT OF COMPUTER ENGINEERING

KWAME NKRUMAH UNIVERSITY OF SCIENCE
AND TECHNOLOGY, KUMASI, GHANA



EE271 Semiconductor Devices

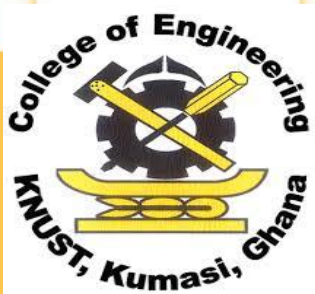
Lecture 1

Introduction to Semiconductors

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Education is what remains after one has forgotten everything he learned in school

Albert Einstein

Overview

Course Info

Course
Outline

Reference
Books

Intro to
Semi-
Conductors

Semi-
Conductor
Devices

Course Information

EE 271 Semiconductor Devices



☐ 2hrs Teaching

☐ 2 Credit Hours

Requirements

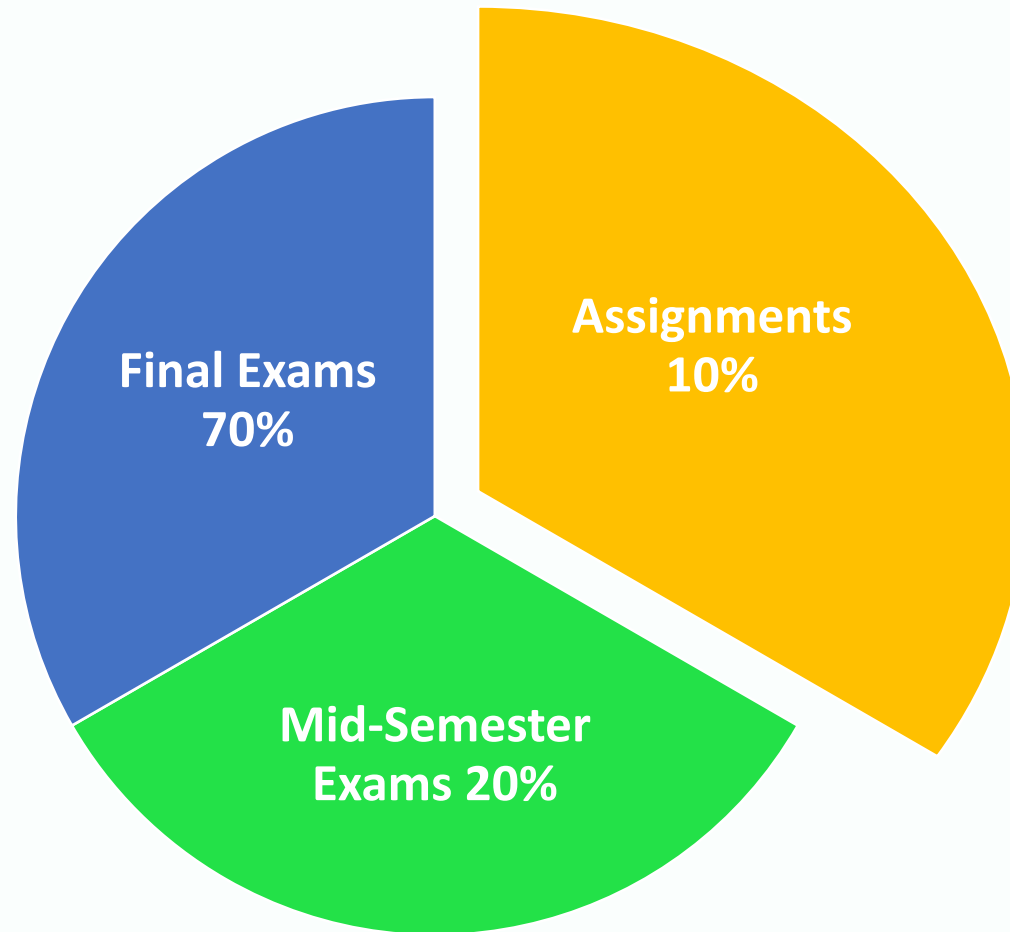


☐ Basic Knowledge in Electronics

☐ Basic Knowledge in Physics of Semiconductors

☐ Basic Knowledge in Mathematics and Physics

Course Grading



Course Outline

Introduction to Semiconductors – Week 1

Physics of Semiconductors – Week 2

Diode Models and Circuits – Week 3

Bipolar Transistors – Week 4

Bipolar Amplifiers – Week 5

Course Outline

Field-Effect Transistors FET -JFET– Week 6

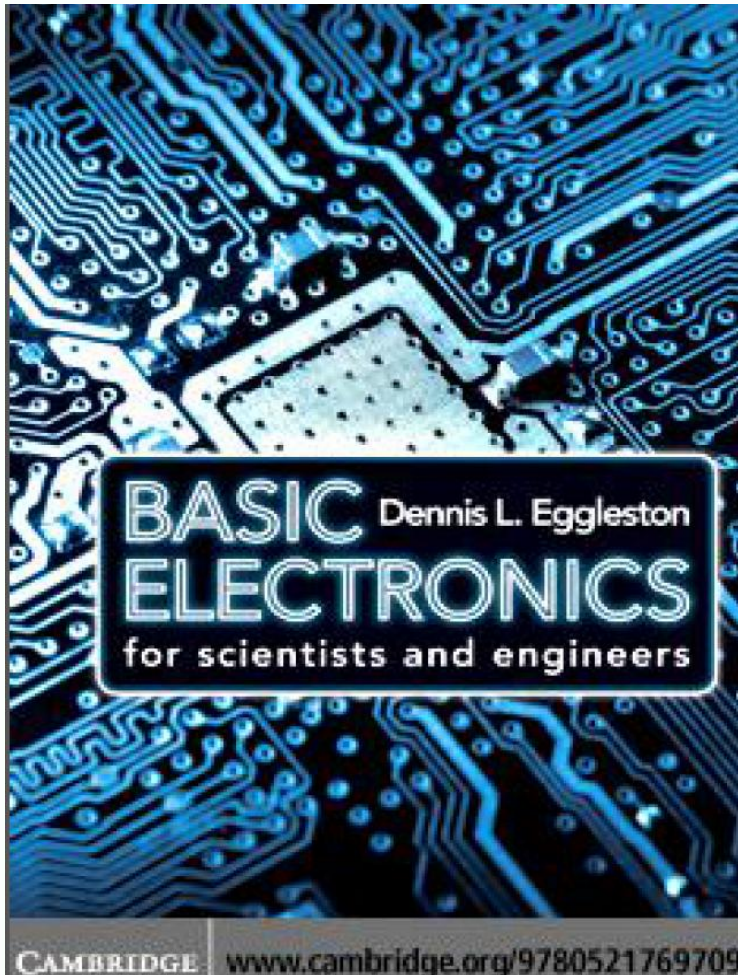
Field-Effect Transistors FET - MOSFET – Week 7

Operational Amplifiers – Week 8

Two Terminal Devices (Solar Cells, Photodiodes) – Week 9

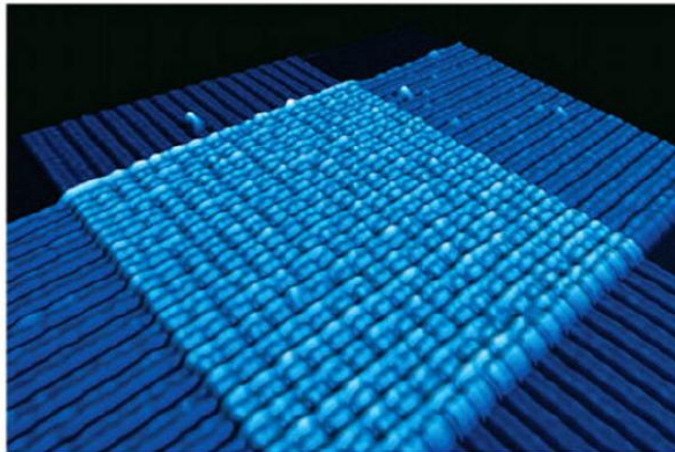
PNPN Devices – Week 10

References Books



electronic devices
and circuit theory

ROBERT L. BOYLESTAD | LOUIS NASHESKY



B. Jayant Baliga

**Fundamentals
of Power
Semiconductor
Devices**

 Springer

Reference Books

BEHZAD
RAZAVI

Fundamentals of Microelectronics



SECOND EDITION

WILEY

Physics and Technology of
Semiconductor Devices

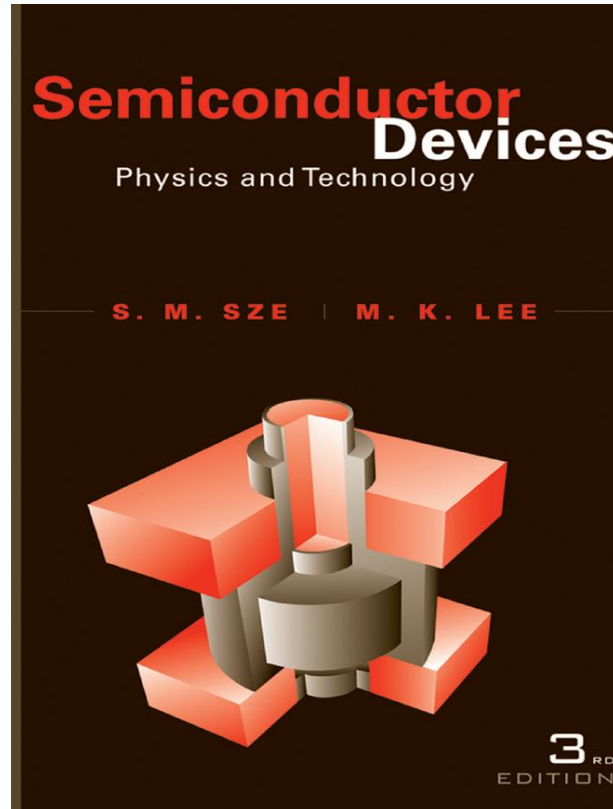
A. S. GROVE

Intel Corporation, Mountain View

University of California, Berkeley

JOHN WILEY & SONS

New York • Chichester • Brisbane • Toronto • Singapore



Semiconductor Physics and Devices

Basic Principles

Third Edition

Donald A. Neamen

University of New Mexico

Mc
Graw
Hill

Boston Burr Ridge, IL Dubuque, IA Madison, WI New York San Francisco St. Louis
Bangkok Bogota Caracas Kuala Lumpur Lisbon London Madrid Mexico City
Milan Montreal New Delhi Santiago Seoul Singapore Sydney Taipei Toronto

Semiconductor Device Applications



Red LED



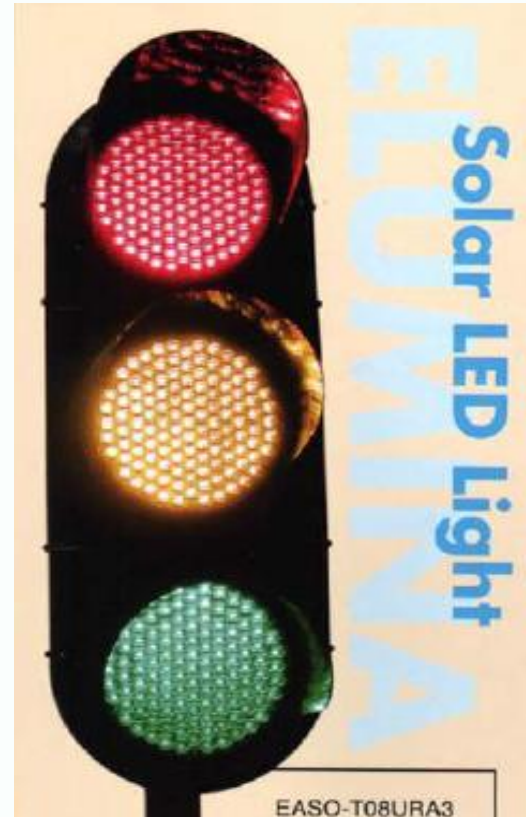
White LED



LED for displays



Blue LED



LED for traffic light

LEDs

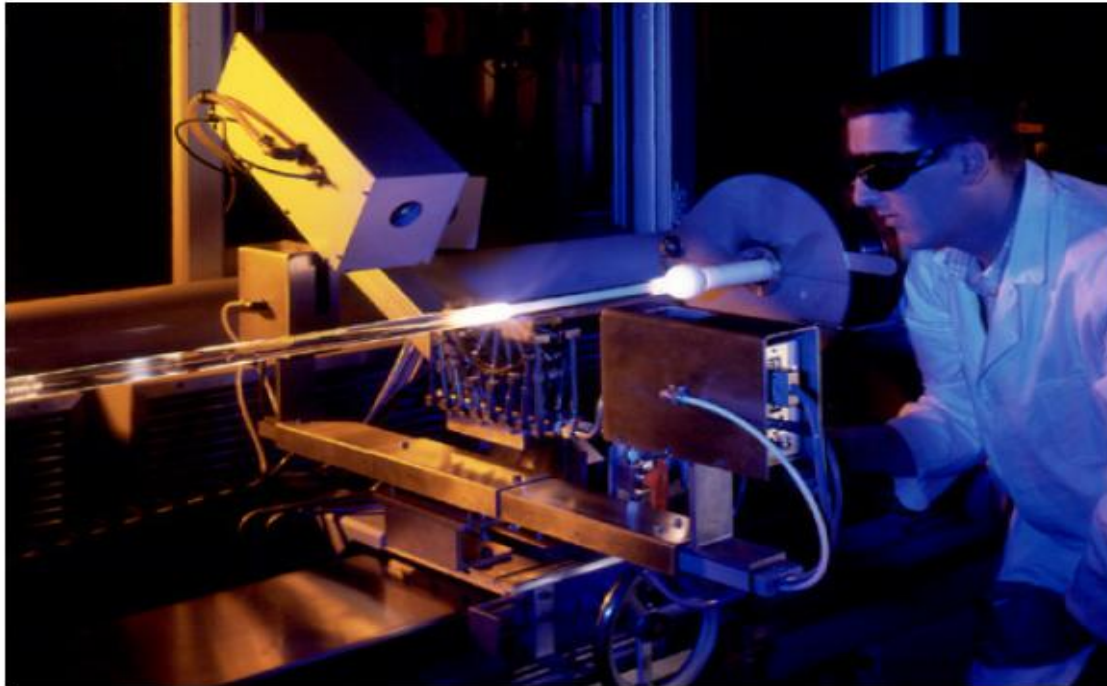


Semiconductor Device Applications

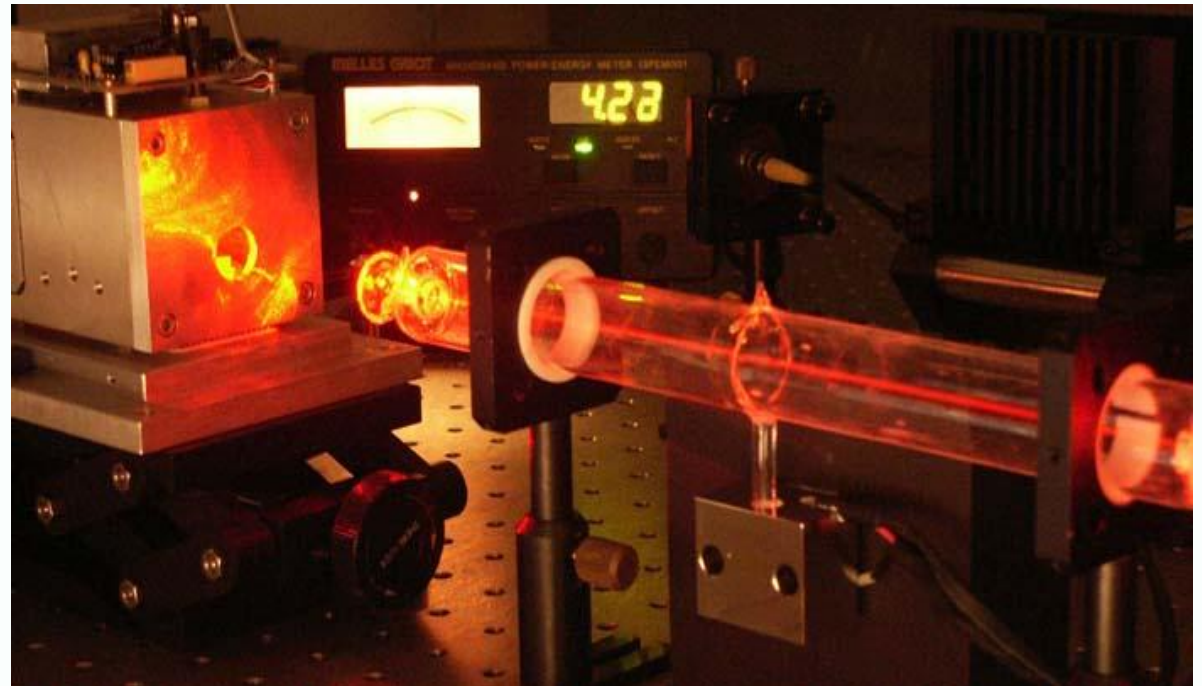
Diode lasers have been used for cutting, surgery, communication (optical fiber), CD writing and reading etc.



Semiconductor Device Applications

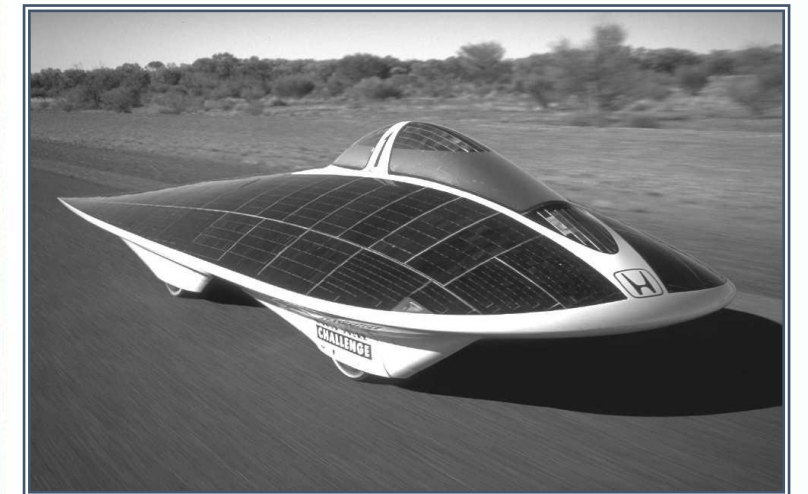
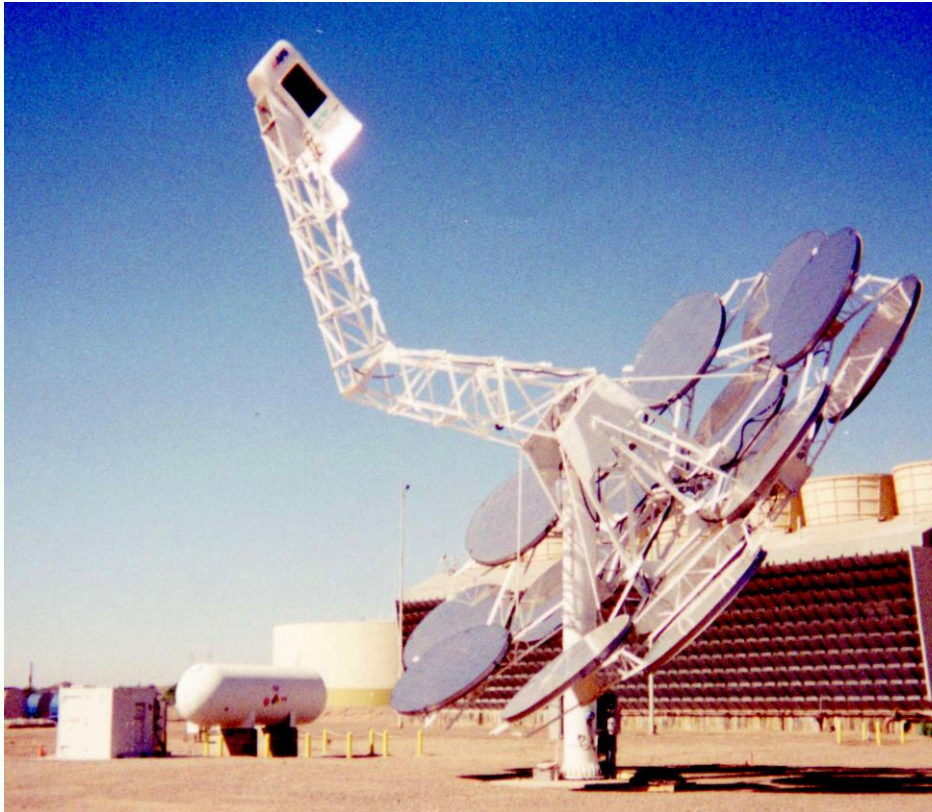


Producing laser in the labs



Semiconductor Device Applications

- Solar cells applications

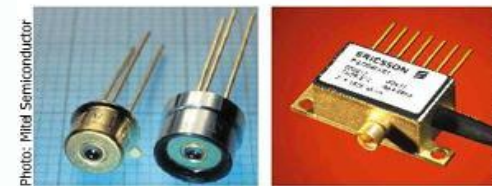
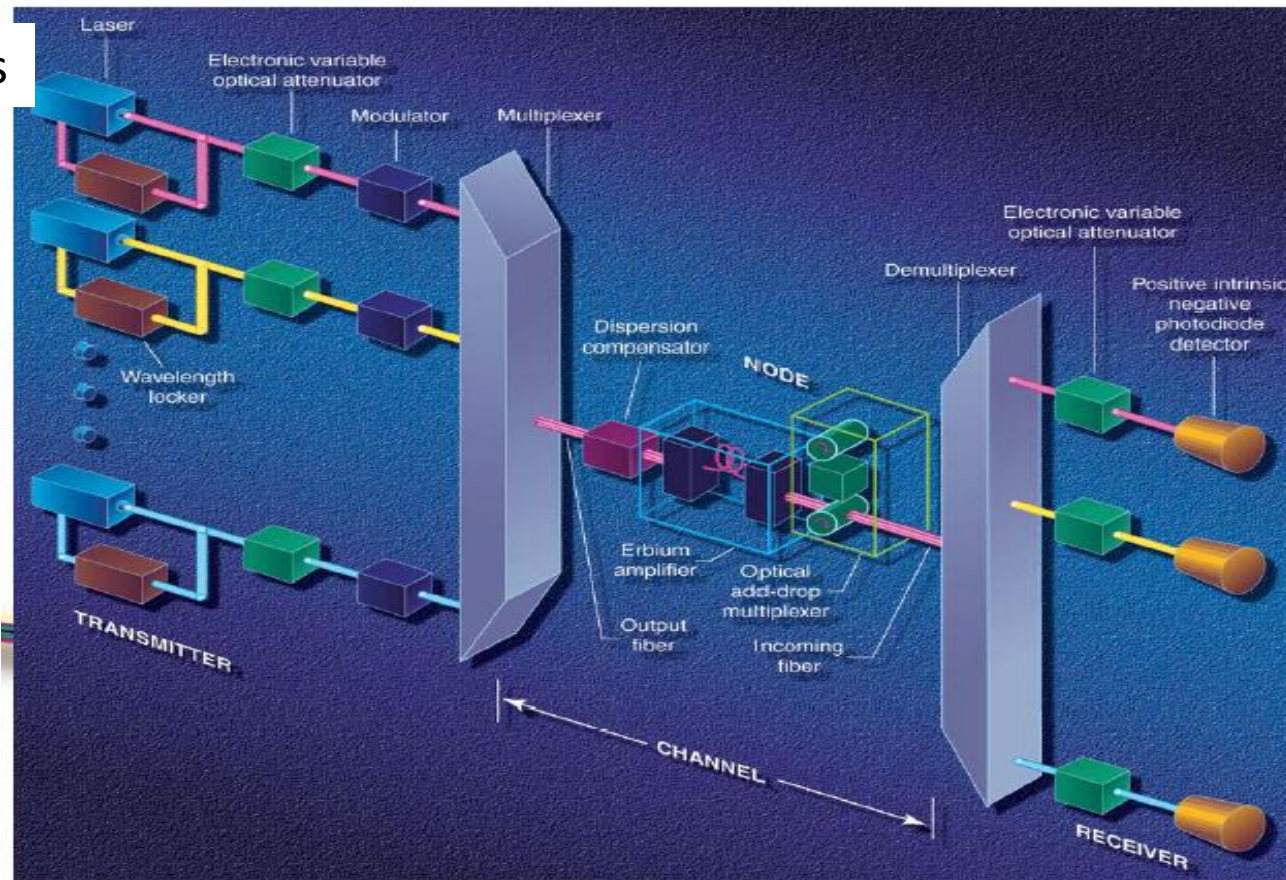


Semiconductor Device Applications

FIBRE OPTICS COMMUNICATION

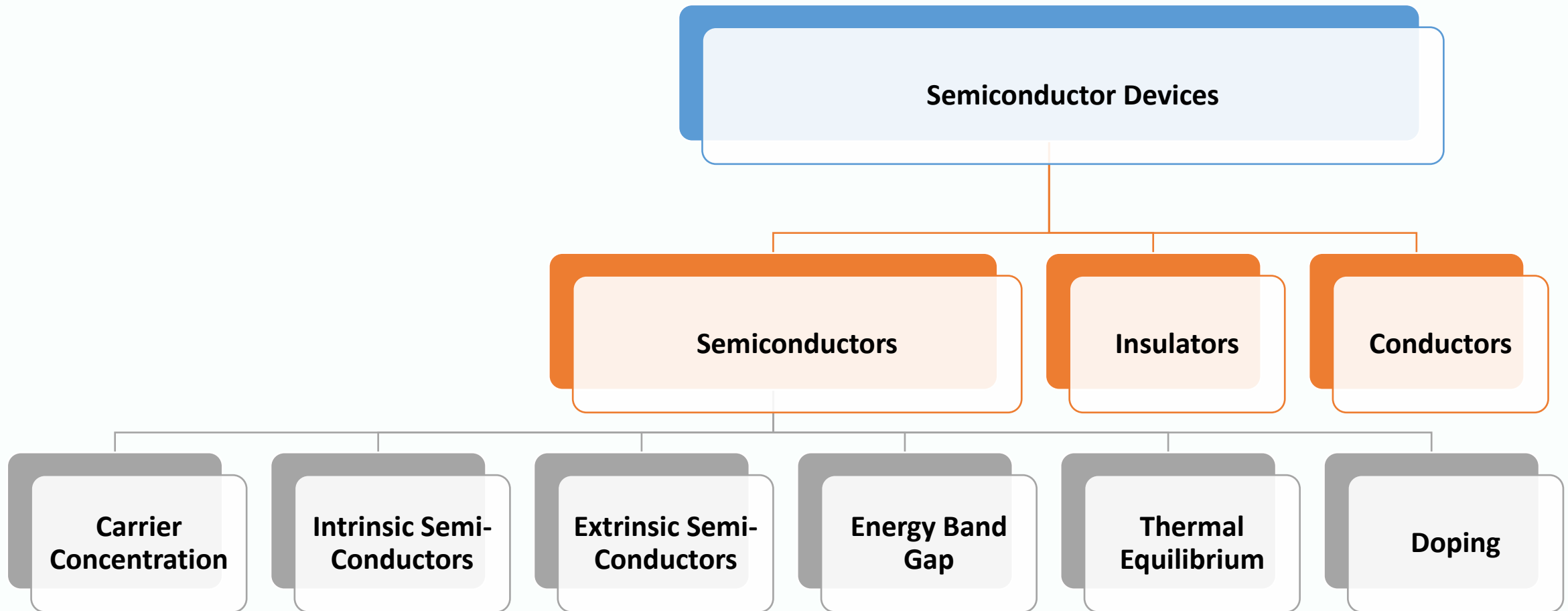
TRANSMITTER → CHANNEL → RECEIVER

IR - Lasers



IR- Photodetector

Overview



Introduction to Semiconductors

- **Electronic Devices**

- Most electronic devices are made out of semiconductors, insulators, and conductors.

Semiconductors

Old Days – Germanium (Ge)

Now – Silicon (Si)

Now – Gallium Arsenide (GaAs) used for high speed and optical devices.

New – Silicon Carbide (SiC) – High voltage Schottky diodes.

Introduction to Semiconductors

Elements in the periodic table are grouped by the number of electrons in their valence shell (most outer shell).

Conductors – Valence shell is mostly empty (1 electron)

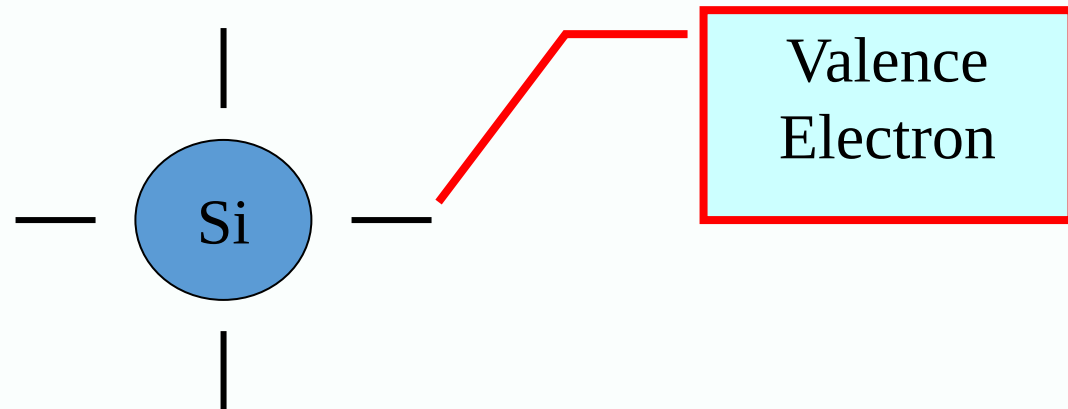
Insulators – Valence shell is mostly full

Semiconductors – Valence shell is half full (Or is it half empty?)

Introduction to Semiconductors

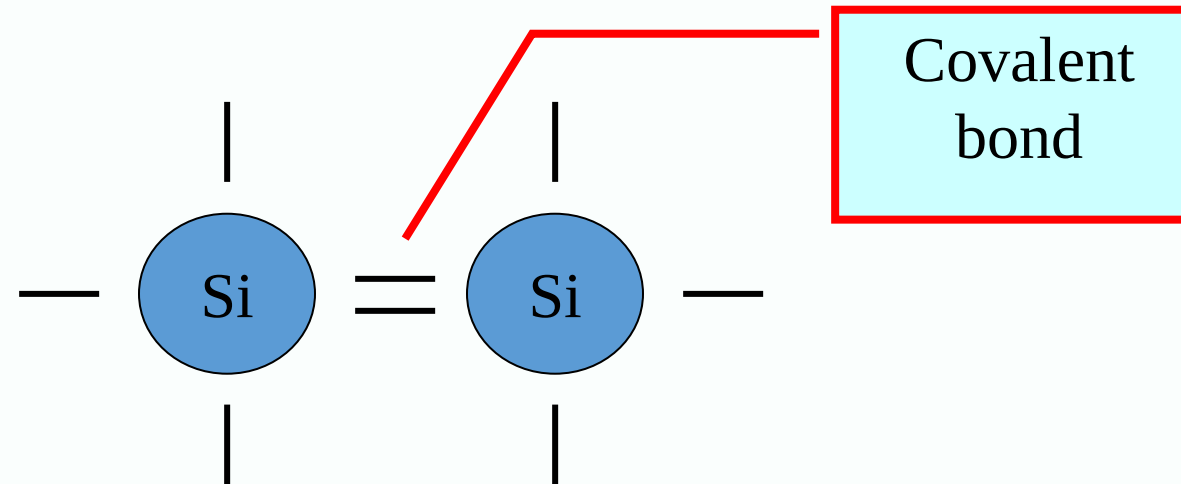
	III	IV	V	VI
	B 5	C 6	N 7	O 8
II	Al 13	Si 14	P 15	S 16
Zn 30	Ga 31	Ge 32	As 33	Se 34
Cd 48	In 49	Sn 50	Sb 51	Te 52
Hg 80	Tl 81	Pb 82	Bi 83	Po 84

Silicon and Germanium are group 4 elements – they have 4 electrons in their valence shell.

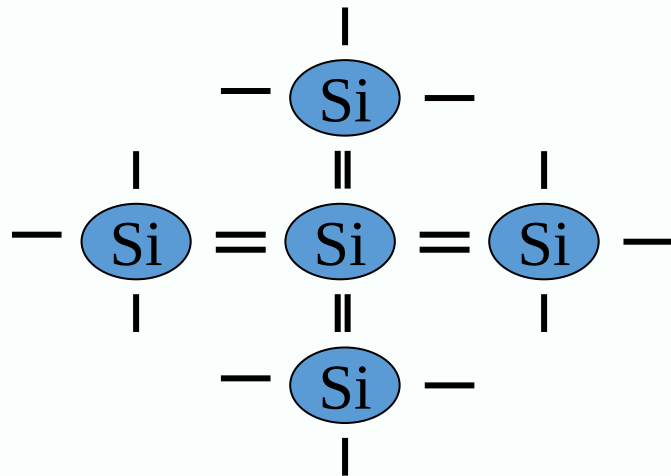


Introduction to Semiconductors

When two silicon atoms are placed close to one another, the valence electrons are shared between the two atoms, forming a covalent bond.



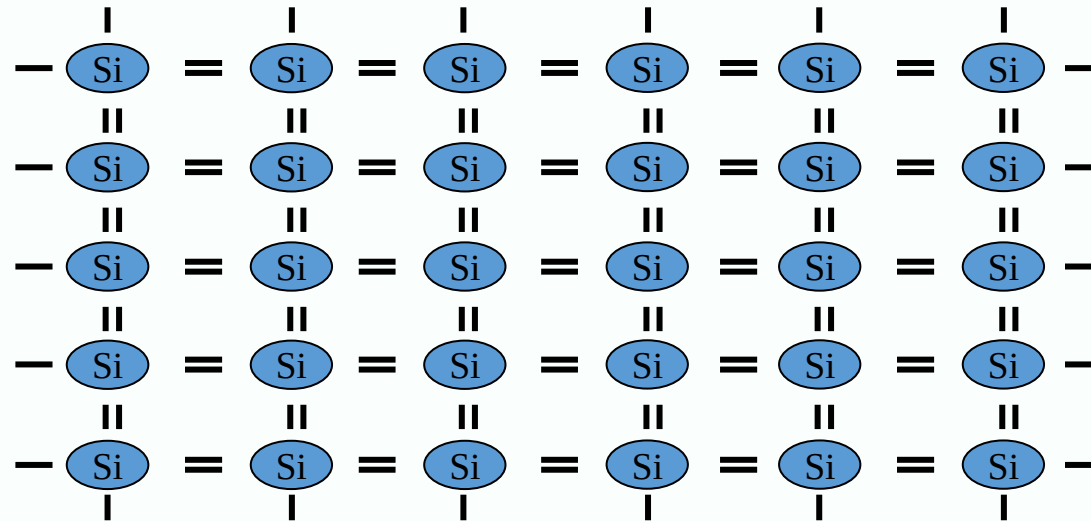
Introduction to Semiconductors



5-atom Silicon Structure

- An important property of the 5-atom silicon lattice structure is that valence electrons are available on the outer edge of the silicon crystal so that other silicon atoms can be added to form a large single silicon crystal.

Introduction to Semiconductors



At 0 °K, each electron is in its lowest energy state so each covalent bond position is filled.

If a small electric field is applied to the material, no electrons will move because they are bound to their individual atoms.

=> At 0 °K, silicon is an insulator.

Introduction to Semiconductors

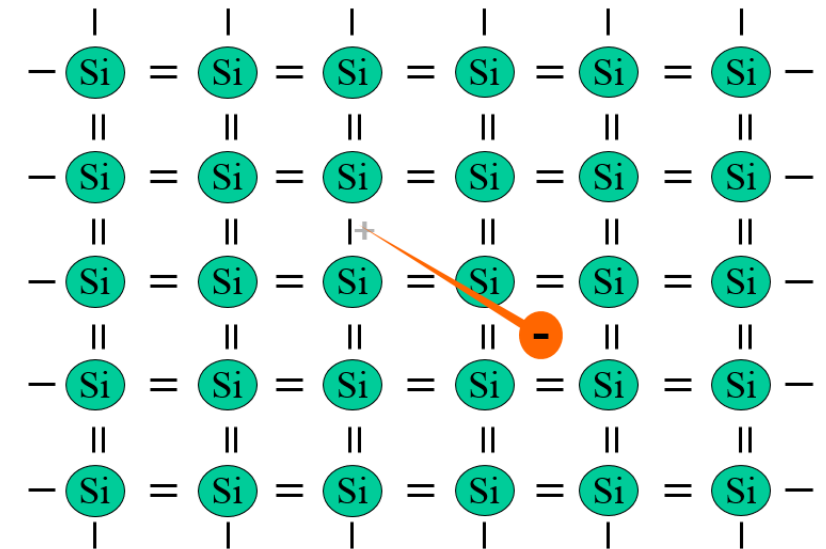
Silicon

As temperature increases, the valence electrons gain thermal energy.

If a valence electron gains enough energy, it may break its covalent bond and move away from its original position.

This electron is free to move within the crystal.

Since the net charge of a crystal is zero, if a negatively (-) charged electron breaks its bond and moves away from its original position, a positively charged “empty state” is left in its original position.



Introduction to Semiconductors

- As temperature increases, more bonds are broken creating more negative free electrons and more positively charged empty states.
- (Number of free electrons is a function of temperature.)
- To break a covalent bond, a valence electron must gain a minimum energy E_g , called the energy band gap.
- (Number of free electrons is a function of E_g .)

Introduction to Semiconductors

Elements that have a large energy band gap of 3 to 6 eV are **insulators** because at room temperature, essentially no free electrons exist.

An electron volt eV is the amount of energy an electron will gain if it is accelerated through a 1 volt potential.

Introduction to Semiconductors

- Elements that have a small energy band gap are **conductors**.
- These elements have a large number of free electrons at room temperature because the electrons need very little energy to escape from their covalent bonds.

Semiconductors have a band gap energy of about 1 eV

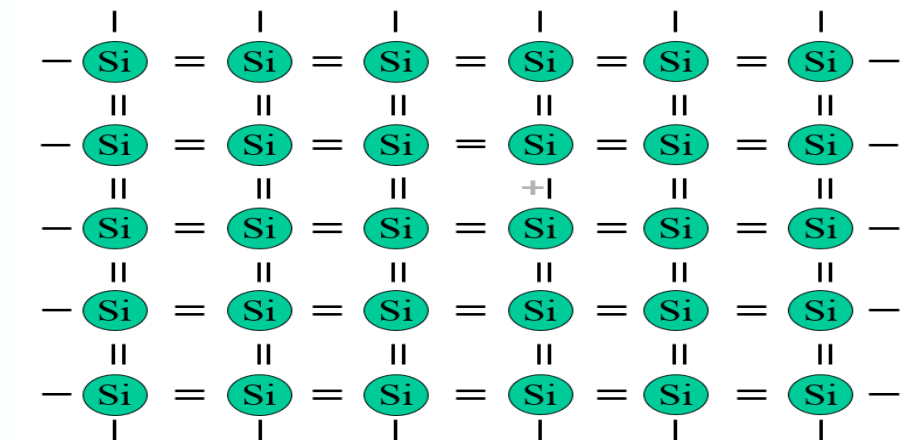
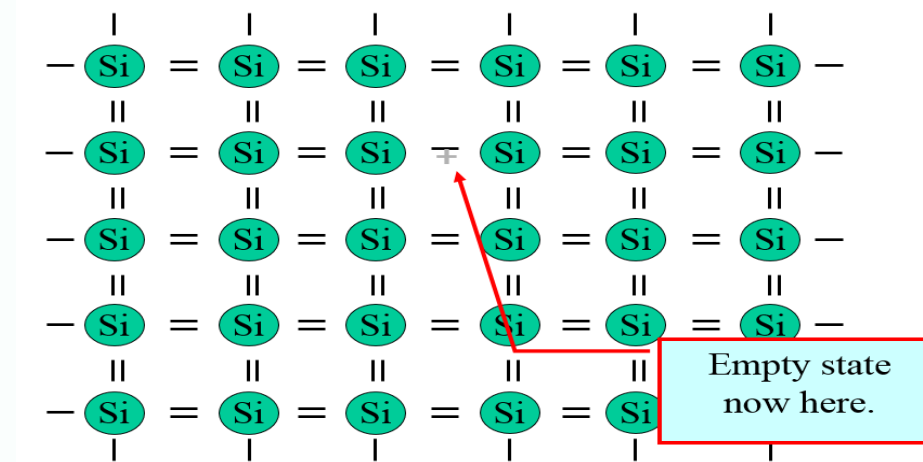
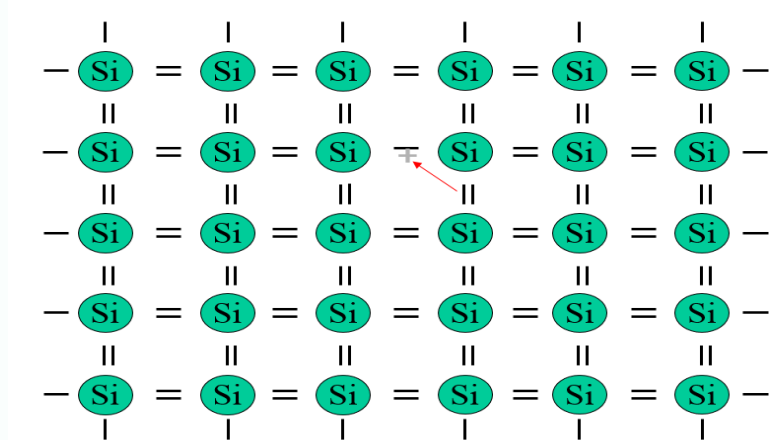
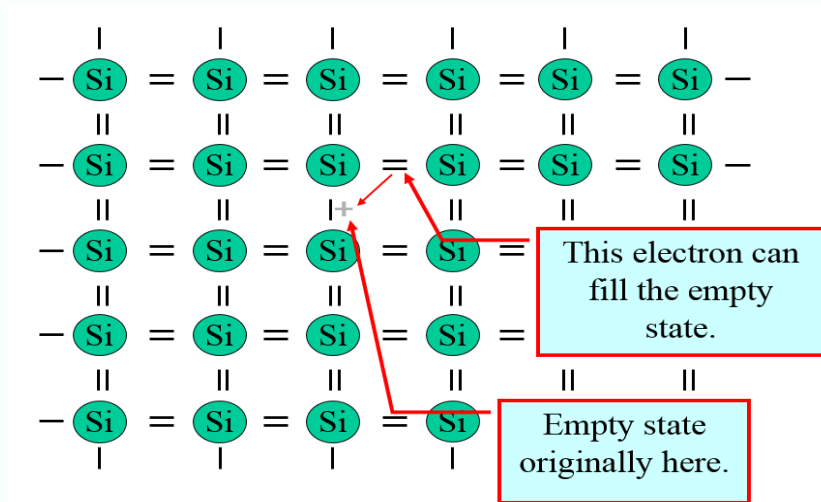
Silicon = 1.1 eV

GaAs = 1.4 eV

Ge = 0.66 eV

An electron that has sufficient energy and is adjacent to an empty state may move into the empty state, leaving an empty state behind.

Introduction to Semiconductors



Introduction to Semiconductors

- Moving empty states can give the appearance that positive charges move through the material.
- This moving empty state is modeled as a positively charged particle called a **hole**.
- In semiconductors, two types of “particles” contribute to the current: **positively charged holes** and **negatively charged electrons**

Carrier Concentrations

The concentrations of holes and free electrons are important quantities in the behavior of semiconductors.

Carrier concentration is given as the number of particles per unit volume, or

Carrier concentration = $\#/cm^3$

Intrinsic Semiconductors

An **intrinsic semiconductor** is a single crystal semiconductor with no other types of atoms in the crystal.

Pure silicon

Pure germanium

Pure gallium arsenide.

Intrinsic Semiconductors

- In an **intrinsic semiconductor**, the number of holes and free electrons are the same because they are thermally generated.
- If an electron breaks its covalent bond we have one free electron and one hole.
- In an **intrinsic semiconductor**, the concentration of holes and free electrons are the same.

 n_i

= the concentration of free electrons in an intrinsic semiconductor.

= the concentration of holes in an intrinsic semiconductor.

Intrinsic Semiconductors

$$n_i = BT^{3/2} \exp\left(\frac{-E_g}{2KT}\right)$$

B and E_g are determined by the properties of the semiconductor.

E_g = band gap energy (eV)

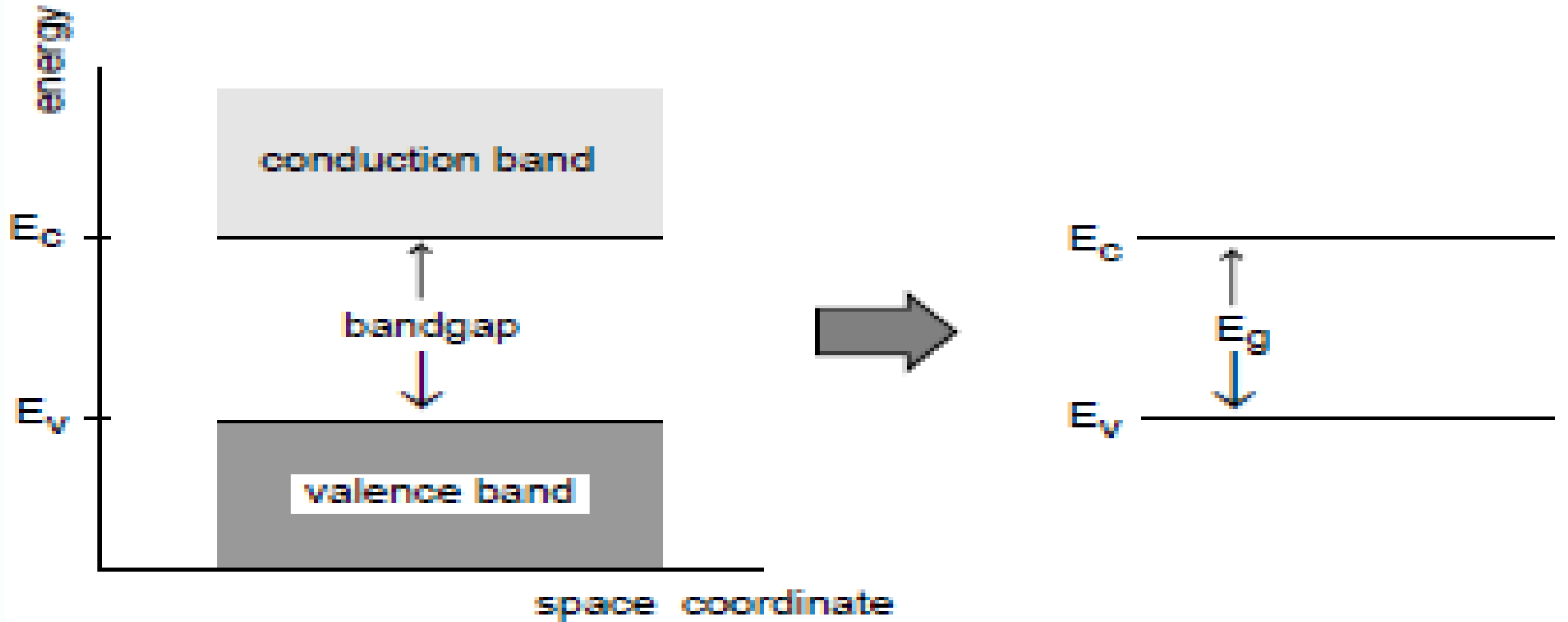
B = material constant

T = temperature ($^{\circ}\text{K}$)

K = Boltzmann's constant = $86.2 \times 10^{-6} \text{ eV}/^{\circ}\text{K}$

$$\left(\frac{\#}{(\text{cm}^3) \cdot (^{\circ}\text{K})^{3/2}} \right)$$

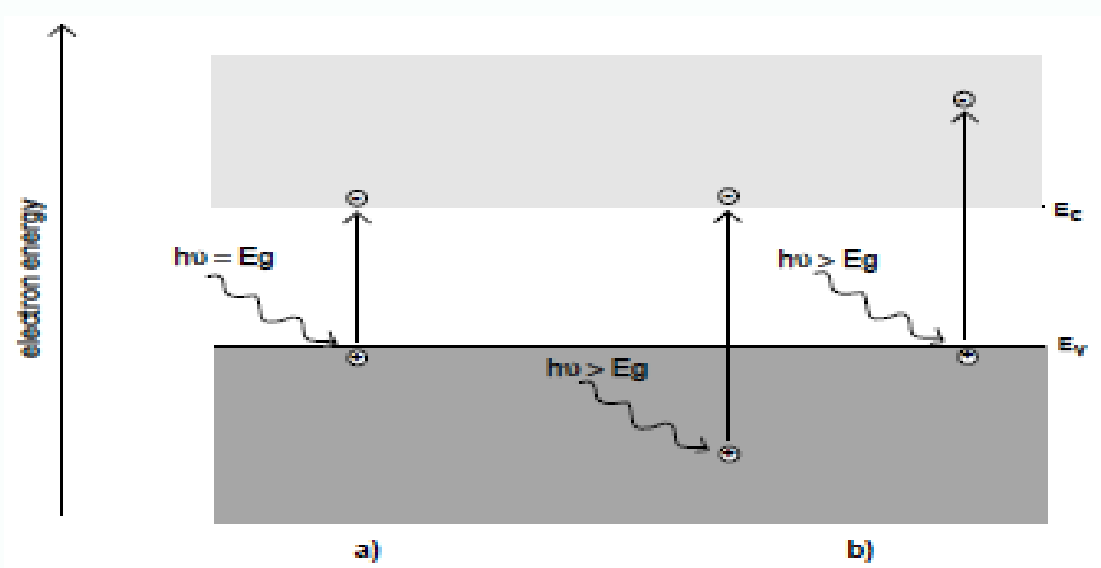
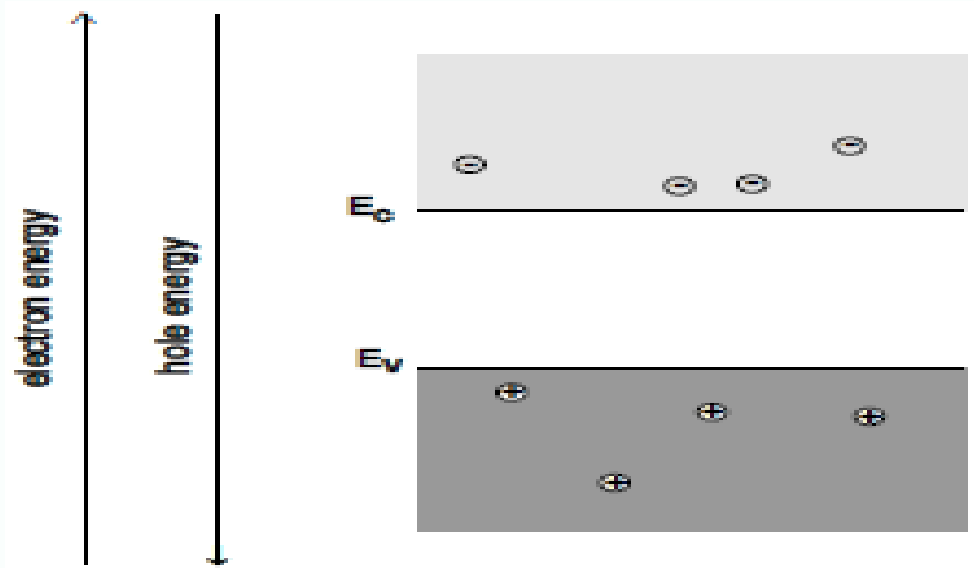
Energy Band Diagram



Energy Band Diagram

- Bonding electrons occupy states in valence band
- Free electrons occupy states in conduction band
- Holes empty states in valence band

Energy Band Diagram



- Elements of energy band diagrams:
 - At edges of the bands, kinetic energy of carriers is zero
 - Electron energies increase upwards
 - Hole energies increase downwards

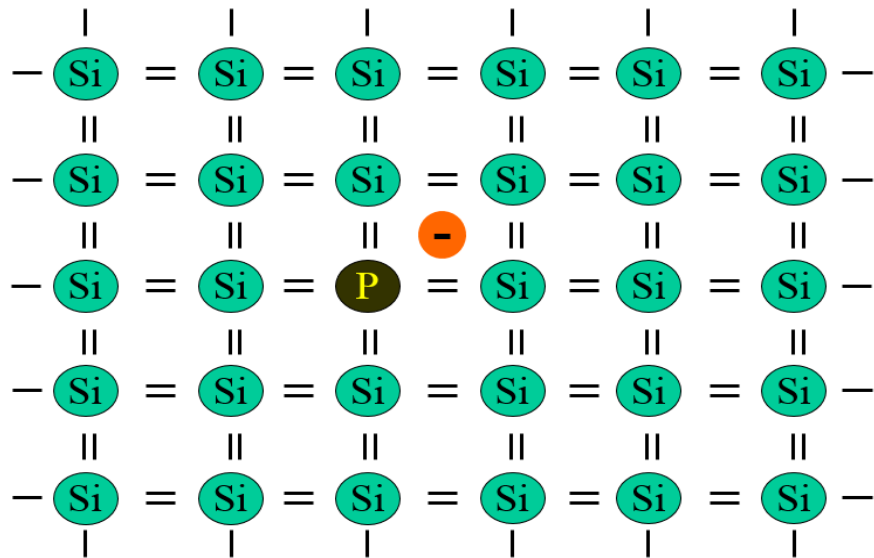
Extrinsic Semiconductors

- Since the concentrations of free electrons and holes is small in an intrinsic semiconductor, only small currents are possible.
- Impurities can be added to the semiconductor to increase the concentration of free electrons and holes
- An impurity would have one less or one more electron in the valance shell than silicon.

Extrinsic Semiconductors

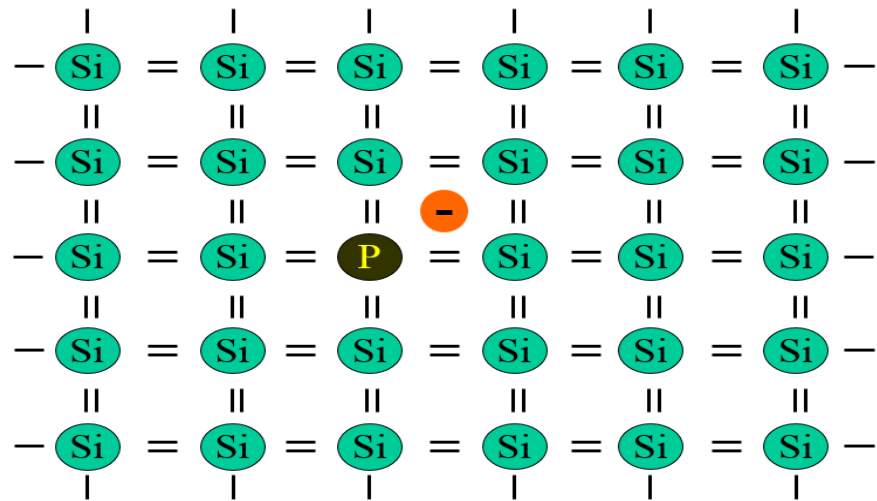
- Impurities for group 4 type atoms (silicon) would come from group 3 or group 5 elements
- The most common group 5 elements are phosphorous and arsenic.
- Group 5 elements have 5 electrons in the valence shell.
- Four of the electrons fill the covalent bonds in the silicon crystal structure.
- The 5th electron is loosely bound to the impurity atom and is a free electron at room temperature

Extrinsic Semiconductors



- The group 5 atom is called a **donor** impurity since it donates a free electron.
- The group 5 atom has a net positive charge that is fixed in the crystal lattice and cannot move.
- With a donor impurity, free electrons are created without adding holes.

Extrinsic Semiconductors

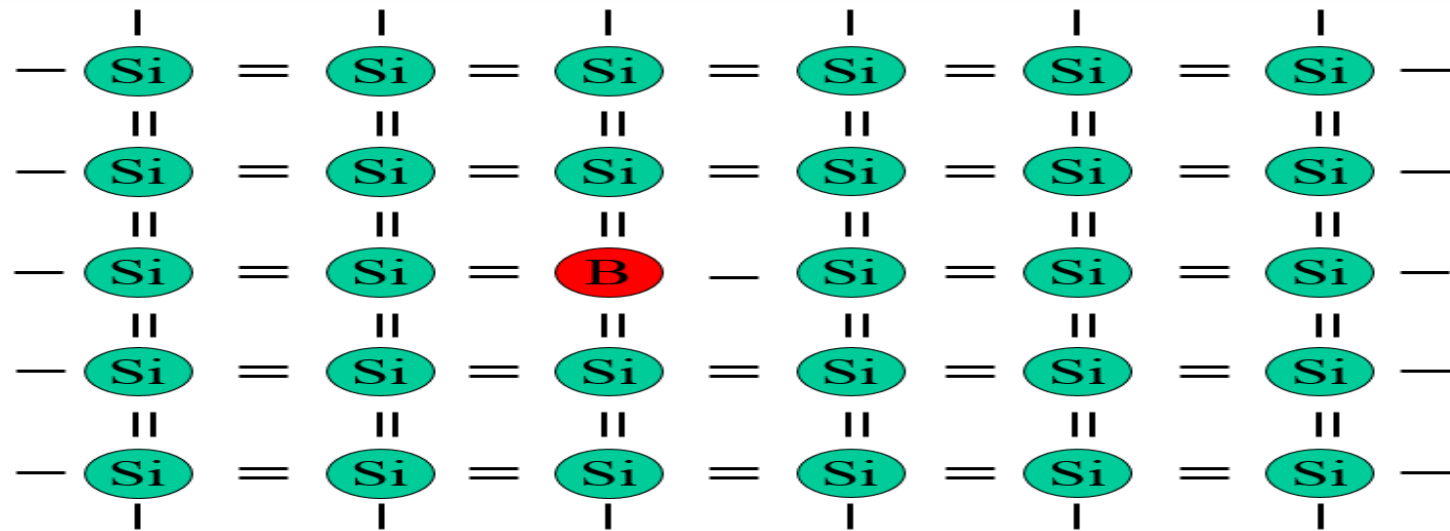


- Adding impurities is called **doping**.
- A semiconductor doped with donor impurities has excess free electron and is called an **n-type** semiconductor

The most common group 3 impurity is boron which has 3 valence electrons.

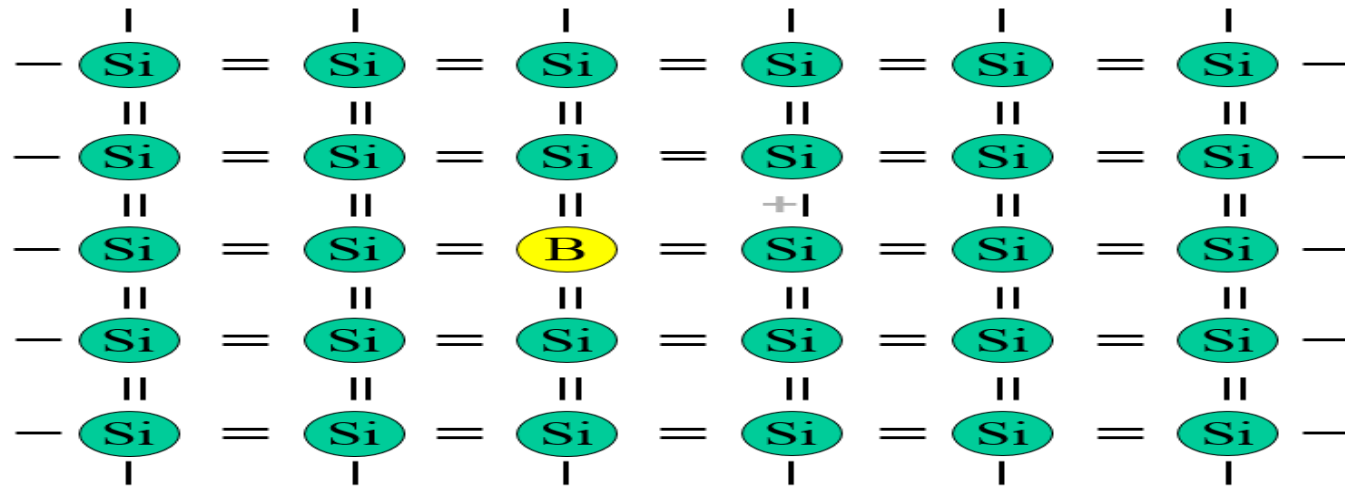
Since boron has only 3 valence electrons, the boron atom can only bond with three of its neighbors leaving one open bond position.

Extrinsic Semiconductors



- At room temperature, silicon has free electrons that will fill the open bond position, creating a hole in the silicon atom from where it came.
- The boron atom has a net negative charge because of the extra electron, but the boron atom cannot move

Extrinsic Semiconductors



Since boron accepts a valence electron, it is called an **acceptor impurity**.

Acceptor impurities create excess holes but do not create free electrons.

A semiconductor doped with an acceptor impurity has extra holes and is called a **p-type** semiconductor.

Extrinsic Carrier Concentrations

- For any semiconductor in thermal equilibrium
- $n_o p_o = n_i^2$,
- where
- n_o = the concentration of free electrons.
- p_o = the concentration of holes.
- n_i = the intrinsic carrier concentration

Extrinsic Carrier Concentrations

$$n_i = BT^{\frac{3}{2}} \exp\left(\frac{-E_g}{2KT}\right)$$

For an n-type semiconductor with donor impurities, the concentration of donor impurities is N_d with units $\#/cm^3$.

If $N_d \gg n_i$, then the concentration of free electrons in the n-type semiconductor is approximately $n_o \approx N_d$.

Extrinsic Carrier Concentrations

Since $n_o p_o = n_i^2$ for any semiconductor in thermal equilibrium, and

For an n-type semiconductor, $n_o \approx N_d$

$$p_o = \frac{n_i^2}{N_d}$$

- Where p_o is the concentration of holes in the n-type semiconductor
- For a p-type semiconductor with acceptor impurities, the concentration of acceptor impurities is N_a with units $\#/cm^3$.
- If $N_a \gg n_i$, then the concentration of holes in the p-type semiconductor is approximately $p_o \approx N_a$

Extrinsic Carrier Concentrations

- Since $n_o p_o = n_i^2$ for any semiconductor in thermal equilibrium, and for a p-type semiconductor, $p_o \approx N_a$

$$n_o = \frac{n_i^2}{N_a}$$

Where n_o is the concentration of free electrons in the p-type semiconductor

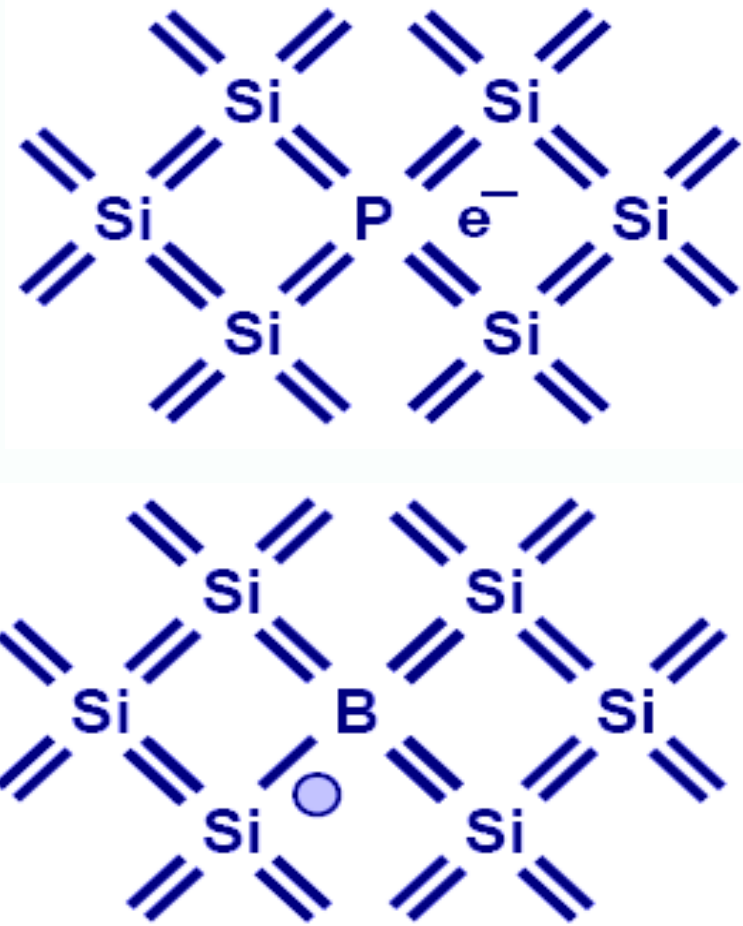
Currents in Semiconductors

The two processes that cause free electrons and holes to move in a semiconductor are **drift** and **diffusion**.

Drift – the movement of holes and electrons due to an electric field

Diffusion – the movement of holes and electrons due to variations in concentrations.

Doping in Semiconductors



- Pure Si can be doped with other elements to change its electrical properties.
- For example, if Si is doped with P (phosphorous), then it has more electrons, or becomes type N (electron).

If Si is doped with B (boron), then it has more holes, or becomes type P.



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Thank You

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COE 271 Semiconductor Devices

Lecture 2

Physics of Semiconductors

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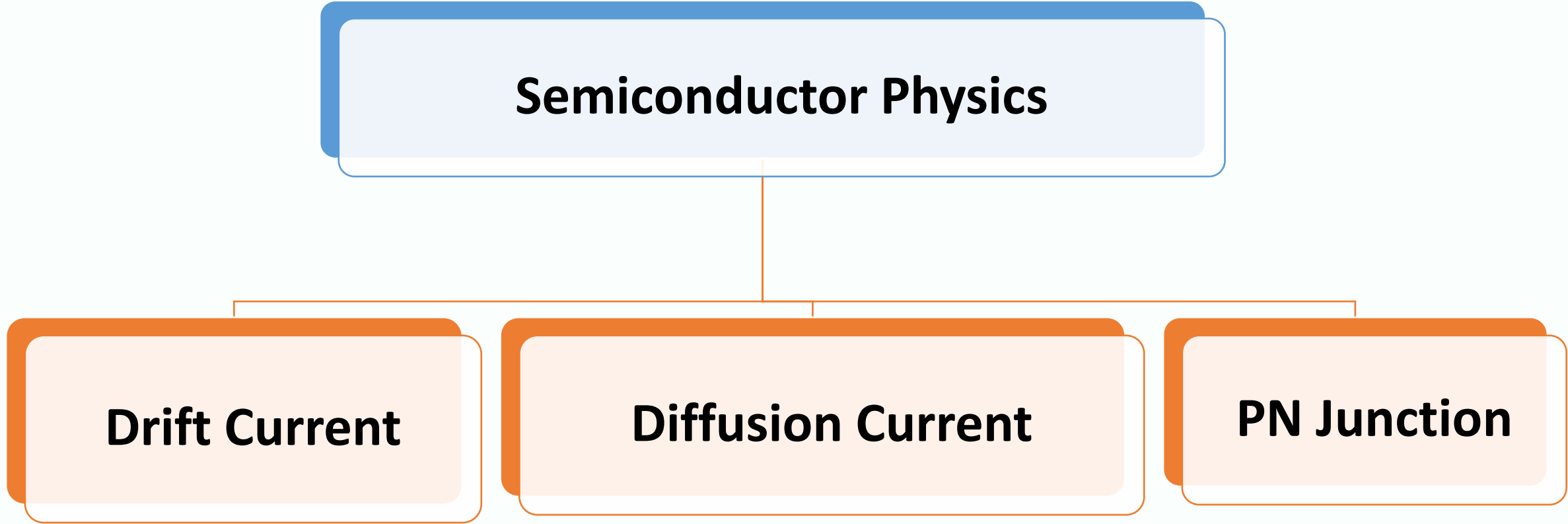


“When you are courting a nice girl an hour seems like a second.
When you sit on a red-hot cinder a second seems like an hour.
That's relativity”

Albert Einstein

Overview

Semiconductor Physics



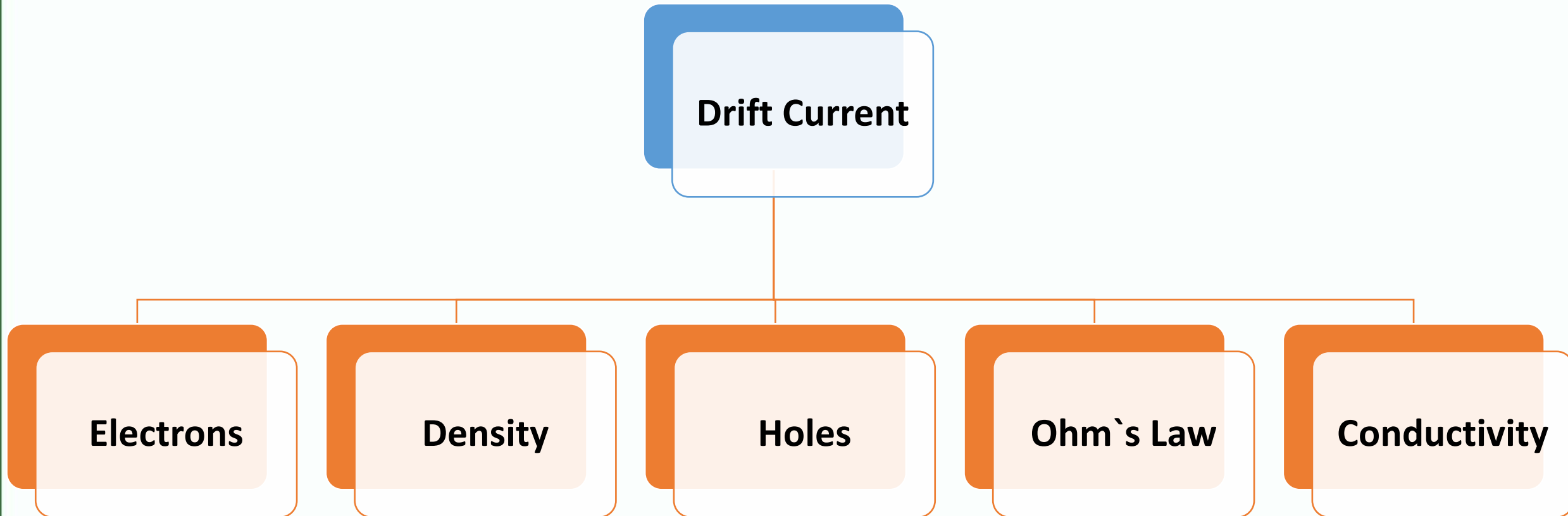
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graph TD; A[Semiconductor Physics] --- B[ ]; B --- C[Drift Current]; B --- D[Diffusion Current]; B --- E[PN Junction];
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Drift Current

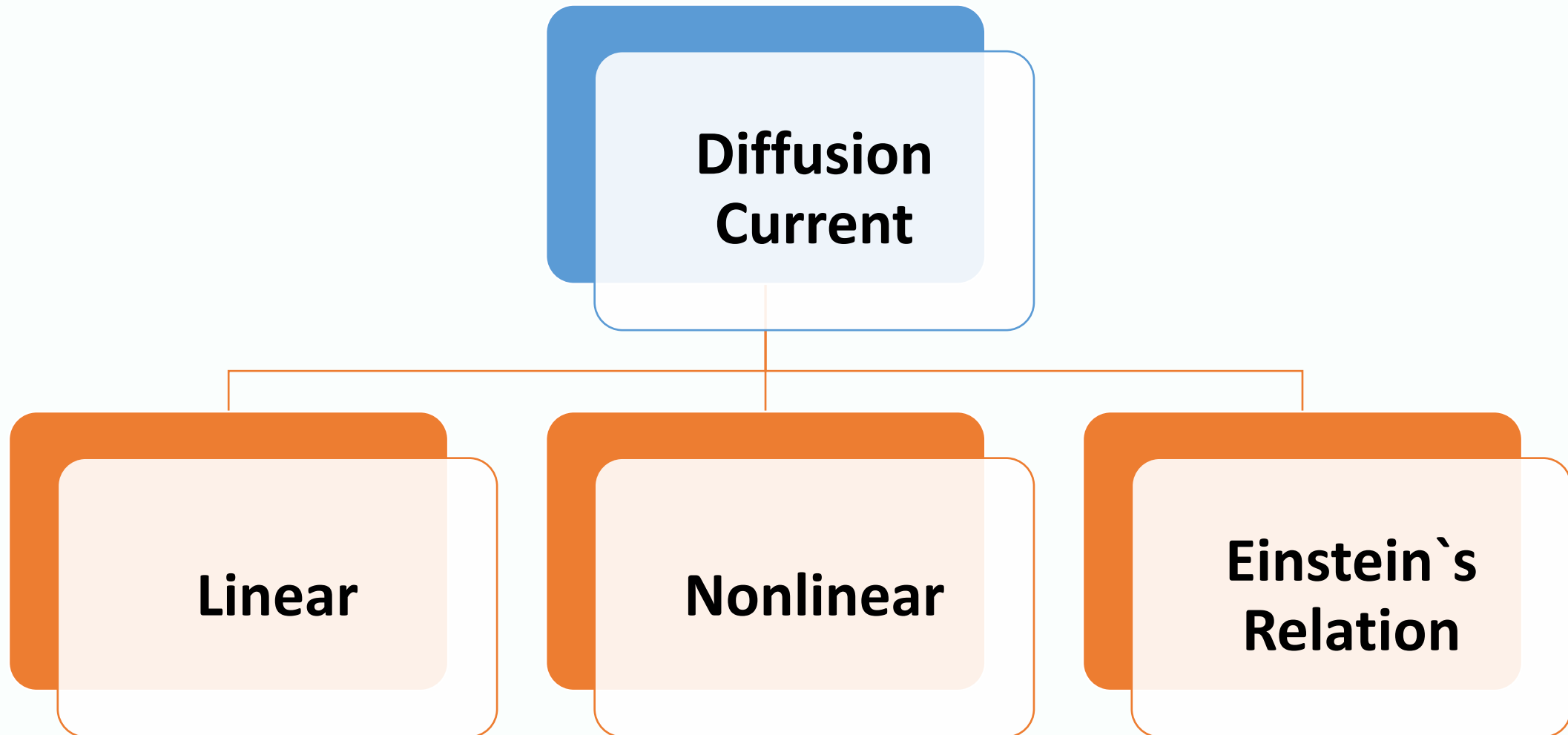
Diffusion Current

PN Junction

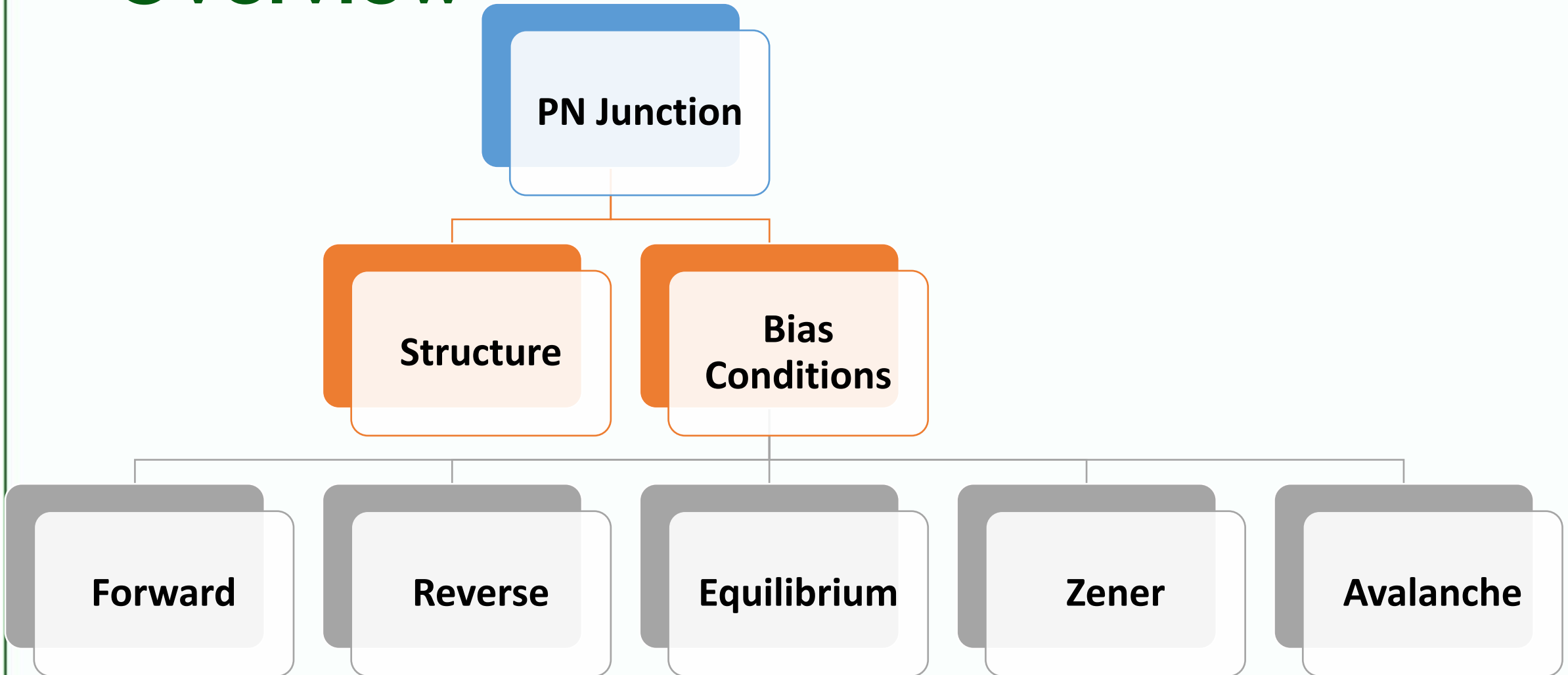
Overview



Overview

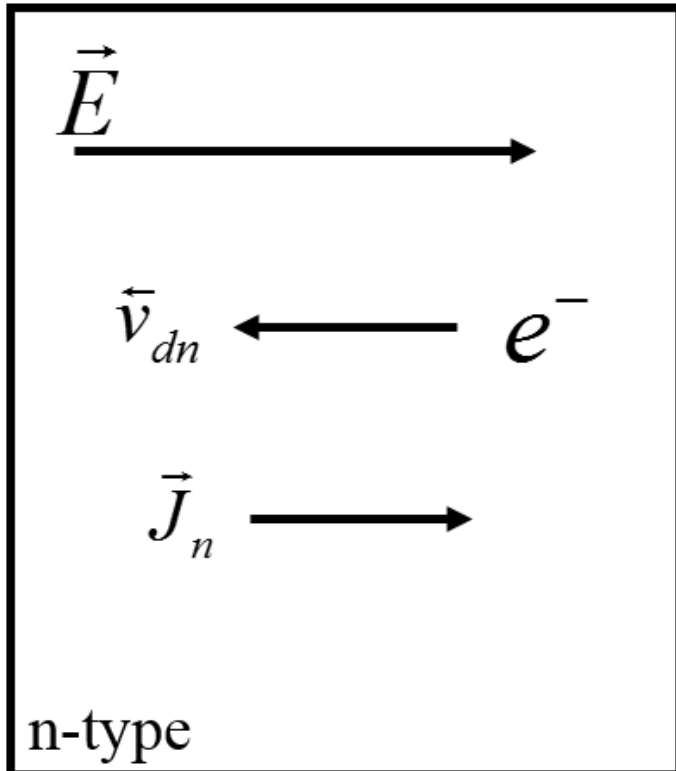


Overview



Physics of Semiconductors

- Assume that an electric field is applied to a semiconductor.
- This field acts on holes and electrons



Electrons – The Electric field creates a force in the opposite direction of the electric field – Attractive.

v_{dn} is the drift velocity of electrons.

J_n is the current density due to electrons.

Drift Current-Electrons

- The electrons acquire a drift velocity of
$$\vec{v}_{dn} = -\mu_n \vec{E}$$

Where μ_n is the mobility of electrons with units of $\text{cm}^2/(\text{volt-sec})$.

The units of v_{dn} are cm/sec .

For low-doped silicon, a typical number is $\mu_n = 1350 \text{ cm}^2/\text{volt-sec}$.

$$\vec{v}_{dn} = -\mu_n \vec{E}$$

The minus sign (-) indicates that the electrons move in the opposite direction of the applied electric field.

Drift Current Density Electrons

- Current = charge per unit time (coul/sec).
- Current density = current flowing through a specific area = amps/unit area = coul/(sec-cm²)

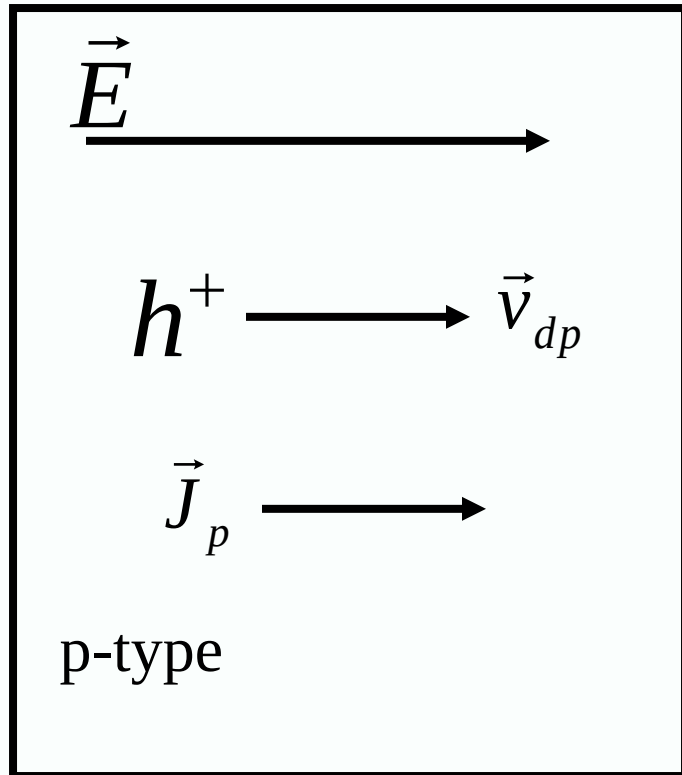
$$J_n = -env_{dn} = en\mu_n E$$

e = the charge on an electron = 1.602×10^{-19} coulombs.

n = concentration of electrons = #/cm³.

$$\rightarrow en = \text{charge/cm}^3. \quad env_{dn} = \frac{\text{charge}}{\text{cm}^3} \frac{\text{cm}}{\text{sec}} = \frac{\text{charge}}{\text{sec} \cdot \text{cm}^2} = \frac{\text{amp}}{\text{cm}^2}$$

Drift Current Holes



- Holes – The Electric field creates a force in the same direction of the electric field.
- v_{dp} is the drift velocity of holes.
- J_p is the current density due to holes.
- The holes acquire a drift velocity of

$$\vec{v}_{dp} = \mu_p \vec{E}$$

Drift Current Holes

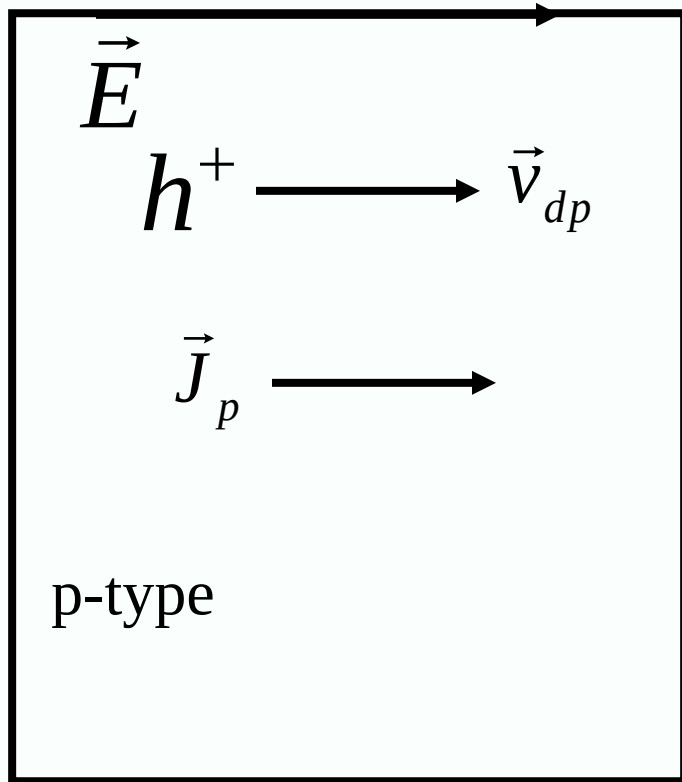
The holes acquire a drift velocity of

$$\vec{v}_{dp} = \mu_p \vec{E}$$

Where μ_p is the mobility of holes with units of $\text{cm}^2/(\text{volt-sec})$.
The units of v_{dp} are cm/sec .

For low-doped silicon, a typical number is $\mu_{dp}=480 \text{ cm}^2/\text{volt-sec}$

Drift Current Holes

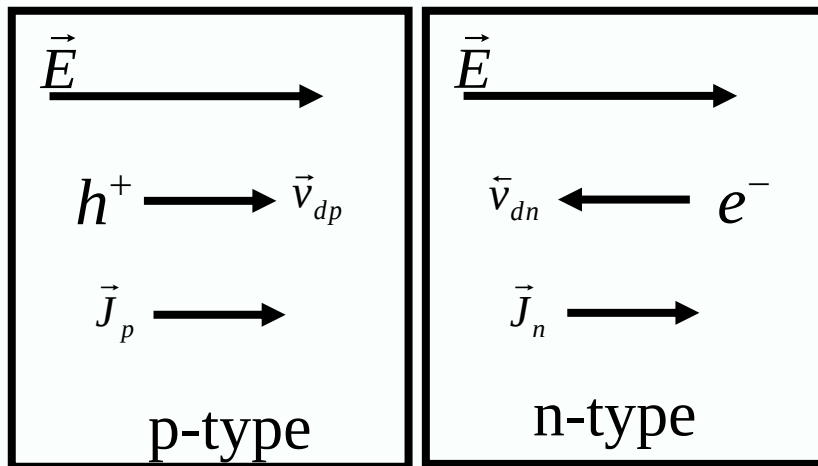


- Note that $\mu_n > \mu_p$.
- Electrons are faster than holes.
- P-type and n-type devices operate the same.
- However, n-type devices are faster

Drift Current Density Holes

$$J_p = epv_{dp} = ep\mu_p E$$

- e = the charge on an electron = 1.602×10^{-19} coulombs.
- p = concentration of holes = $\#/cm^3$.
- $\rightarrow ep$ = charge/ cm^3



$$env_{dp} = \frac{\text{charge}}{cm^3} \frac{cm}{sec} = \frac{\text{charge}}{sec \cdot cm^2} = \frac{amp}{cm^2}$$

Drift current due to holes and electrons is in the same direction.

Total Drift Current

- Since the hole current and the electron current are in the same direction, the currents add.

$$\vec{J} = en\mu_n\vec{E} + ep\mu_p\vec{E}$$

- The total drift current is:

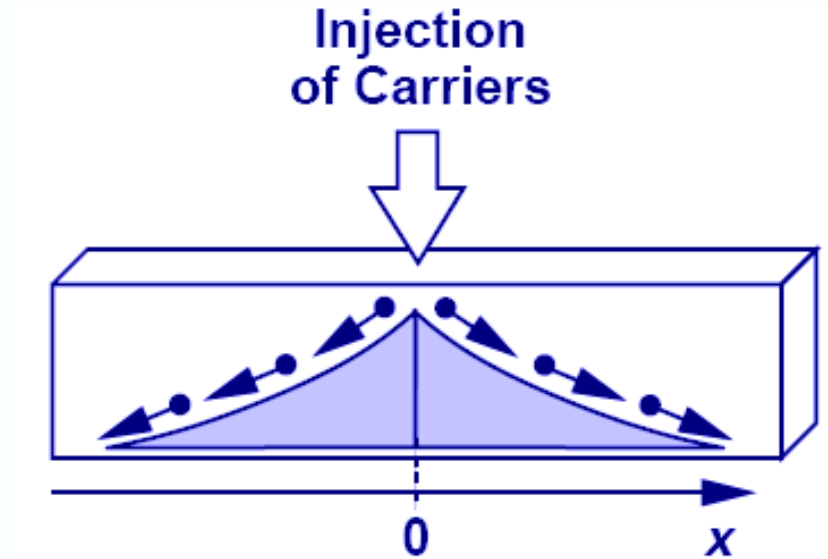
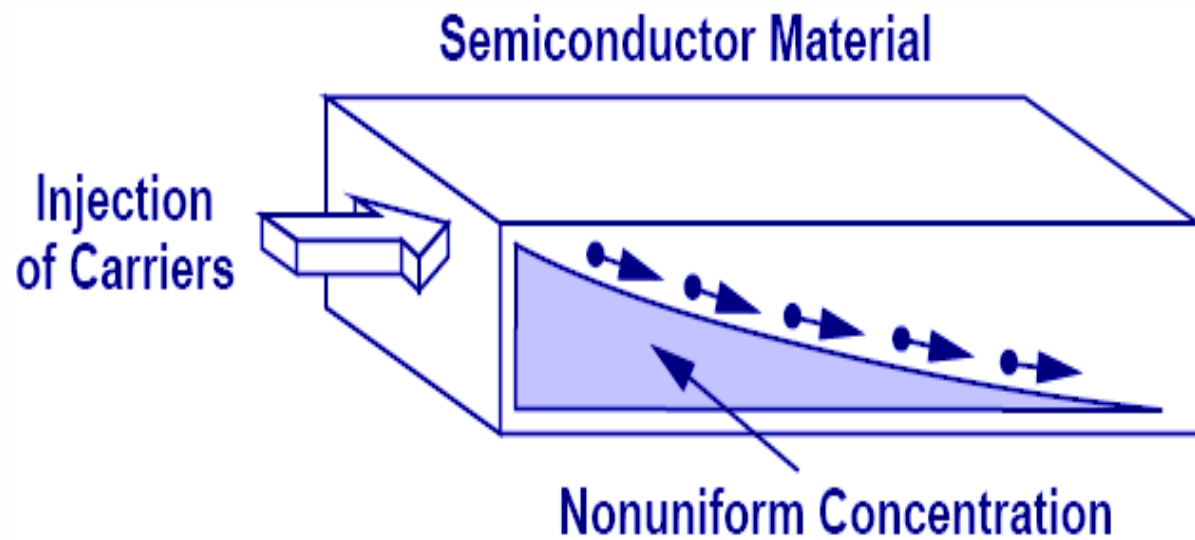
$$\vec{J} = \sigma\vec{E} \quad \text{another form of Ohm's law}$$

σ is the conductivity of the material.

We can find the conductivity of a semiconductor as

$$\sigma = en\mu_n + ep\mu_p$$

Diffusion Current



- Charge particles move from a region of high concentration to a region of low concentration.
- It is analogous to an every day example of an ink droplet in water.

Diffusion Current

- Diffusion current is proportional to the gradient of charge (dn/dx) along the direction of current flow.
- Its total current density consists of both electrons and holes.

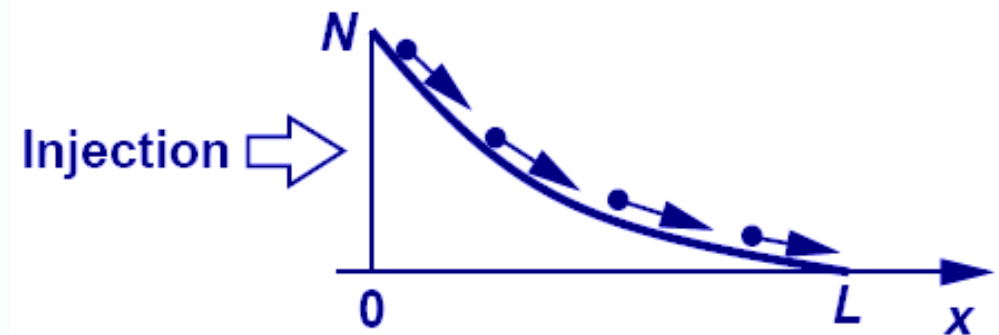
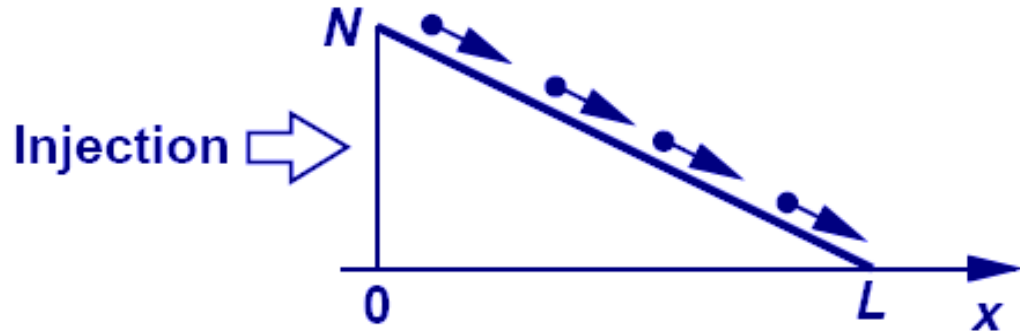
$$I_d = AqD_n \frac{dn}{dx}$$

$$J_n = qD_n \frac{dn}{dx}$$

$$J_p = -qD_p \frac{dp}{dx}$$

$$J_{\text{tot}} = q \left(D_n \frac{dn}{dx} - D_p \frac{dp}{dx} \right)$$

Linear and Nonlinear Charge Density Profile

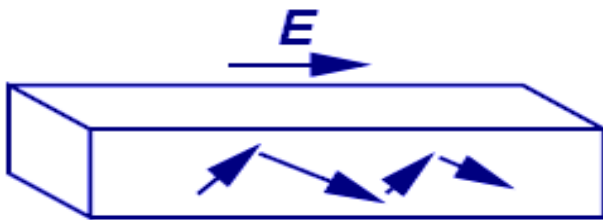


- Linear charge density profile means constant diffusion current, whereas nonlinear charge density profile means varying diffusion current.

$$J_n = qD_n \frac{dn}{dx} = -qD_n \cdot \frac{N}{L} \quad J_n = qD \frac{dn}{dx} = \frac{-qD_n N}{L_d} \exp \frac{-x}{L_d}$$

Einstein's Relation

Drift Current



$$J_n = q \mu_n E$$

$$J_p = q \mu_p E$$

Diffusion Current



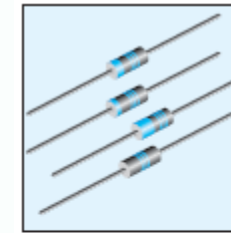
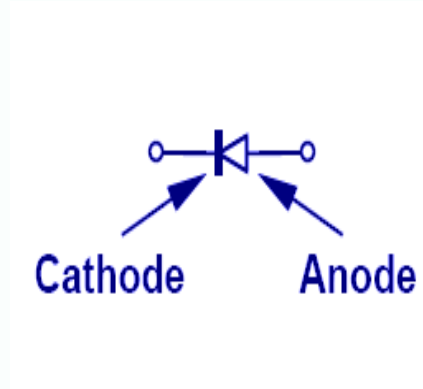
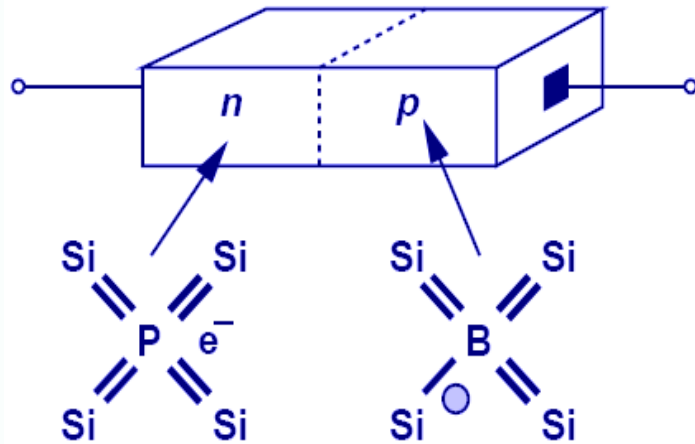
$$J_n = q D_n \frac{dn}{dx}$$

$$J_p = -q D_p \frac{dp}{dx}$$

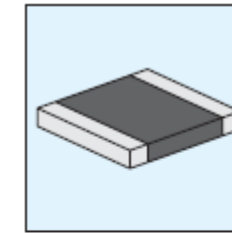
$$\frac{D}{\mu} = \frac{kT}{q}$$

- While the underlying physics behind drift and diffusion currents are totally different, Einstein's relation provides a mysterious link between the two.

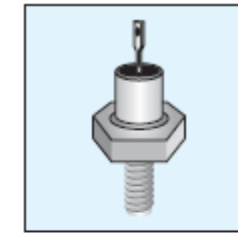
PN Junction (Diode)



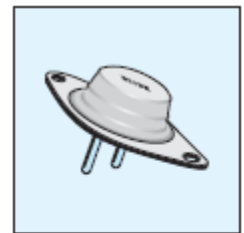
General purpose diode



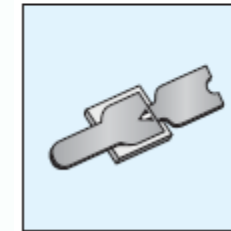
Surface mount high-power PIN diode



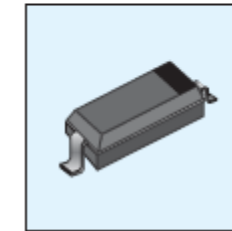
Power (stud) diode



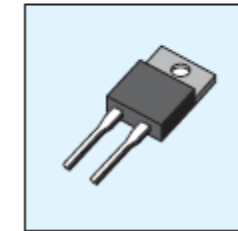
Power (planar) diode



Beam lead pin diode



Flat chip surface mount diode



Power diode



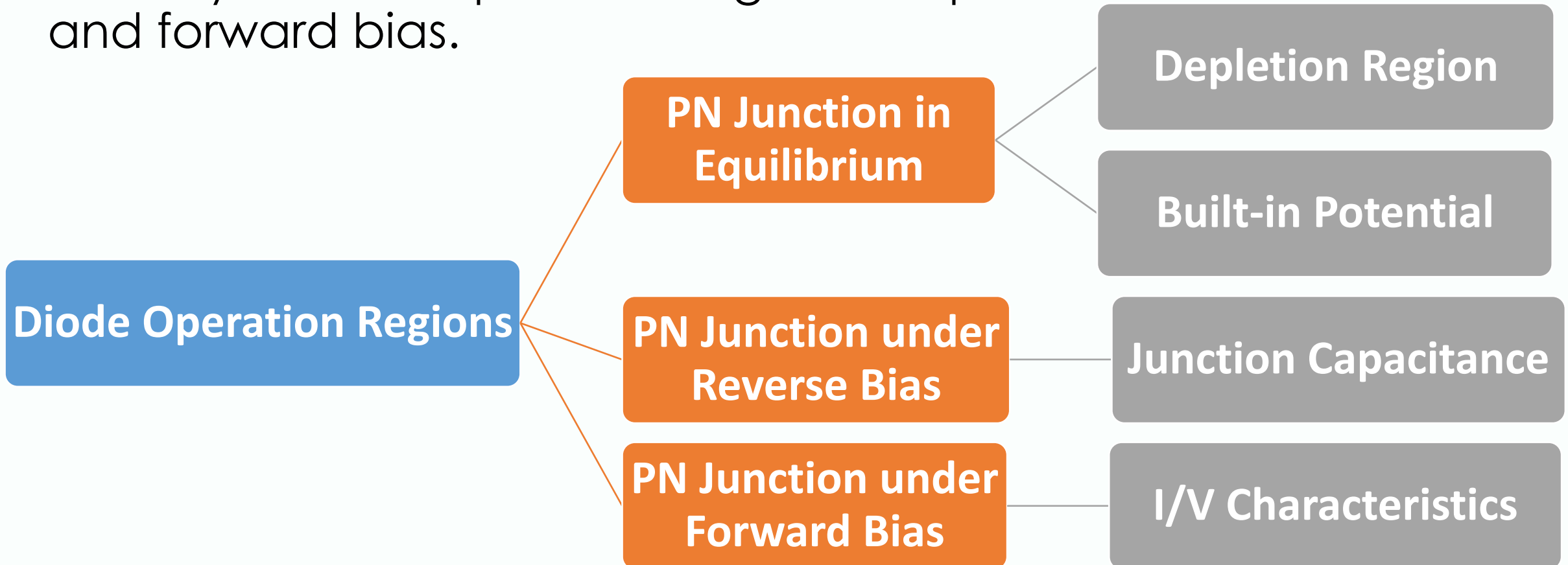
Power (disc, puck) diode

Different diodes

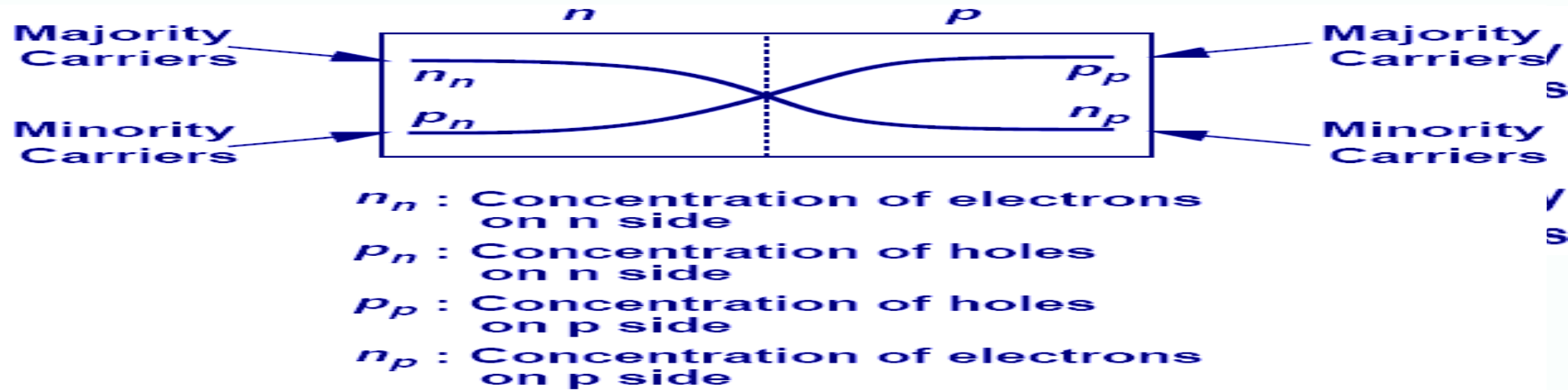
- When N-type and P-type dopants are introduced side-by-side in a semiconductor, a PN junction or a diode is formed.

Diode's Three Operation Regions

- In order to understand the operation of a diode, it is necessary to study its three operation regions: equilibrium, reverse bias, and forward bias.

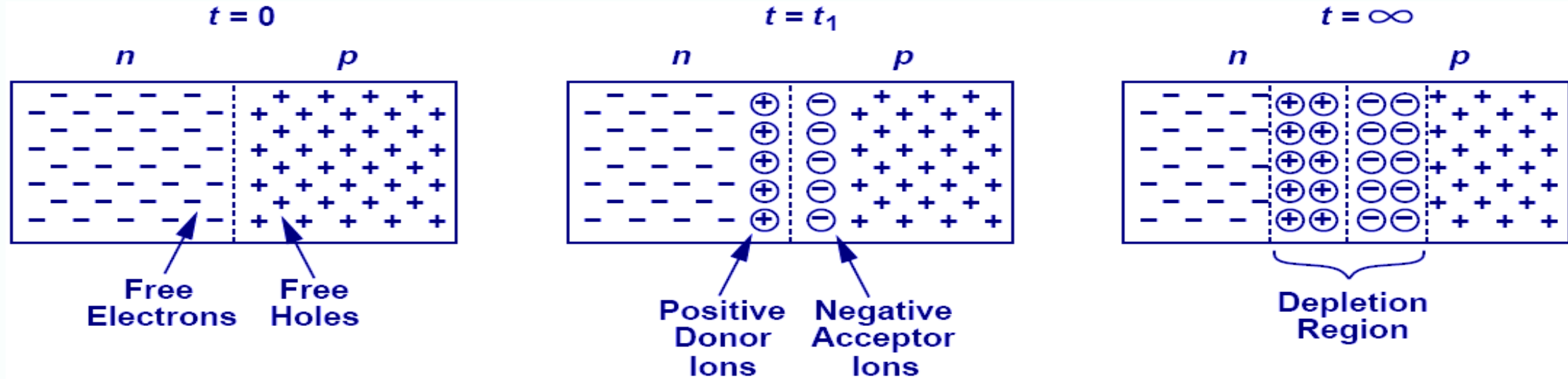


Current Flow Across Junction: Diffusion



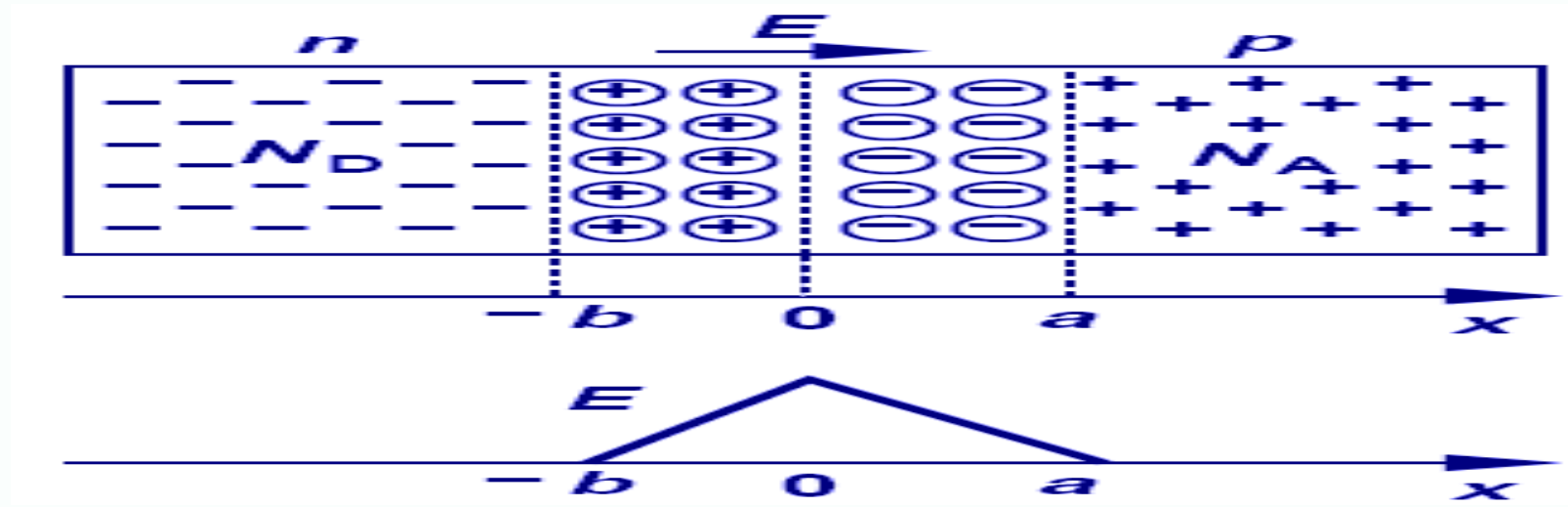
- Because each side of the junction contains an excess of holes or electrons compared to the other side, there exists a large concentration gradient.
- Therefore, a diffusion current flows across the junction from each side.

Depletion Region



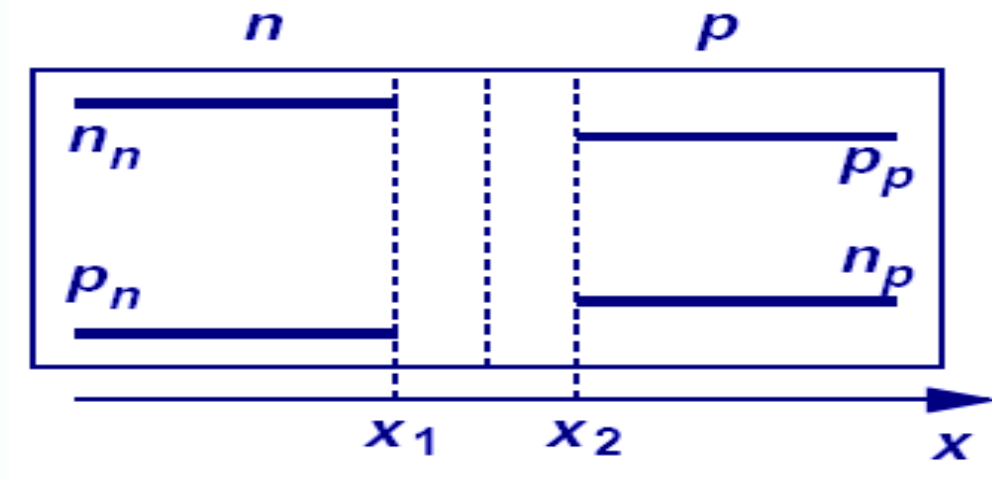
- As free electrons and holes diffuse across the junction, a region of fixed ions is left behind.
- This region is known as the “depletion region.”

Current Flow Across Junction: Drift



- The fixed ions in depletion region create an electric field that results in a drift current.

Current Flow Across Junction: Equilibrium



$$I_{drift,p} = I_{diff,p}$$

$$I_{drift,n} = I_{diff,n}$$

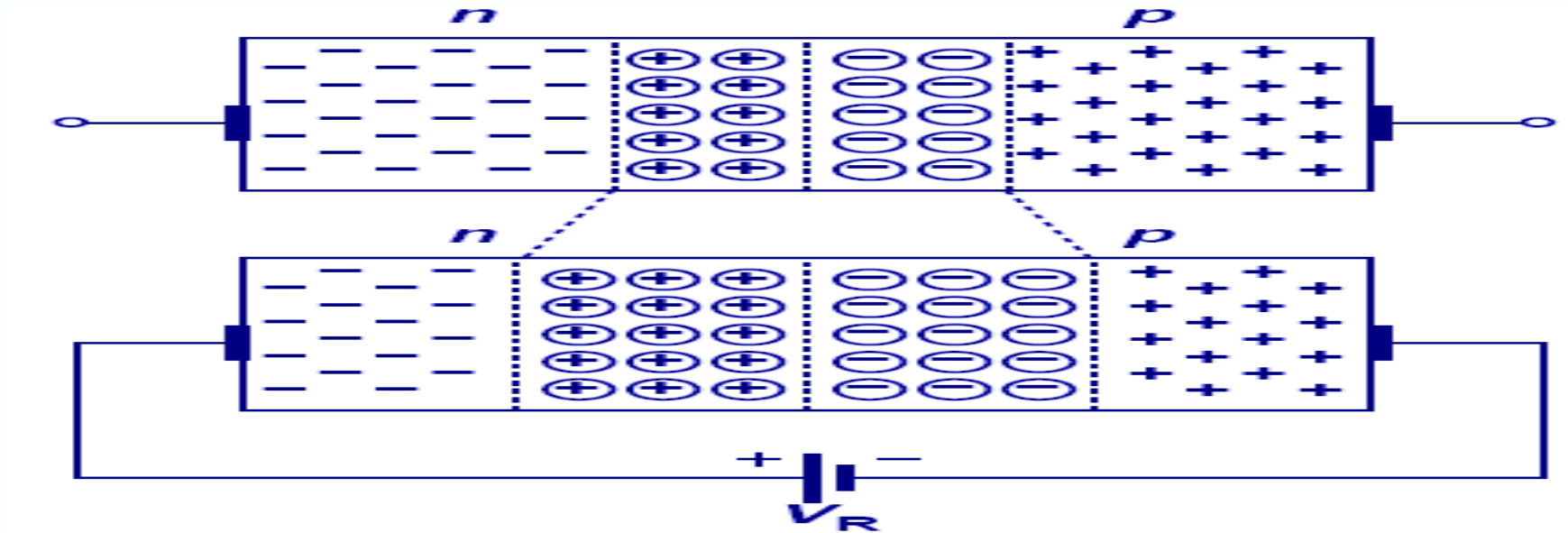
- At equilibrium, the drift current flowing in one direction cancels out the diffusion current flowing in the opposite direction, creating a net current of zero.
- The figure shows the charge profile of the PN junction.

Built-in Potential

$$\begin{aligned}
 q\mu_p pE &= -qD_p \frac{dp}{dx} & -\mu_p p \frac{dV}{dx} &= -D_p \frac{dp}{dx} \\
 \mu_p \int_{x_1}^{x_2} dV &= D_p \int_{p_p}^{p_n} \frac{dp}{p} & V(x_2) - V(x_1) &= \frac{D_p}{\mu_p} \ln \frac{p_p}{p_n} \\
 V_0 &= \frac{kT}{q} \ln \frac{p_p}{p_n}, & V_0 &= \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}
 \end{aligned}$$

- Because of the electric field across the junction, there exists a built-in potential.
- Its derivation is shown above.

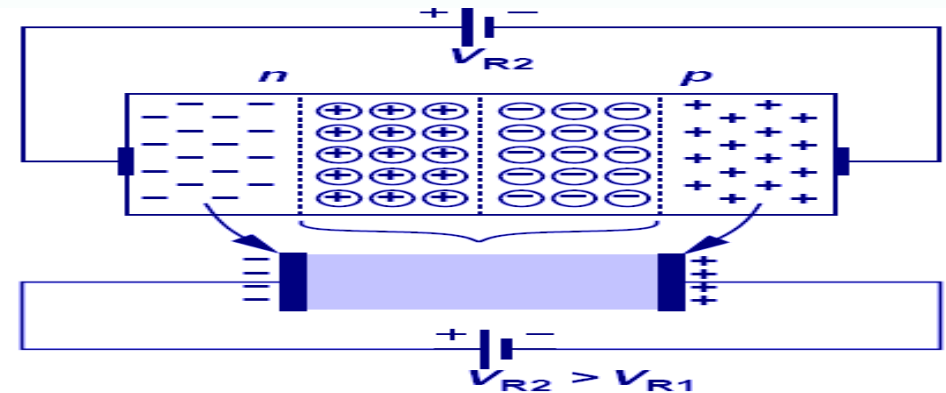
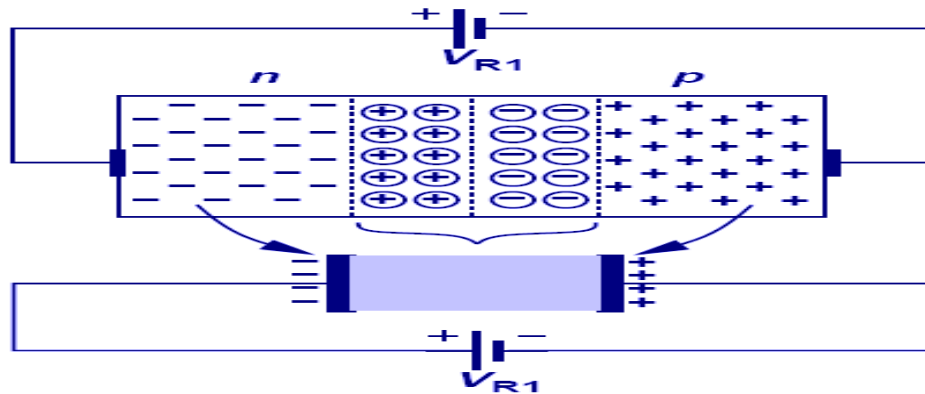
Diode in Reverse Bias



- When the N-type region of a diode is connected to a higher potential than the P-type region, the diode is under reverse bias, which results in wider depletion region and larger built-in electric field across the junction.

Reverse Biased Diode's Application

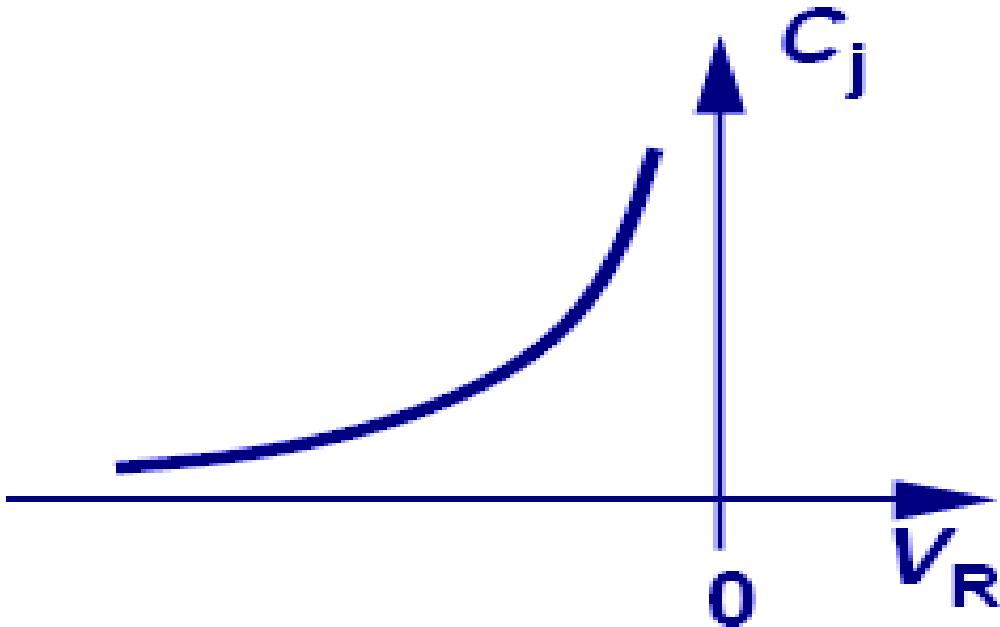
Voltage-Dependent Capacitor



- The PN junction can be viewed as a capacitor.
- By varying V_R , the depletion width changes, changing its capacitance value;
- therefore, the PN junction is actually a voltage-dependent capacitor.

Voltage-Dependent Capacitance

- Equations that describe the voltage-dependent

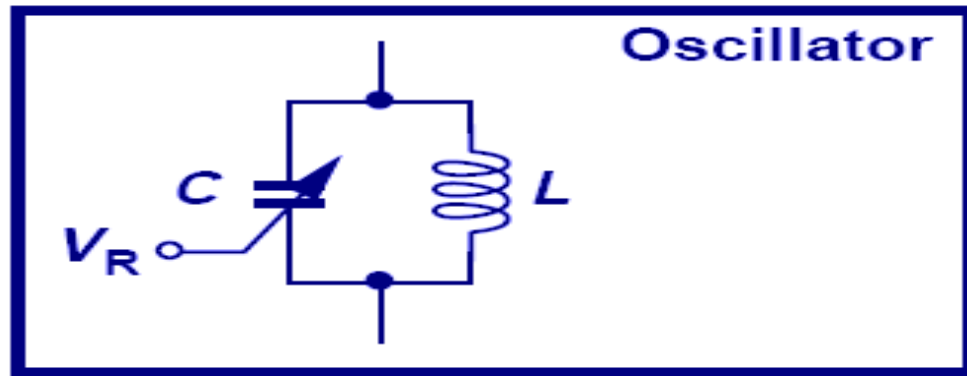


Voltage-dependent capacitance curve

$$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

$$C_{j0} = \sqrt{\frac{\epsilon_{si} q}{2} \frac{N_A N_D}{N_A + N_D} \frac{1}{V_0}}$$

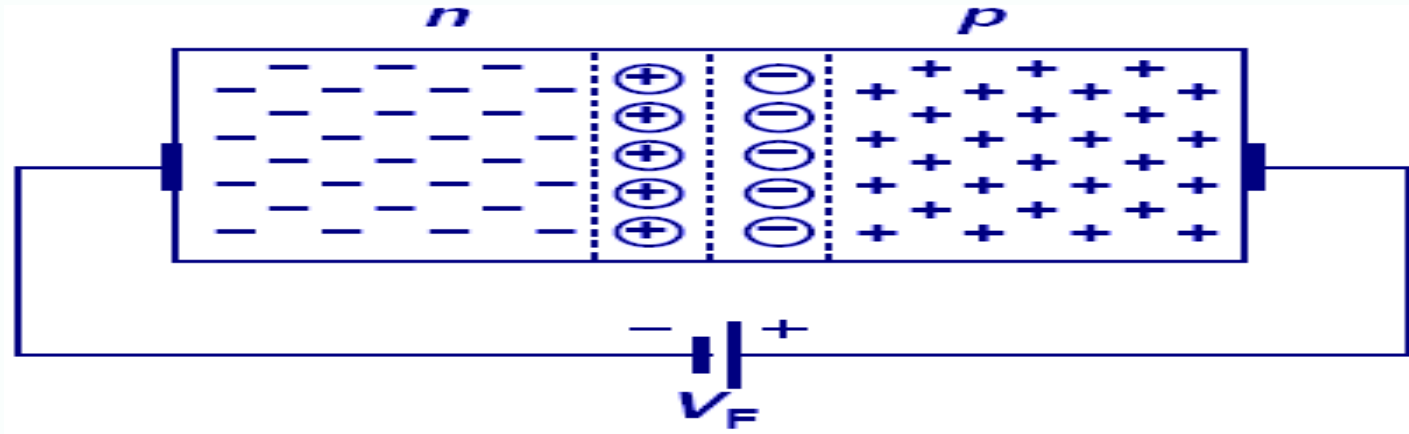
Voltage-Controlled Oscillator



$$f_{res} = \frac{1}{2\pi} \frac{1}{\sqrt{LC}}$$

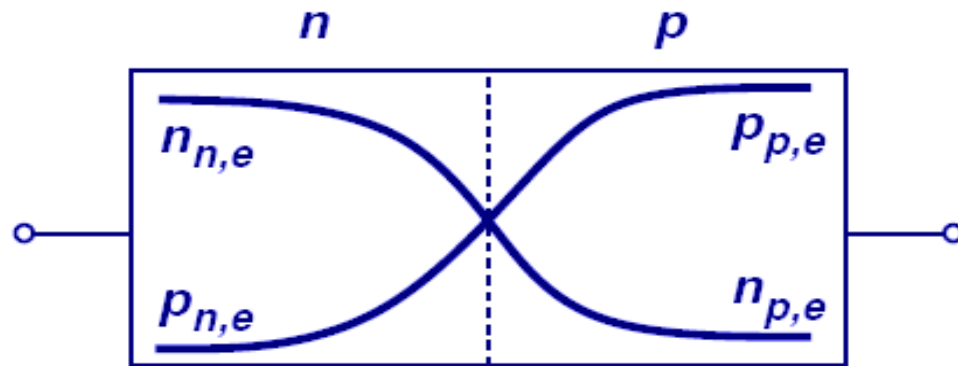
- A very important application of a reverse-biased PN junction is VCO, in which an LC tank is used in an oscillator.
- By changing V_R , we can change C , which also changes the oscillation frequency.

Diode in Forward Bias

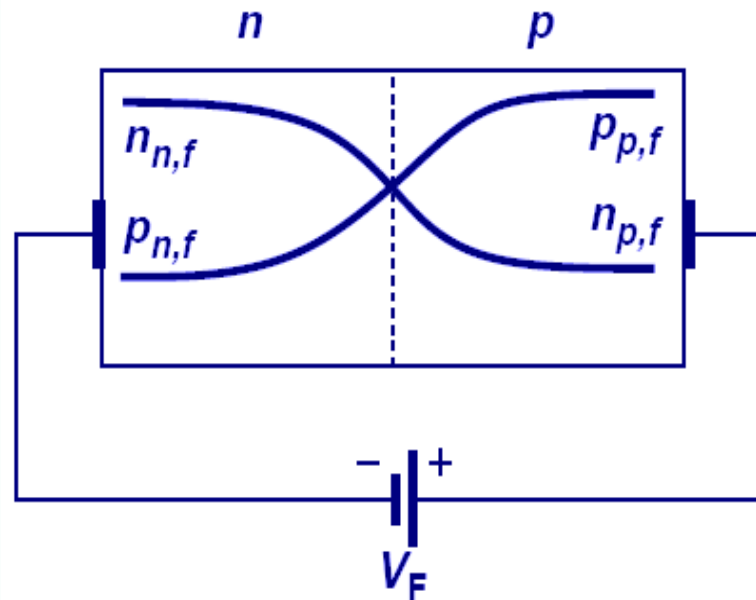


- When the N-type region of a diode is at a lower potential than the P-type region, the diode is in forward bias.
- The depletion width is shortened and the built-in electric field decreased.

Minority Carrier Profile in Forward Bias



$$p_{n,e} = \frac{p_{p,e}}{\exp \frac{V_o}{V_T}}$$



$$p_{n,f} = \frac{p_{p,f}}{\exp \frac{V_o - V_F}{V_T}}$$

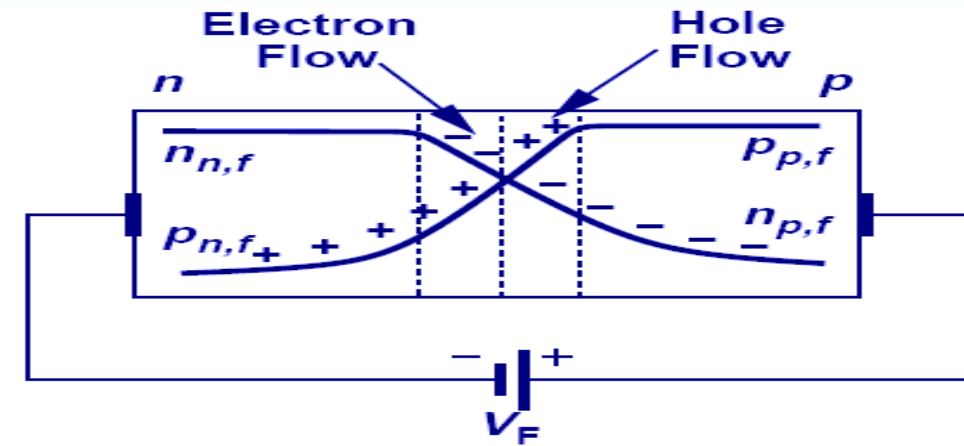
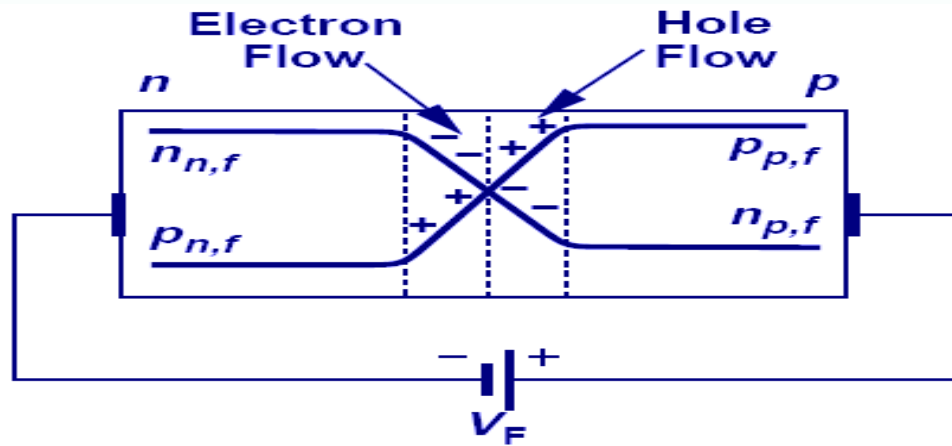
- Under forward bias, minority carriers in each region increase due to the lowering of built-in field/potential.
- Therefore, diffusion currents increase to supply these minority carriers.

Diffusion Current in Forward Bias

$$\begin{aligned}\Delta n_p &\approx \frac{N_D}{V_0} \left(\exp \frac{V_F}{V_T} - 1 \right) & \Delta p_n &\approx \frac{N_A}{V_0} \left(\exp \frac{V_F}{V_T} - 1 \right) \\ I_{tot} &\propto \frac{N_A}{V_0} \left(\exp \frac{V_F}{V_T} - 1 \right) + \frac{N_D}{V_0} \left(\exp \frac{V_F}{V_T} - 1 \right) \\ I_{tot} &= I_s \left(\exp \frac{V_F}{V_T} - 1 \right) & I_s &= A q n_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right)\end{aligned}$$

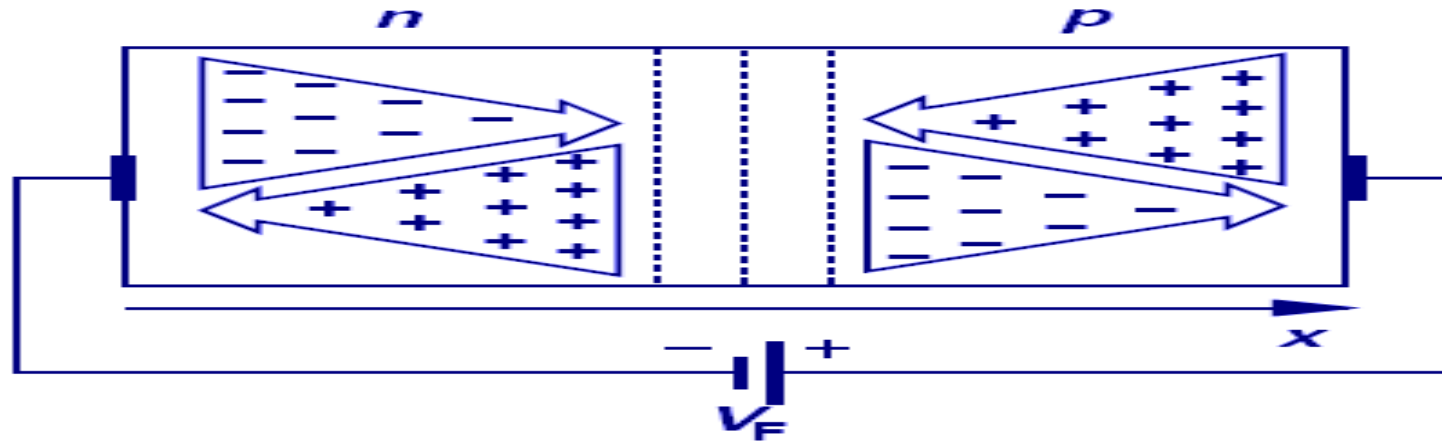
- Diffusion current will increase in order to supply the increase in minority carriers.
- The mathematics are shown above.

Minority Charge Gradient



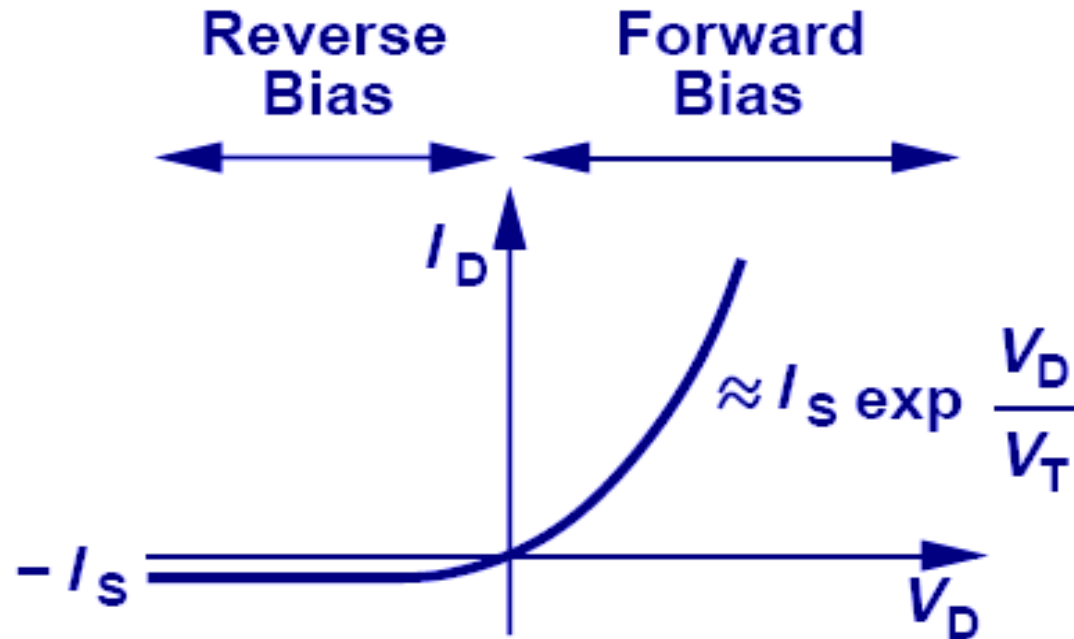
- Minority charge profile should not be constant along the x-axis; otherwise, there is no concentration gradient and no diffusion current.
- Recombination of the minority carriers with the majority carriers accounts for the dropping of minority carriers as they go deep into the P or N region.

Forward Bias Condition: Summary



- In forward bias, there are large diffusion currents of minority carriers through the junction.
- However, as we go deep into the P and N regions, recombination currents from the majority carriers dominate.
- These two currents add up to a constant value.

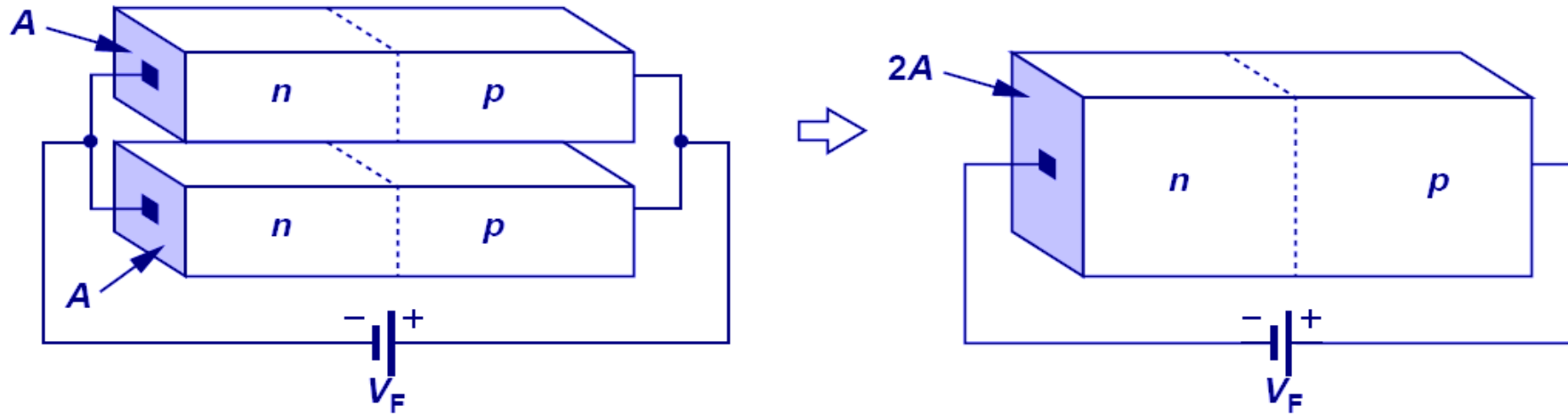
IV Characteristic of PN Junction



$$I_D = I_S \left(\exp \frac{V_D}{V_T} - 1 \right)$$

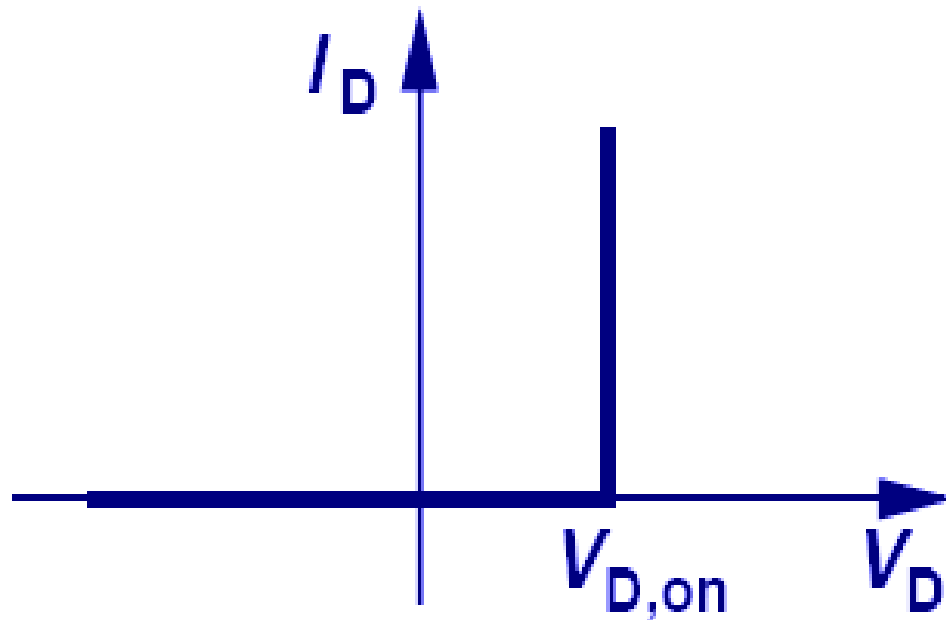
- The current and voltage relationship of a PN junction is exponential in forward bias region, and relatively constant in reverse bias region.

Parallel PN Junctions



- Since junction currents are proportional to the junction's cross-section area.
- Two PN junctions put in parallel are effectively one PN junction with twice the cross-section area, and hence twice the current.

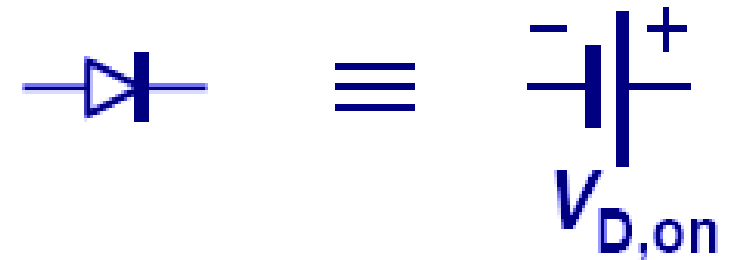
Constant-Voltage Diode Model



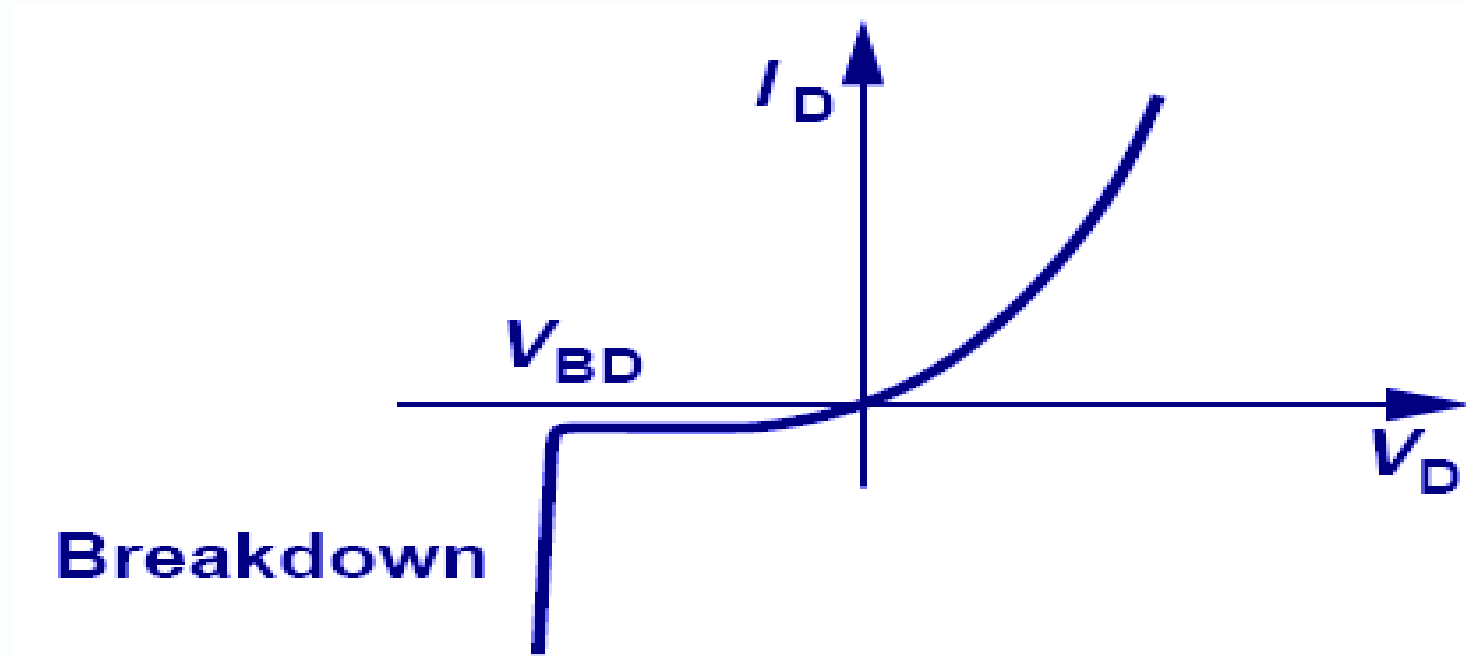
$$V_D < V_{D,on}$$



$$V_D > V_{D,on}$$

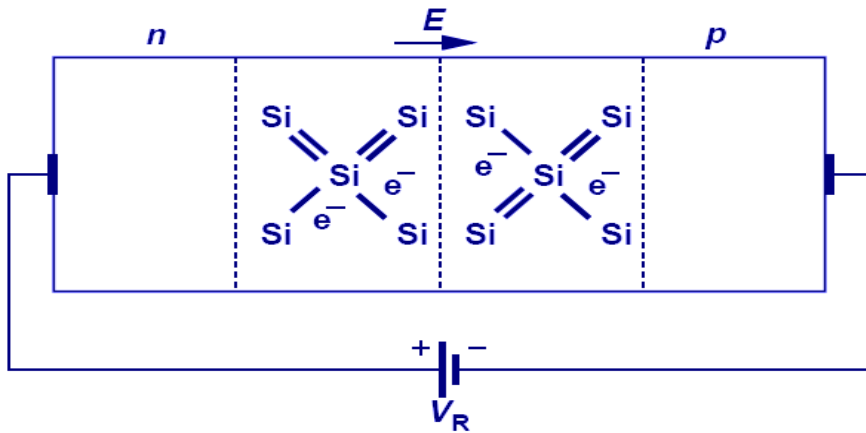


Reverse Breakdown

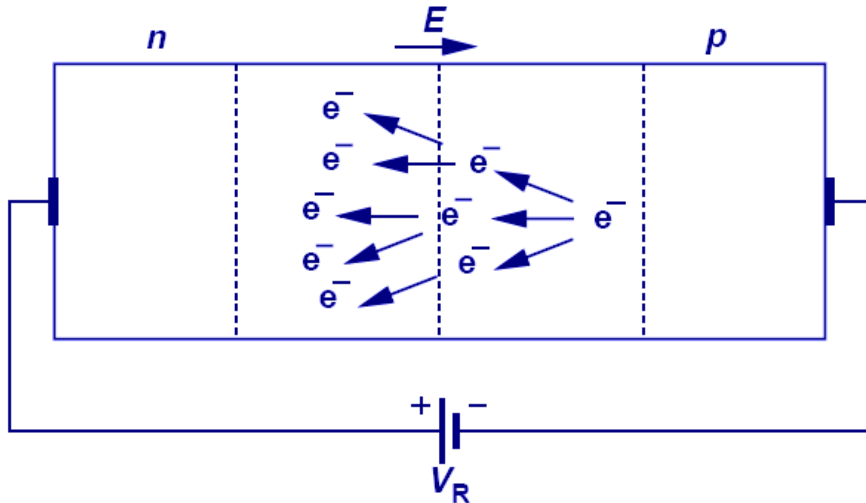


- When a large reverse bias voltage is applied, breakdown occurs and an enormous current flows through the diode.

Zener vs. Avalanche Breakdown

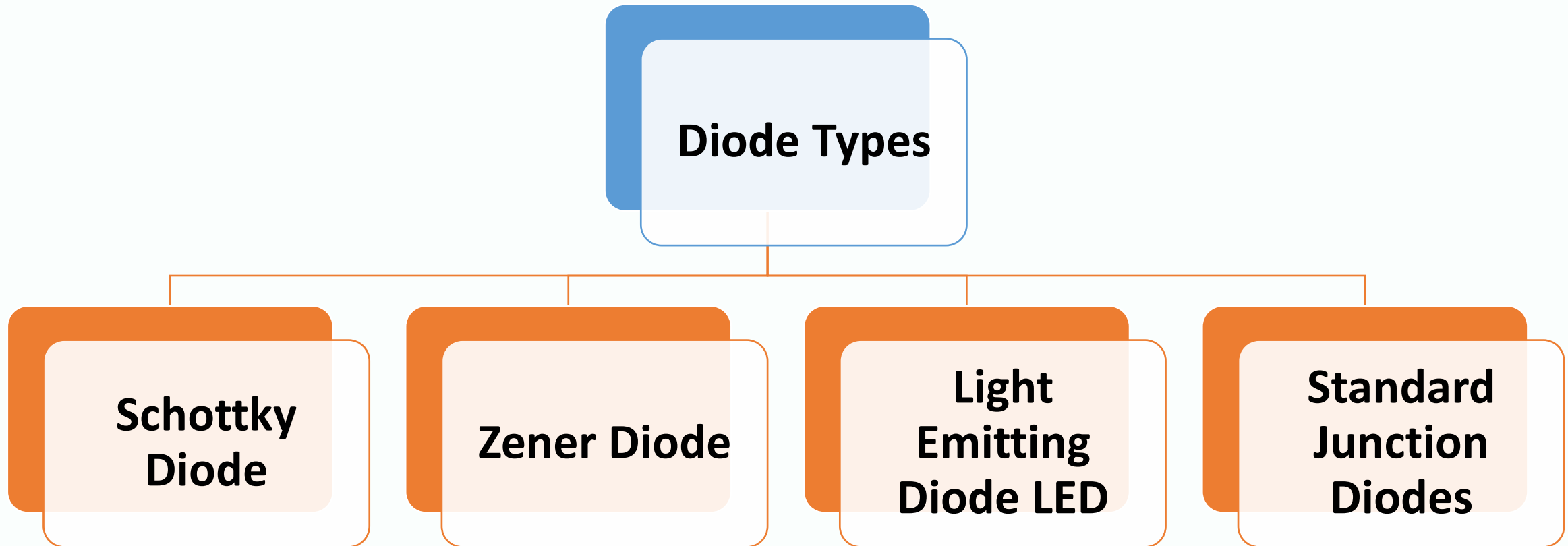


- **Avalanche breakdown** is a result of electrons or holes colliding with the fixed ions inside the depletion region.



Zener breakdown is a result of the large electric field inside the depletion region that breaks electrons or holes off their covalent bonds.

Types of Diodes



Schottky Diode

- Schottky diodes are manufactured by bonding a metal conductor to an N-type semiconductor
- Electron from the N-type material migrate into the metal
- This creates a potential barrier across the boundary, which then behaves in a similar fashion to a PN Junction
- Schottky diodes are used in applications requiring high speed and low capacitance
- Diodes are commonly rated by their switching speed, maximum power dissipation, maximum forward current, maximum forward voltage at a specified forward current, and reverse breakdown voltage

Zener Diode

- Zener diodes are manufactured with controlled reverse-breakdown properties.
- Their forward characteristics are similar to those of junction diodes;
- however, Zener diodes are used in reverse-biased mode.
- Zener diodes are designed to operate at and around their reverse-breakdown('Zener') voltage.
- The Zener voltage is determined during the manufacturing process by adjusting the semiconductor doping.
- Typical Zener voltages range from 3.3 to 75 volts.

Light-Emitting Diode -LED

- LEDs are junction diodes typically made from gallium arsenide phosphide (GaAsP).
- LEDs act very much like silicon junction diodes except that they emit light when conducting forward current
- LEDs have forward voltage drops about twice as large as silicon diodes.
- LED Types
- Infrared, red, orange, yellow, green, and blue



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Thank You

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COE 271 Semiconductor Devices

Lecture 3

PN Devices- Diodes and Circuits

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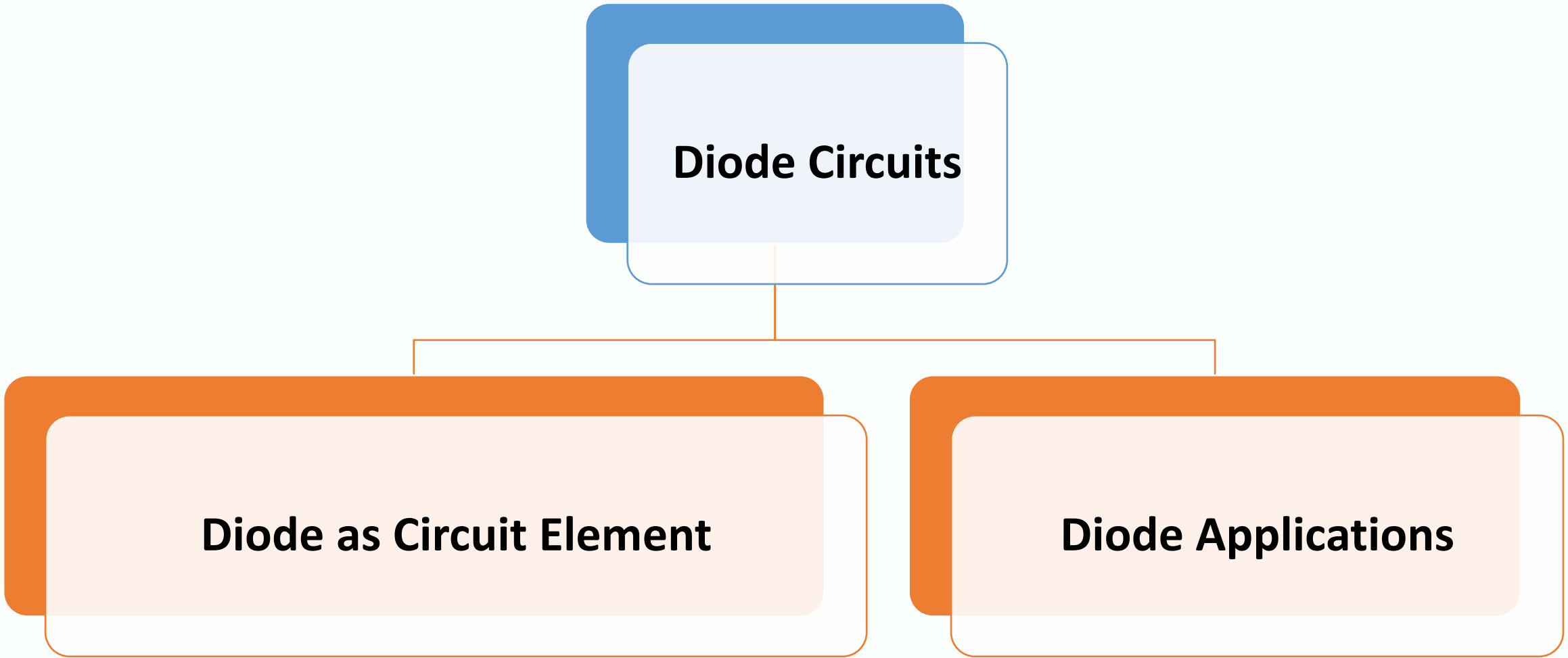
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AND TECHNOLOGY, KUMASI, GHANA



“Men marry women with the hope they
will never change.
Women marry men with the hope they
will change.
Invariably they are both disappointed.”

Albert Einstein

Overview



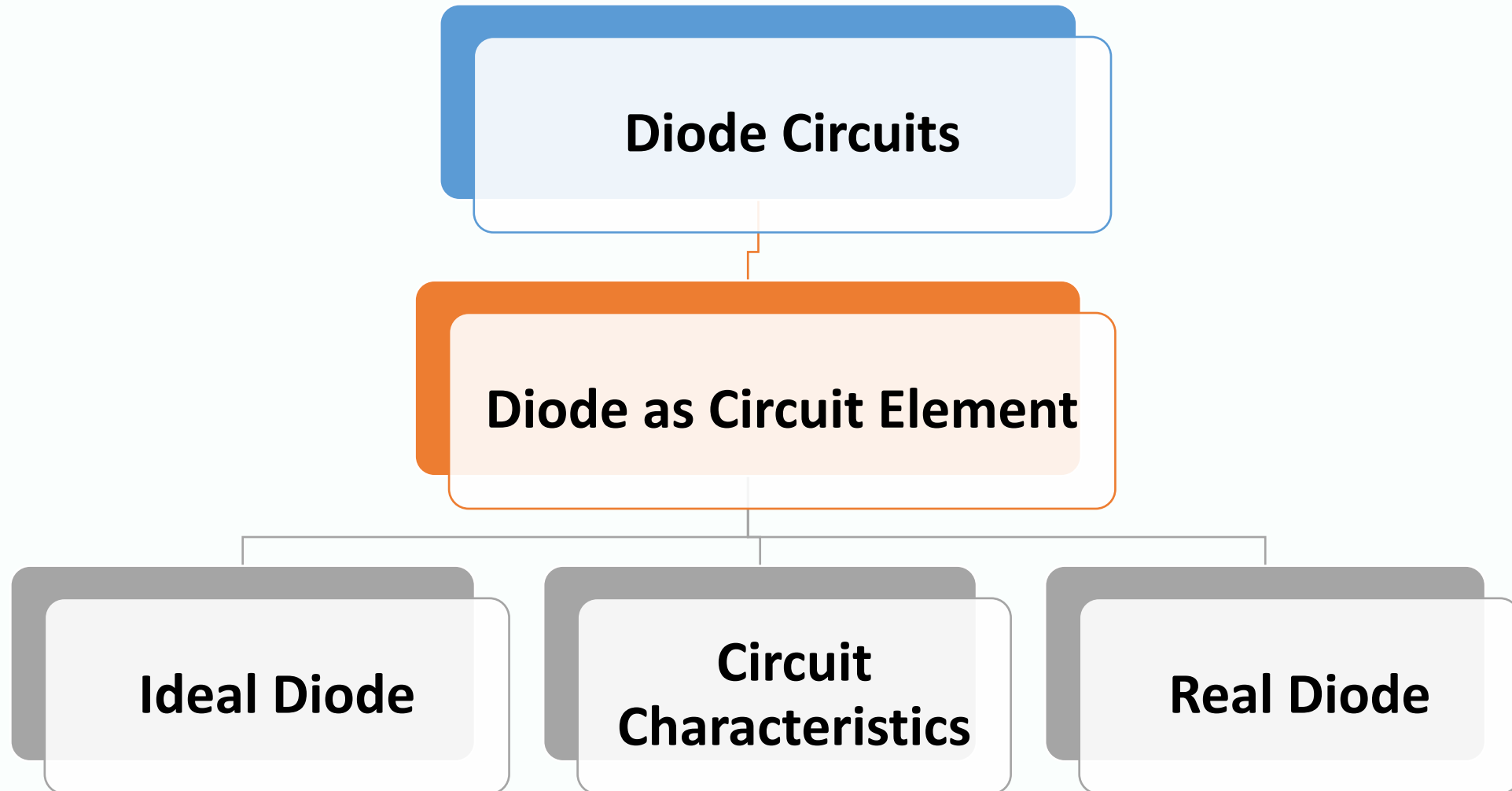
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graph TD; A[Diode Circuits] --> B[Diode as Circuit Element]; A --> C[Diode Applications];
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Diode Circuits

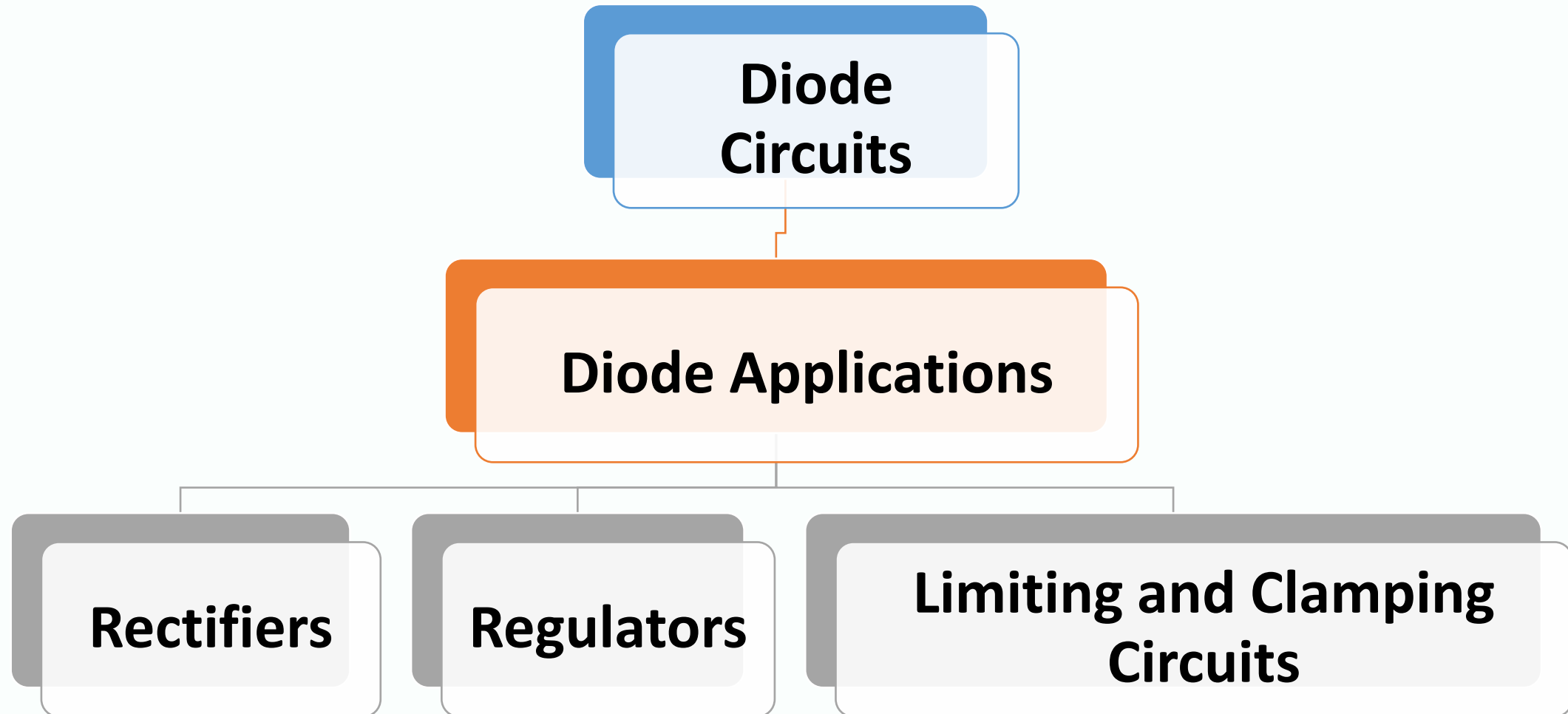
Diode as Circuit Element

Diode Applications

Overview



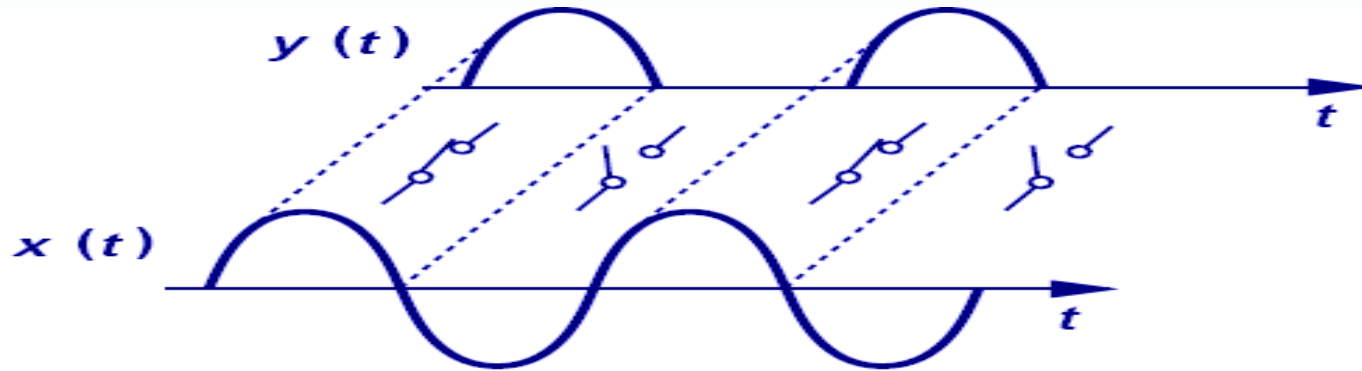
Overview



Diode - Recap

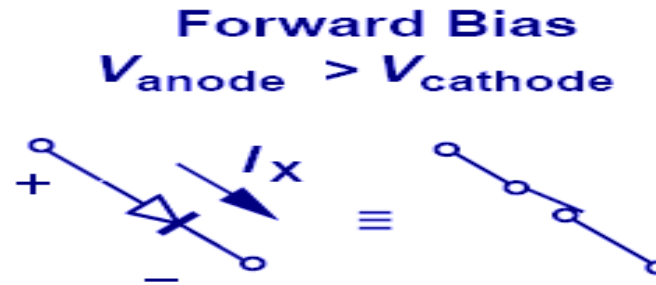
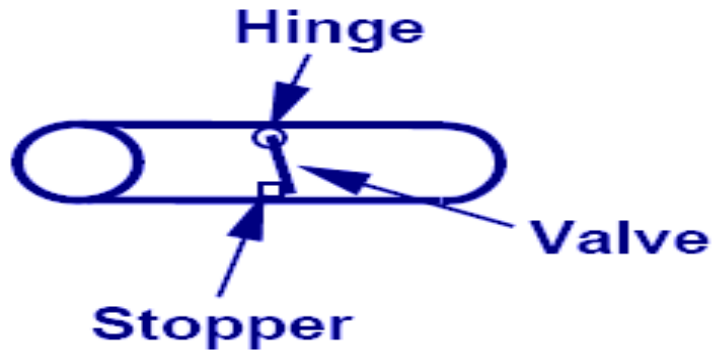
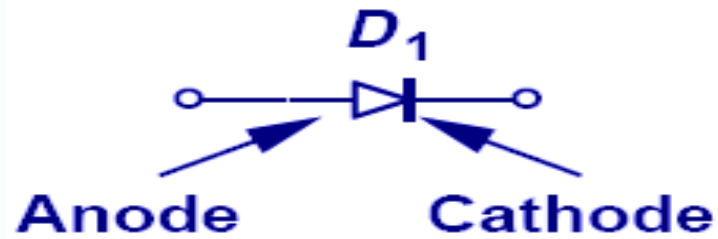
- The diode is a two terminal non-linear device
- Its I-V characteristics besides exhibiting non-linear behavior is also polarity dependent
- Meaning that the behavior of a diode depends on the relative polarity of its terminals
- Depending on the polarity of the voltage v_d the diode is said to be:
- Forward bias $v_d > 0$; Anode voltage > cathode voltage
- Reverse bias $v_d < 0$; Cathode voltage > anode voltage

Ideal Diode

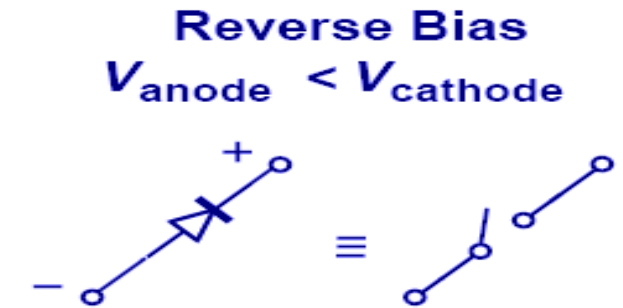
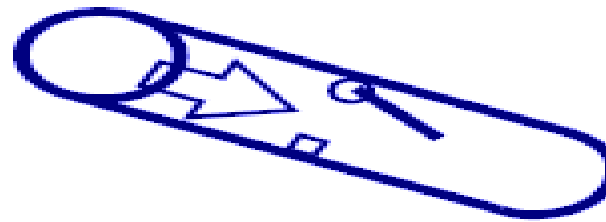


- The diode behaves as a short circuit during the positive half cycle (voltage across it tends to exceed zero),
- and an open circuit during the negative half cycle
- (voltage across it is less than zero).

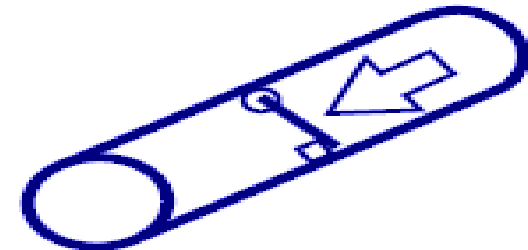
Ideal Diode



Forward Bias

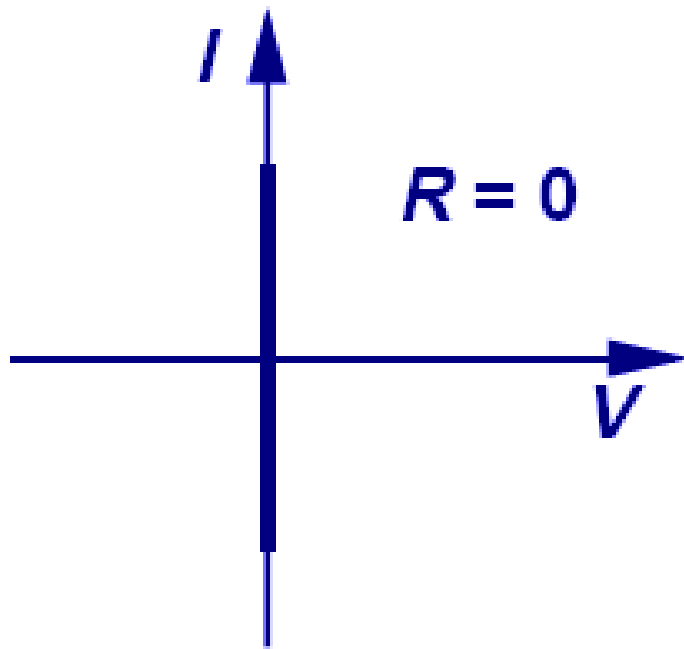


Reverse Bias



- In an ideal diode, if the voltage across it tends to exceed zero, current flows.
- It is analogous to a water pipe that allows water to flow in only one direction.

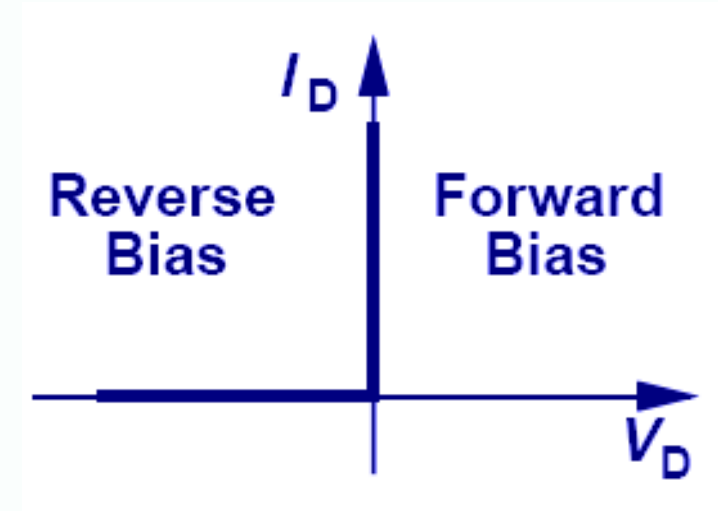
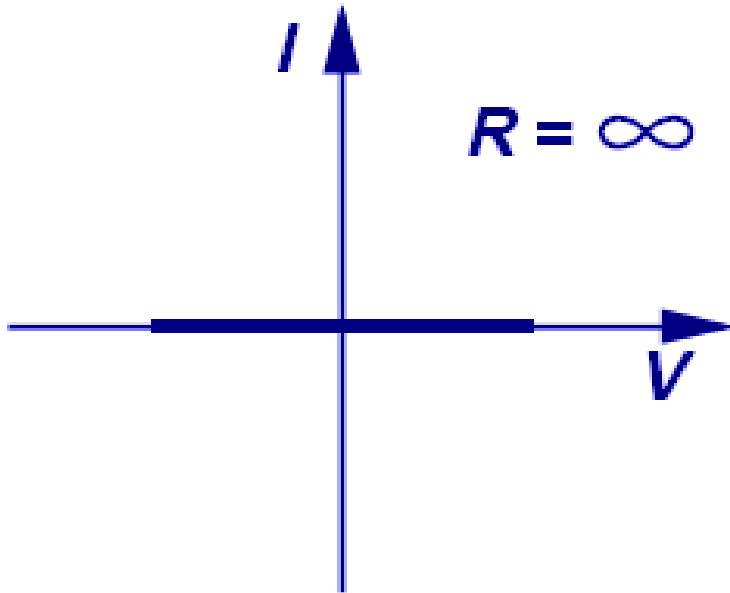
IV Characteristics of an Ideal Diode



- If the voltage across anode and cathode is greater than zero, the resistance of an ideal diode is zero and current becomes infinite.

IV Characteristics of an Ideal Diode

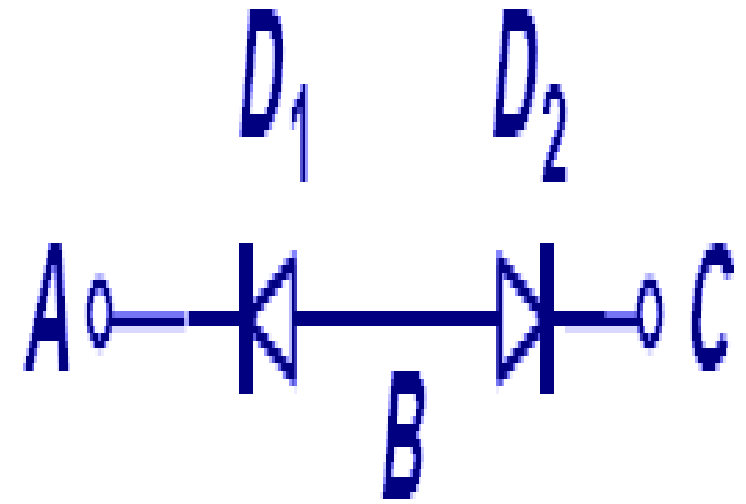
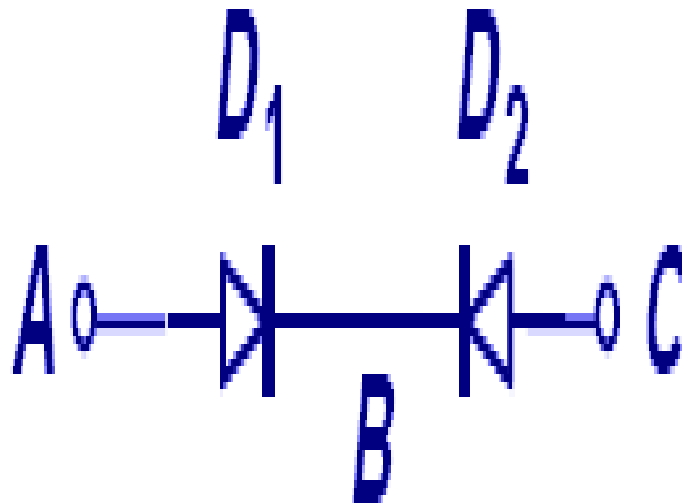
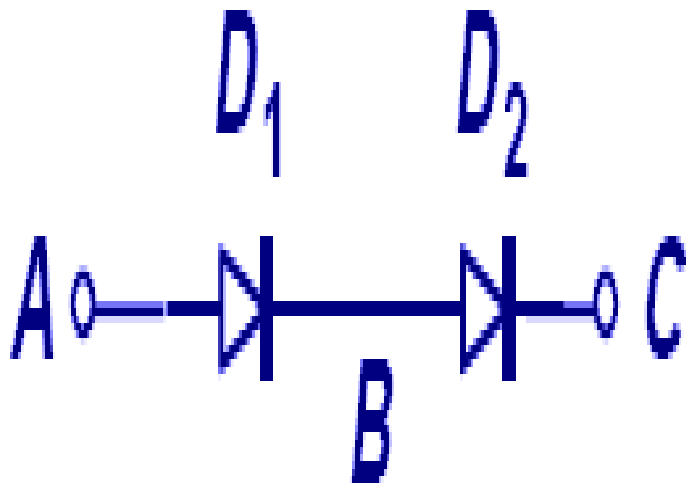
However, if the voltage is less than zero, the resistance becomes infinite and current is zero.



$$R = 0 \Rightarrow I = \frac{V}{R} = \infty$$

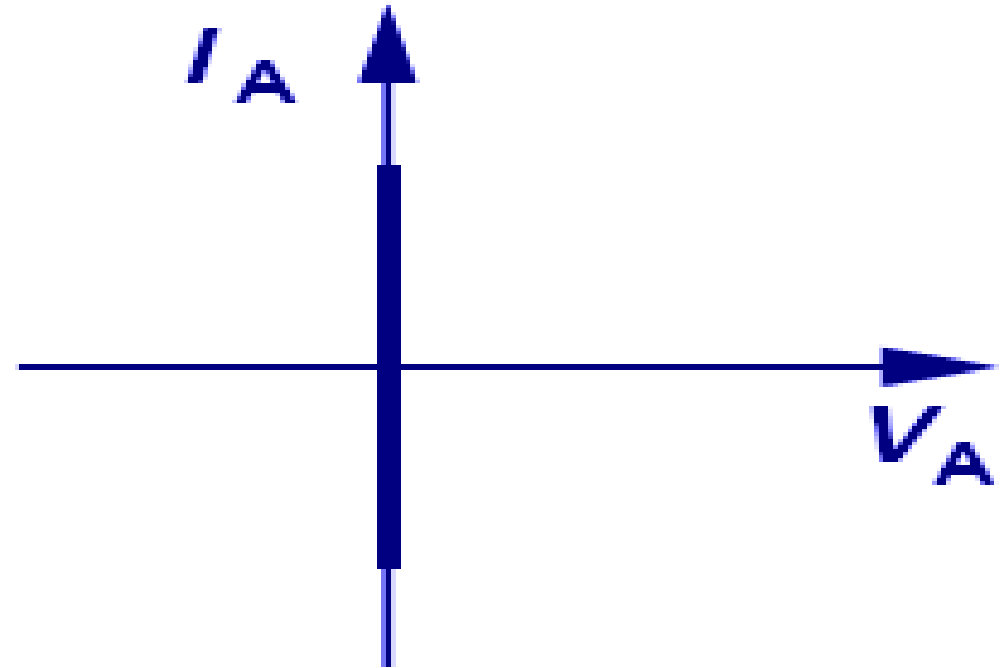
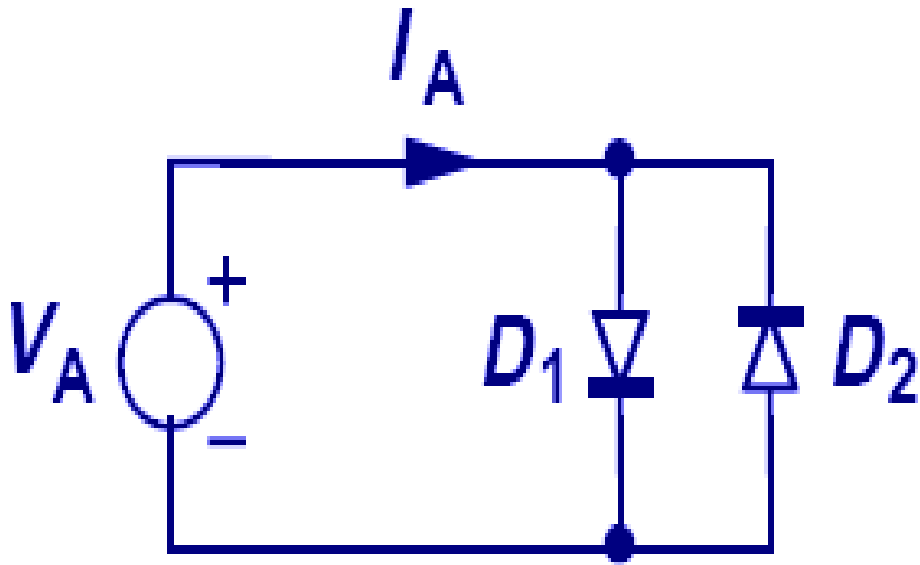
$$R = \infty \Rightarrow I = \frac{V}{R} = 0$$

Diodes in Series



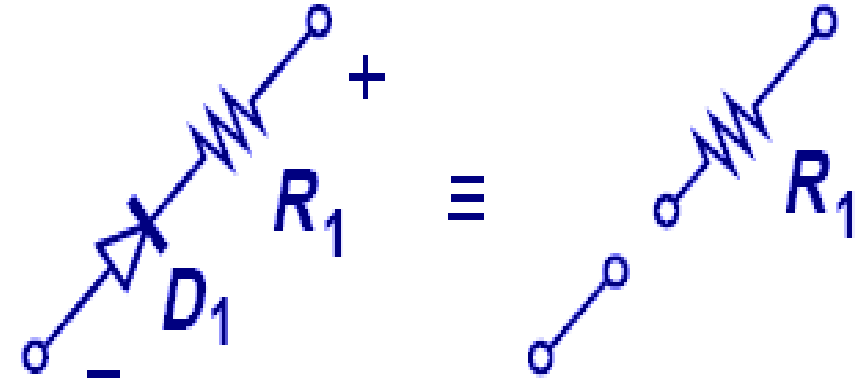
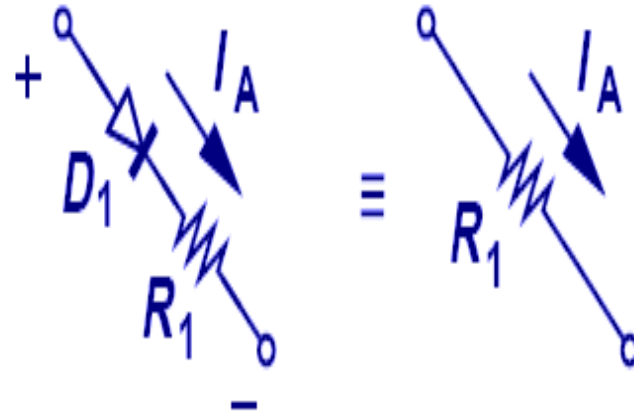
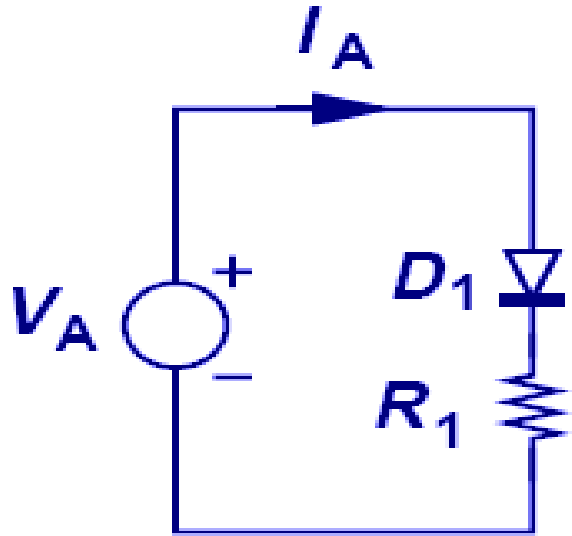
- Diodes cannot be connected in series randomly.
- For the circuits above, only a) can conduct current from A to C.

Anti-Parallel Ideal Diodes

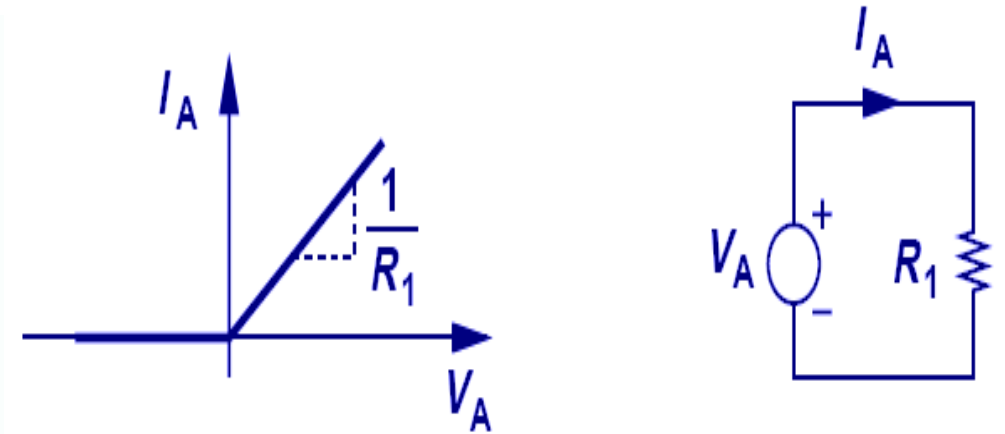


- If two diodes are connected in anti-parallel, it acts as a short for all voltages.

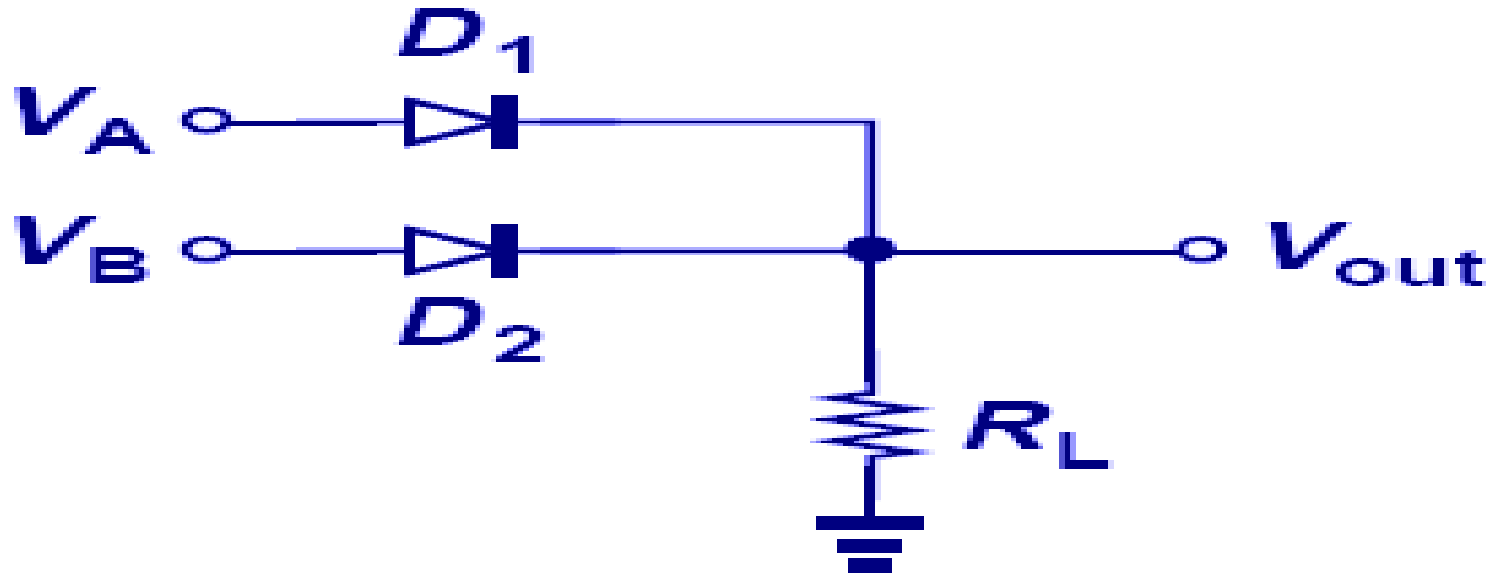
Diode-Resistor Combination



- The IV characteristic of this diode-resistor combination is zero for negative voltages and Ohm's law for positive voltages.

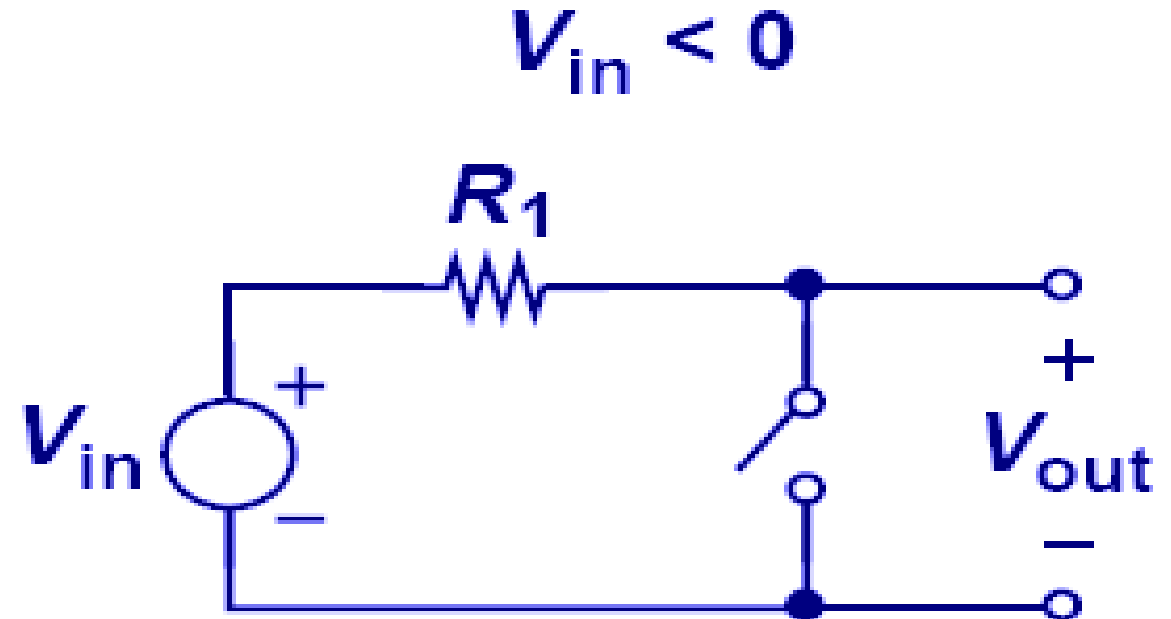
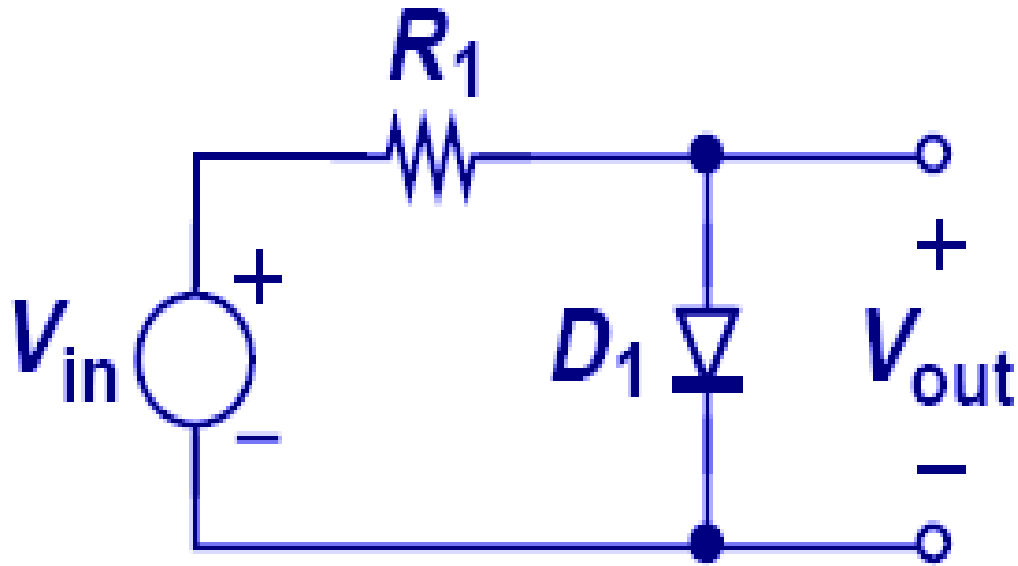


Diode Implementation of OR Gate



- The circuit shows an example of diode-implemented as OR gate.
- V_{out} can only be either V_A or V_B , not both.

Input / Output Characteristics

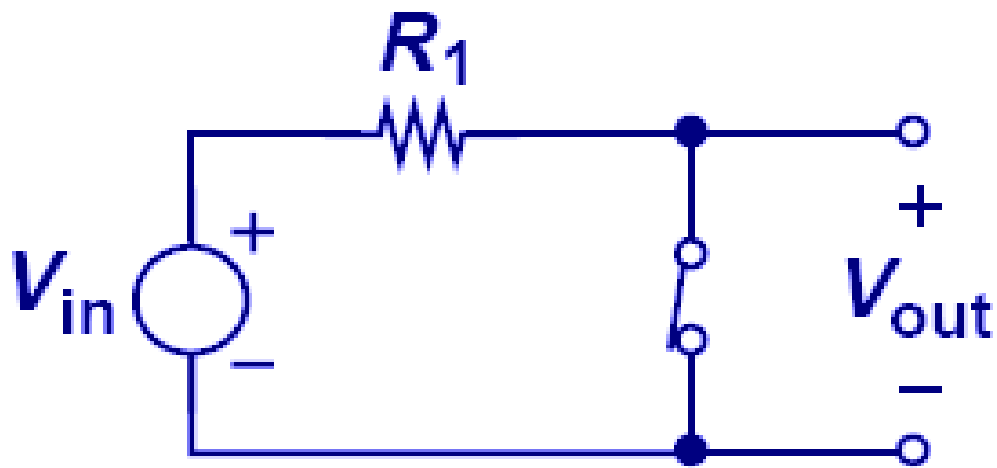


- When V_{in} is less than zero, the diode opens, so $V_{out} = V_{in}$.

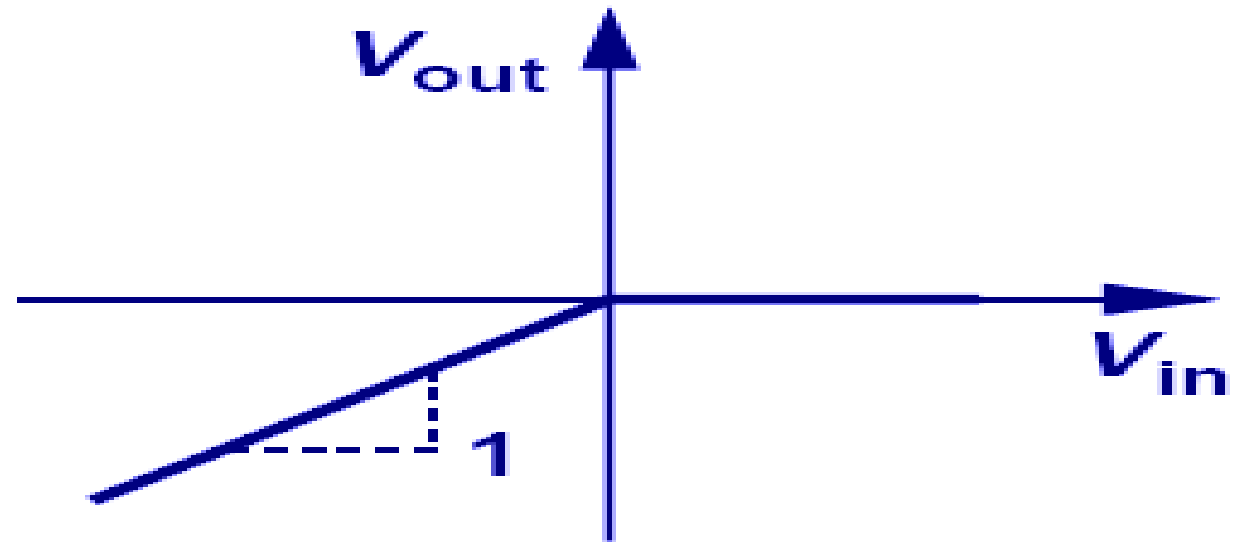
Input / Output Characteristics

When V_{in} is greater than zero, the diode shorts,
so $V_{out} = 0$.

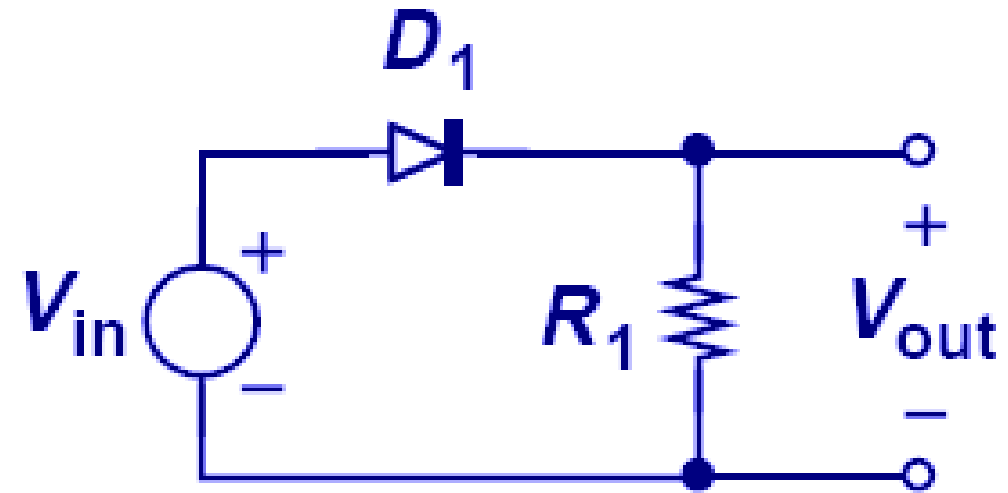
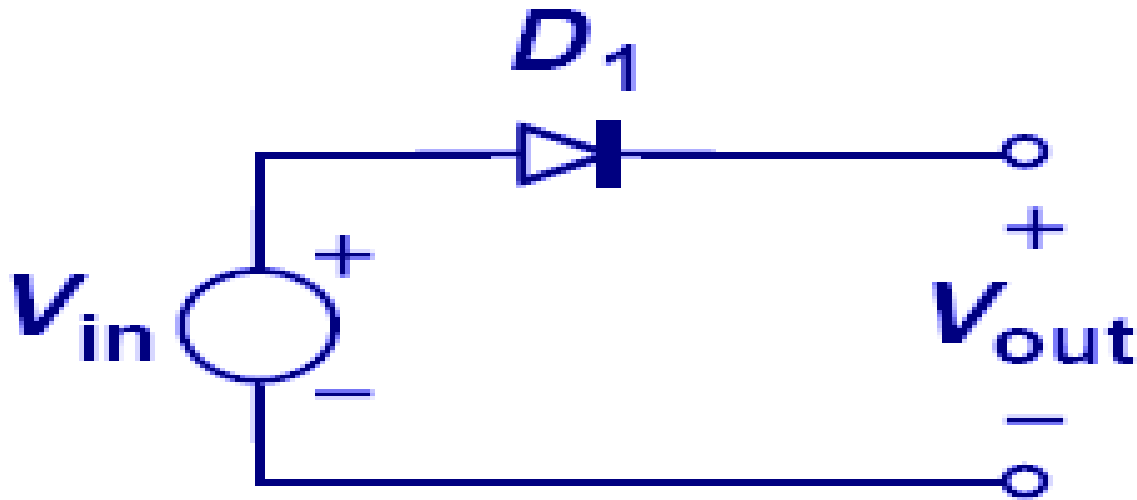
$$V_{in} > 0$$



IO Characteristics



Diode's Application: Rectifier

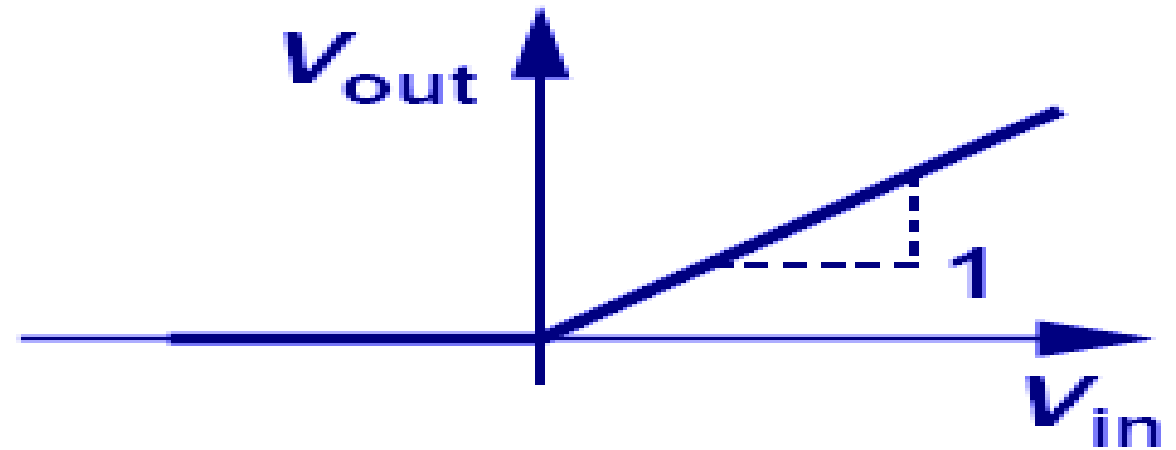
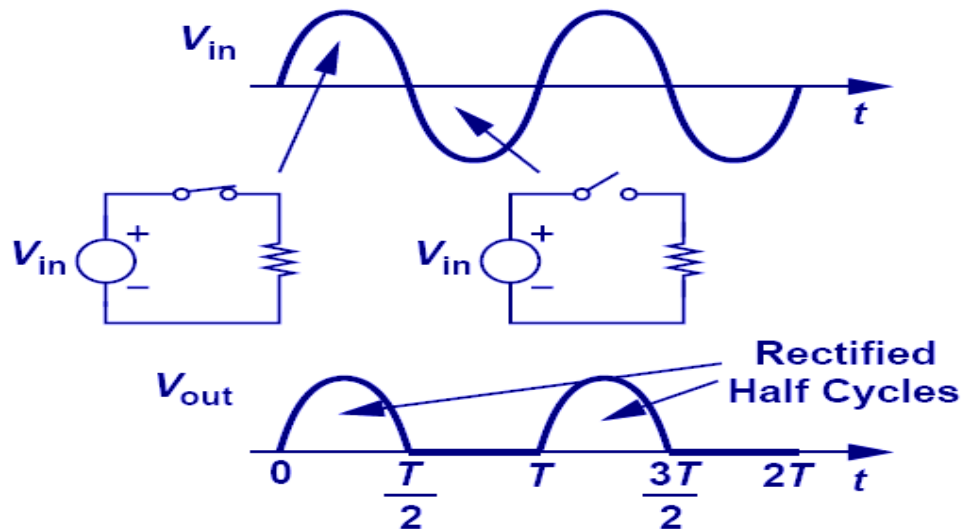


A rectifier is a device that converts AC to DC by blocking the flow of current in one direction

A rectifier passes positive-half cycle of a sinusoid and blocks the negative half-cycle or vice versa.

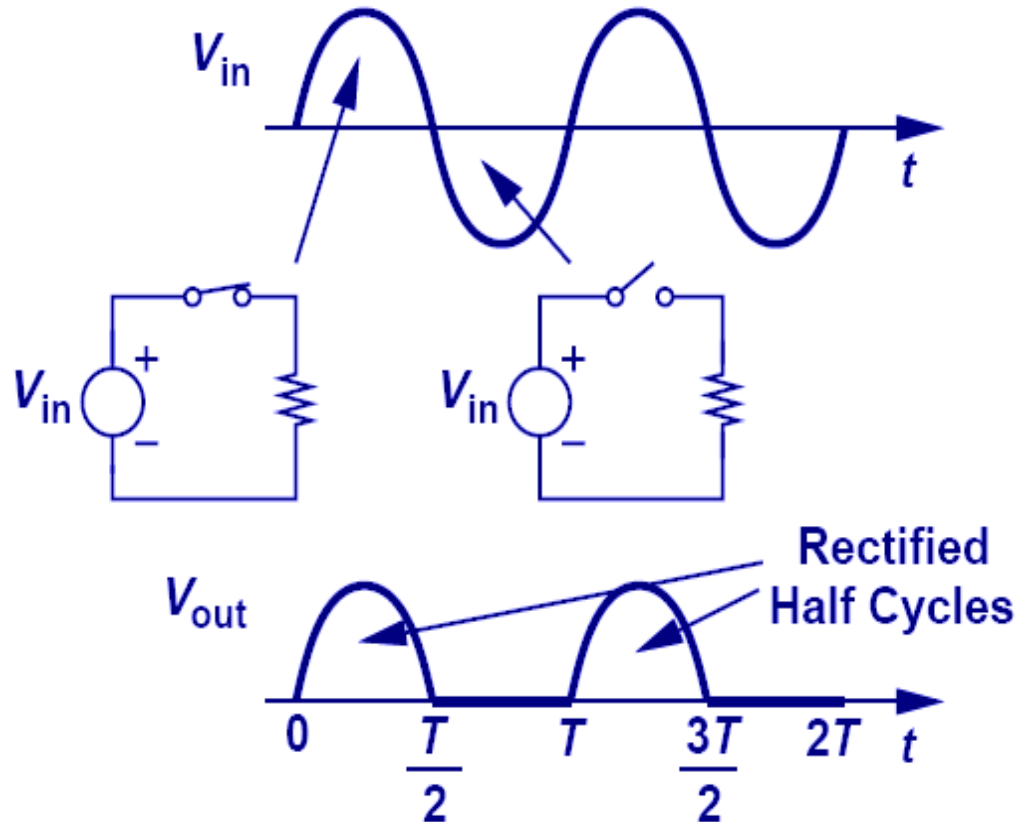
Diode's Application: Rectifier

- When V_{in} is greater than 0, diode shorts, so $V_{out} = V_{in}$;
- however, when V_{in} is less than 0, diode opens, no current flows thru R_1 , $V_{out} = I_{R1}R_1 = 0$.



Signal Strength Indicator SSI

$$V_{out} = V_p \sin \omega t = 0 \quad \text{for} \quad 0 \leq t \leq \frac{T}{2}$$



$$\begin{aligned} V_{out,avg} &= \frac{1}{T} \int_0^T V_{out}(t) dt = \frac{1}{T} \int_0^{T/2} V_p \sin \omega t dt \\ &= \frac{1}{T} \frac{V_p}{\omega} [-\cos \omega t]_0^{T/2} = \frac{V_p}{\pi} \end{aligned}$$

$$\text{for} \quad \frac{T}{2} \leq t \leq T$$

Signal Strength Indicator SSI

for

$$\frac{T}{2} \leq t \leq T$$

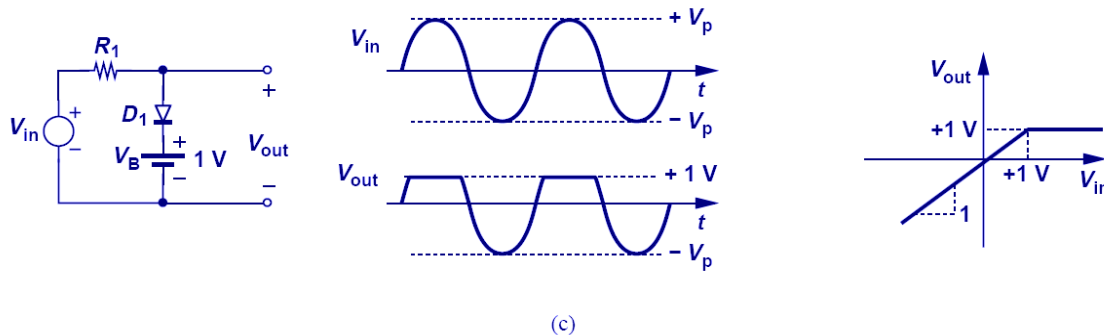
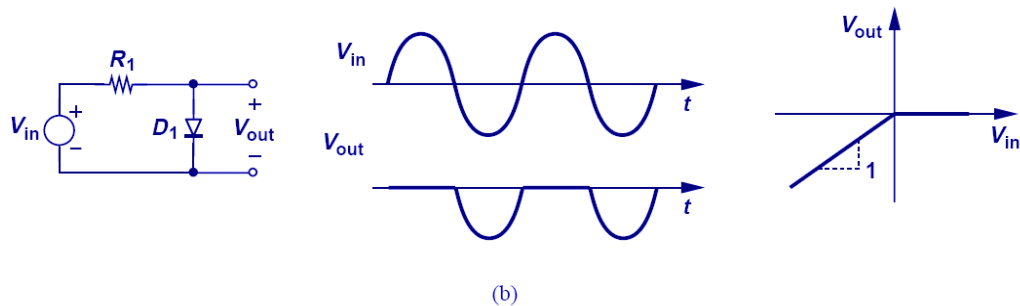
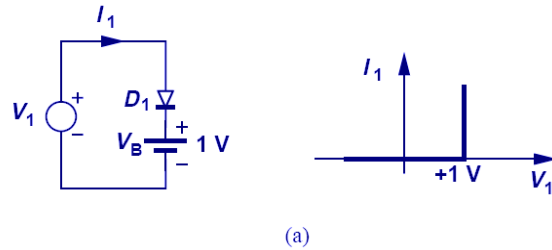
$$\begin{aligned} V_{out,avg} &= \frac{1}{T} \int_0^T V_{out}(t) dt = \frac{1}{T} \int_0^{T/2} V_p \sin \omega t dt \\ &= \frac{1}{T} \frac{V_p}{\omega} [-\cos \omega t]_0^{T/2} = \frac{V_p}{\pi} \end{aligned}$$

- The averaged value of a rectifier output can be used as a signal strength indicator for the input,
- since $V_{out,avg}$ is proportional to V_p , the input signal's amplitude.

Diode's Application: Limiter

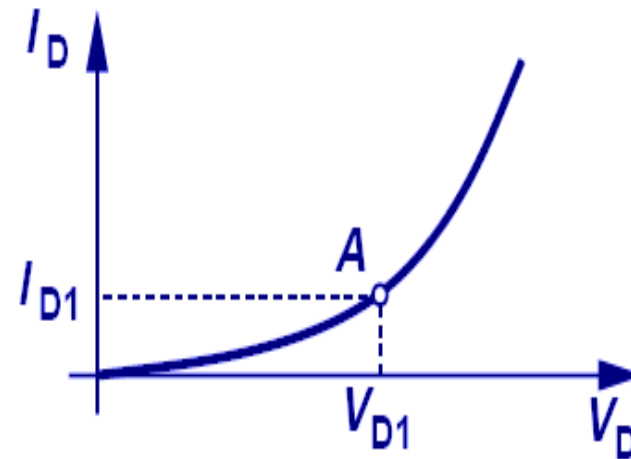
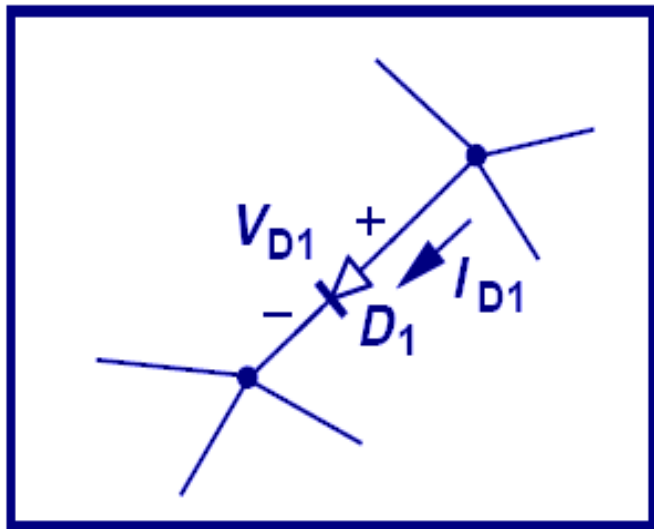
- The purpose of a limiter is to force the output to remain below certain value.

In a), the addition of a 1 V battery forces the diode to turn on after V_1 has become greater than 1 V.

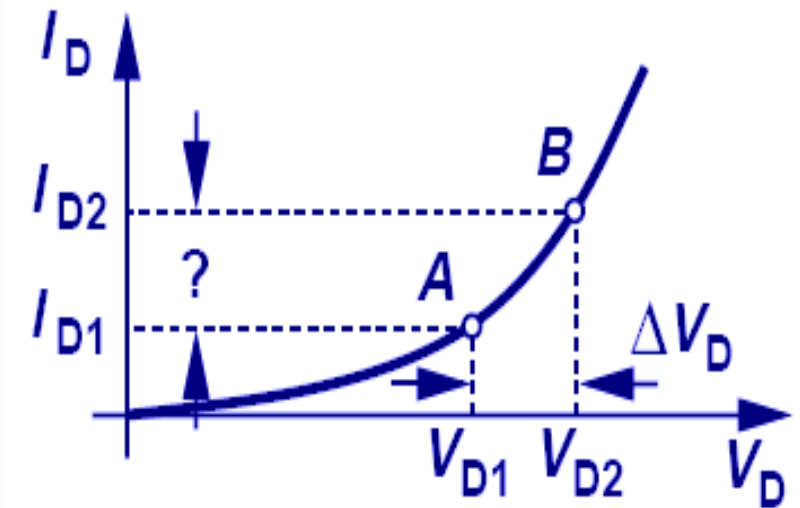


Small-Signal Analysis

- Small-signal analysis is performed around a bias point by perturbing the voltage by a small amount and observing the resulting linear current perturbation.

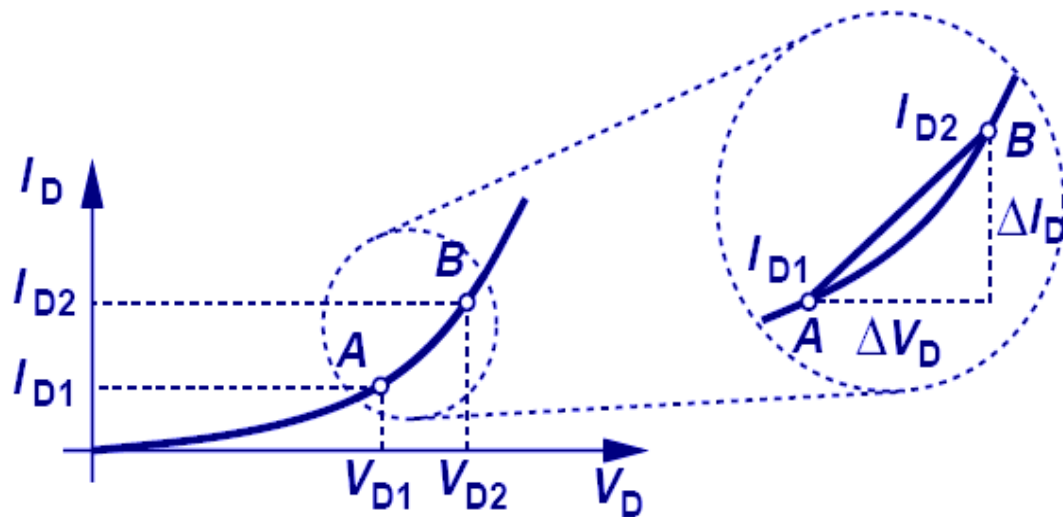


$$\Delta I_D = \frac{\Delta V}{V_T} I_{D1}$$



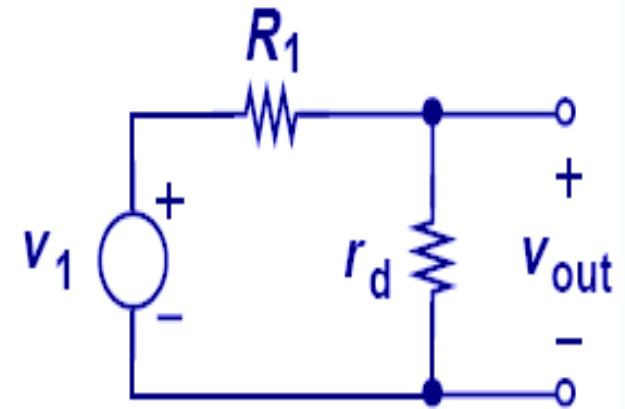
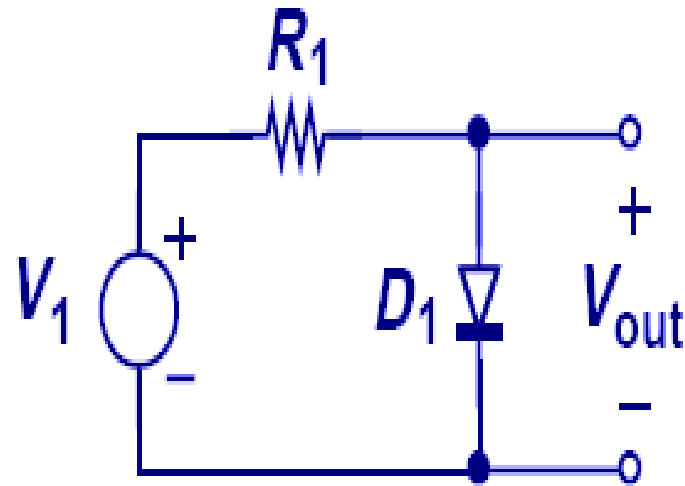
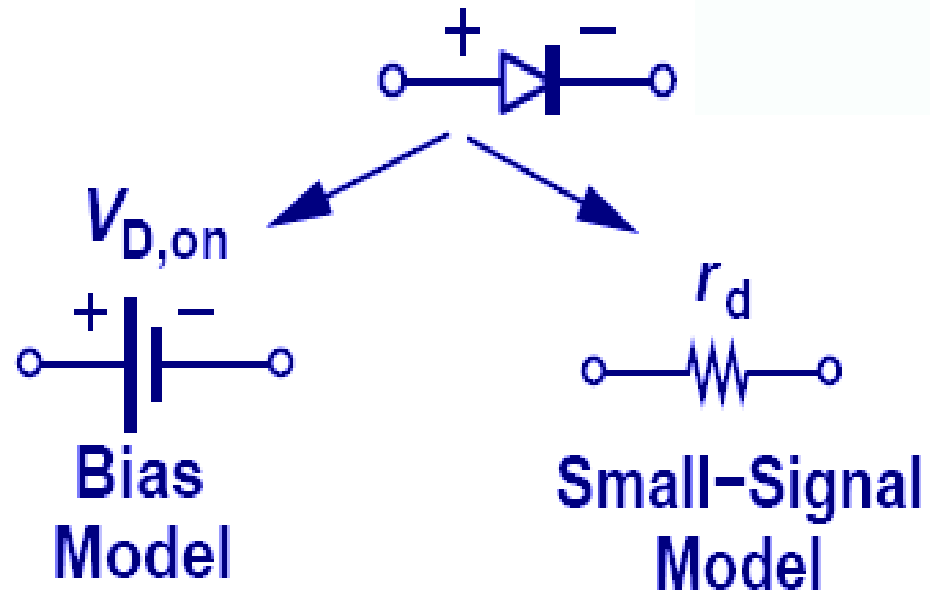
Small-Signal Analysis in Detail

- If two points on the IV curve of a diode are close enough, the trajectory connecting the first to the second point is like a line, with the slope being the proportionality factor between change in voltage and change in current.



$$\begin{aligned} \frac{\Delta I_D}{\Delta V_D} &= \left. \frac{dI_D}{dV_D} \right|_{V_D=V_{D1}} \\ &= \frac{I_s}{V_T} \exp \frac{V_{D1}}{V_T} \\ &= \frac{I_{D1}}{V_T} \end{aligned}$$

Small-Signal Incremental Resistance

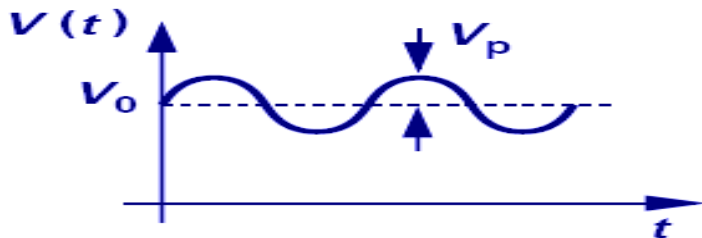


$$r_d = \frac{V_T}{I_D}$$

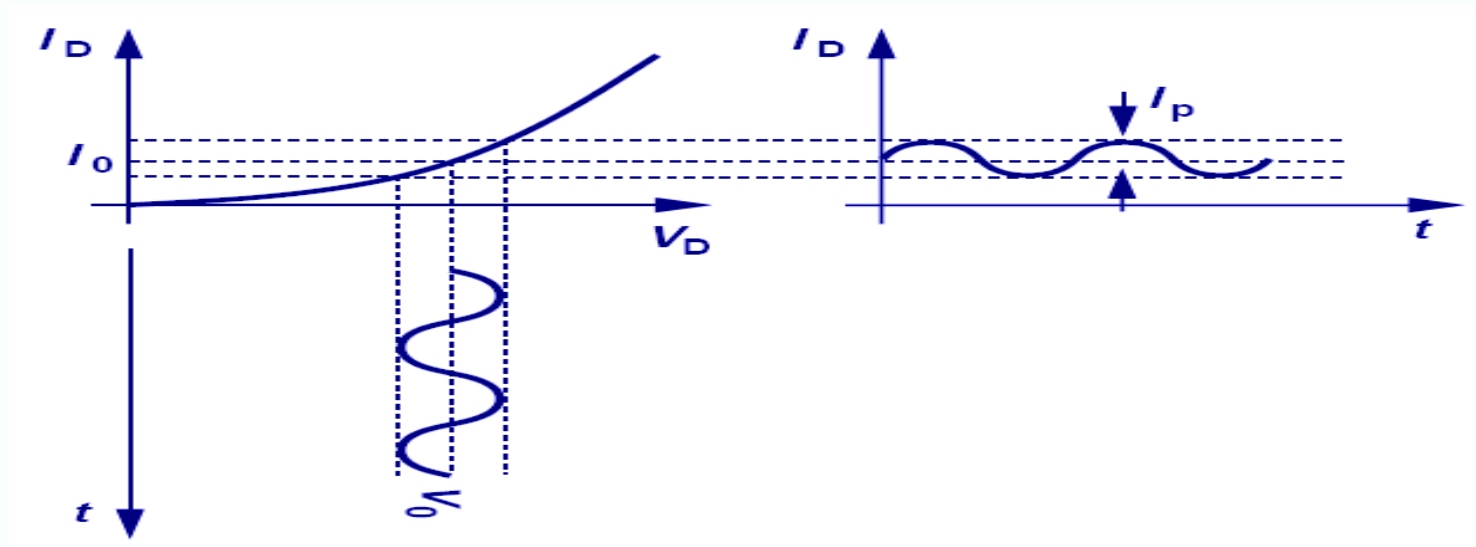
- Since there's a linear relationship between the small signal current and voltage of a diode, the diode can be viewed as a linear resistor when only small changes are of interest.

Small Sinusoidal Analysis

- If a sinusoidal voltage with small amplitude is applied, the resulting current is also a small sinusoid around a DC value.

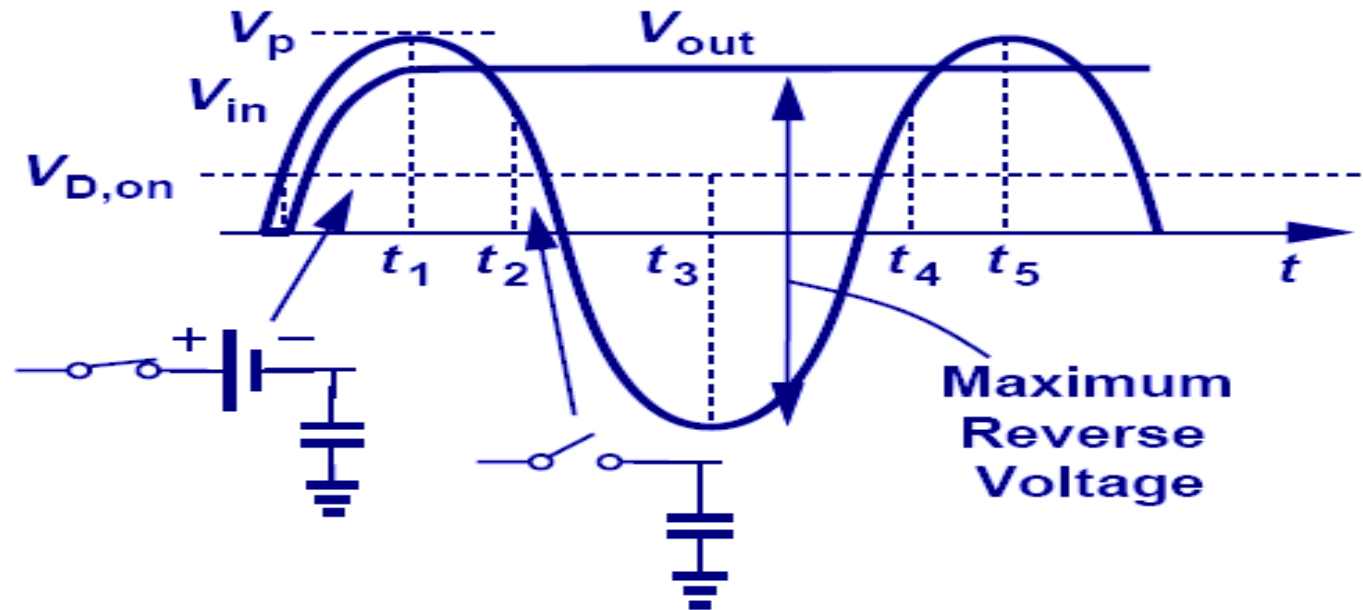
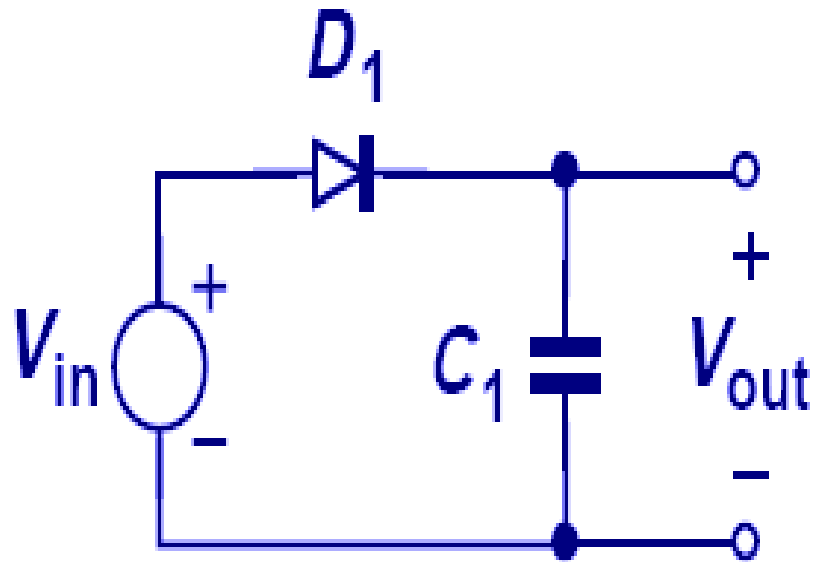


$$V(t) = V_0 + V_p \cos \omega t$$



$$I_D(t) = I_0 + I_p \cos \omega t = I_s \exp \frac{V_0}{V_T} + \frac{V_T}{I_0} V_p \cos \omega t$$

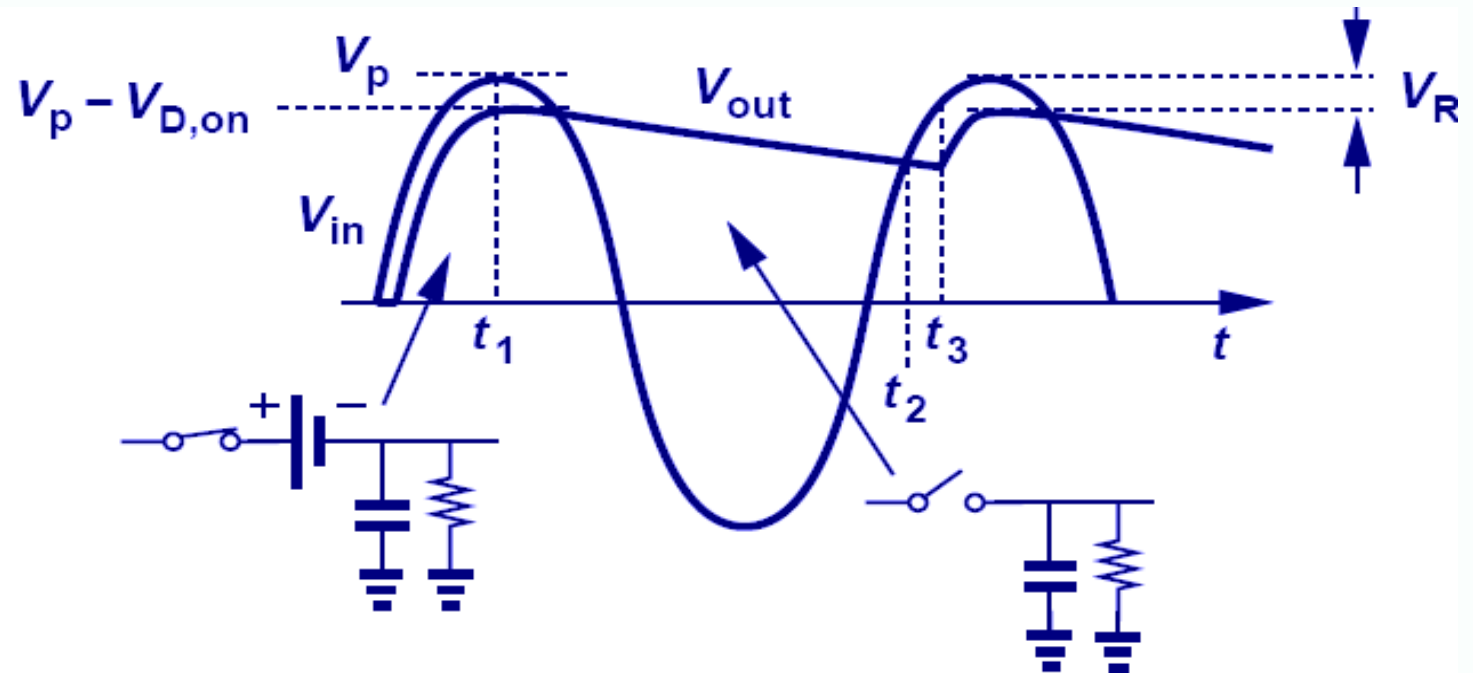
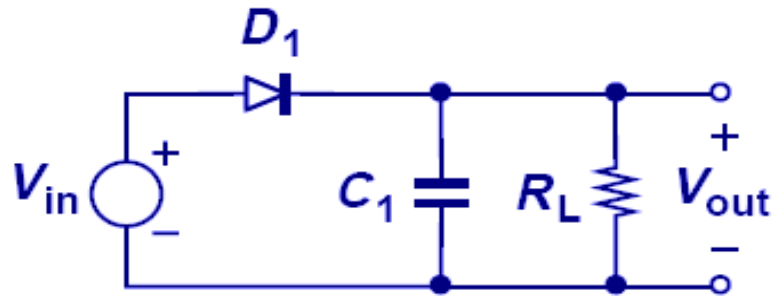
Diode-Capacitor Circuit: Constant Voltage Model



- If the resistor in half-wave rectifier is replaced by a capacitor, a fixed voltage output is obtained since the capacitor (assumed ideal) has no path to discharge.

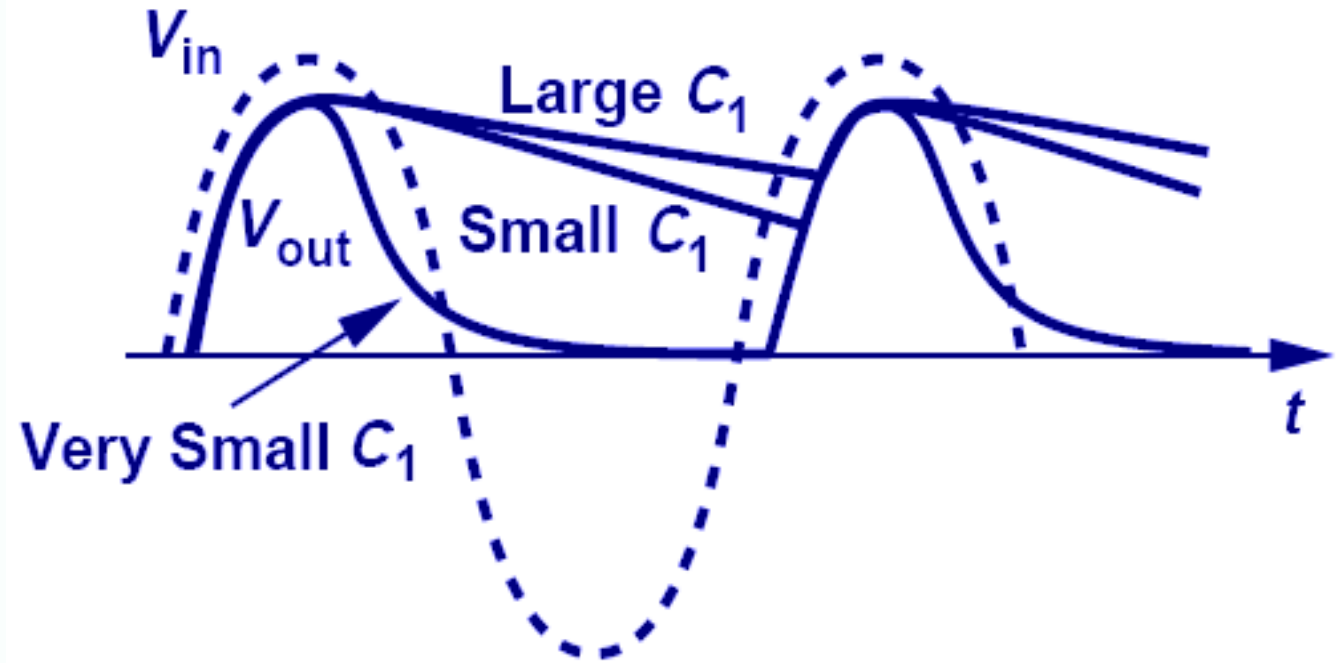
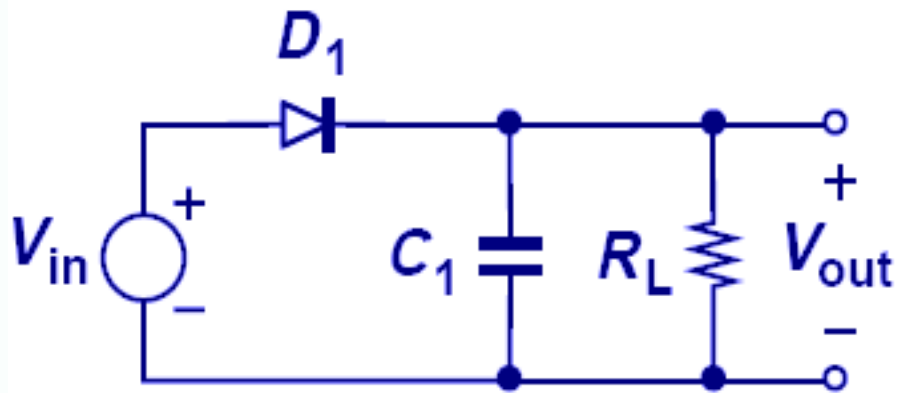
Diode-Capacitor With Load Resistor

- A path is available for capacitor to discharge.
- Therefore, V_{out} will not be constant and a ripple exists.



Behavior for Different Capacitor Values

- For large C_1 , V_{out} has small ripple.



Peak to Peak Amplitude of Ripple

$$V_{out}(t) = (V_p - V_{D,on}) \exp \frac{-t}{R_L C_1} \quad 0 \leq t \leq T_{in}$$

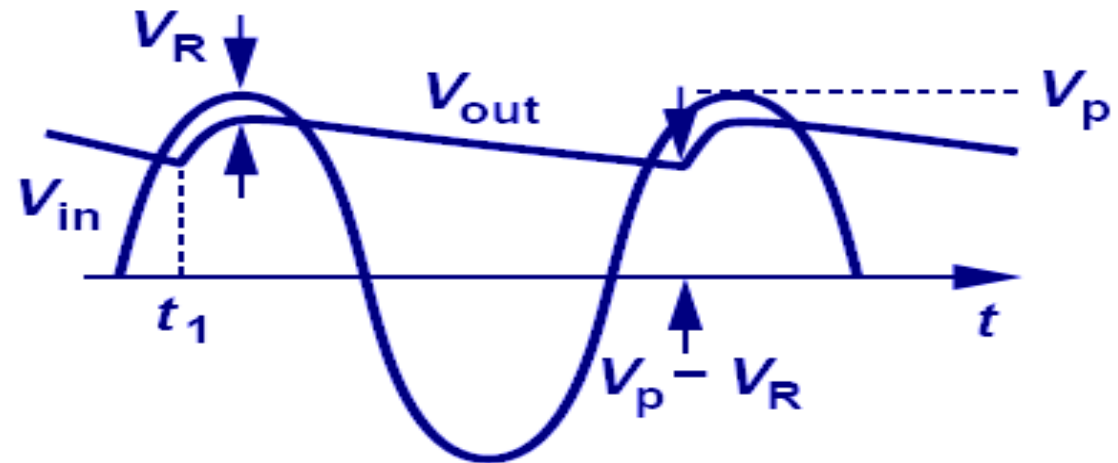
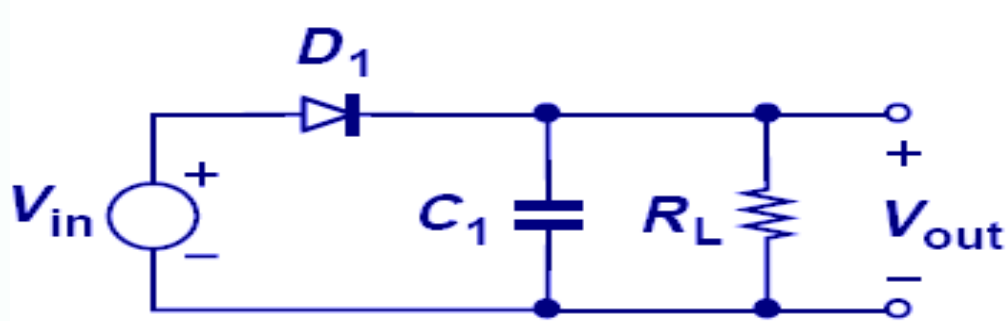
$$V_{out}(t) \approx (V_p - V_{D,on}) \left(1 - \frac{t}{R_L C_1}\right) \approx (V_p - V_{D,on}) - \frac{V_p - V_{D,on}}{R_L} \frac{t}{C_1}$$

$$V_R \approx \frac{V_p - V_{D,on}}{R_L} \cdot \frac{T_{in}}{C_1} \approx \frac{V_p - V_{D,on}}{R_L C_1 f_{in}}$$

- The ripple amplitude is the decaying part of the exponential.
- Ripple voltage becomes a problem if it goes above 5 to 10% of the output voltage.

Maximum Diode Current

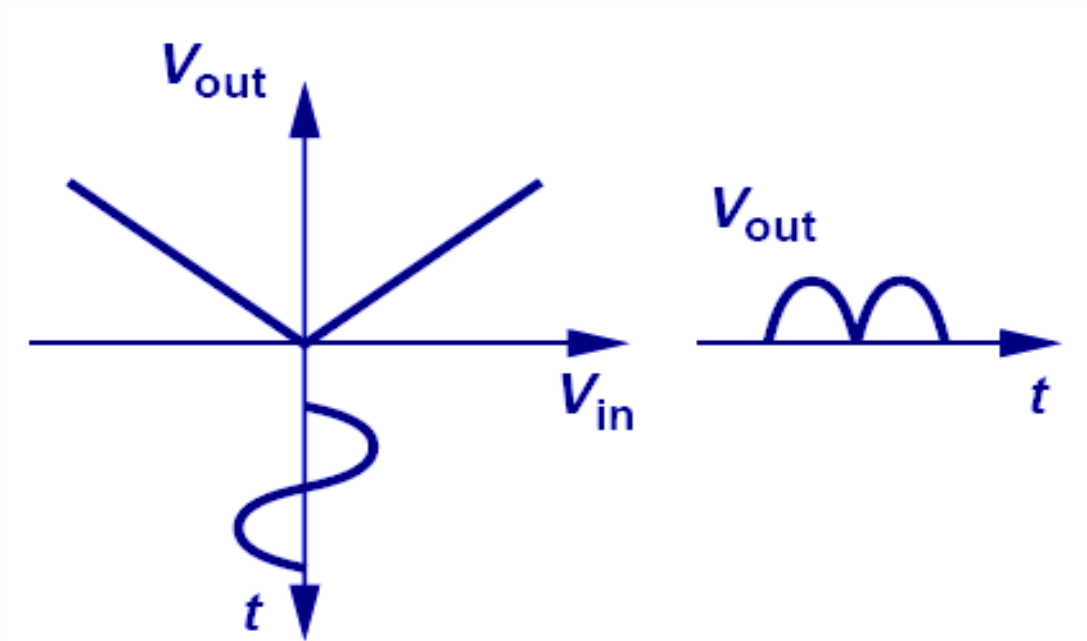
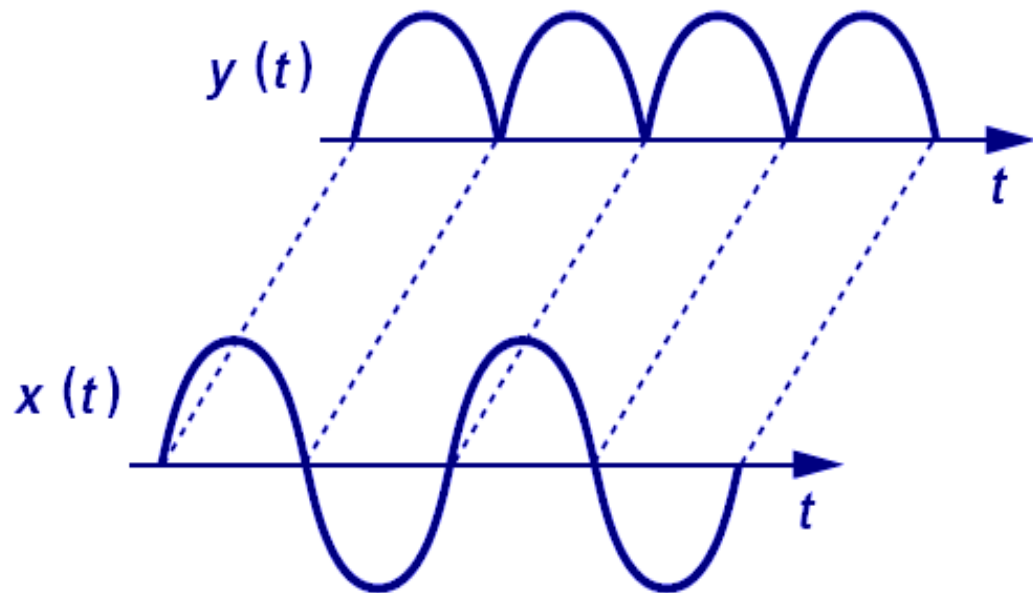
- The diode has its maximum current at t_1 , since that's when the slope of V_{out} is the greatest.
- This current has to be carefully controlled so it does not damage the device.



$$I_p \approx C_1 \omega_{in} V_p \sqrt{\frac{2V_R}{V_p} + \frac{V_p}{R_L}} \approx \frac{V_p}{R_L} (R_L C_1 \omega_{in} \sqrt{\frac{2V_R}{V_p} + 1})$$

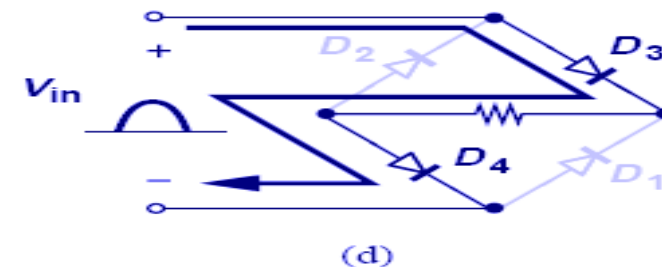
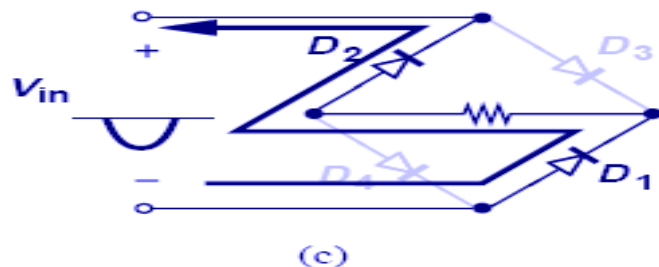
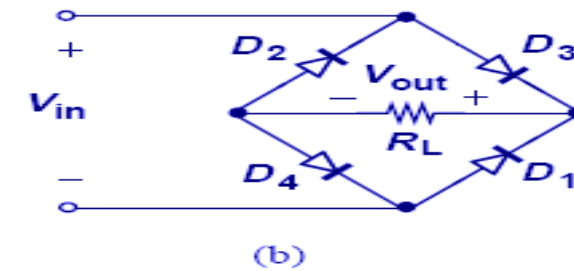
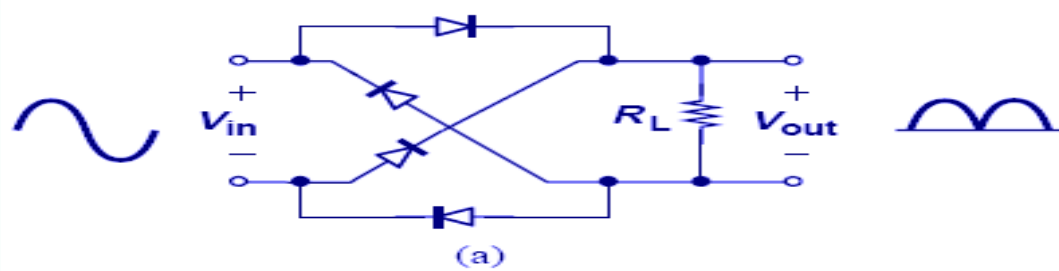
Full-Wave Rectifier

- A full-wave rectifier passes both the negative and positive half cycles of the input, while inverting the negative half of the input.
- A full-wave rectifier reduces the ripple by a factor of two.



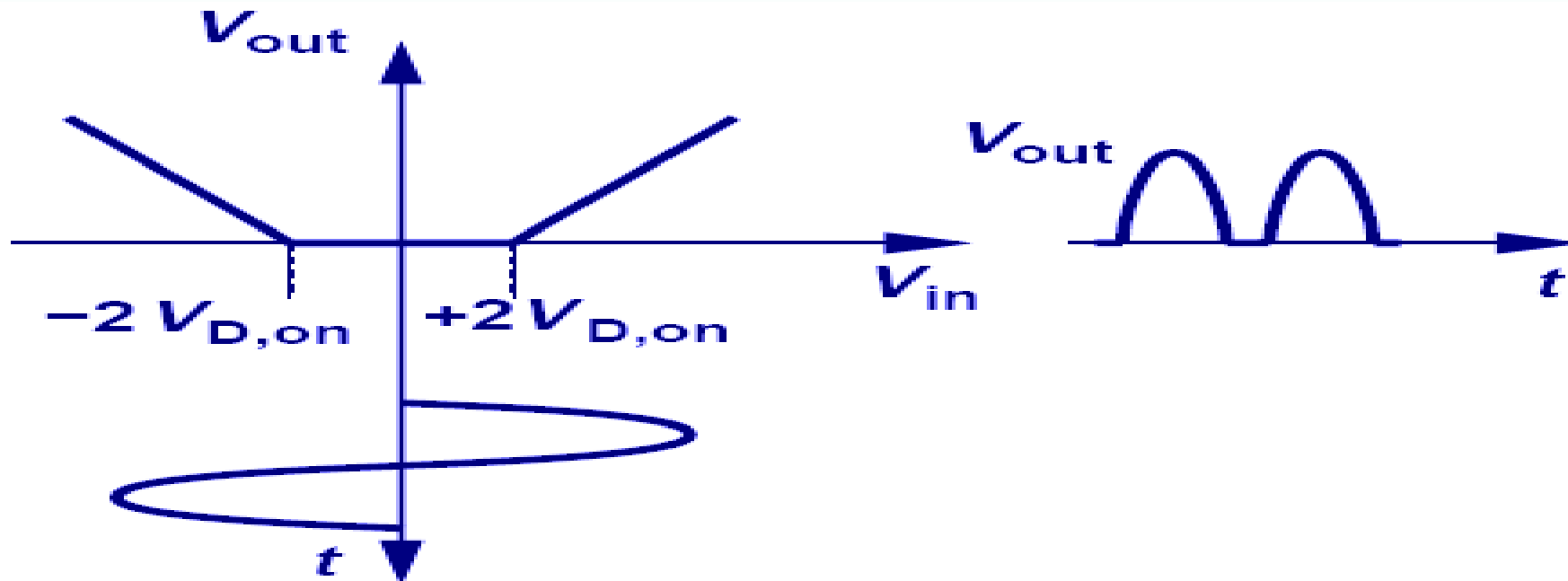
Full-Wave Rectifier: Bridge Rectifier

- The figure above shows a full-wave rectifier, where D_1 and D_2 pass/invert the negative half cycle of input and D_3 and D_4 pass the positive half cycle.



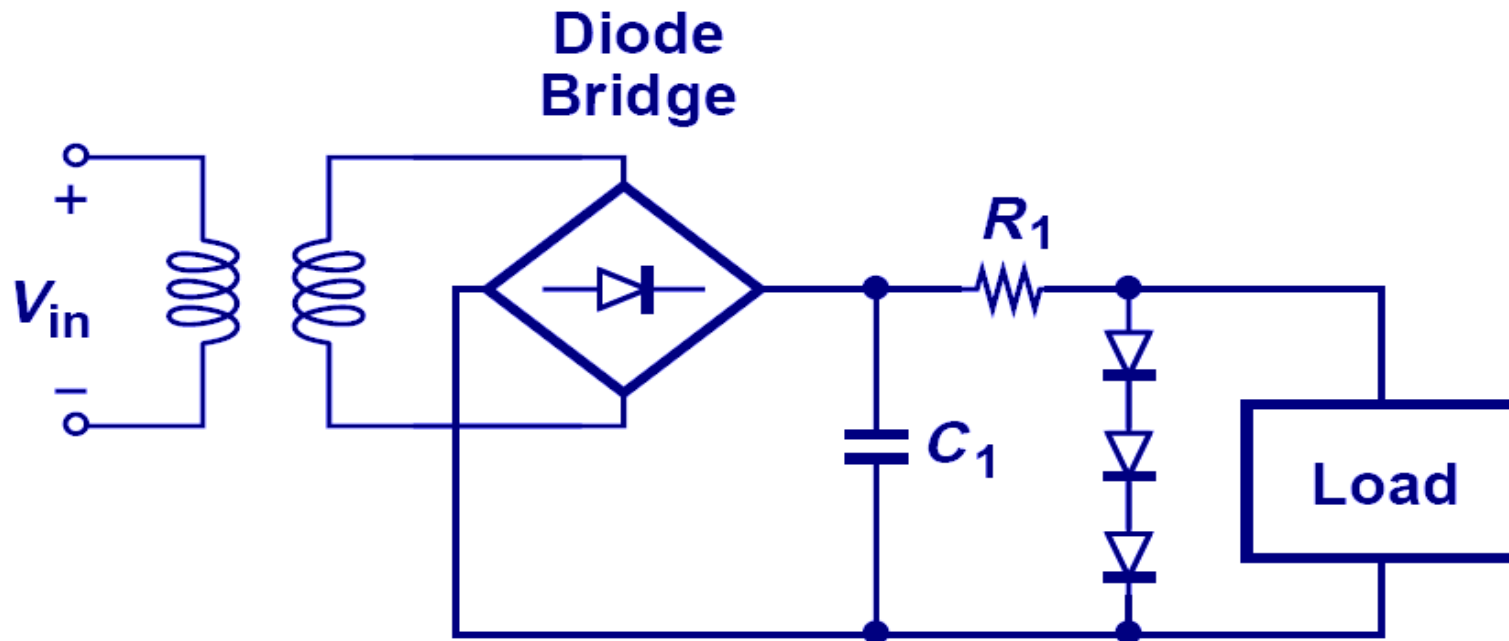
Input/Output Characteristics of a Full-Wave Rectifier (Constant-Voltage Model)

- The dead-zone around V_{in} arises because V_{in} must exceed $2 V_{D,ON}$ to turn on the bridge.



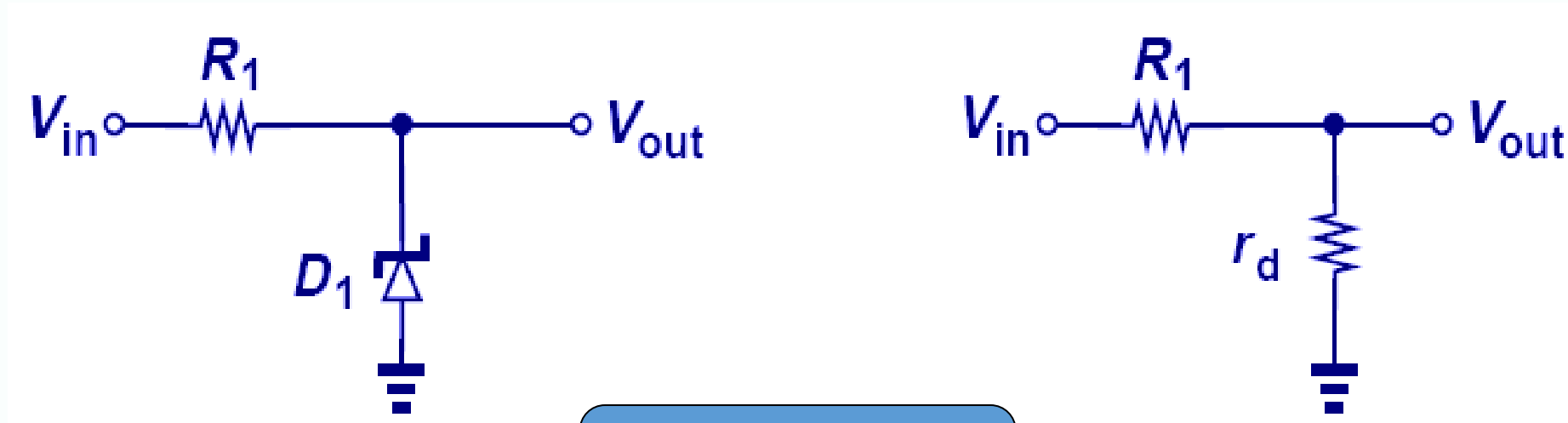
Voltage Regulator

- The ripple created by the rectifier can be unacceptable to sensitive load; therefore, a regulator is required to obtain a very stable output.
- Three diodes operate as a primitive regulator.



Voltage Regulation With Zener Diode

- Voltage regulation can be accomplished with Zener diode.

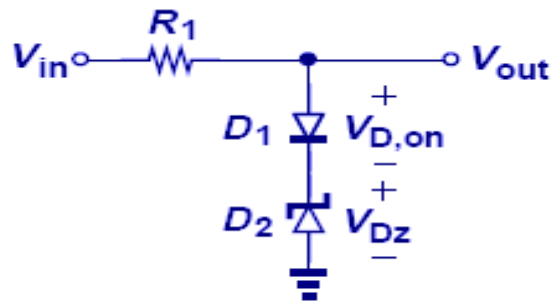


$$V_{out} = \frac{r_D}{r_D + R_1} V_{in}$$

Since r_d is small, large change in the input will not be reflected at the output.

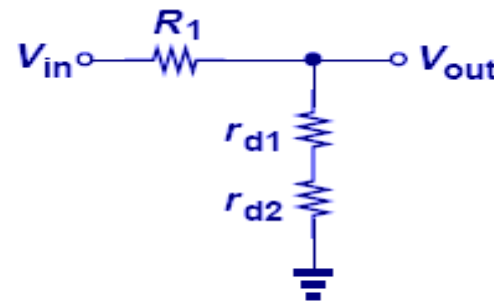
Line Regulation VS. Load Regulation

- Line regulation is the suppression of change in V_{out} due to change in V_{in} (b).

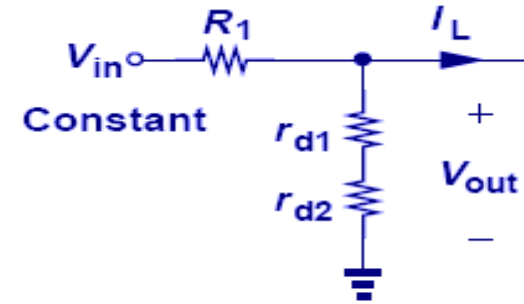


(a)

$$\frac{V_{out}}{V_{in}} = \frac{r_{D1} + r_{D2}}{r_{D1} + r_{D2} + R_1}$$



(b)



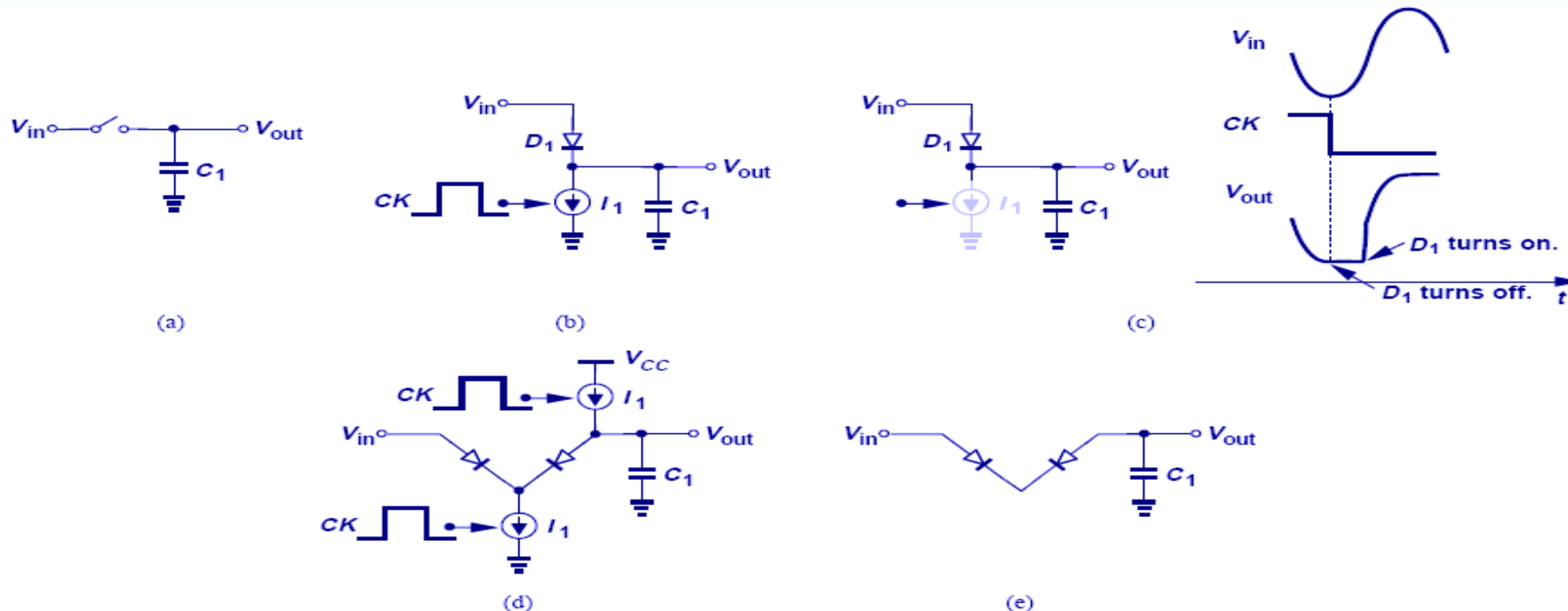
(c)

$$\left| \frac{V_{out}}{I_L} \right| = (r_{D1} + r_{D2}) \parallel R_1$$

Load regulation is the suppression of change in V_{out} due to change in load current (c).

Diode as Electronic Switch

- Diode as a switch finds application in logic circuits and data converters.





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Thank You

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COE 271 Semiconductor Devices

Lecture 4

Bipolar Junction Transistors BJT

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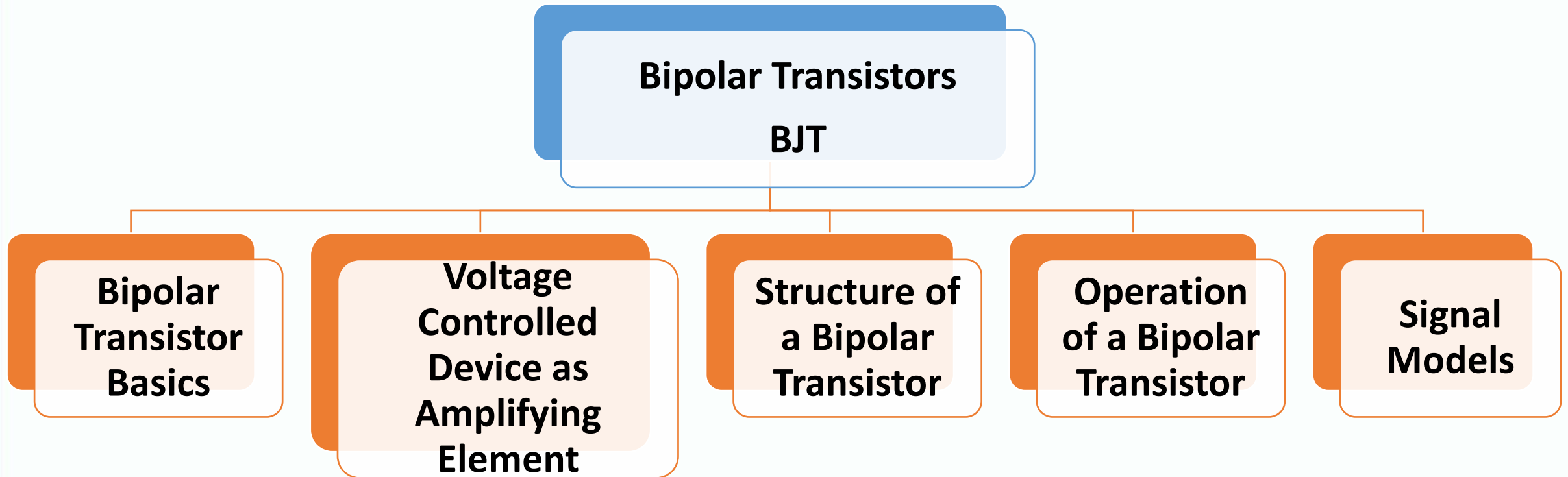
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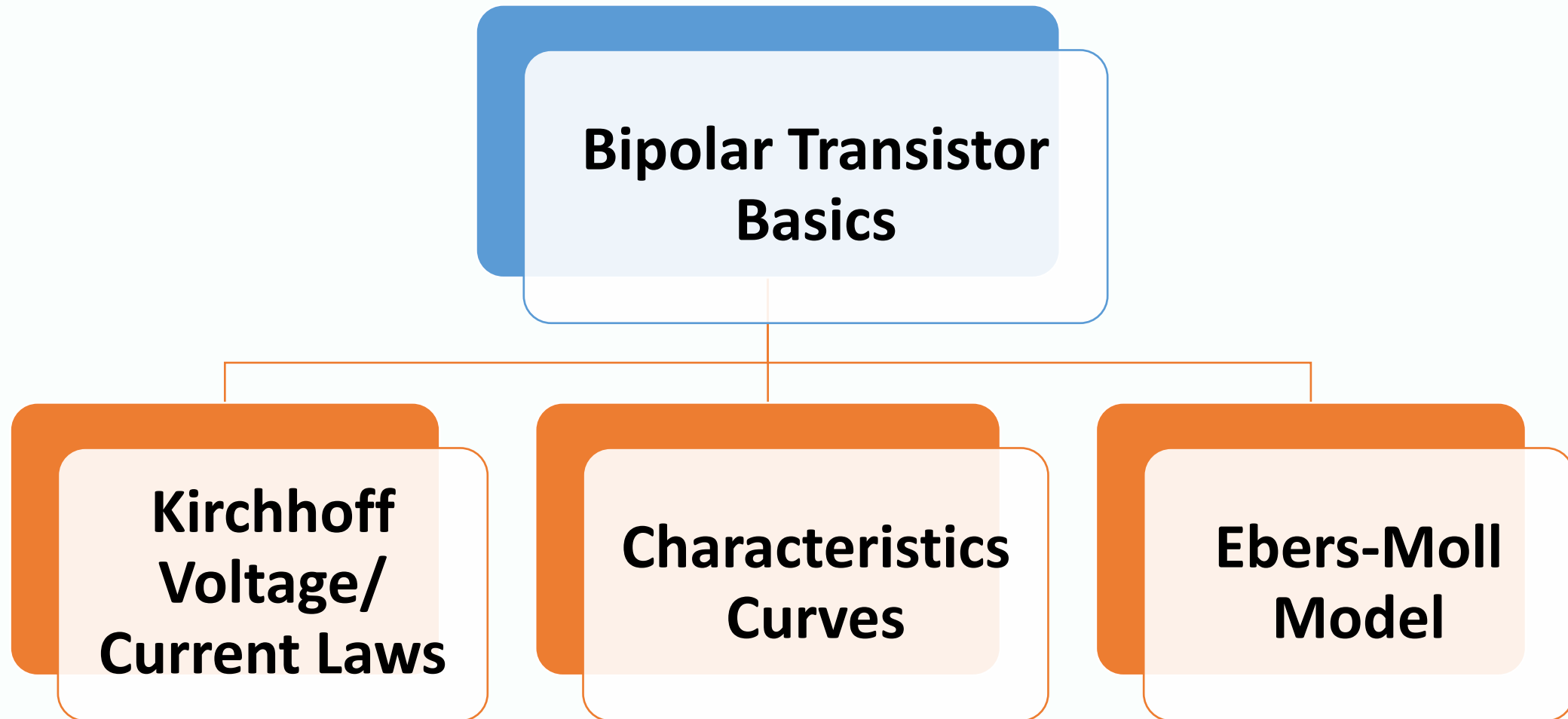
“A person starts to live when he can live
outside himself”

Albert Einstein

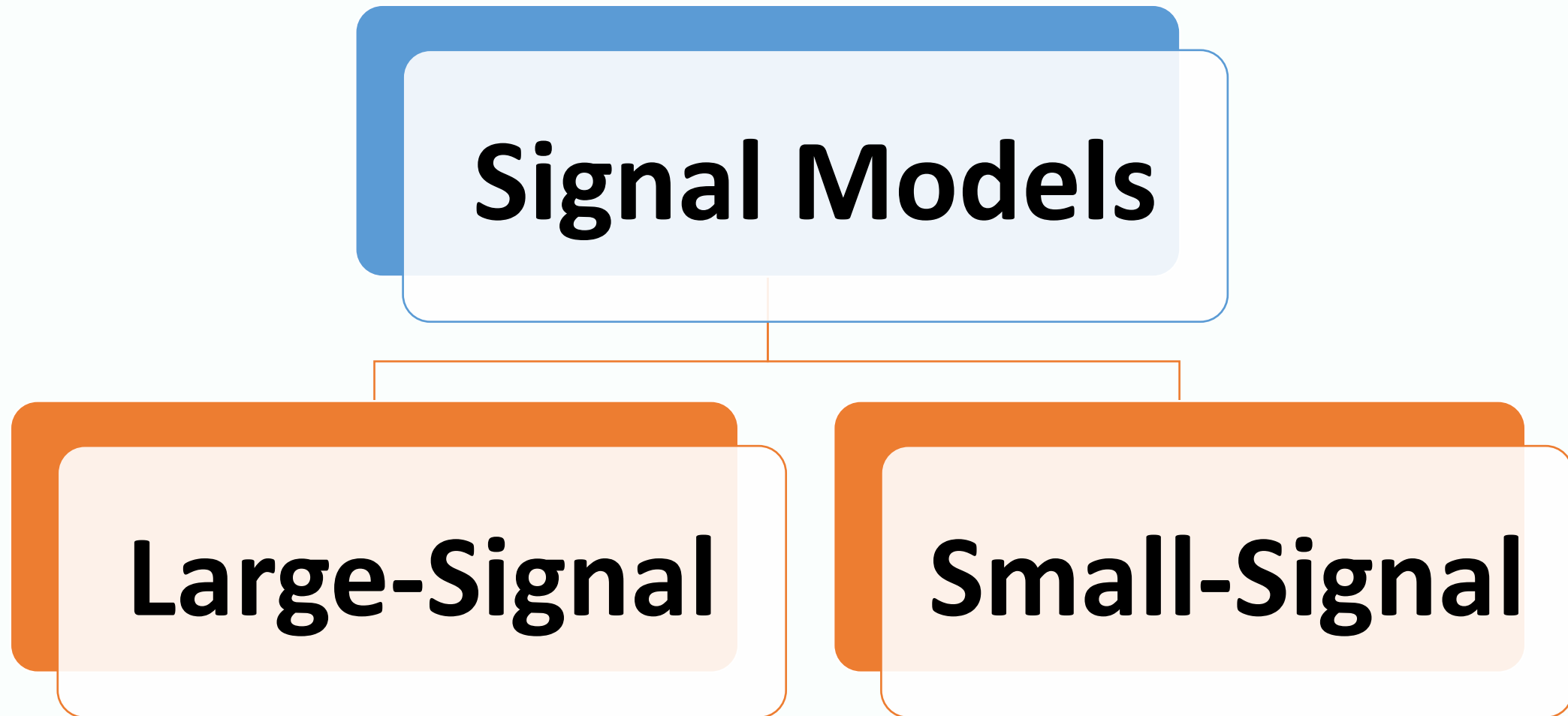
Overview



Overview



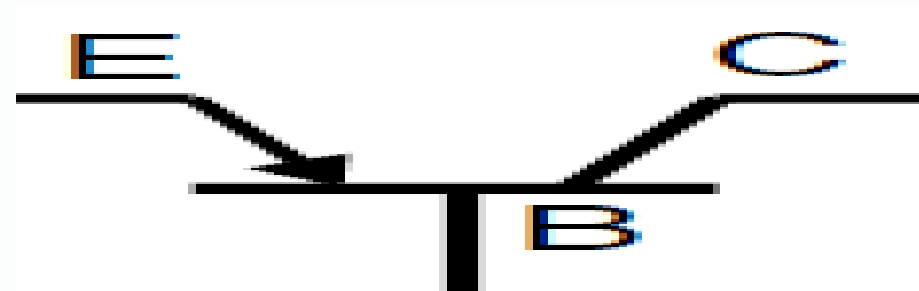
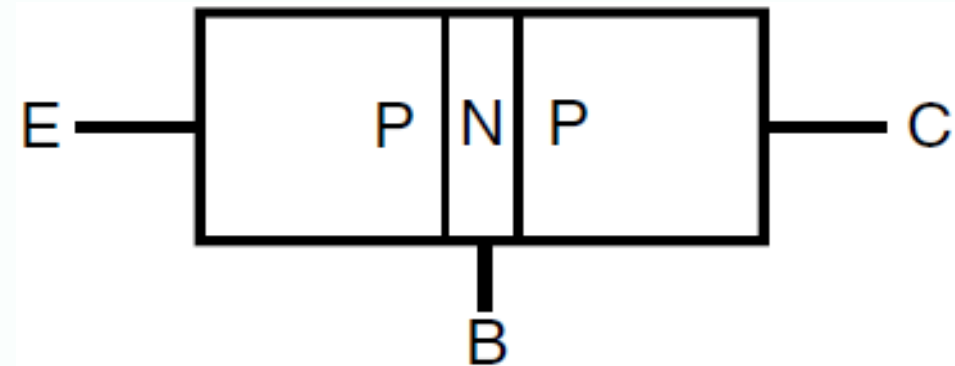
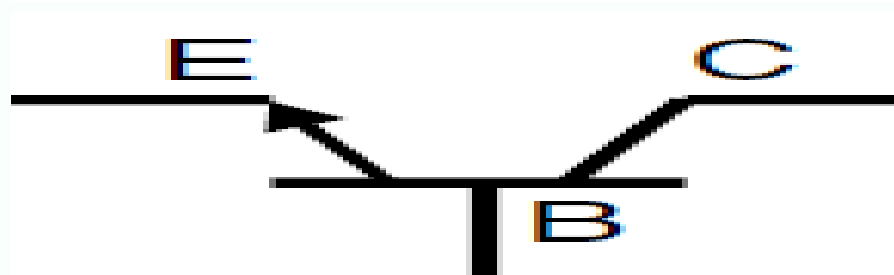
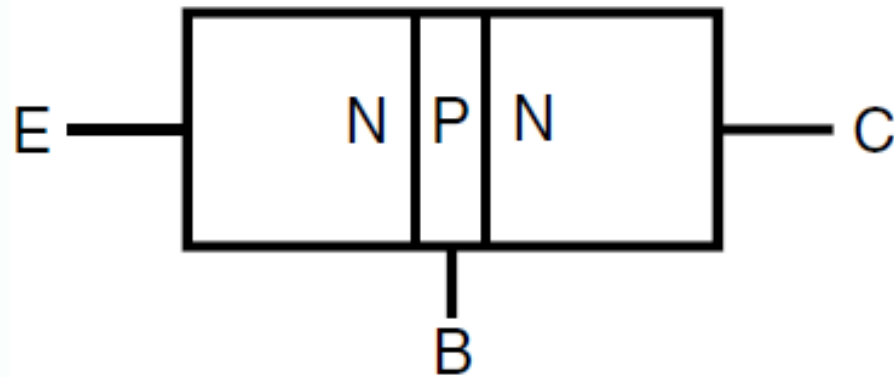
Overview



Bipolar Transistors

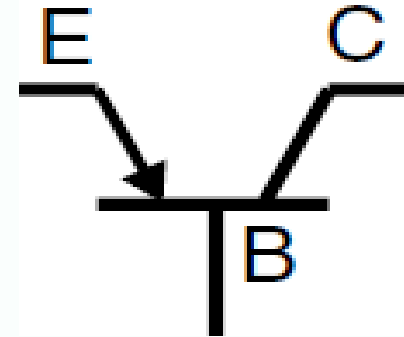
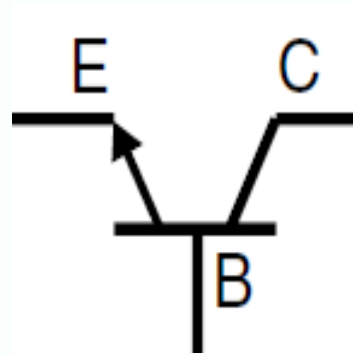
- A bipolar junction transistor consists of two PN junctions sandwiched very close together within a single crystal of semiconductor
- The region common to the two junctions, called the *base*, may be of either N-type or P-type material.
- This thin region is surrounded by material of the opposite type, in regions known as the *emitter* and *collector*.

Bipolar Transistors



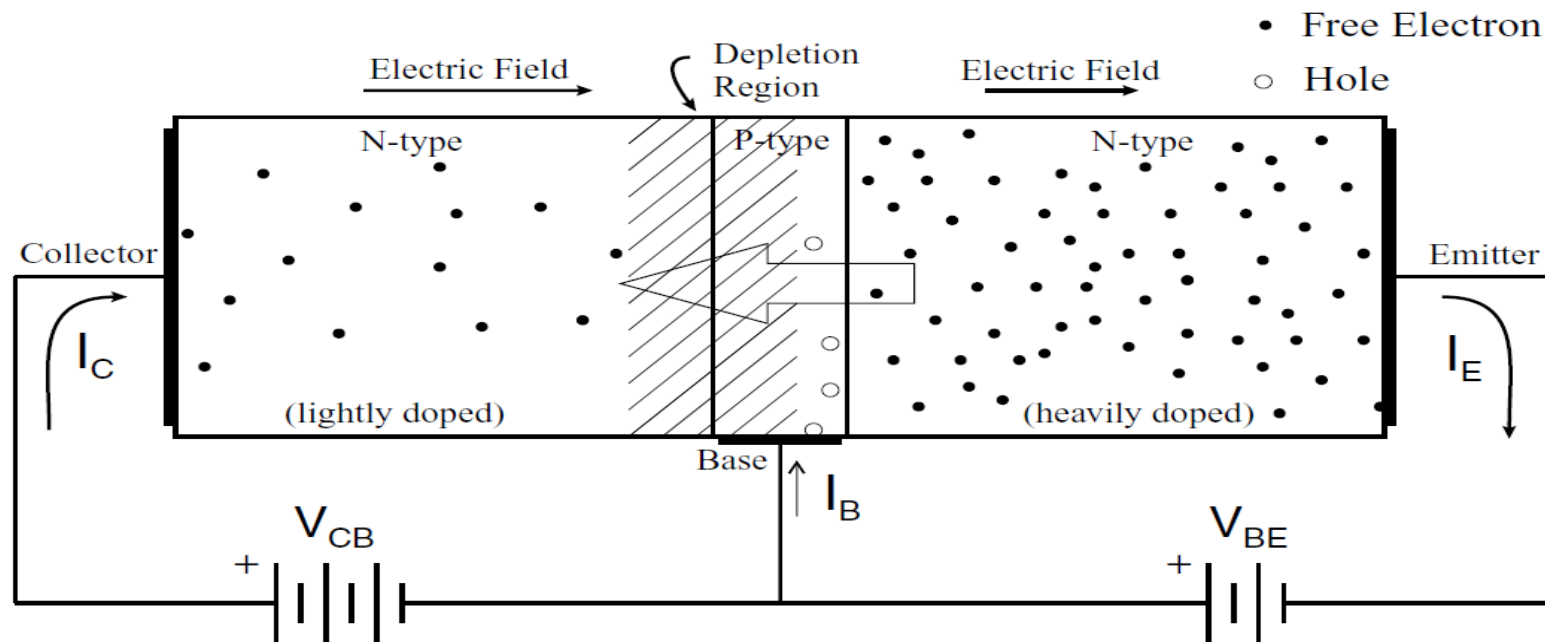
The arrow on the emitter lead points in the direction of positive current flow. You can tell whether a transistor in a schematic diagram is PNP or NPN by the direction of the arrow

Bipolar Transistors



- A small current flowing *out* from the base controls a large current flowing *out* from the collector, with both currents flowing *in* through the emitter.
- The ratio of collector current to base current is called β (or h_{fe}) (range 20 to 300)

Bipolar Transistors



$$\beta = hfe = \frac{I_c}{I_b}$$

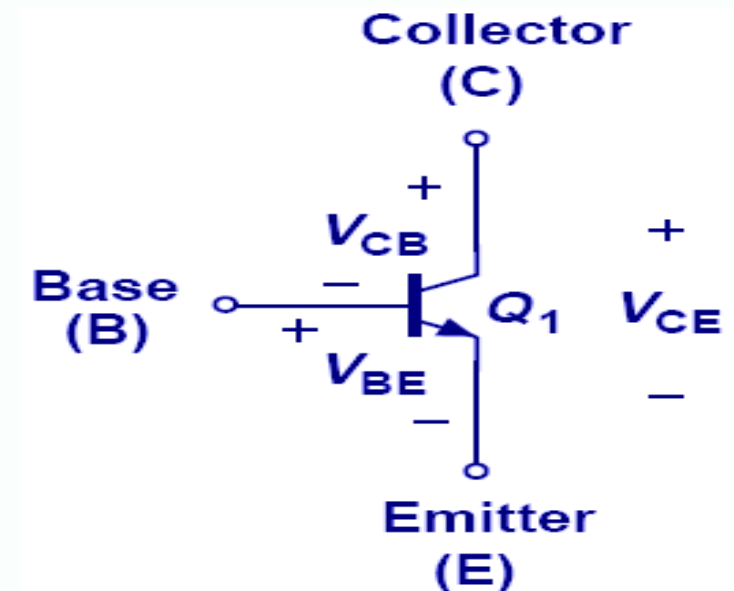
- A transistor is a voltage-controlled current source:
- small changes in the base voltage cause large changes in collector current.

Bipolar Transistors

- $V_{BE} \equiv V_B - V_E$ = potential of base relative to emitter,
- $V_{CB} \equiv V_C - V_B$ = potential of collector relative to base
- $V_{CE} \equiv V_C - V_E$ = potential of collector relative to emitter,
-

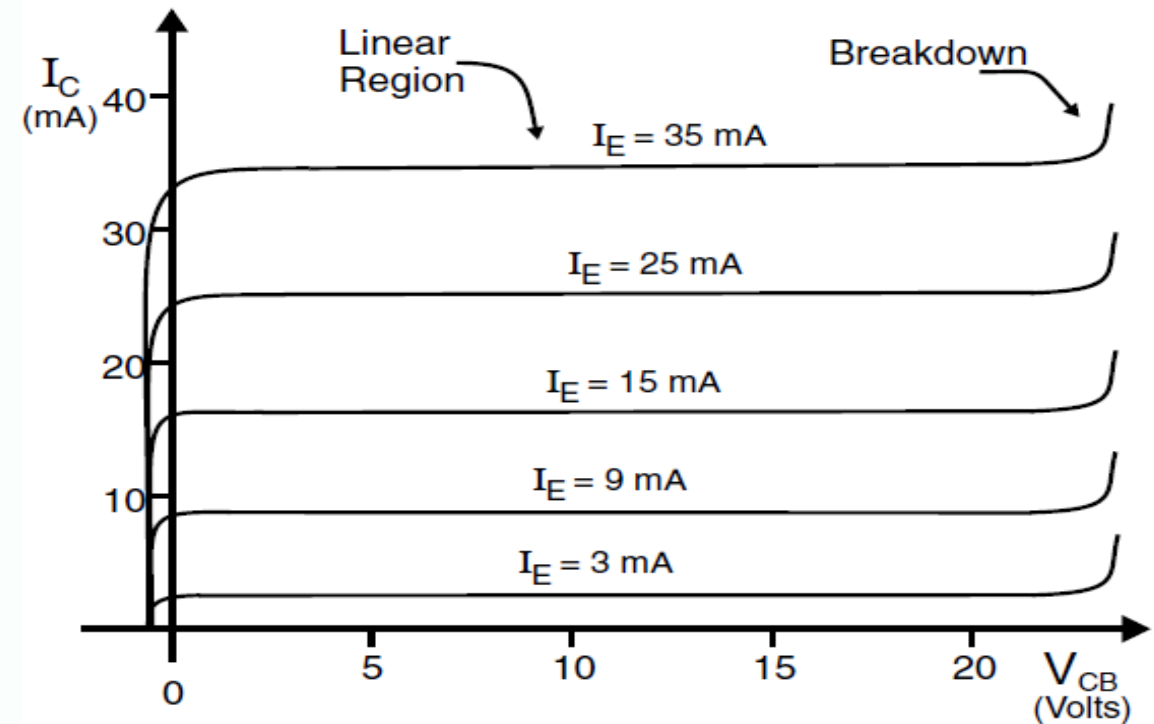
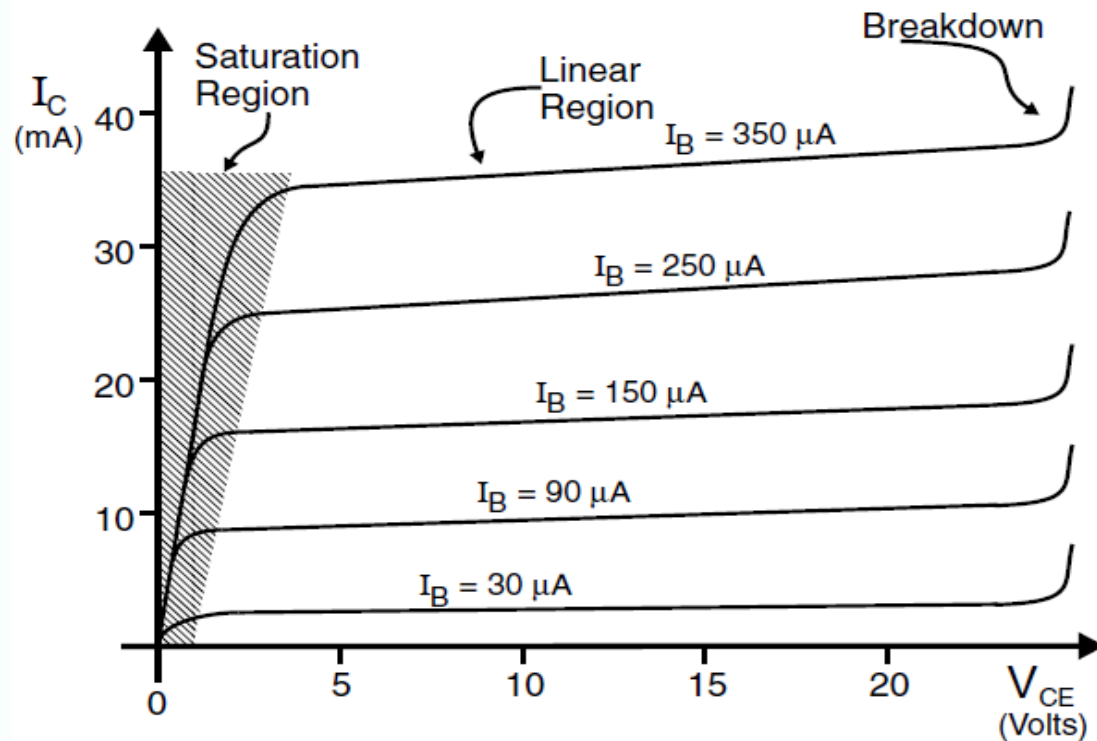
• **Kirchhoff's voltage law:** $V_{CE} = V_{BE} + V_{CB}$.

- I_C = current flowing into collector,
- I_E = current flowing out of emitter,
- I_B = current flowing into base,
- **Kirchhoff's current law:** $I_E = I_C + I_B$.



Bipolar Transistors

- The relationships between these voltages and currents are usually expressed in terms of characteristic curves

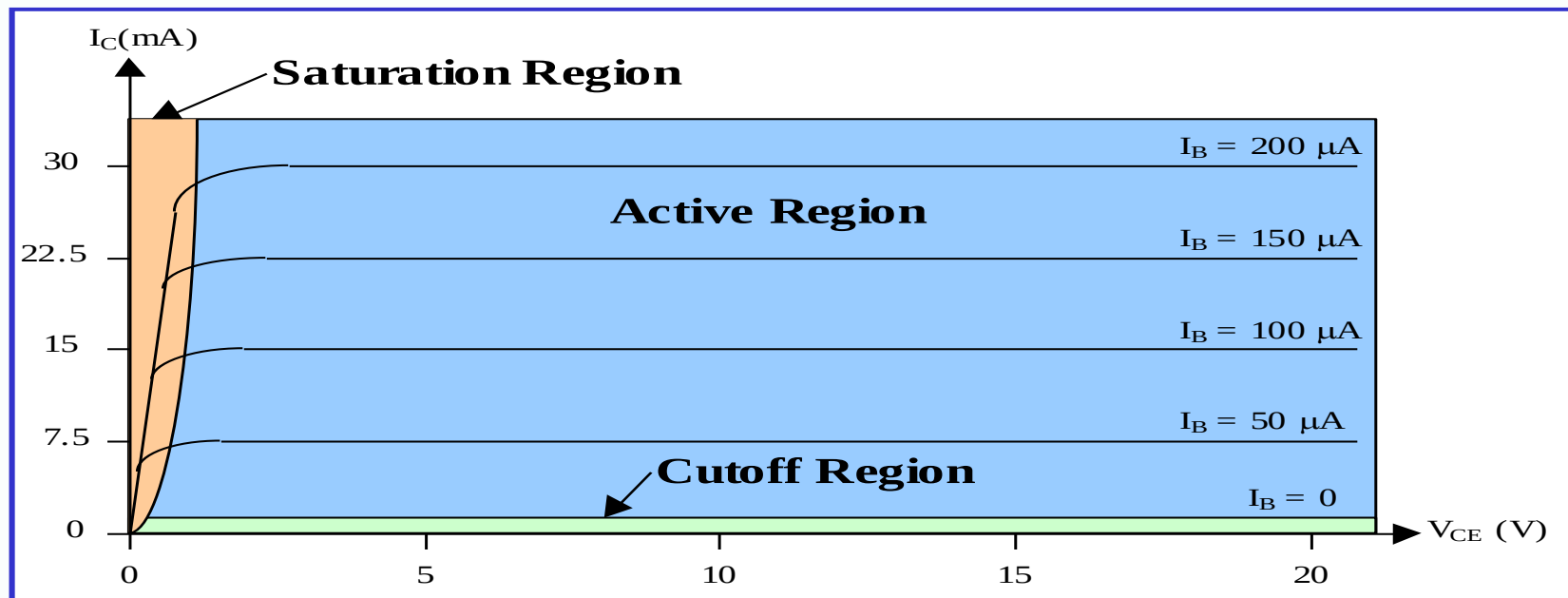


Characteristic curves for an npn bipolar transistor

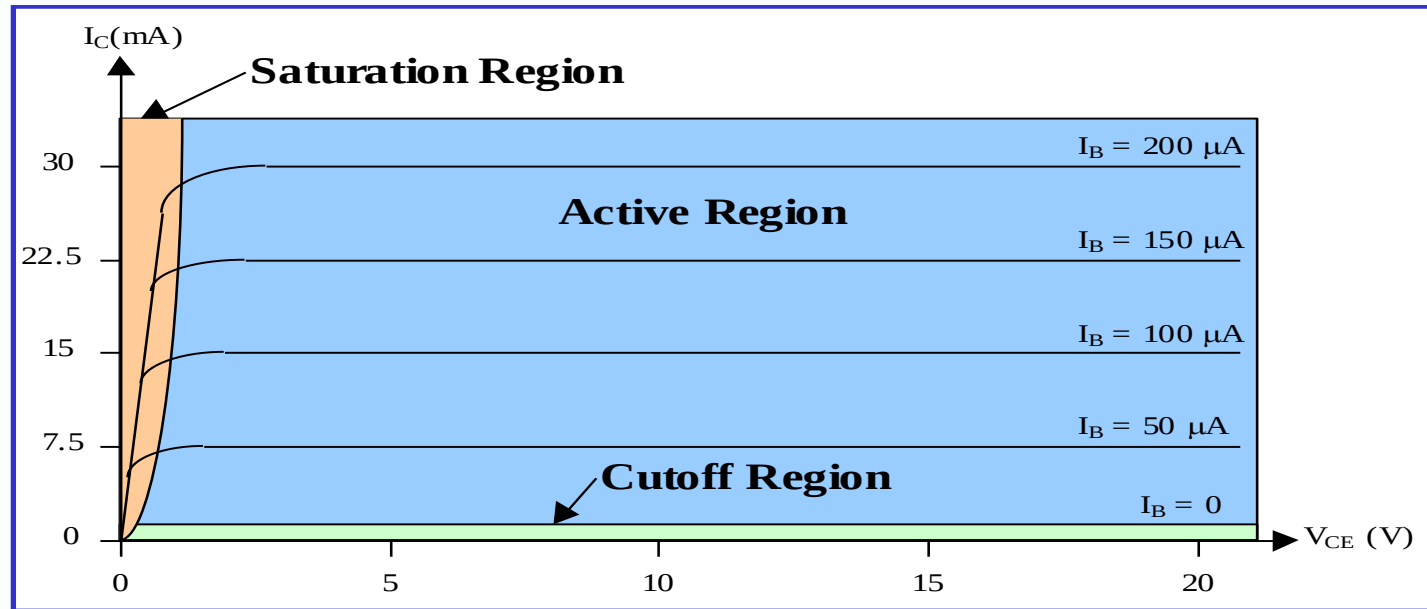
Regions (Modes) of Operation of BJT

Active:

- **Most important mode of operation**
- **Central to amplifier operation**
- **The region where current curves are practically flat**



Regions (Modes) of Operation of BJT



Saturation:

- Barrier potential of the junctions cancel each other out causing a virtual short (behaves as on state Switch)

Cutoff:

- Current reduced to zero
- Ideal transistor behaves like an open switch

Ebers-Moll Transistor Model

- In the Ebers-Moll model, the amount of collector current that flows is determined by the amount of forward-bias that is applied to the base-emitter diode.

$$I_C = I_s \left(e^{eV_{BE}/kT} - 1 \right)$$

I_s is the reverse saturation current

The base current is related to the collector current by

$$I_B = \frac{I_C}{\beta}.$$

Ebers-Moll Transistor Model

- **Dynamic resistance of emitter**
- The dynamic resistance tells us how, for fixed base voltage, the emitter voltage would change in response to a change in emitter current
- – for example, how the emitter voltage would differ if the emitter were driving a small load resistance as opposed to a large one

$$r_e \equiv \frac{\partial V_E}{\partial I_E} \quad r_e = \frac{kT}{e} \frac{1}{I_C}$$

Ebers-Moll Transistor Model

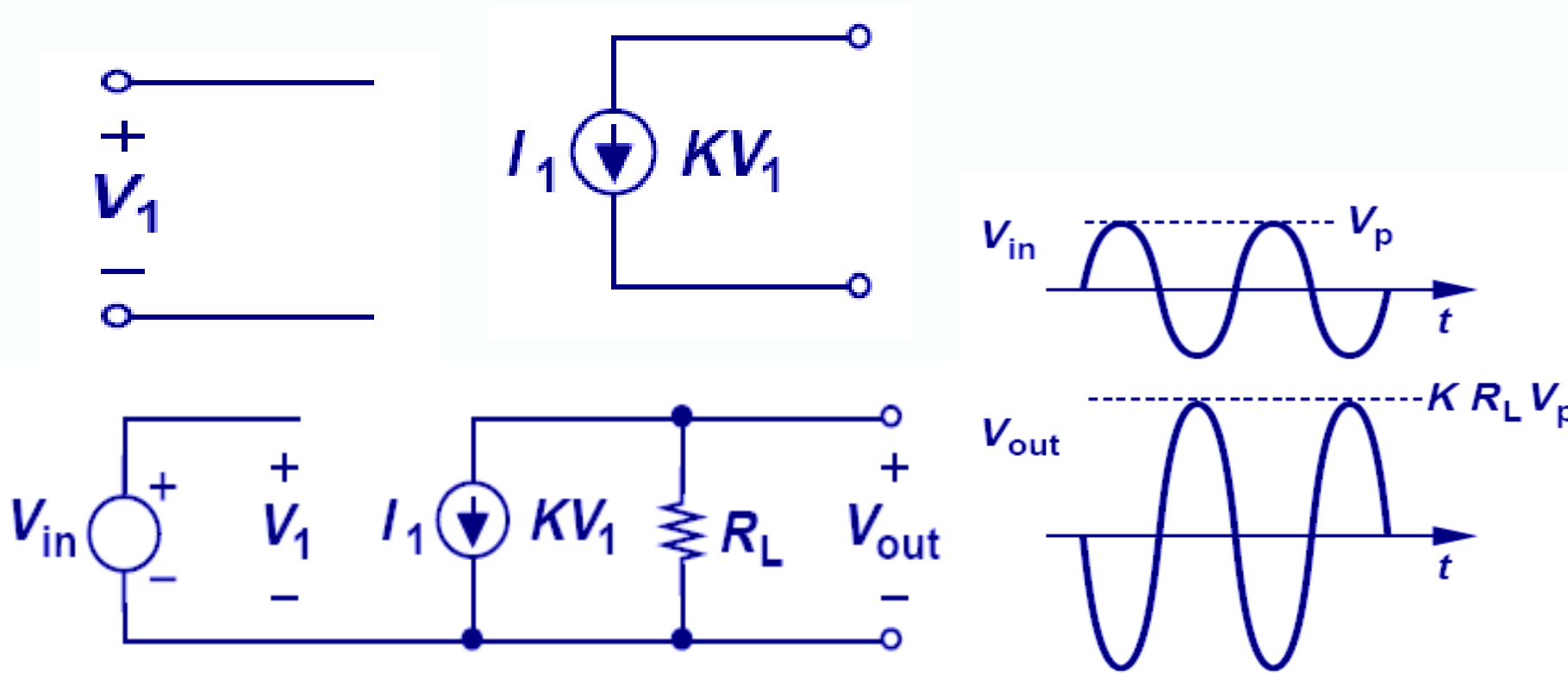
Dynamic resistance of base

The dynamic resistance of the base tells us how, the base loads the current that is driving it

$$r_{BE} = \frac{\partial V_B}{\partial I_B} = \beta \frac{kT}{e} \frac{1}{I_C}$$

Voltage-Dependent Current Source

- A voltage-dependent current source can act as an amplifier.

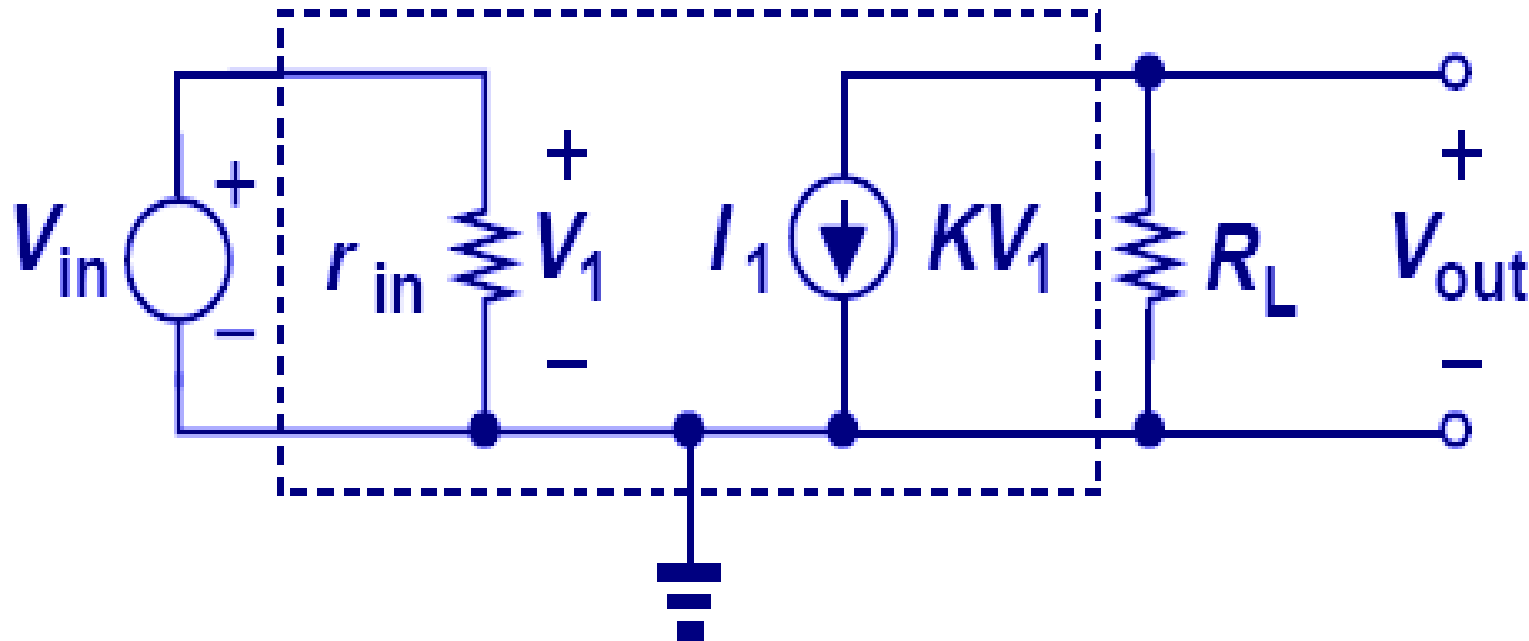


$$A_V = \frac{V_{out}}{V_{in}} = -K R_L$$

If $K R_L$ is greater than 1, then the signal is amplified.

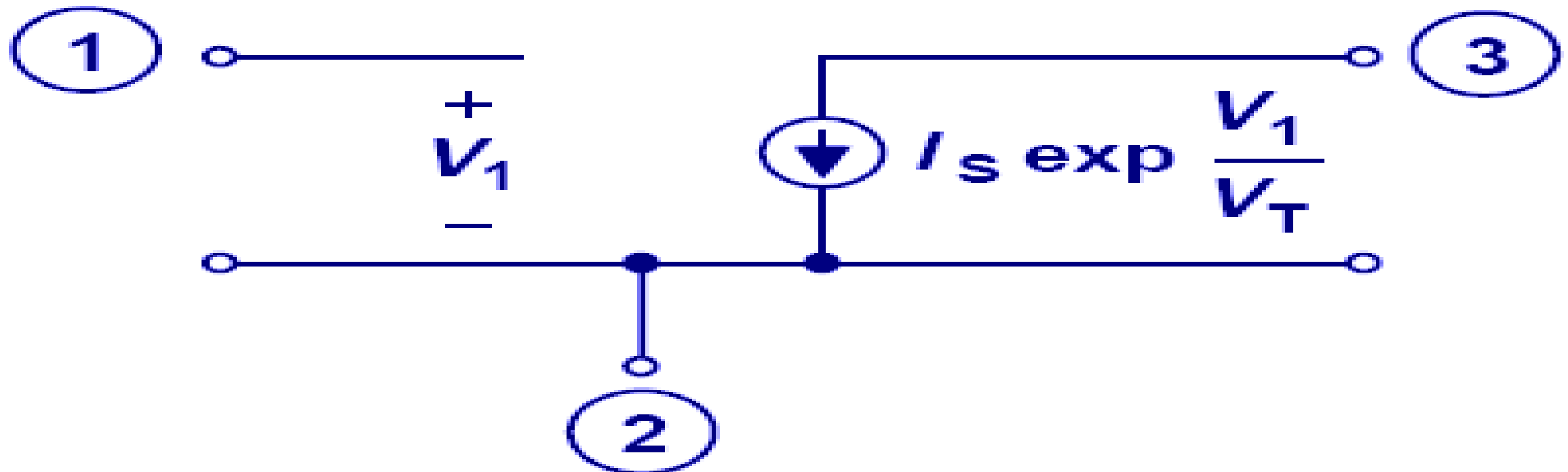
Voltage-Dependent Current Source with Input Resistance

- Regardless of the input resistance, the magnitude of amplification remains unchanged.



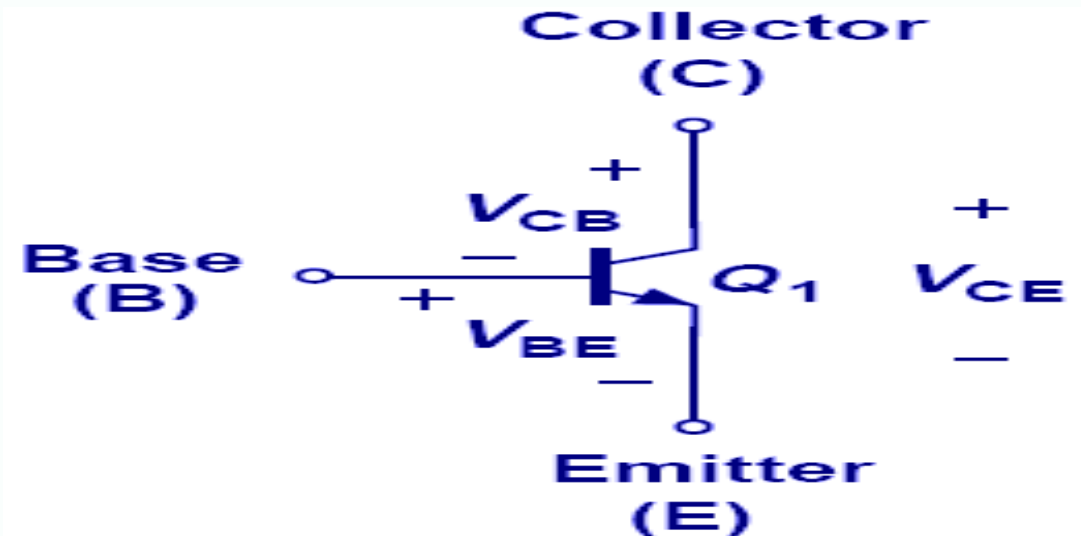
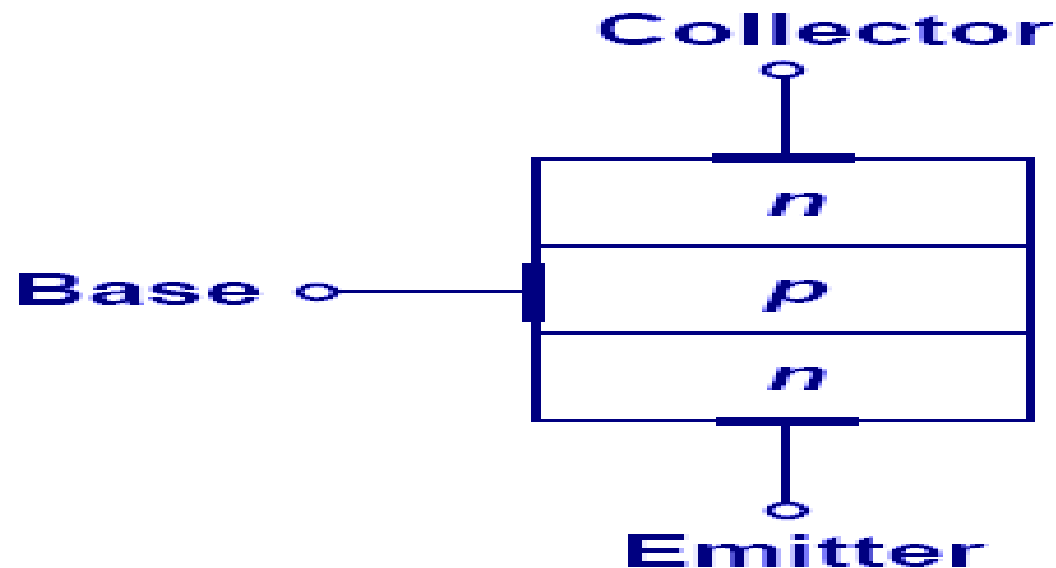
Exponential Voltage-Dependent Current Source

- Ideally, bipolar transistor can be modeled as a three-terminal exponential voltage-dependent current source

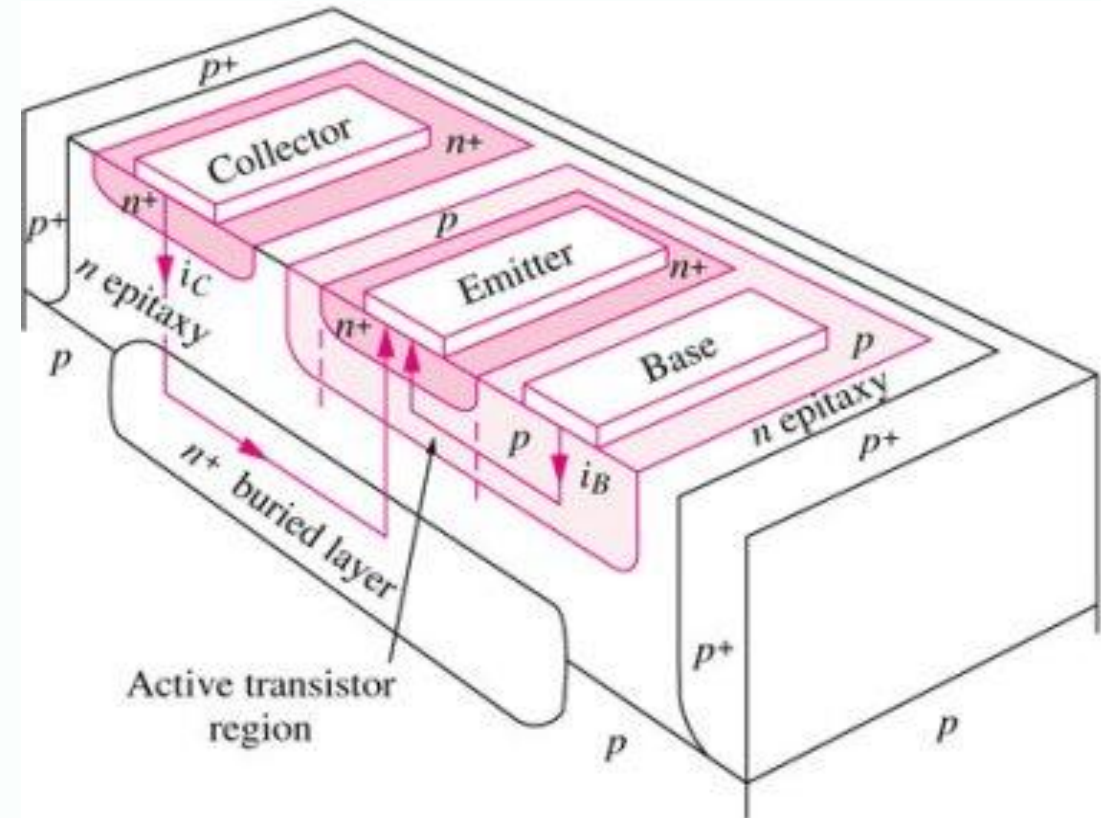
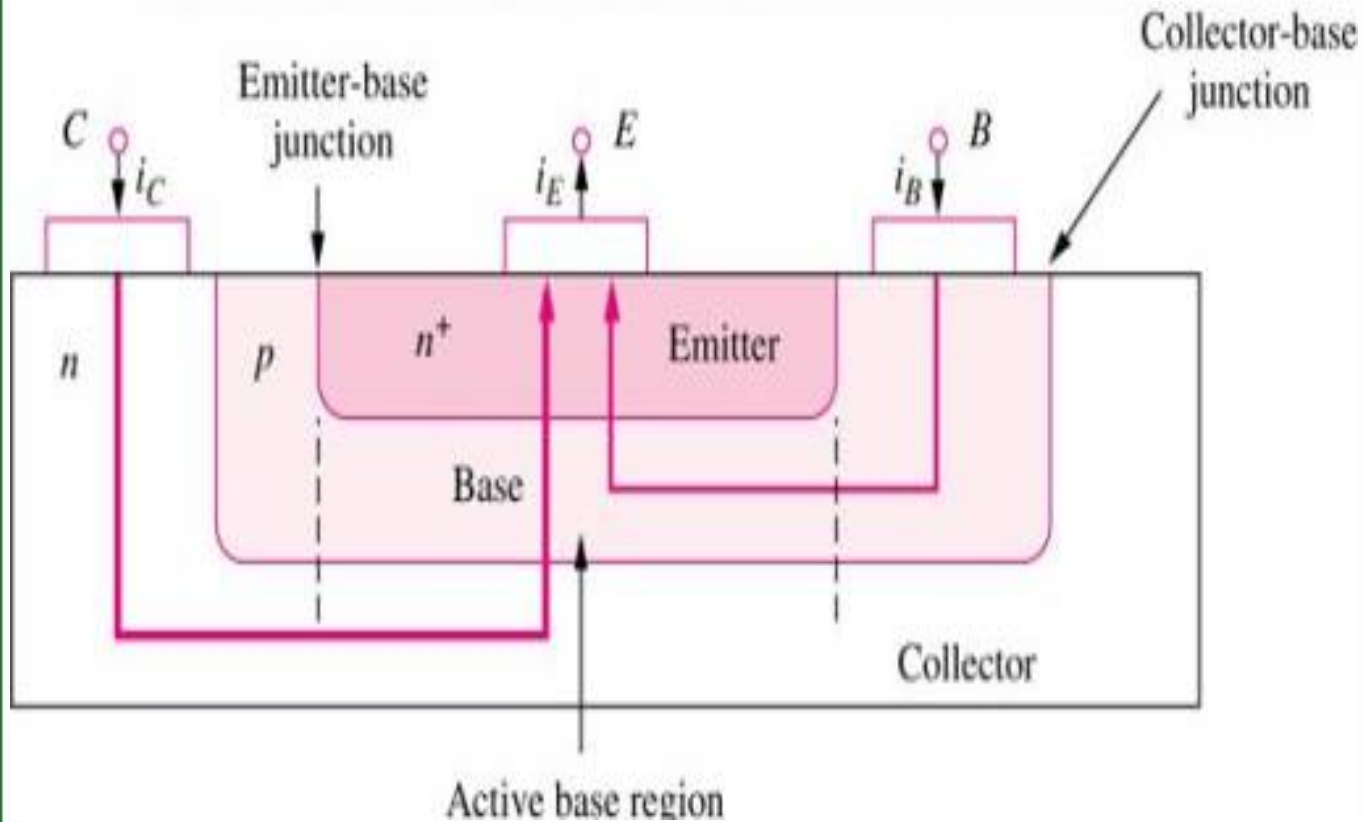


Structure and Symbol of Bipolar Transistor

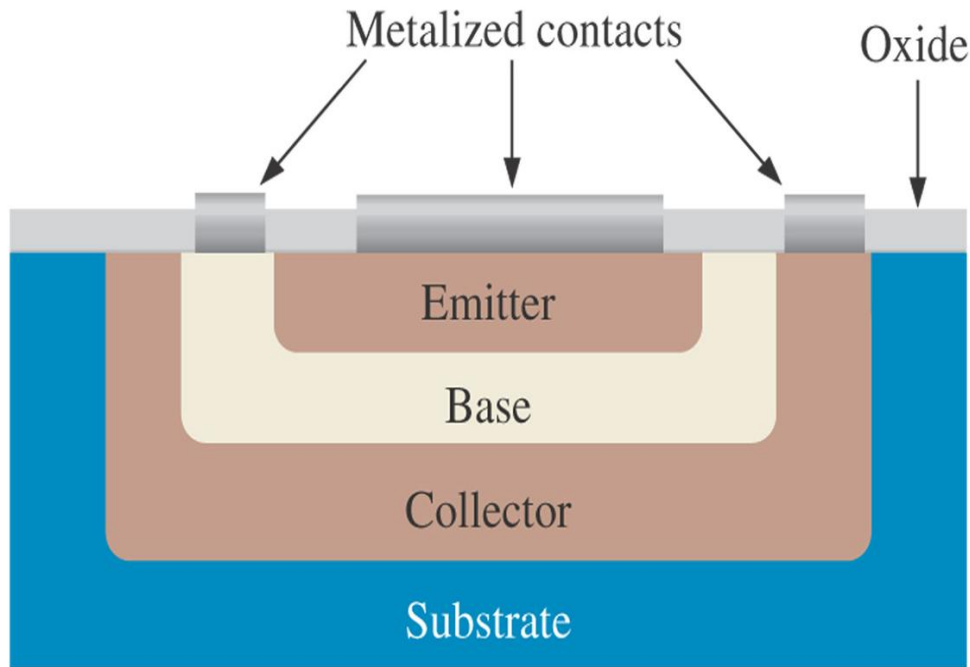
- Bipolar transistor can be thought of as a sandwich of three doped Si regions.
- The outer two regions are doped with the same polarity, while the middle region is doped with opposite polarity.



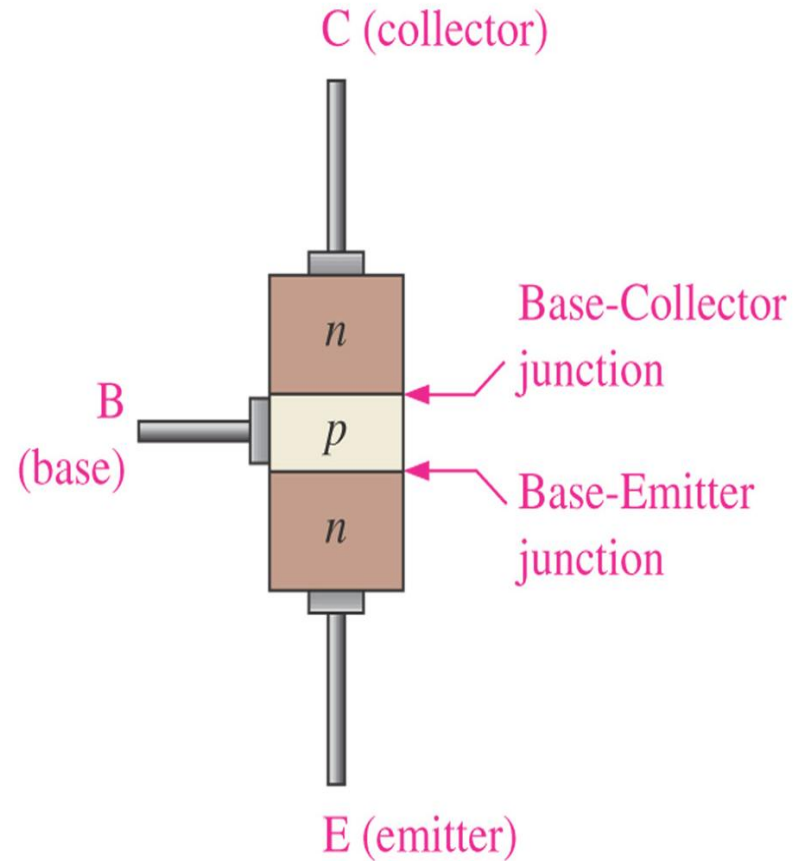
Structure and Symbol of Bipolar Transistor



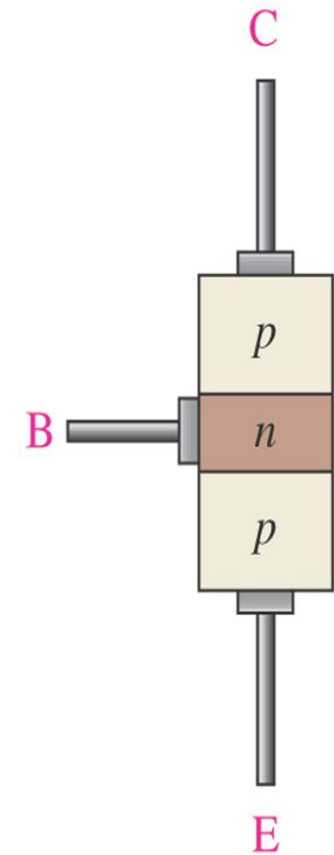
Structure and Symbol of Bipolar Transistor



(a) Basic epitaxial planar structure



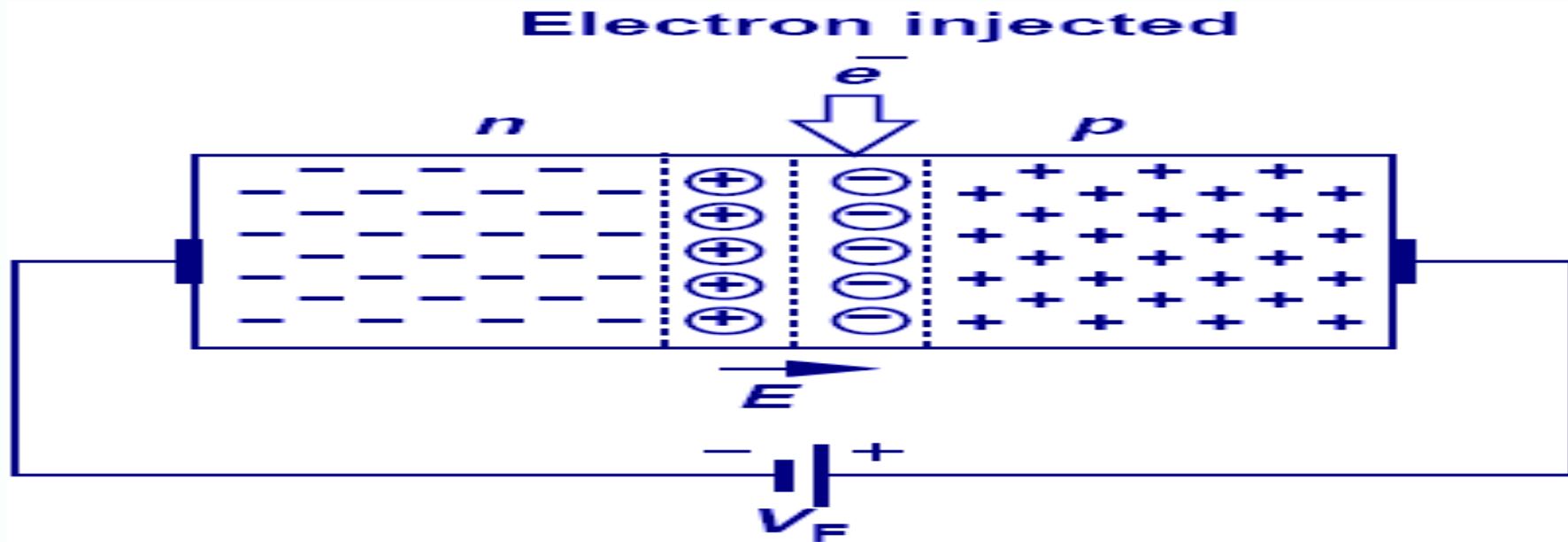
(b) npn



(c) pnp

Injection of Carriers

- Reverse biased PN junction creates a large electric field that sweeps any injected minority carriers to their majority region.
- This ability proves essential in the proper operation of a bipolar transistor.



Collector Current

- Applying the law of diffusion, we can determine the charge flow across the base region into the collector.

$$I_C = \frac{A_E q D_n n_i^2}{N_E W_B} \left(\exp \frac{V_{BE}}{V_T} - 1 \right)$$

$$I_C = I_S \exp \frac{V_{BE}}{V_T}$$

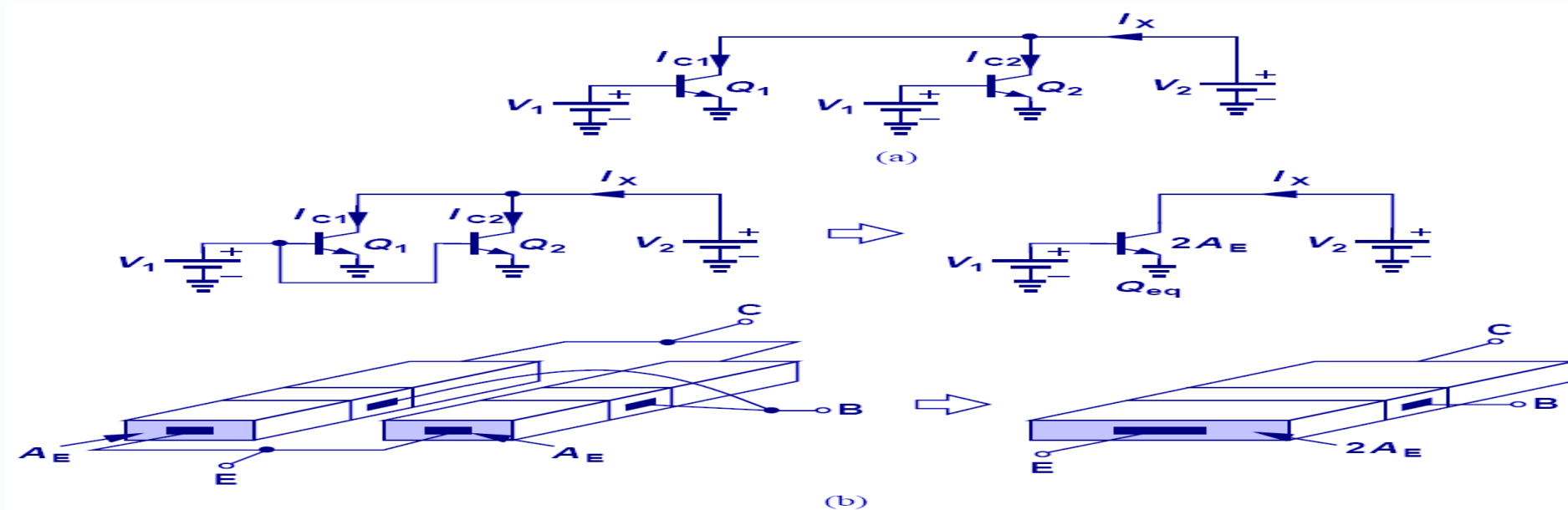
$$I_S = \frac{A_E q D_n n_i^2}{N_E W_B}$$

$$V_T = \frac{KT}{q}$$

The equation above shows that the transistor is indeed a voltage-controlled element, thus a good candidate as an amplifier.

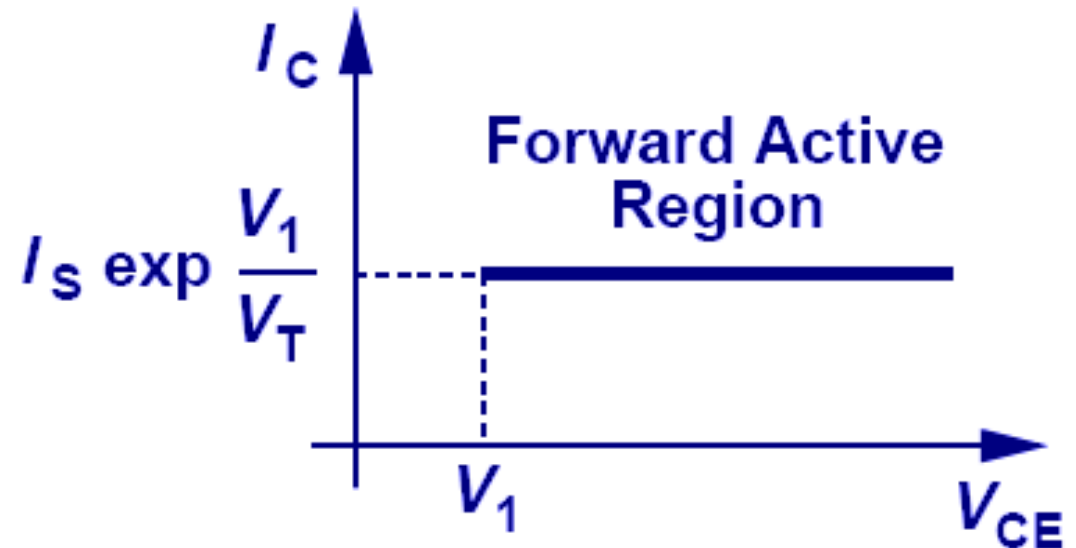
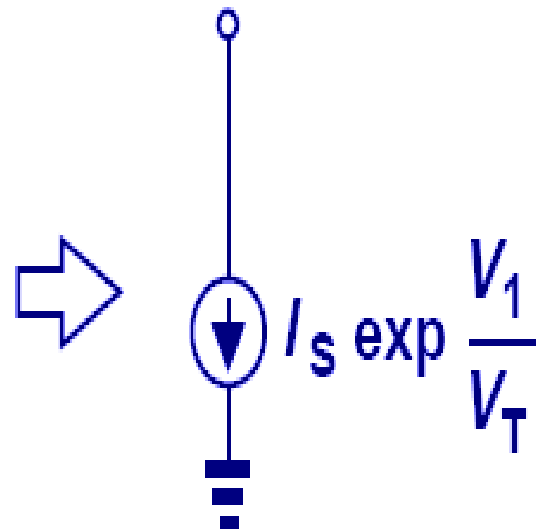
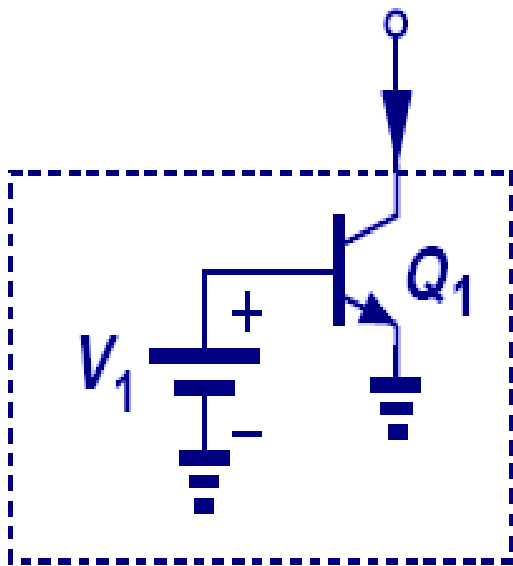
Parallel Combination of Transistors

- When two transistors are put in parallel and experience the same potential across all three terminals, they can be thought of as a single transistor with twice the emitter area.



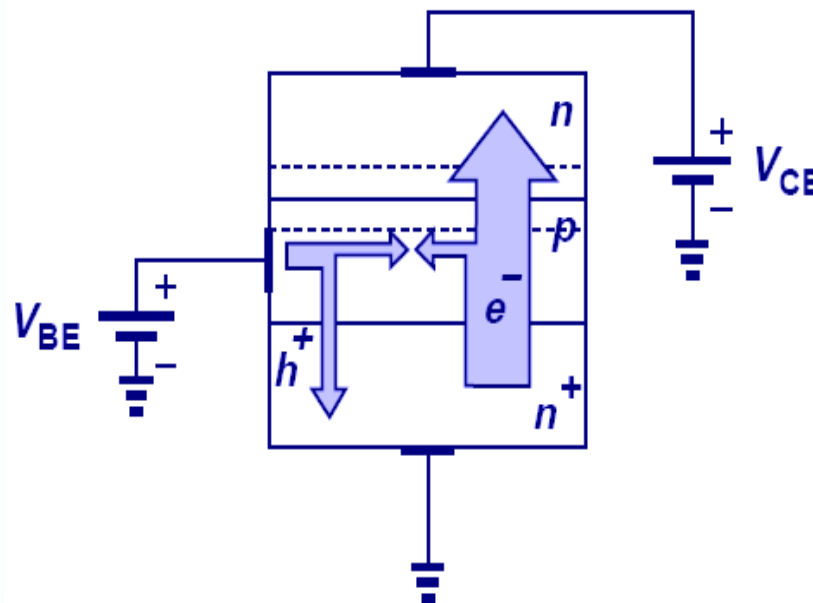
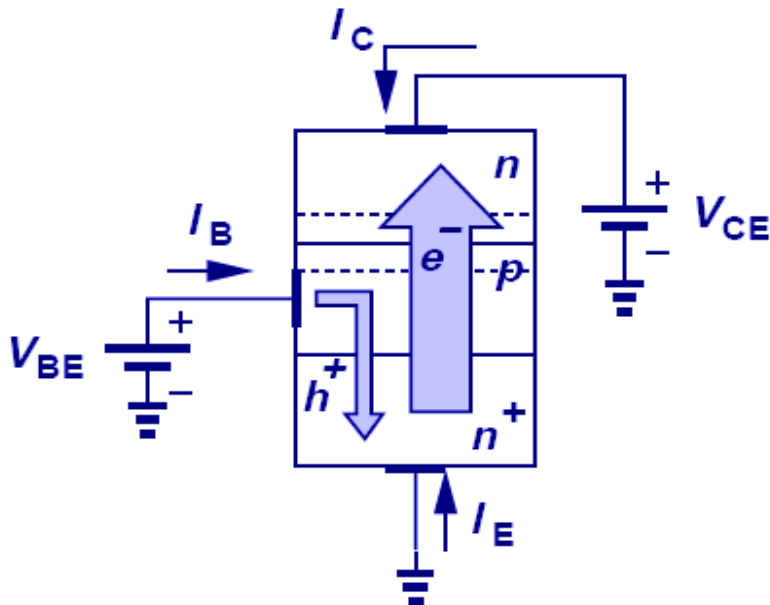
Constant Current Source

- Ideally, the collector current does not depend on the collector to emitter voltage.
- This property allows the transistor to behave as a constant current source when its base-emitter voltage is fixed.



Base Current

- Base current consists of two components:
- 1) Reverse injection of holes into the emitter and
- 2) recombination of holes with electrons coming from the emitter.



$$I_C = \beta I_B$$

Emitter Current

- Applying Kirchhoff's current law to the transistor, we can easily find the emitter current.

$$I_E = I_C + I_B$$

$$I_E = I_C \left(1 + \frac{1}{\beta} \right)$$

$$\beta = \frac{I_C}{I_B}$$

DC β and DC α

β = Common-emitter current gain

α = Common-base current gain

$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{I_C}{I_E}$$

The relationships between the two parameters are:

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Note: α and β are sometimes referred to as α_{dc} and β_{dc} because the relationships being dealt with in the BJT are DC.

Summary of Currents

$$I_C = I_s \exp \frac{V_{BE}}{V_T}$$

$$I_B = \frac{1}{\beta} I_s \exp \frac{V_{BE}}{V_T}$$

$$I_E = \frac{\beta + 1}{\beta} I_s \exp \frac{V_{BE}}{V_T}$$

$$\frac{\beta}{\beta + 1} = \alpha$$

Three Possible Configurations of BJT

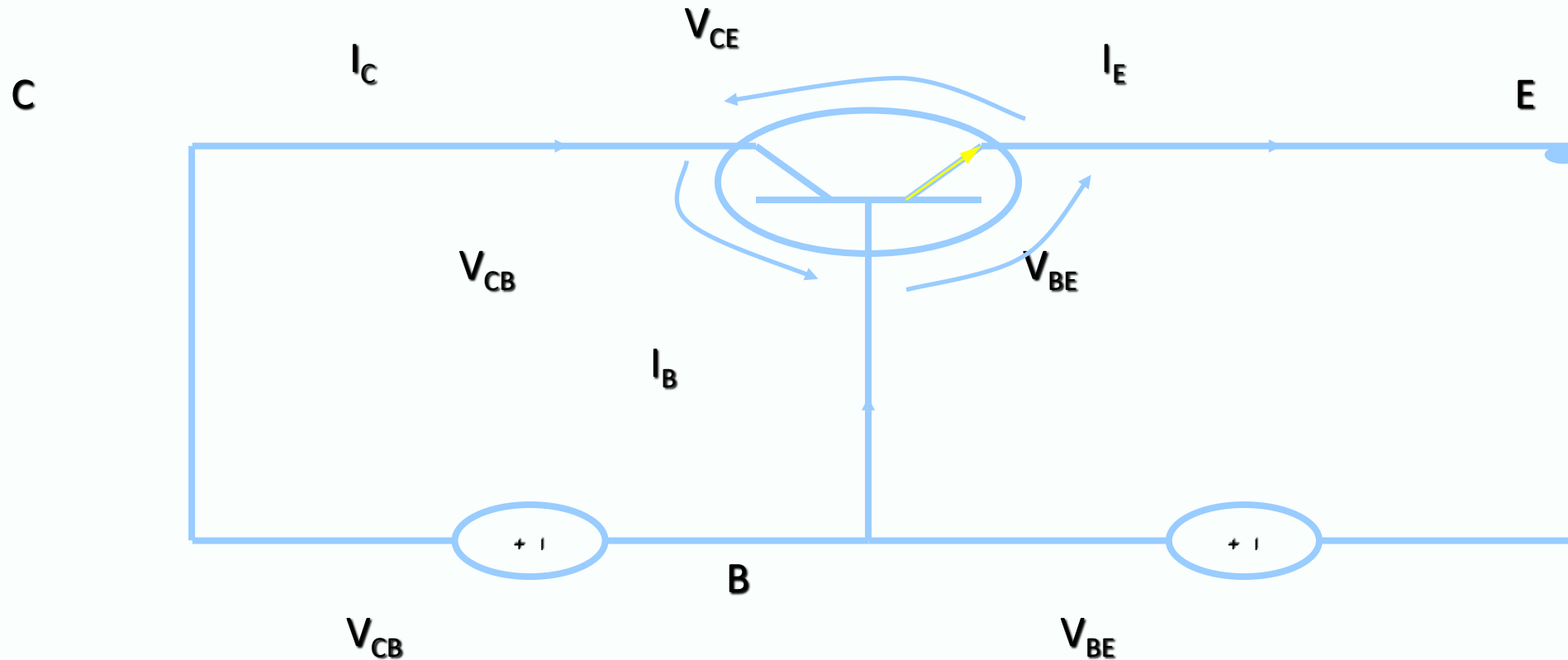
Biasing the transistor refers to applying voltages to the transistor to achieve certain operating conditions.

1. Common-Base Configuration (CB) : input = V_{EB} & I_E
output = V_{CB} & I_C

2. Common-Emitter Configuration (CE): input = V_{BE} & I_B
output = V_{CE} & I_C

3. Common-Collector Configuration (CC) : input = V_{BC} & I_B
(Also known as Emitter follower) output = V_{EC} & I_E

Common-Base BJT Configuration

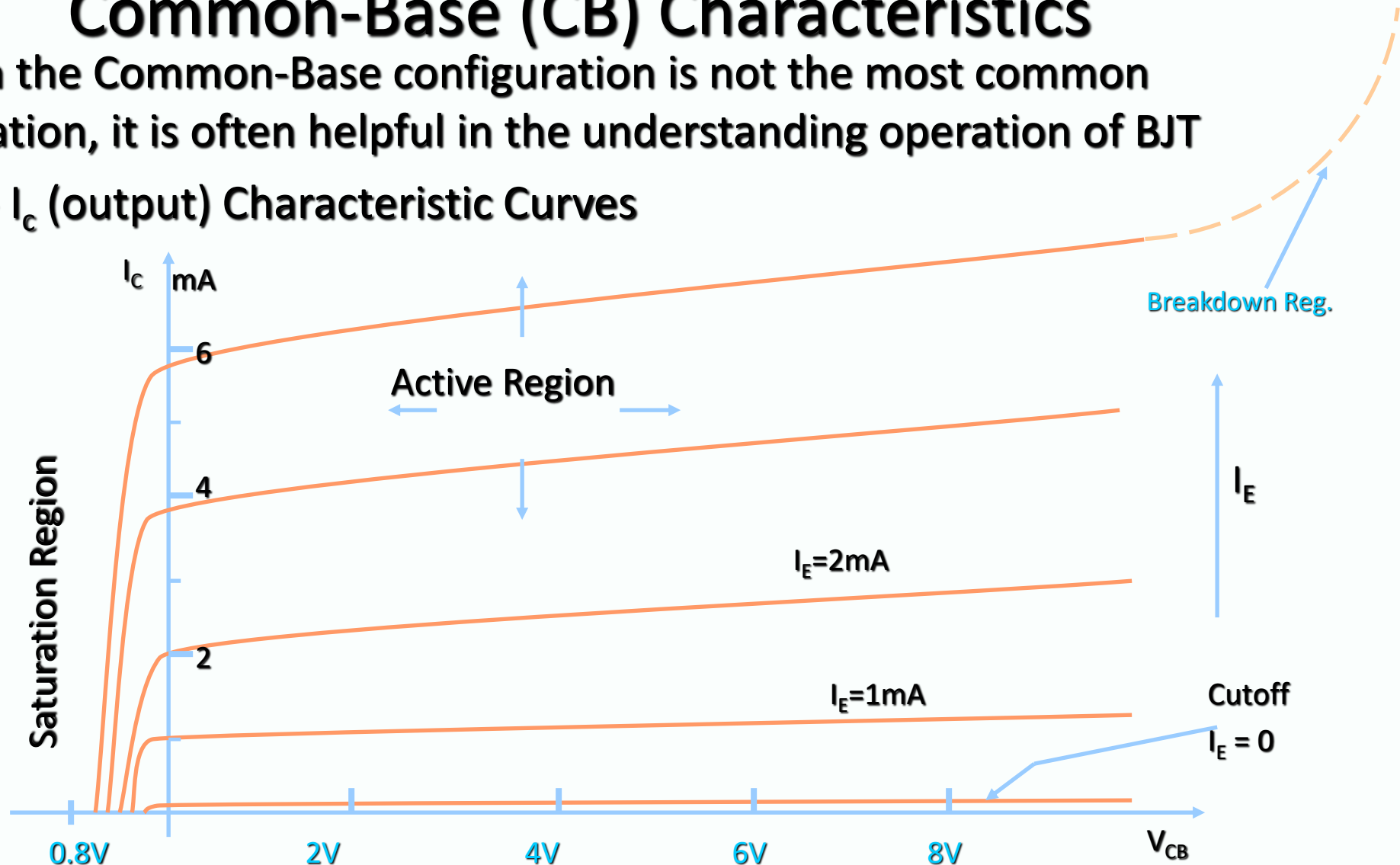


Circuit Diagram: NPN Transistor

Common-Base (CB) Characteristics

Although the Common-Base configuration is not the most common configuration, it is often helpful in the understanding operation of BJT

$V_c - I_c$ (output) Characteristic Curves

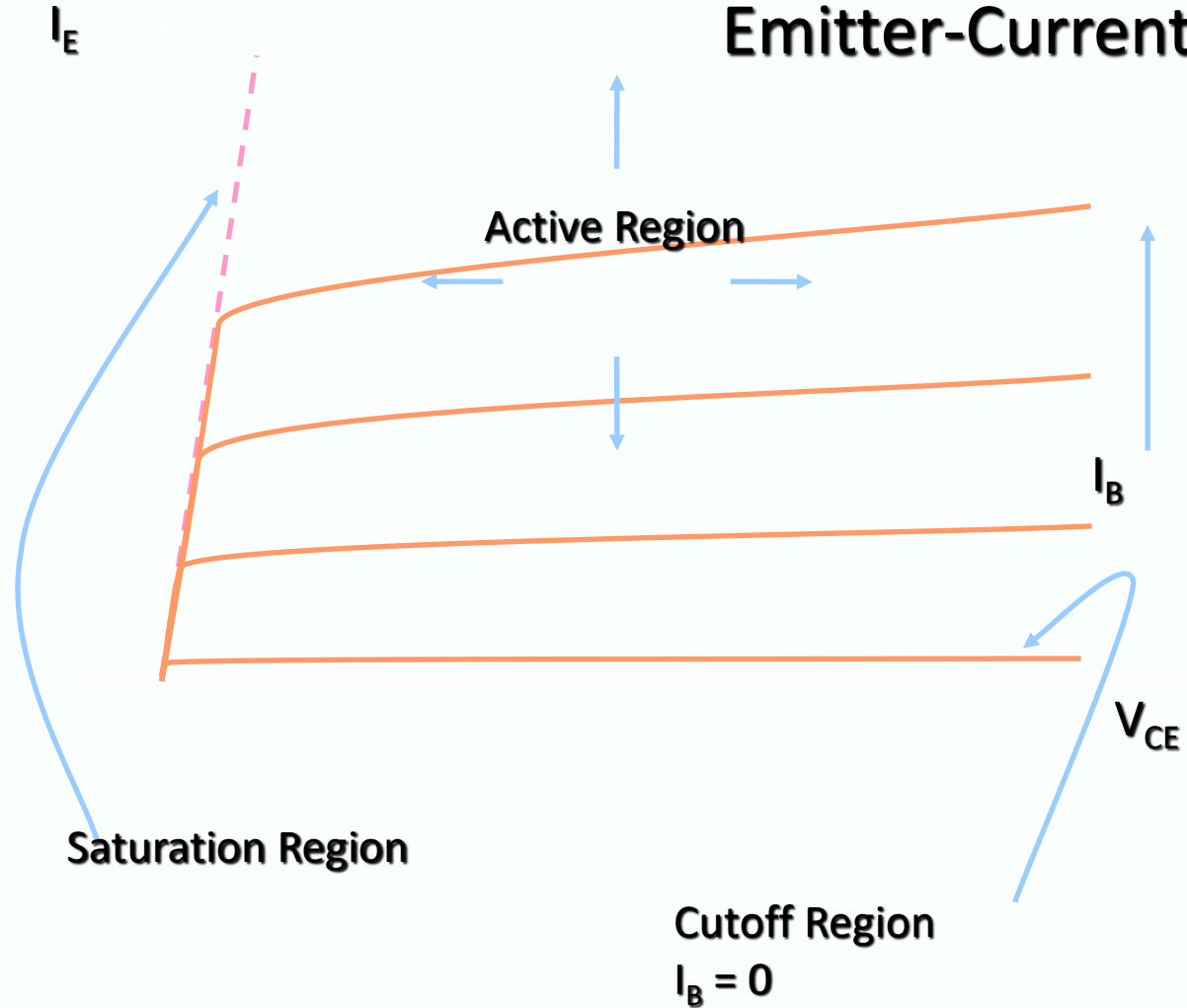


Common-Collector BJT Characteristics

Emitter-Current Curves

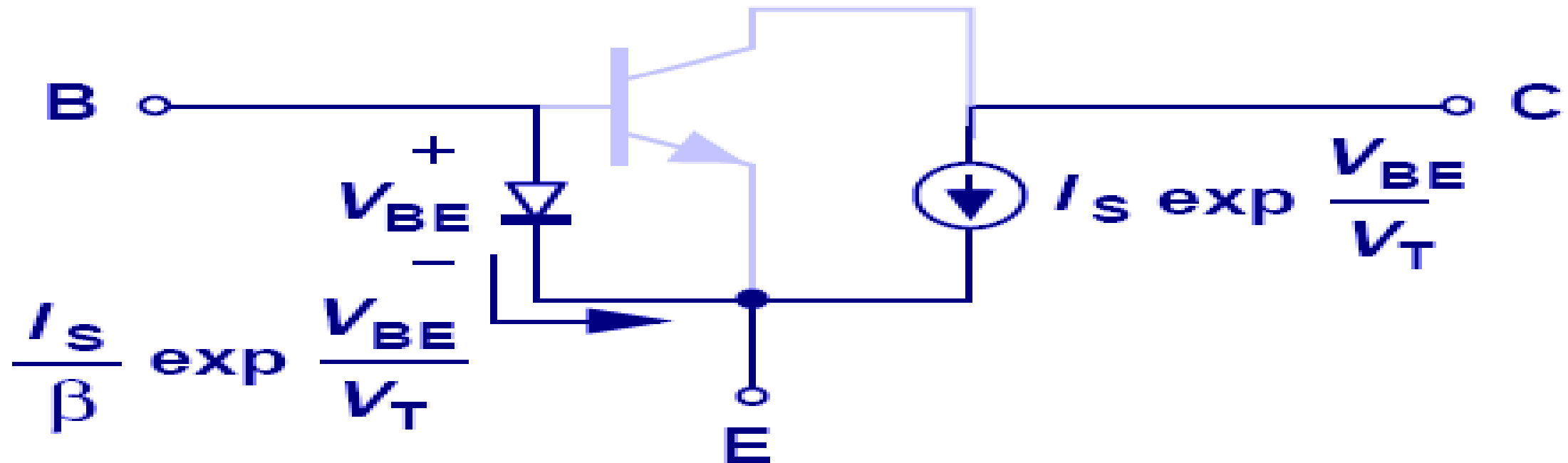
The Common-Collector biasing circuit is basically equivalent to the common-emitter biased circuit except instead of looking at I_C as a function of V_{CE} and I_B we are looking at I_E .

Also, since $\alpha \sim 1$, and $\alpha = I_C/I_E$ that means $I_C \sim I_E$



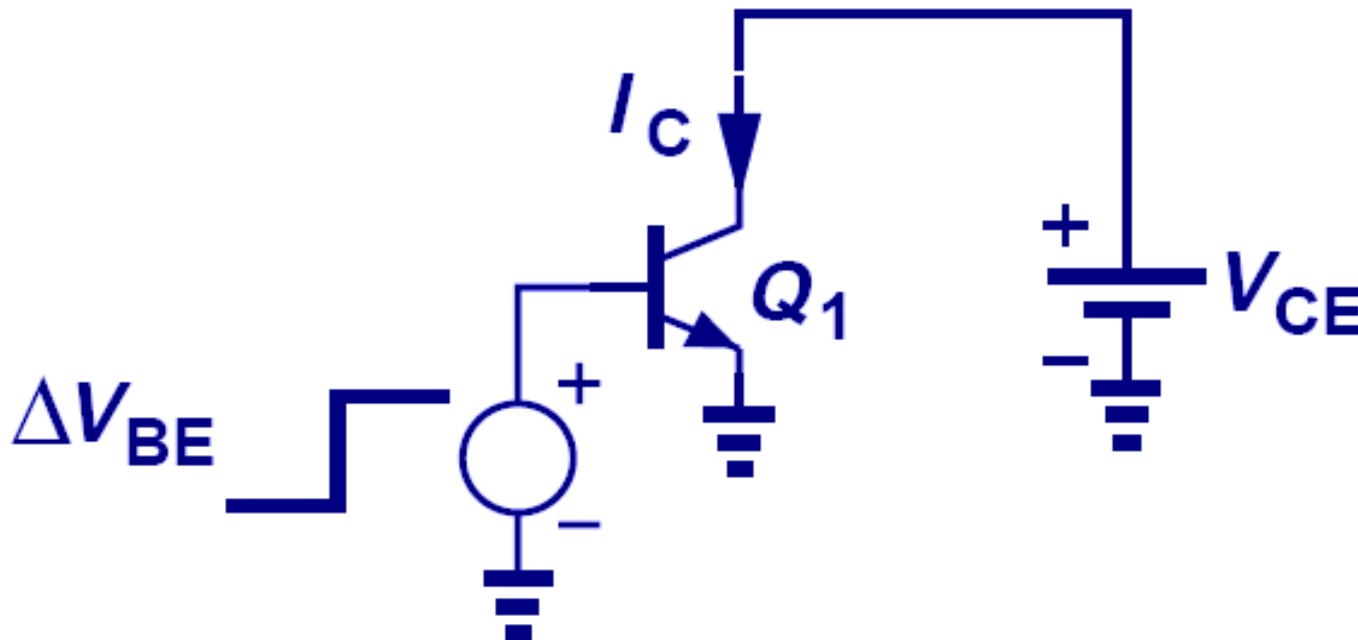
Bipolar Transistor Large Signal Model

- A diode is placed between base and emitter
- and a voltage controlled current source is placed between the collector and emitter.



Transconductance

- Transconductance, g_m shows a measure of how well the transistor converts voltage to current.
- g_m is one of the most important parameters in circuit design.



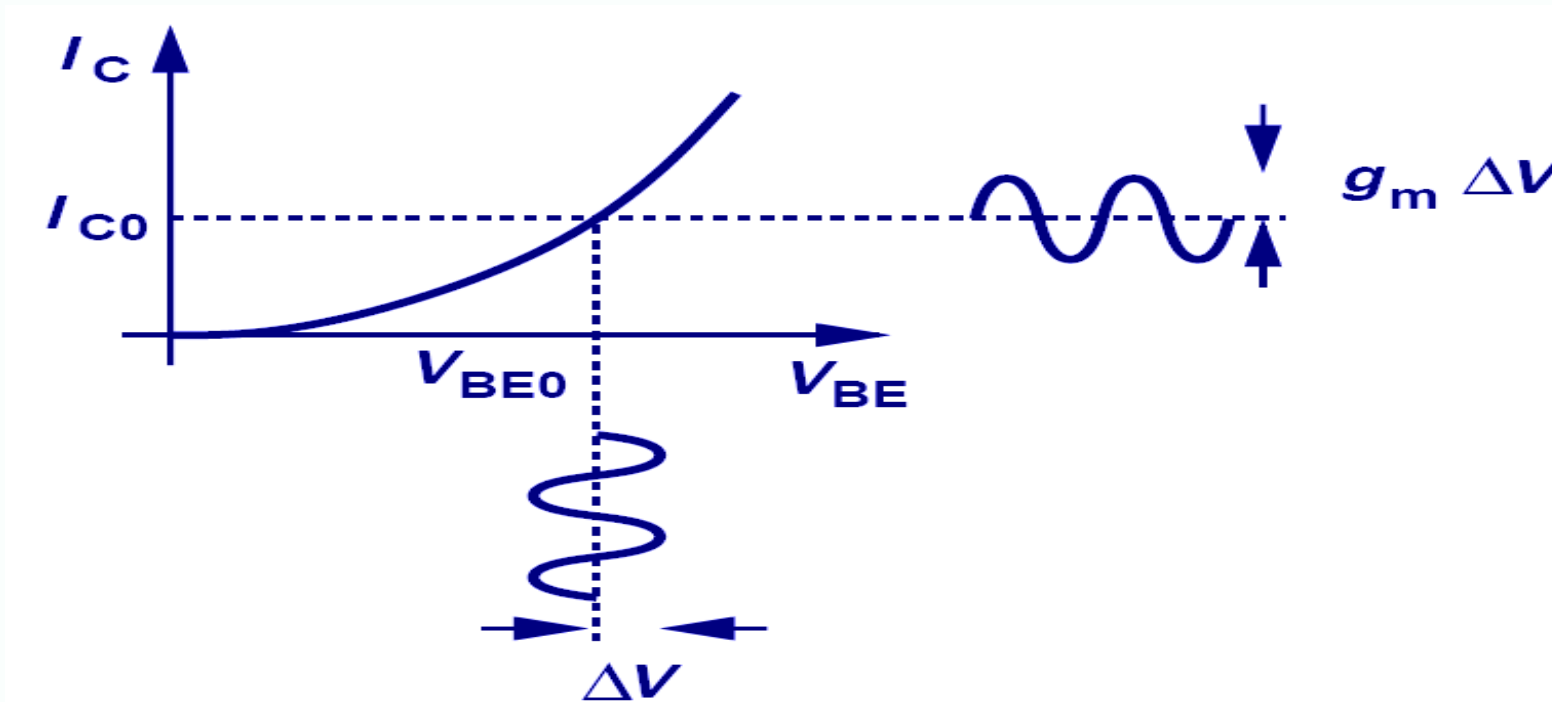
$$g_m = \frac{d}{dV_{BE}} \left(I_S \exp \frac{V_{BE}}{V_T} \right)$$

$$g_m = \frac{1}{V_T} I_S \exp \frac{V_{BE}}{V_T}$$

$$g_m = \frac{I_C}{V_T}$$

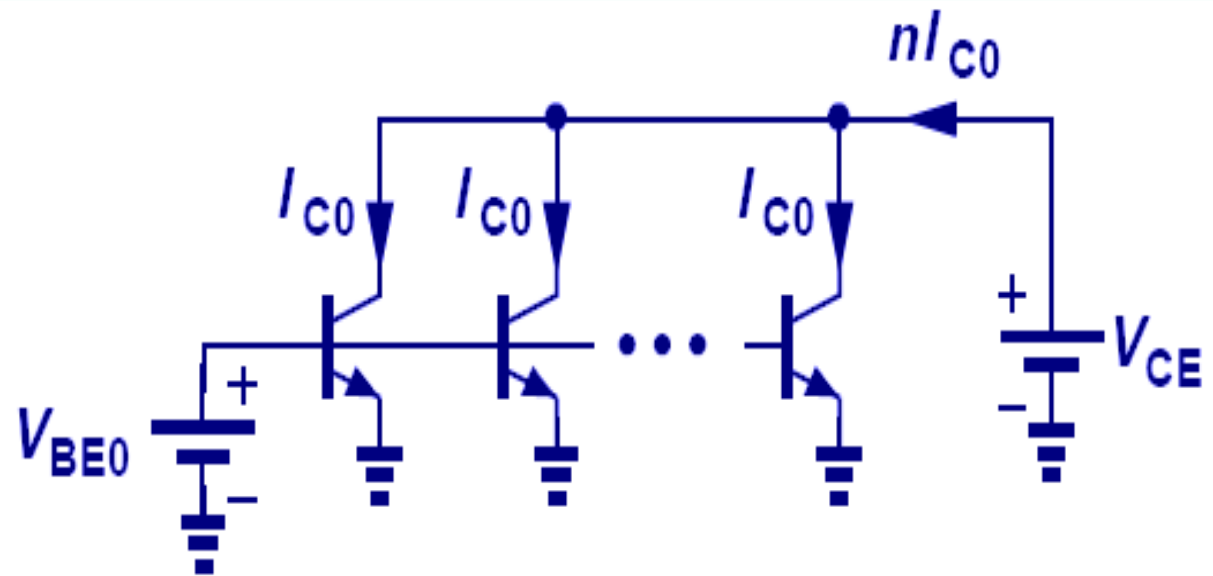
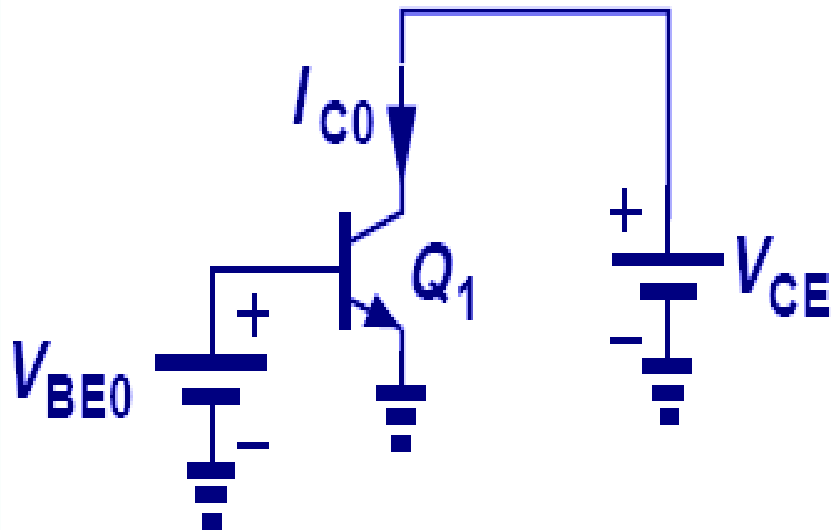
Visualization of Transconductance

- g_m can be visualized as the slope of I_C versus V_{BE} .
- A large I_C has a large slope and therefore a large g_m .



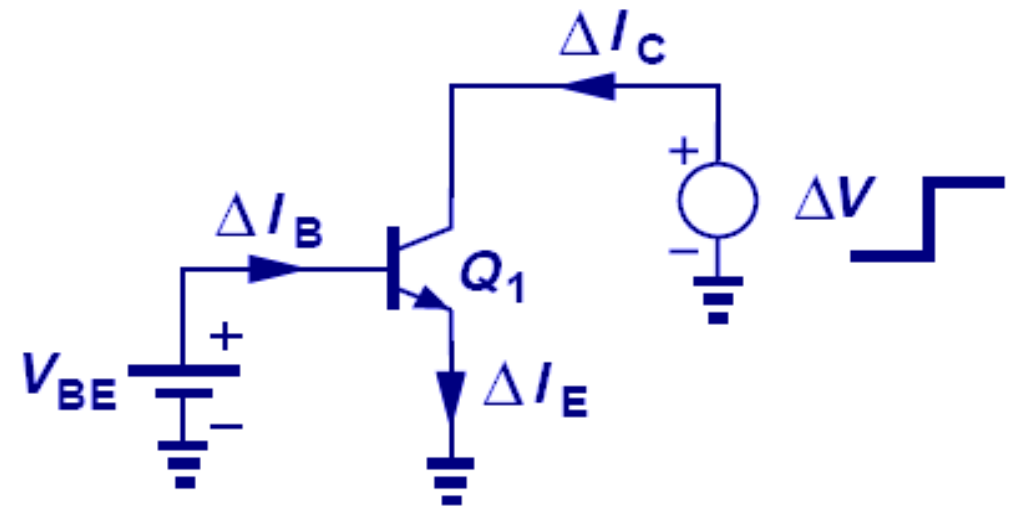
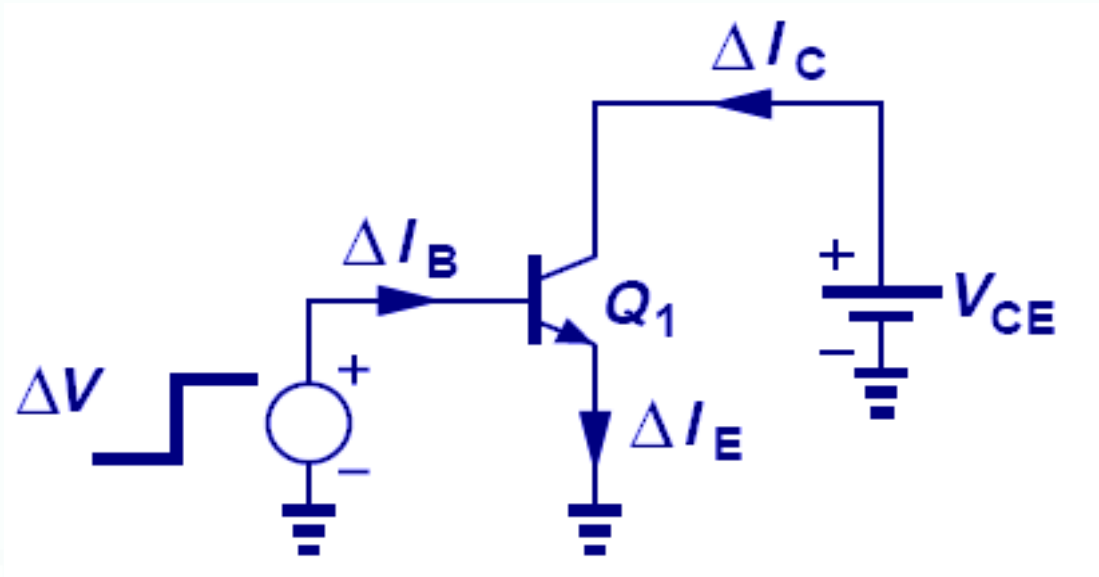
Transconductance and Area

- When the area of a transistor is increased by n , I_S increases by n .
- For a constant V_{BE} , I_C and hence g_m increases by a factor of n .

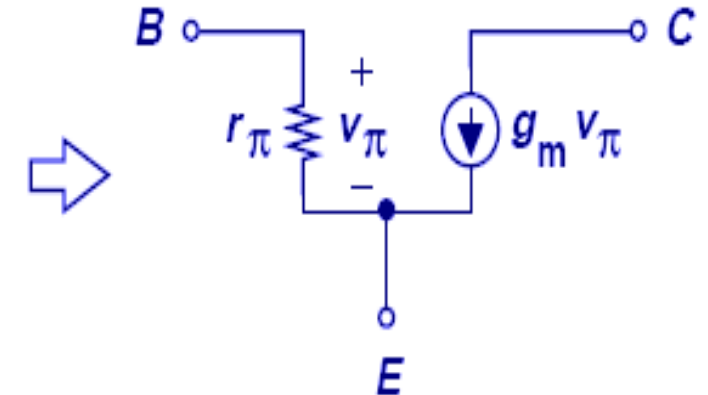
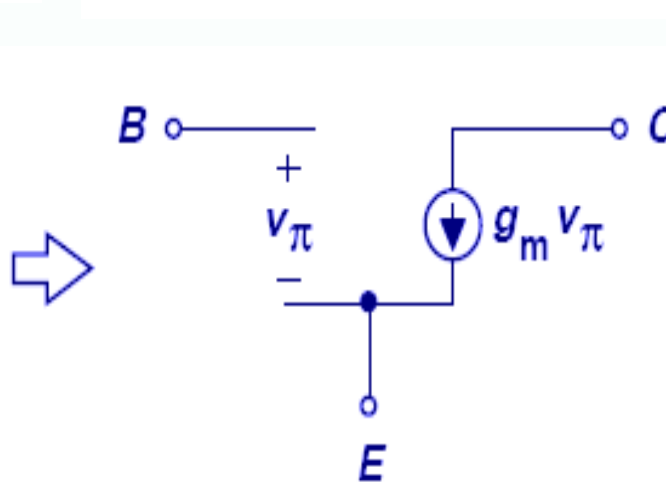
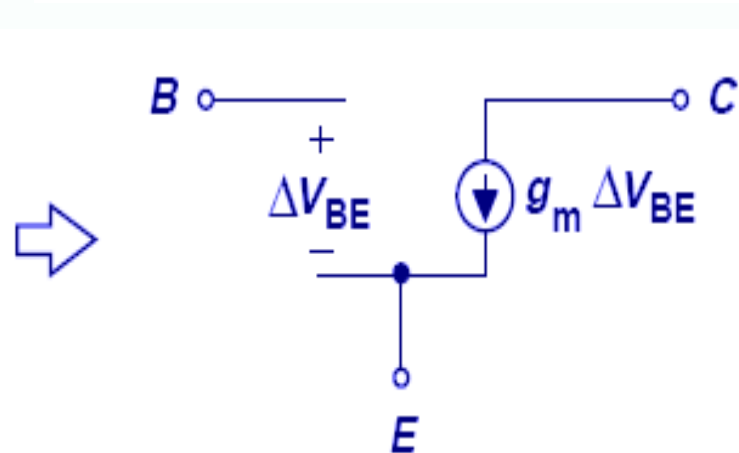
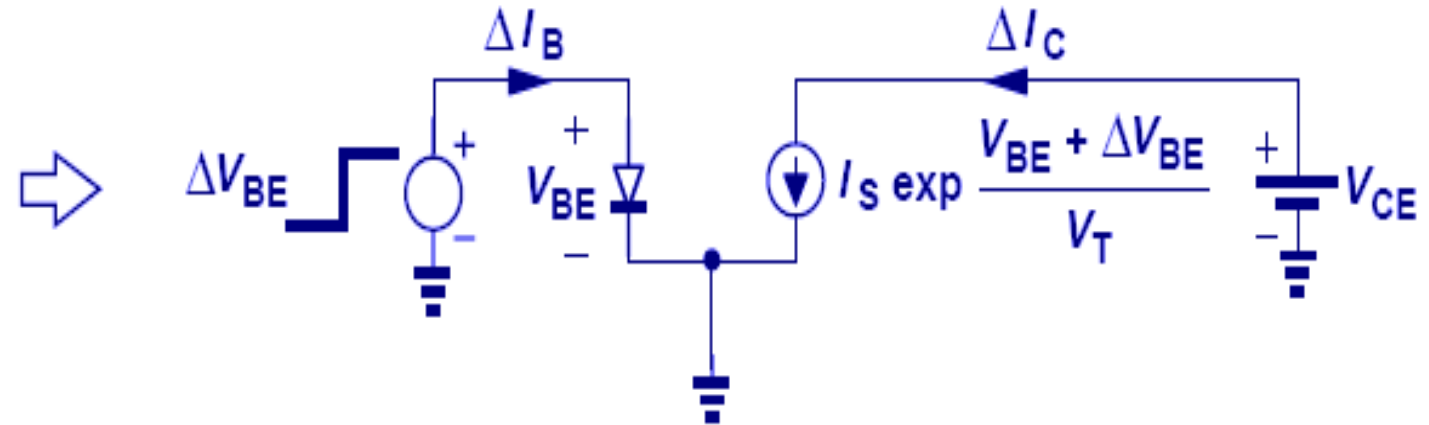
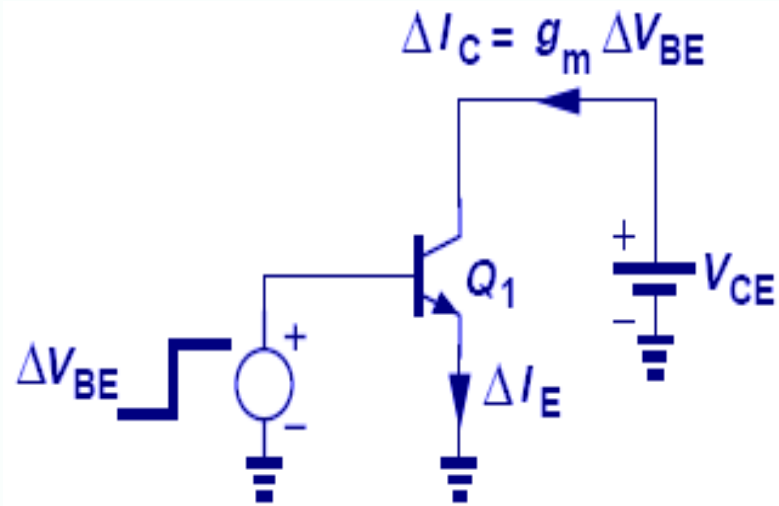


Small-Signal Model: Derivation

- Small signal model is derived by perturbing voltage difference every two terminals while fixing the third terminal and analyzing the change in current of all three terminals.
- We then represent these changes with controlled sources or resistors.

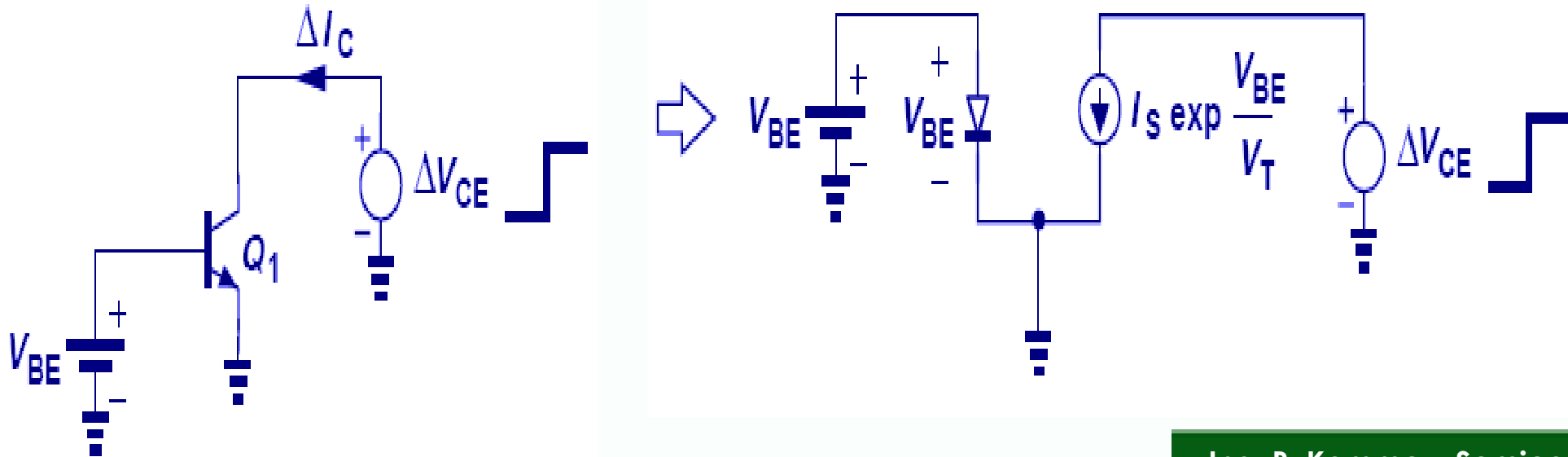


Small-Signal Model: V_{BE} Change



Small-Signal Model: V_{CE} Change

- Ideally, V_{CE} has no effect on the collector current.
- Thus, it will not contribute to the small signal model.
- It can be shown that V_{CB} has no effect on the small signal model, either.

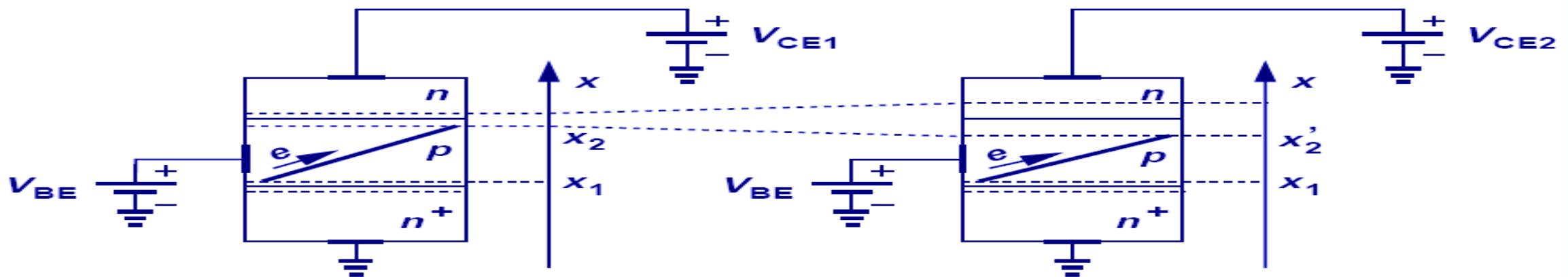


AC Ground

- Since the power supply voltage does not vary with time, it is regarded as a ground in small-signal analysis.

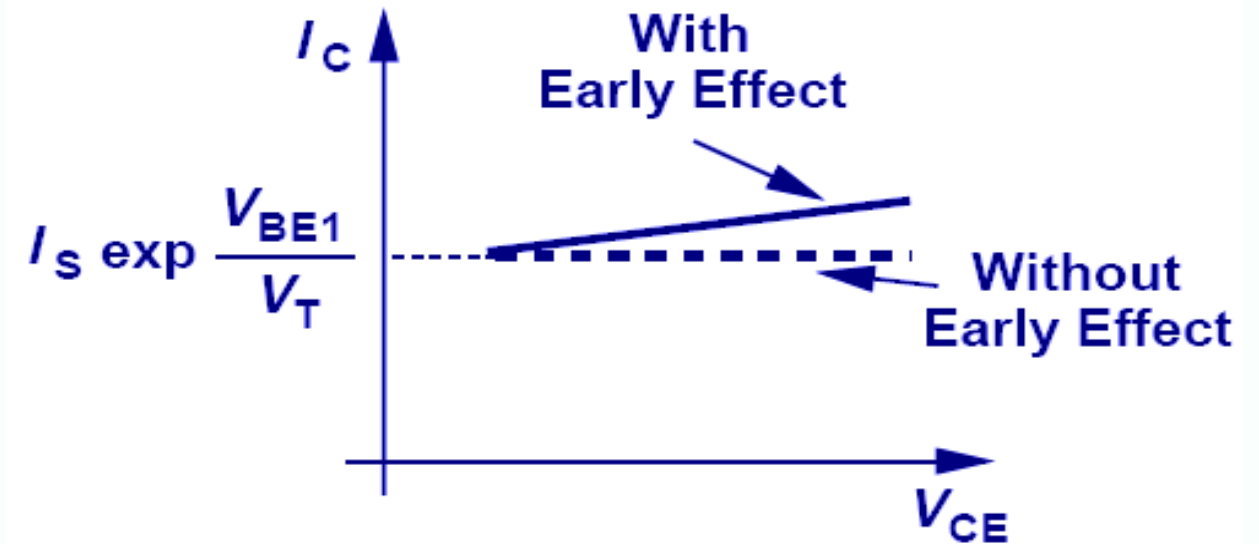
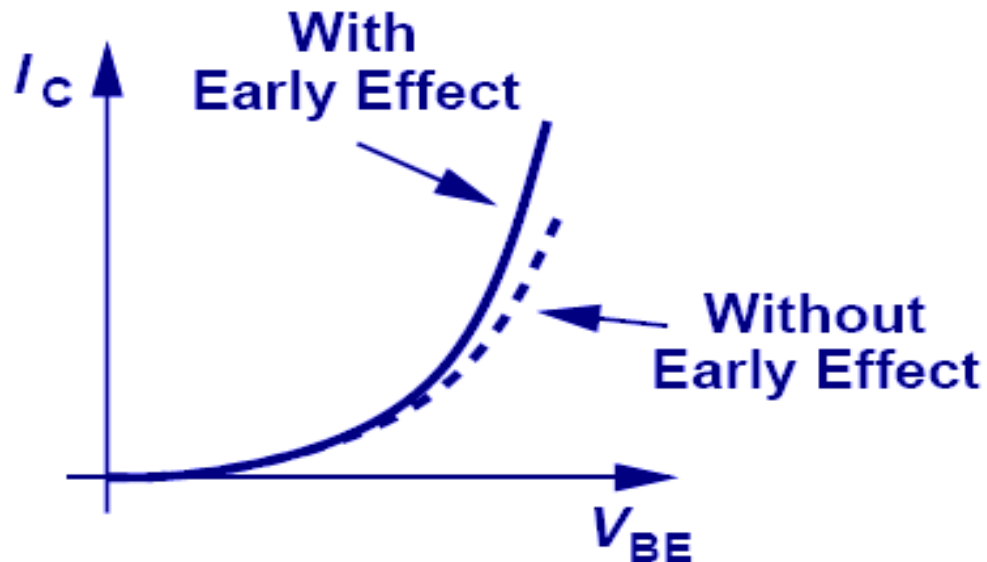
Early Effect

- The claim that collector current does not depend on V_{CE} is not accurate.
- As V_{CE} increases, the depletion region between base and collector increases. Therefore, the effective base width decreases, which leads to an increase in the collector current.

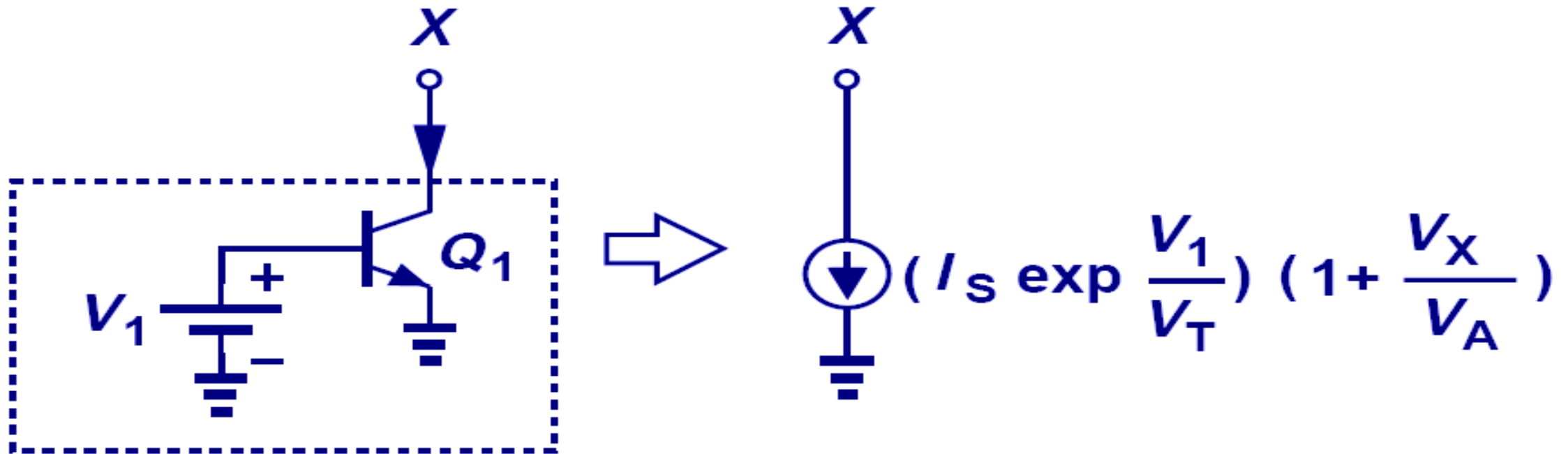


Early Effect Illustration

- Early effect is the variation in width of the base in a BJT due to a variation in the applied base-to-collector voltage
- With Early effect, collector current becomes larger than usual and I_c is a function of V_{CE} .

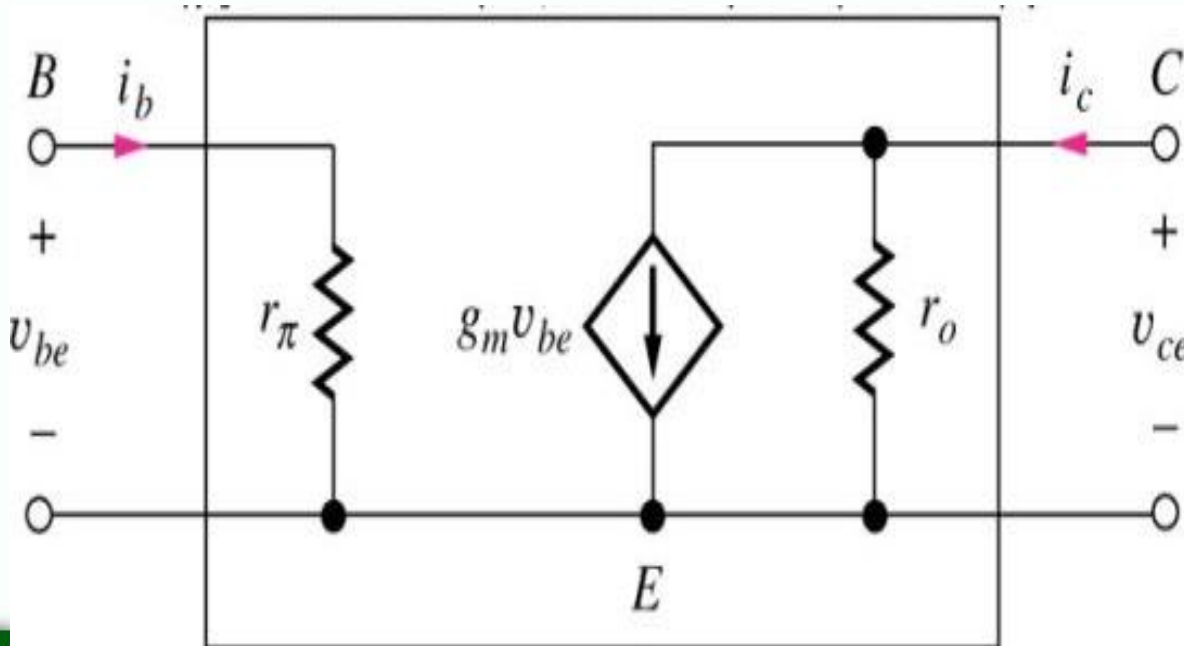


Early Effect Representation



Hybrid-Pi Model for the BJT

- The hybrid-pi small-signal model is the intrinsic low-frequency representation of the BJT.
- The small-signal parameters are controlled by the Q-point and are independent of the geometry of the BJT.



Transconductance: $g_m = \frac{I_C}{V_T}, V_T = \frac{KT}{q}$

Input resistance: R_{in}

$$r_\pi = \frac{\beta_o V_T}{I_C} = \frac{\beta_o}{g_m}$$

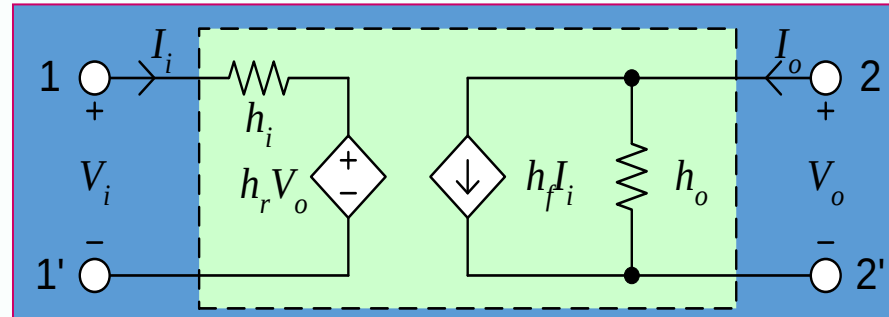
Output resistance:

$$r_o = \frac{V_A + V_{CE}}{I_C}$$

Where, V_A is Early Voltage
($V_A = 100V$ for npn)

• Hybrid-Pi Model for The BJT

Hybrid Parameter Model



$$V_i = h_{11} I_i + h_{12} V_o = h_i I_i + h_r V_o$$

$$I_o = h_{21} I_i + h_{22} V_o = h_f I_i + h_o V_o$$

h -Parameters

$$h_{11} = \left. \frac{V_i}{I_i} \right|_{V_o = 0}$$

$$h_{12} = \left. \frac{V_i}{V_o} \right|_{I_i = 0}$$

$$h_{21} = \left. \frac{I_o}{I_i} \right|_{V_o = 0}$$

$$h_{22} = \left. \frac{I_o}{V_o} \right|_{I_i = 0}$$

$h_{11} = h_i$ = Input Resistance

$h_{12} = h_r$ = Reverse Transfer Voltage Ratio

$h_{21} = h_f$ = Forward Transfer Current Ratio

$h_{22} = h_o$ = Output Admittance



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Thank You

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COE 271 Semiconductor Devices

Lecture 5

Bipolar Amplifiers

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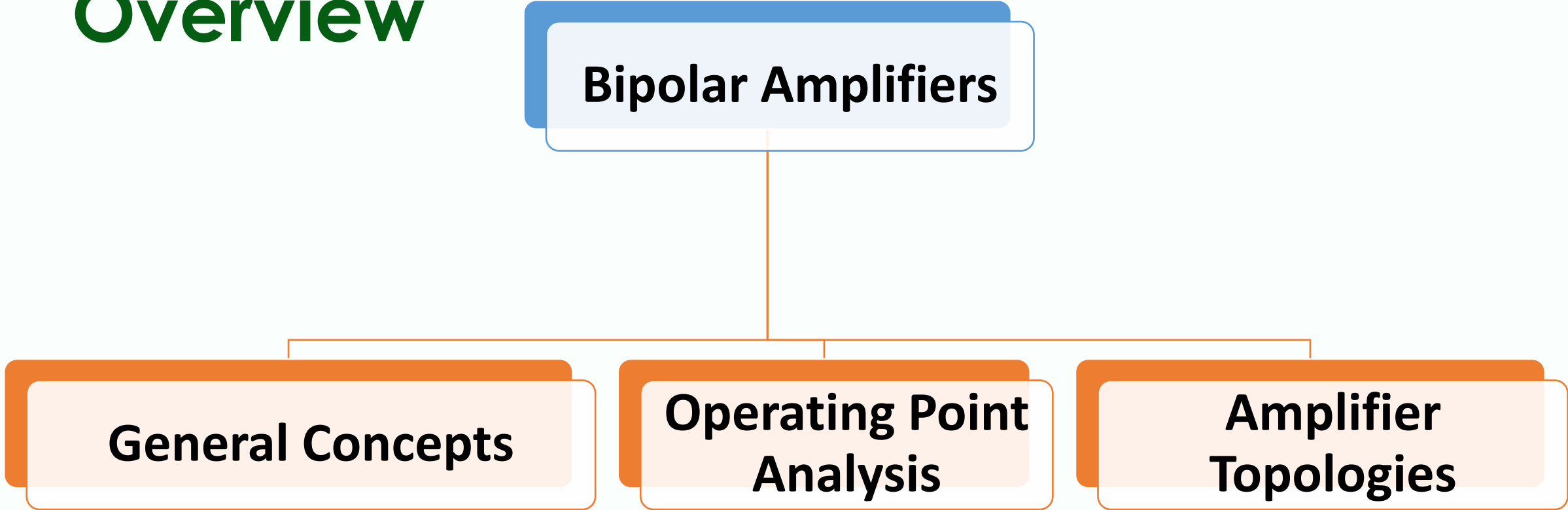


“Paper is to write things down that we need to remember. Our brains are used to think”

Albert Einstein

Overview

Bipolar Amplifiers



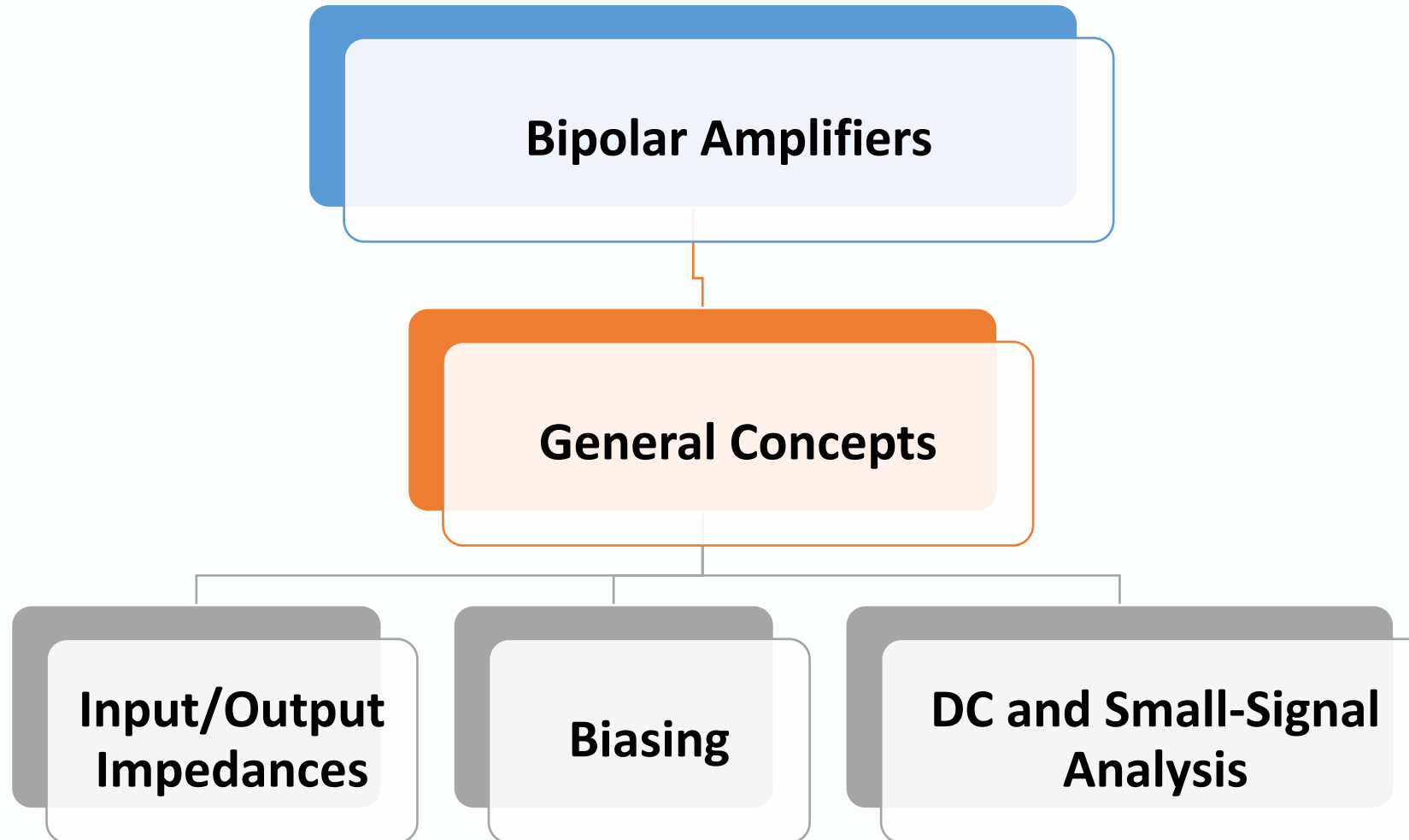
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graph TD; A[Bipolar Amplifiers] --> B[General Concepts]; A --> C[Operating Point Analysis]; A --> D[Amplifier Topologies];
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General Concepts

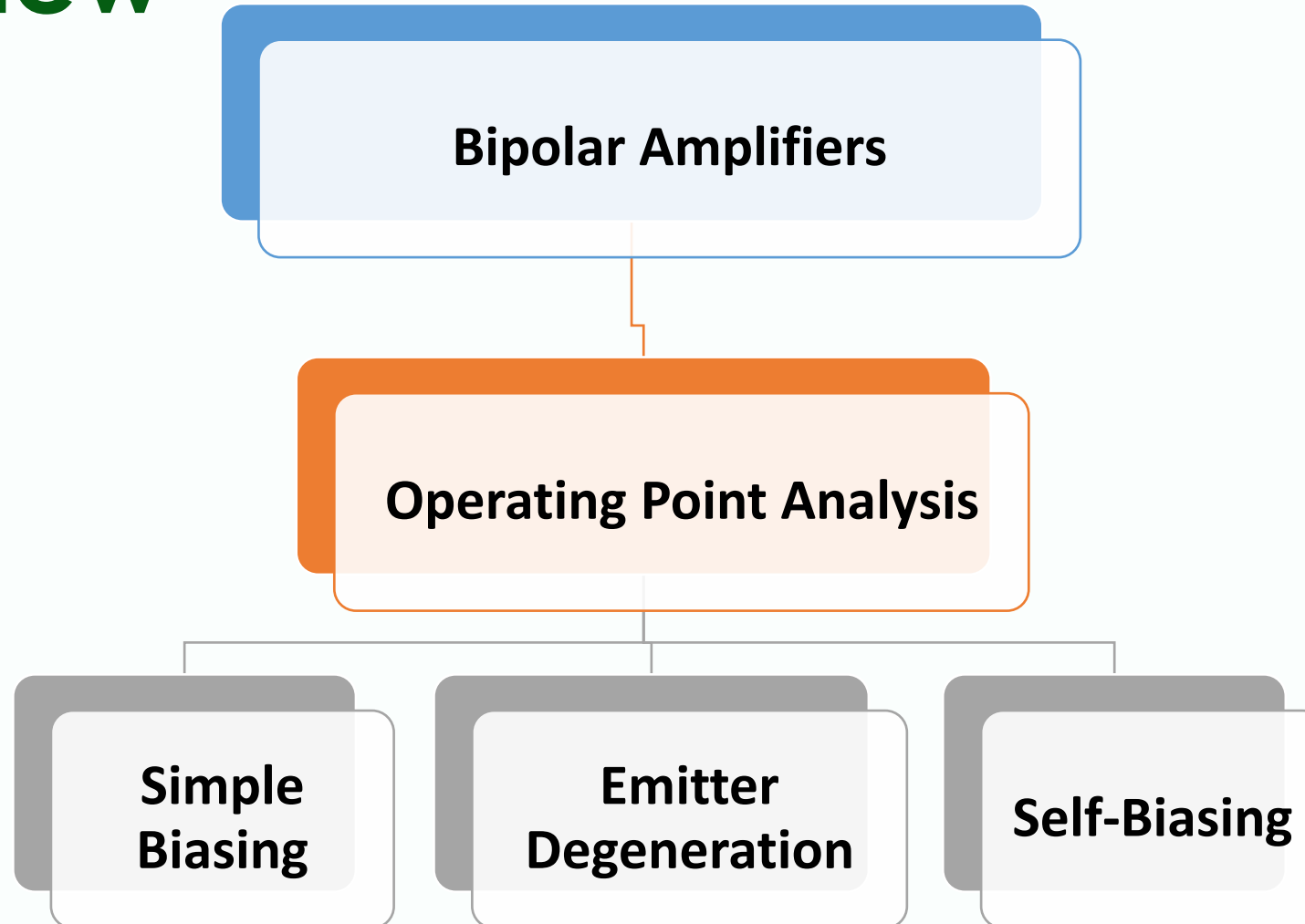
**Operating Point
Analysis**

**Amplifier
Topologies**

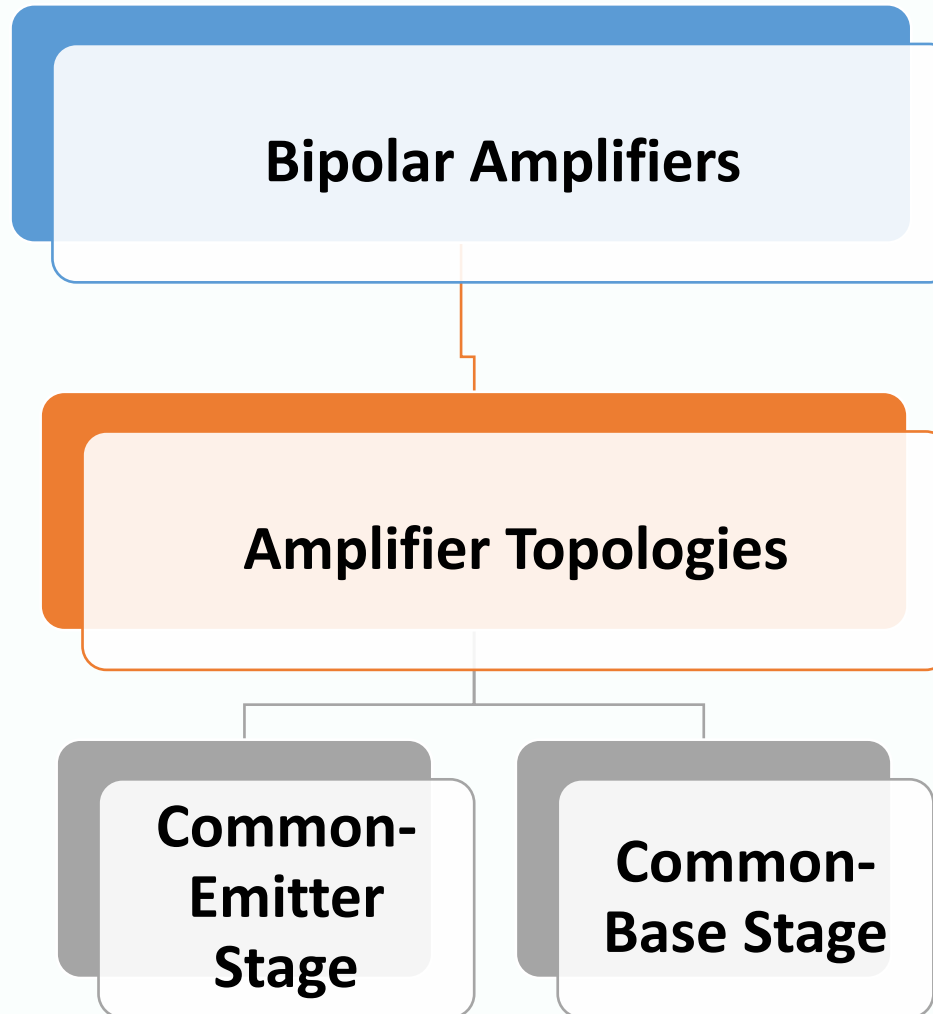
Overview



Overview



Overview



Introduction to Amplifiers

- The BJT is an excellent amplifier when biased in the forward-active region.
- The FET can be used as an amplifier if operated in the saturation region.
- In these regions, the transistors can provide high voltage, current and power gains.
- DC bias is provided to stabilize the operating point in the desired operation region.

The **dc operating point** between saturation and cutoff is called the Q-point. The goal is to set the Q-point such that it does not go into saturation or cutoff when an ac signal is applied

Introduction to Amplifiers

- The DC Q-point also determines

The small-signal parameters of the transistor

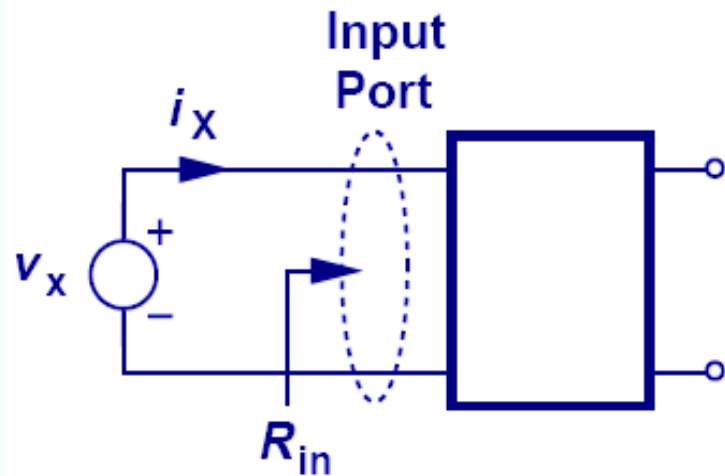
The voltage gain, input resistance, and output resistance

The maximum input and output signal amplitudes

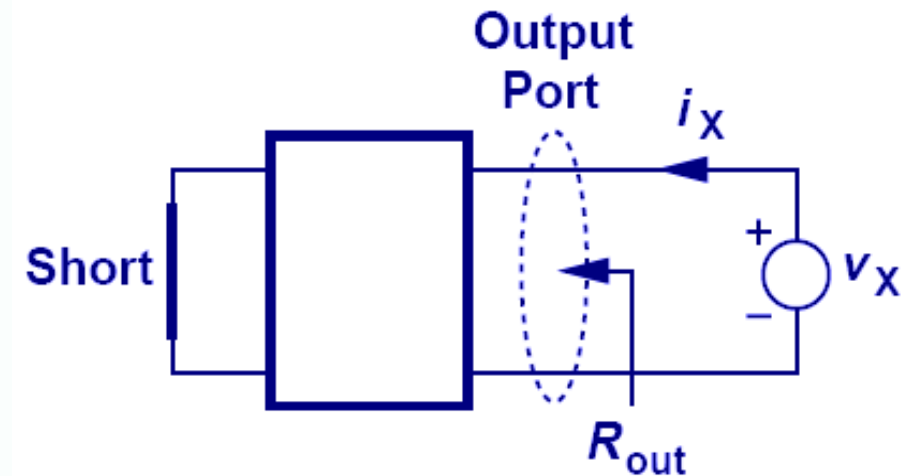
The overall power consumption of the amplifier

Input/Output Impedances

- The techniques of measuring input and output impedances.

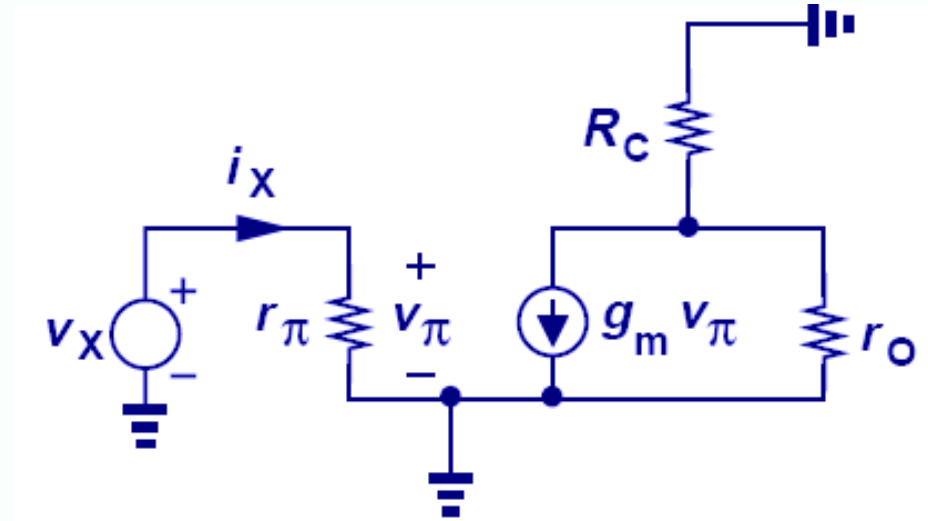
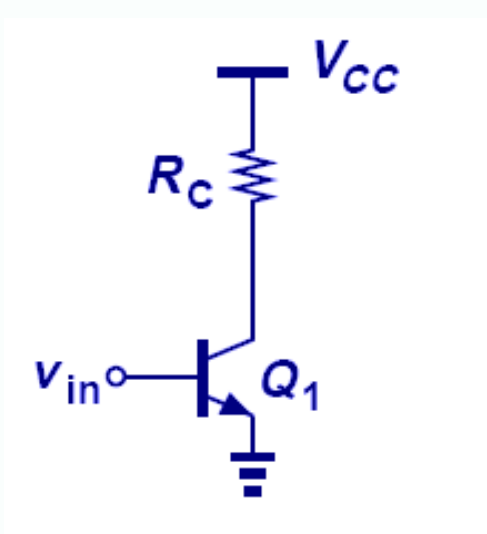


$$R_x = \frac{V_x}{i_x}$$



Input Impedance Example

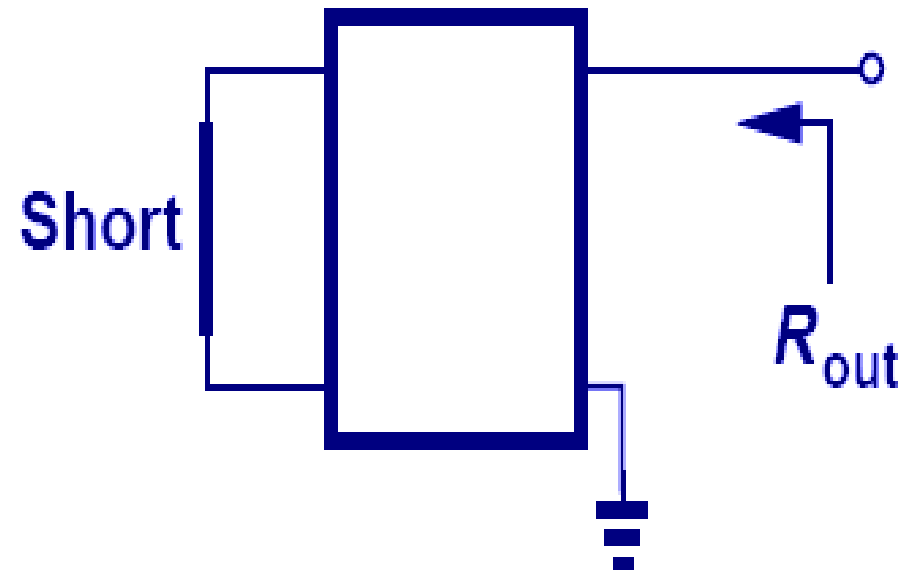
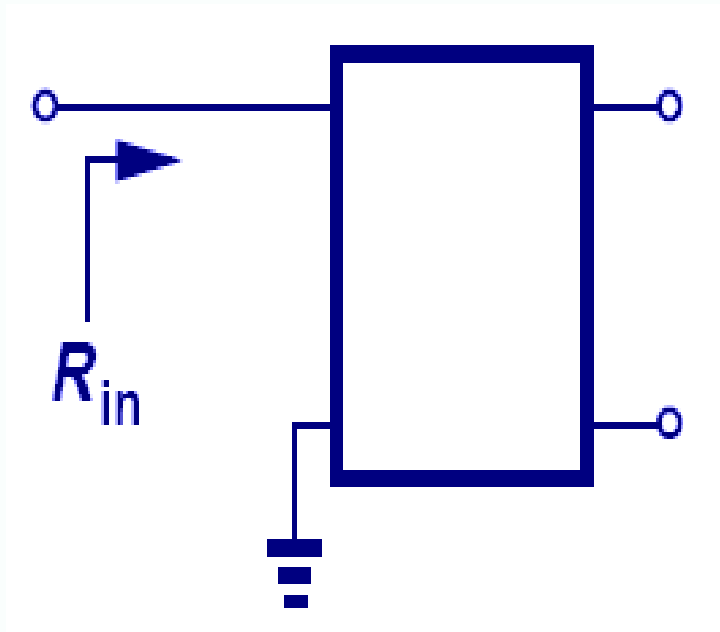
- When calculating input/output impedance, small-signal analysis is assumed.



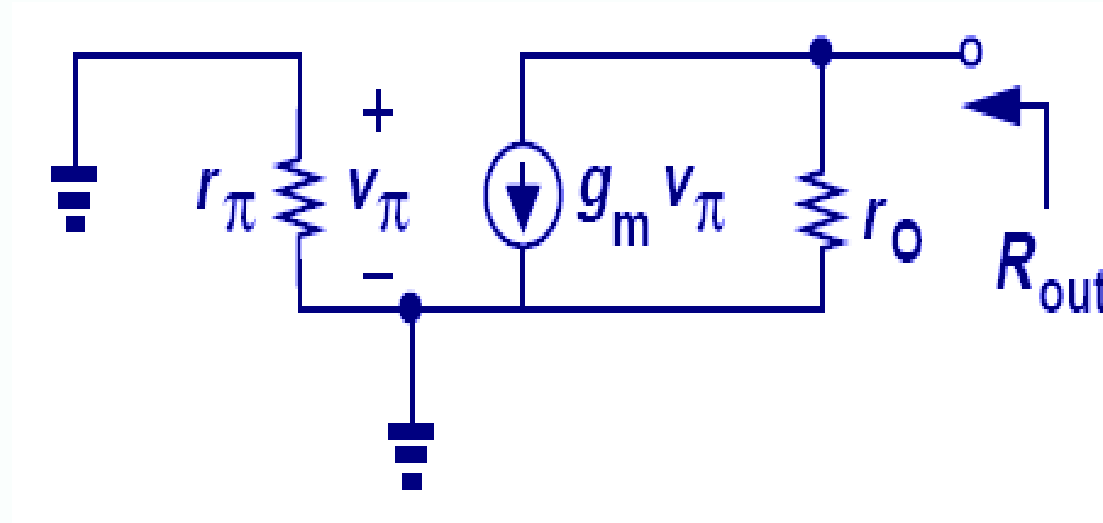
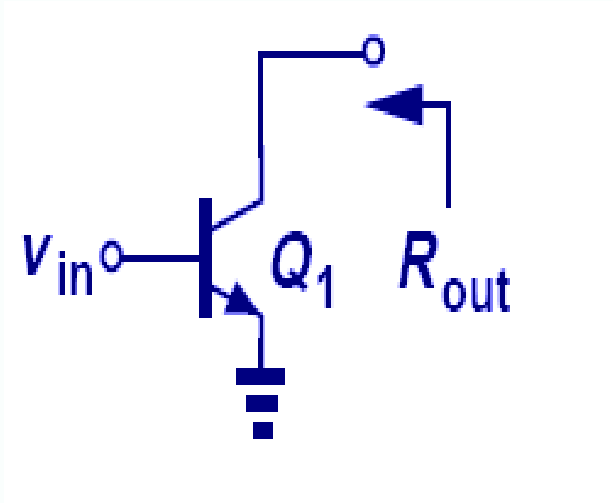
$$\frac{v_x}{i_x} = r_{\pi}$$

Impedance at a Node

- When calculating I/O impedances at a port, we usually ground one terminal while applying the test source to the other terminal of interest.



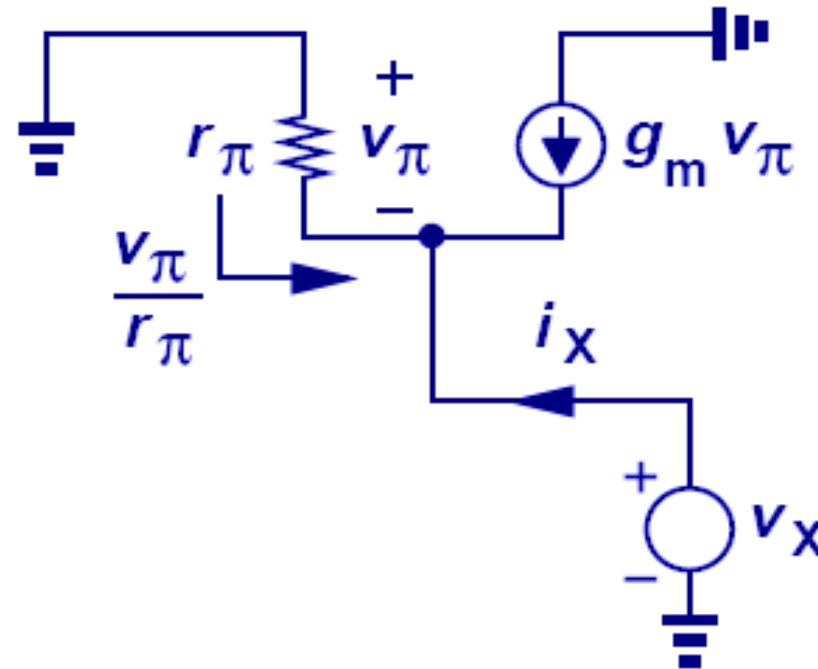
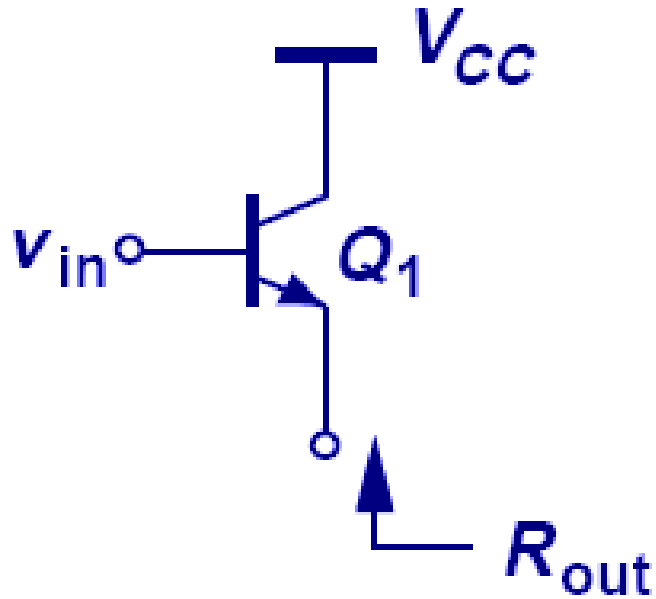
Impedance at Collector



$$R_{out} = r_o$$

- Early effect is the variation in width of the base in a BJT due to a variation in the applied base-to-collector voltage
- With Early effect, the impedance seen at the collector is equal to the intrinsic output impedance of the transistor (if emitter is grounded).

Impedance at Emitter



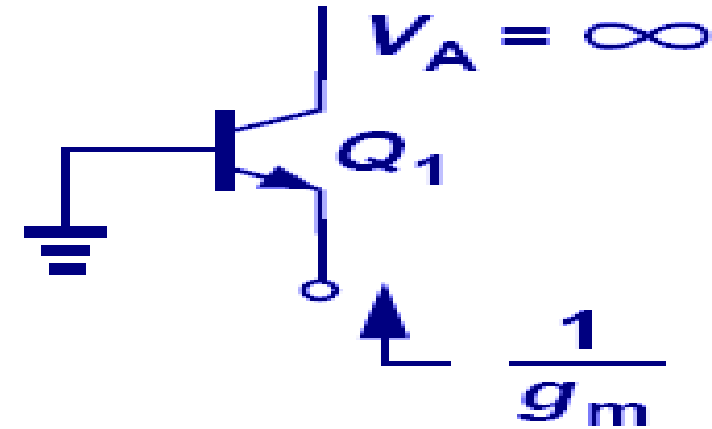
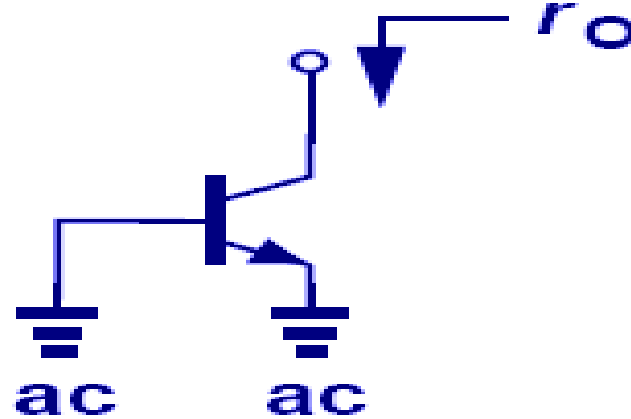
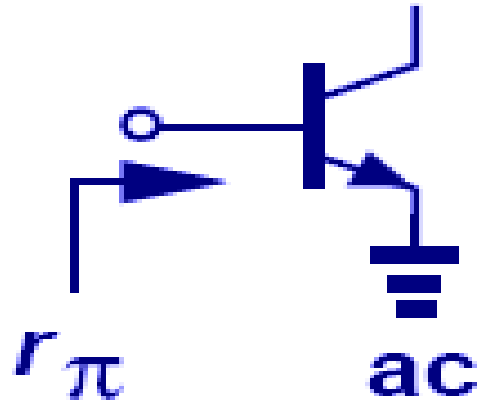
$$\frac{v_x}{i_x} = \frac{1}{g_m + \frac{1}{r_\pi}}$$

$$R_{out} \approx \frac{1}{g_m}$$

$$(V_A = \infty)$$

- The impedance seen at the emitter of a transistor is approximately equal to one over its transconductance (if the base is grounded).

Three Master Rules of Transistor Impedances



- Rule # 1:

looking into the base, the impedance is r_{π} if emitter is (ac) grounded.

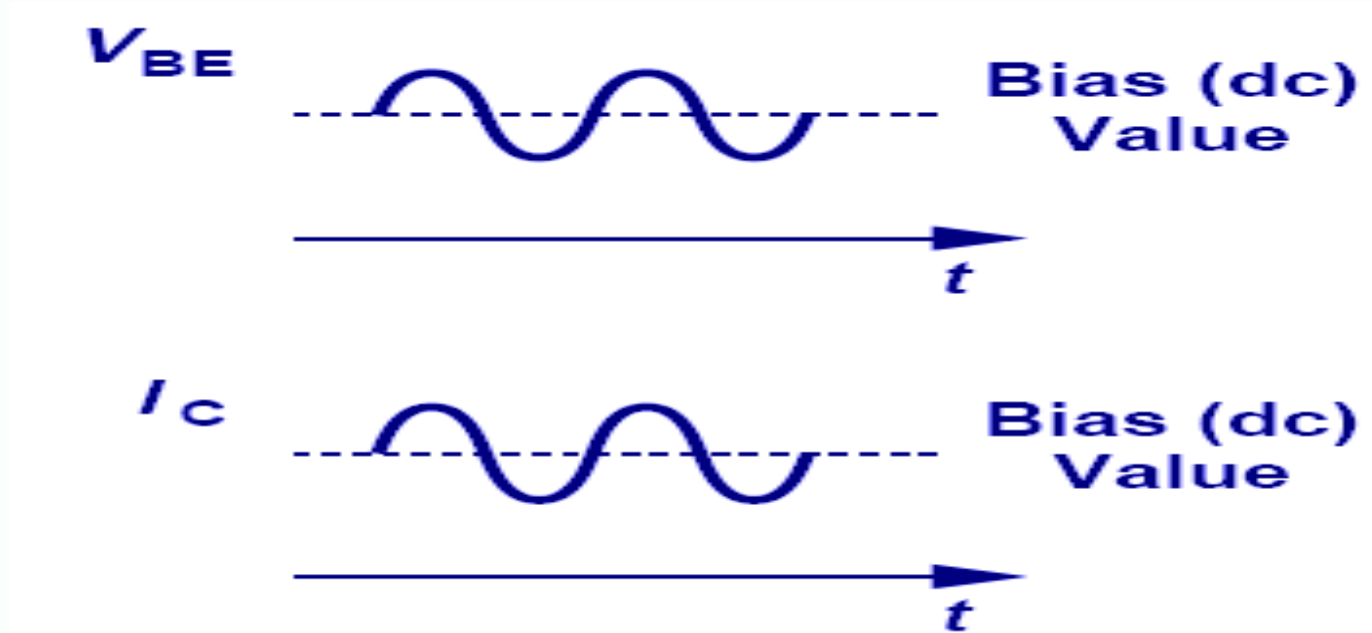
Rule # 2:

looking into the collector, the impedance is r_o if emitter is (ac) grounded.

Rule # 3:

looking into the emitter, the impedance is $1/g_m$ if base is (ac) grounded and Early effect is neglected.

Biasing of BJT

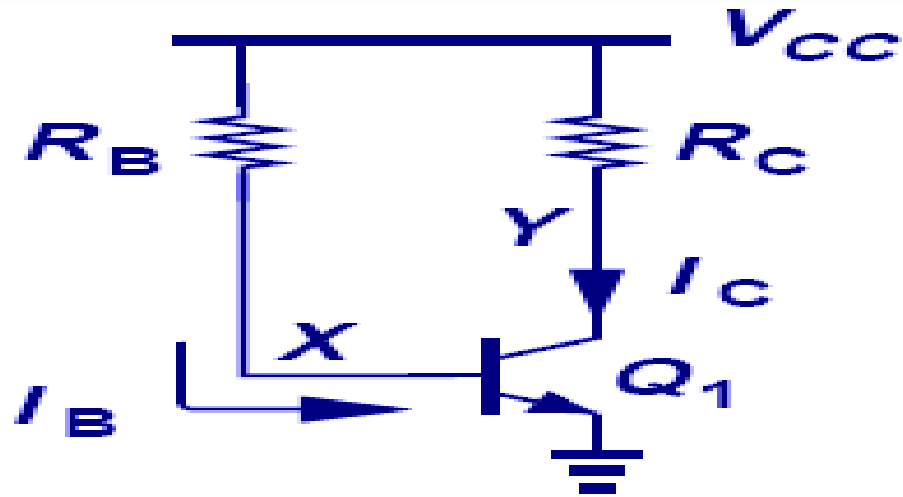


- Transistors and circuits must be biased because
 - (1) transistors must operate in the active region,
 - (2) their small-signal parameters depend on the bias conditions.

Methods For BJT Biasing

- Base resistor method
- Emitter bias method
- Biasing with collector-feedback resistor
- Voltage-divider bias

Biassing with Base Resistor



$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta \frac{V_{CC} - V_{BE}}{R_B}$$

- Assuming a constant value for V_{BE} , one can solve for both I_B and I_C and determine the terminal voltages of the transistor.

Bias point is sensitive to β variations.

Biassing with Base Resistor

- This method is also called fixed bias method

The stability factor $S = \frac{\beta + 1}{1 - \beta \left(\frac{dI_B}{dI_C} \right)}$

Since I_B is independent of I_C , $dI_B/dI_C = 0$, hence $S = \beta + 1$

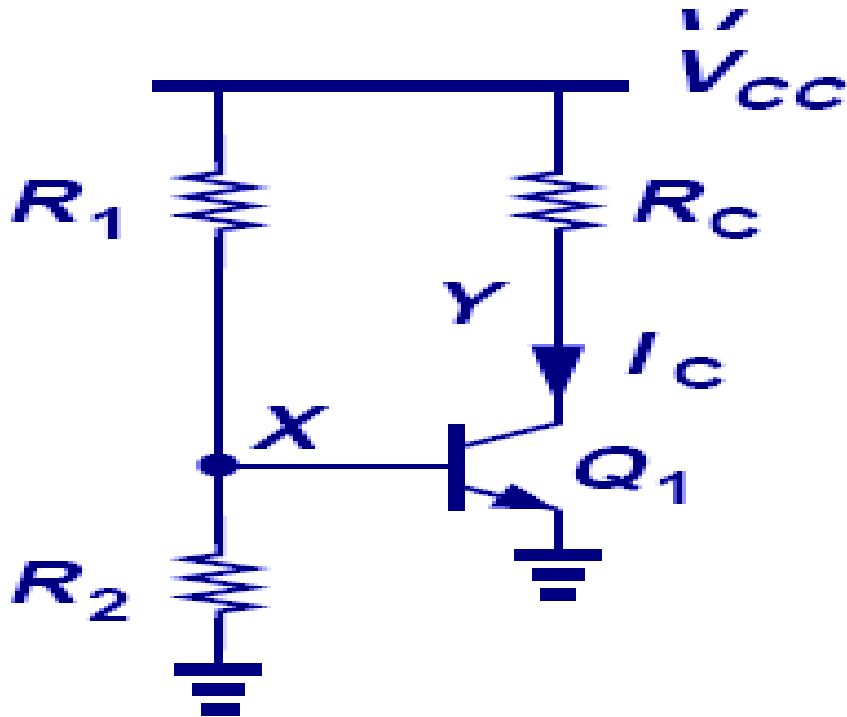
Biassing with Base Resistor

- Advantages
 - The biasing circuit is simple, since one resistance R_B is required
 - Biasing conditions can easily be set and the calculations are simple
 - There is no loading of the source by the biasing circuit, since no resistor is employed across base-emitter junction

Biasing with Base Resistor

- Disadvantages
 - The biasing provides poor stabilization
 - The stability factor is very high and hence there are strong chances of thermal runaway. Poor thermal stability

Improved Biasing: Resistive Divider

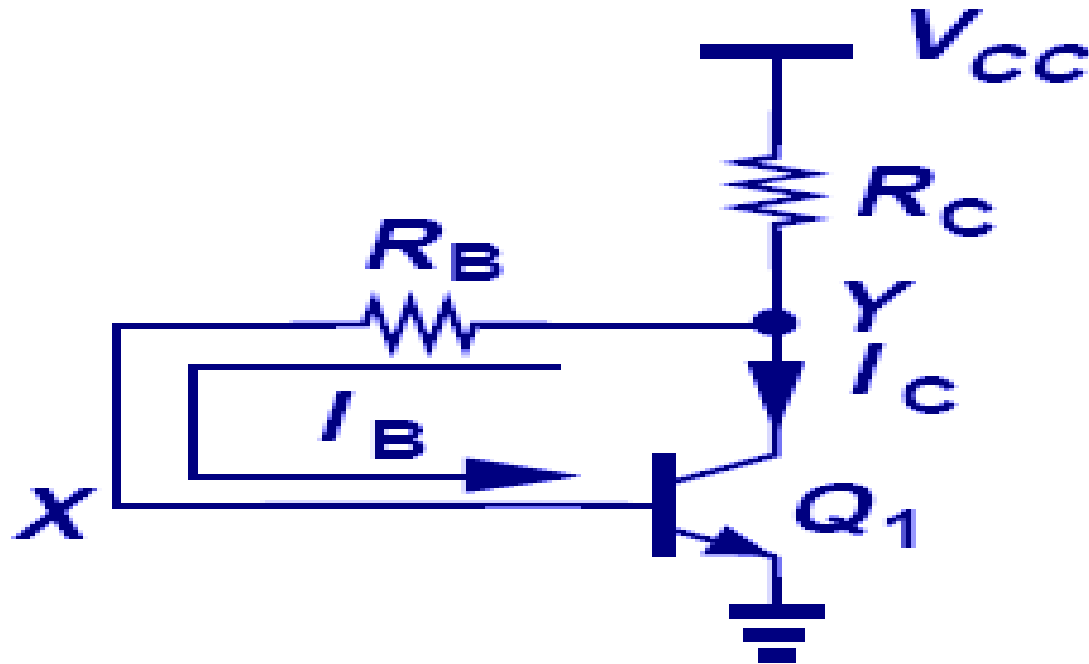


$$V_X = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$I_C = I_S \exp\left(\frac{R_2}{R_1 + R_2} \frac{V_{CC}}{V_T}\right)$$

- Using resistor divider to set V_{BE} , it is possible to produce a collector current I_C that is relatively independent of β if base current is small.

Self-Biasing Technique (Collector-



Disadvantage

The circuit provides a negative feedback which reduces the gain of the amplifier

- This bias method utilizes the collector voltage to provide the necessary V_x and I_B .
- One important characteristic of this technique is that collector has a higher potential than the base, thus guaranteeing active operation of the transistor.

Self-Biasing Design Guidelines

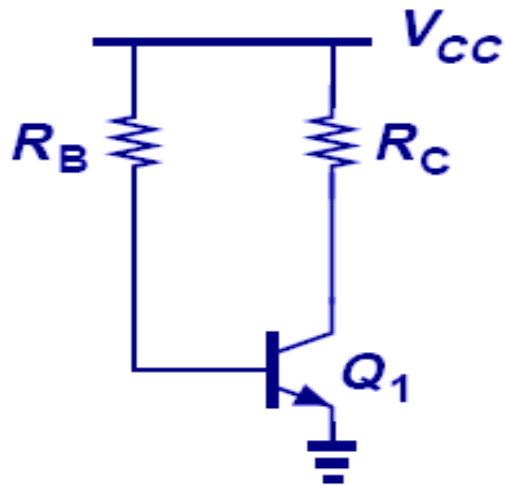
$$R_C \gg \frac{R_B}{\beta}$$

(1) provides insensitivity to β .

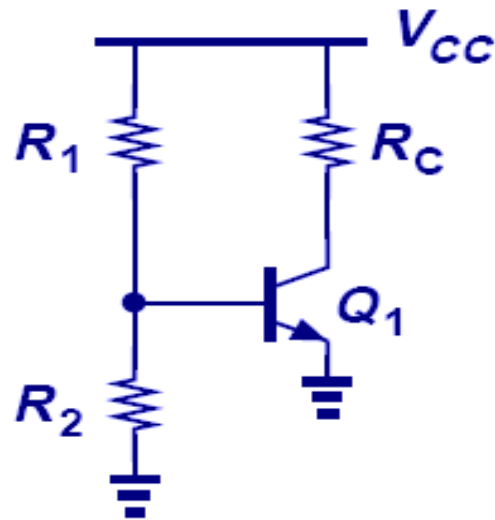
$$\Delta V_{BE} \ll V_{CC} - V_{BE}$$

(2) provides insensitivity to variation in V_{BE} .

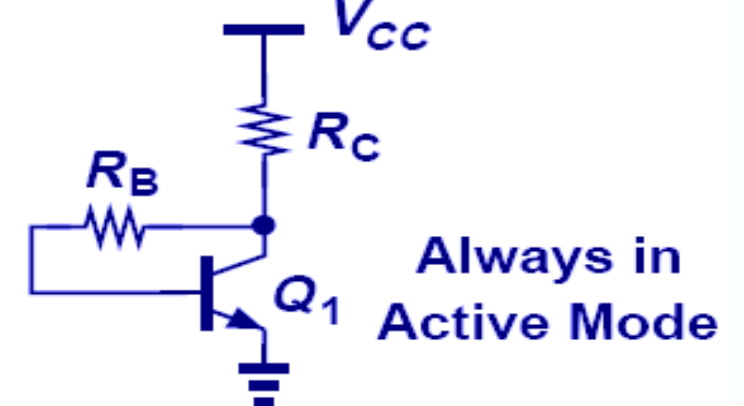
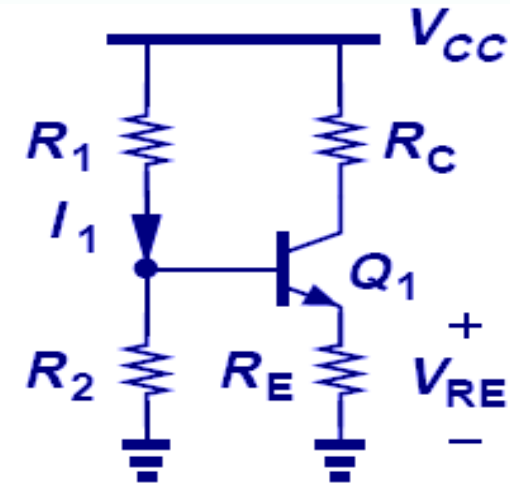
Summary of Biasing Techniques



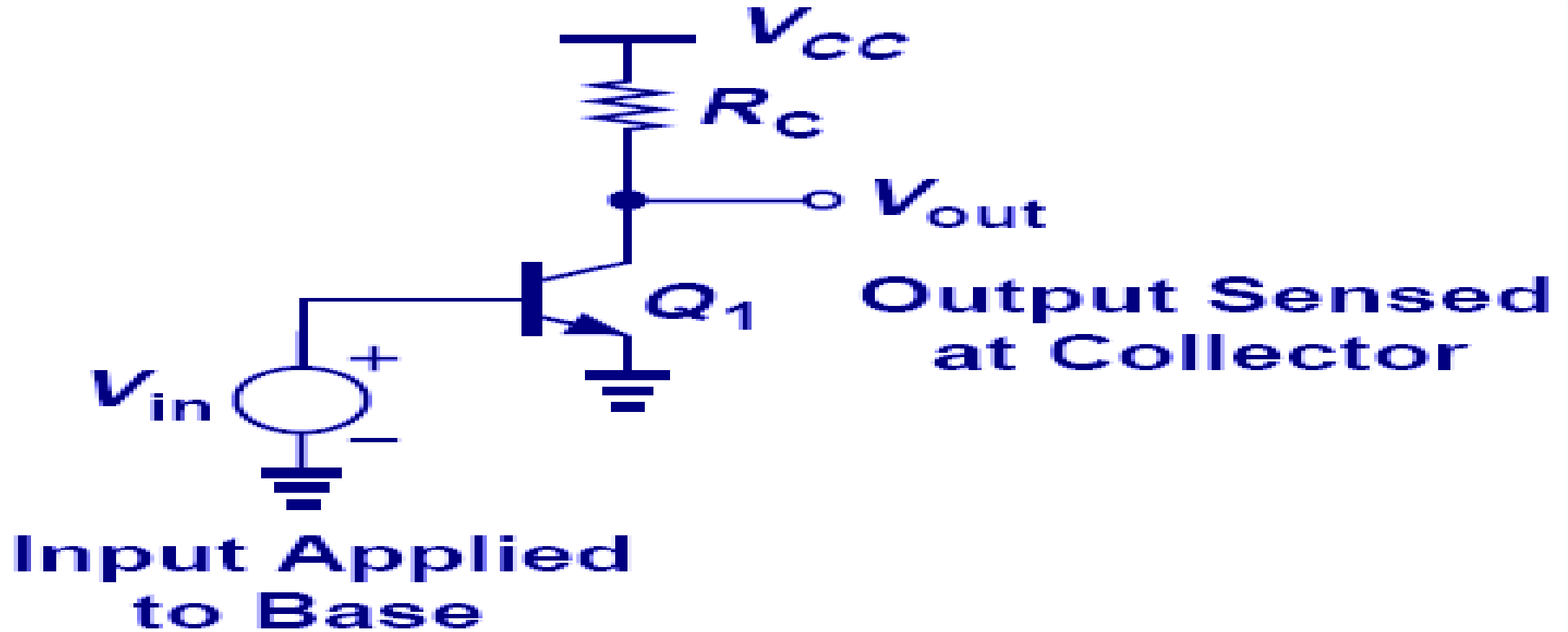
Sensitive
to β



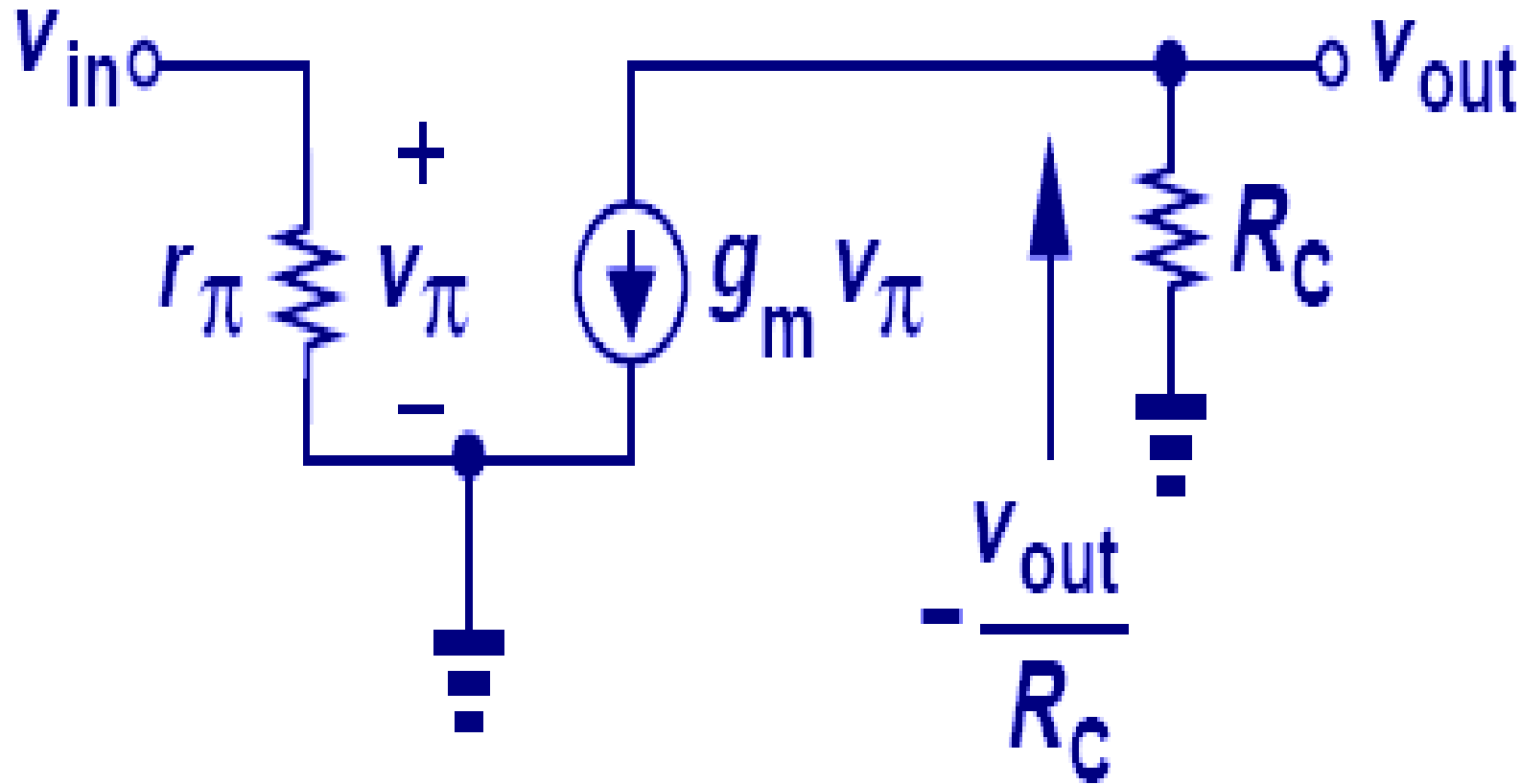
Sensitive
to Resistor Error



Common-Emitter Topology



Small Signal of CE Amplifier

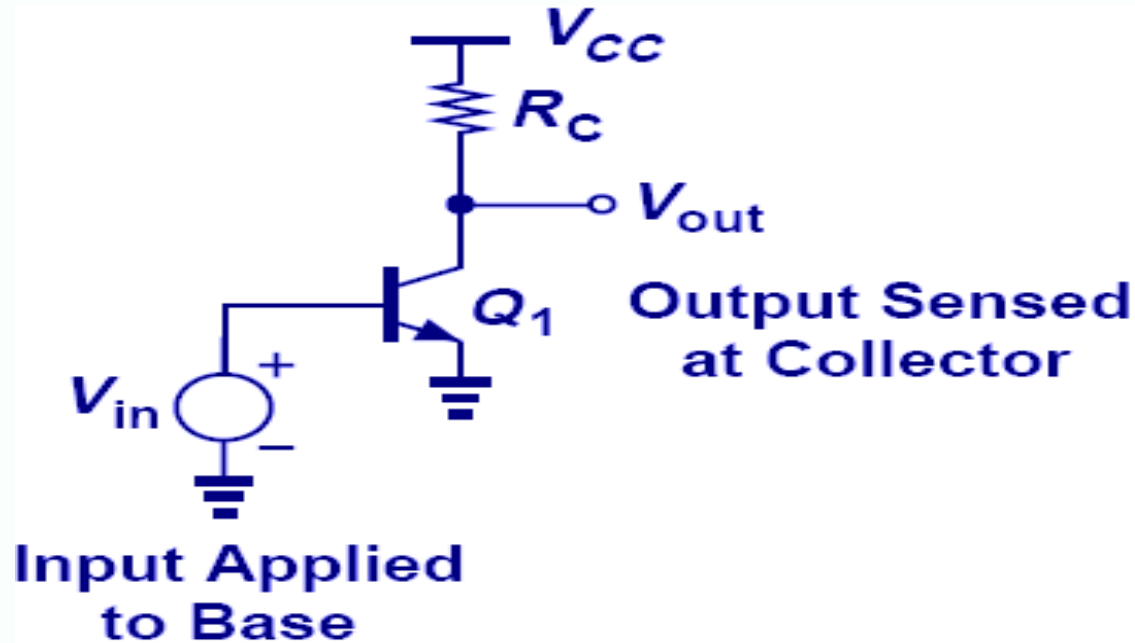


$$A_v = \frac{V_{out}}{V_{in}}$$

$$-\frac{V_{out}}{R_C} = g_m v_\pi = g_m v_{in}$$

$$A_v = -g_m R_C$$

Limitation on CE Voltage Gain

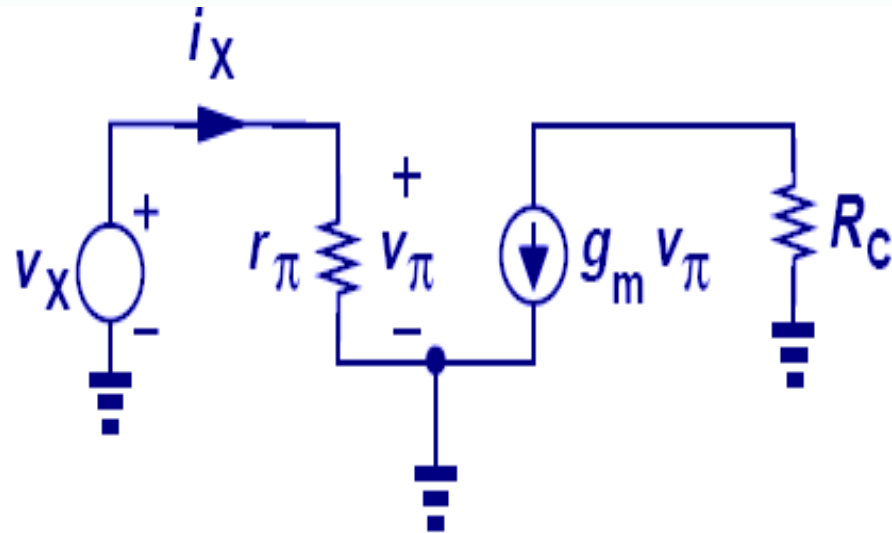


$$|A_v| = \frac{I_C R_C}{V_T} \quad |A_v| = \frac{V_{RC}}{V_T} \quad |A_v| < \frac{V_{CC} - V_{BE}}{V_T}$$

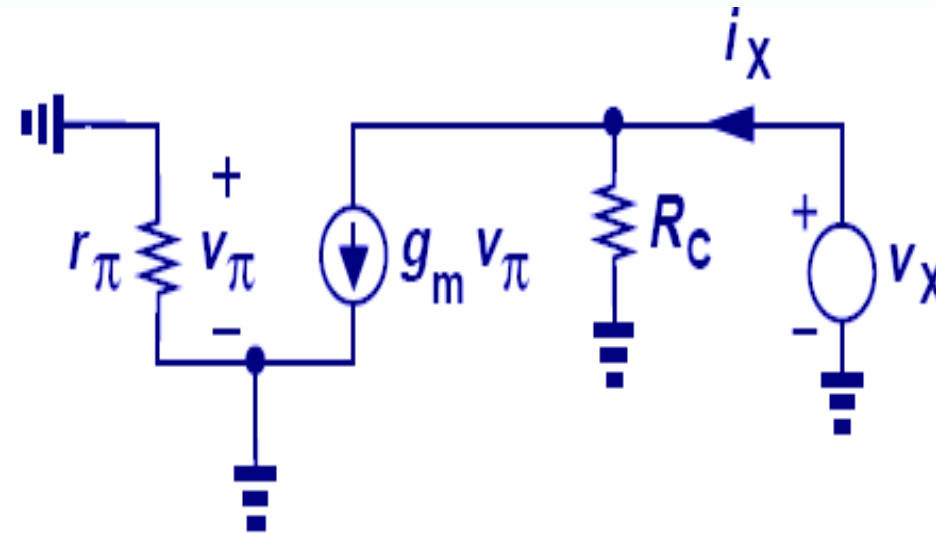
- Since g_m can be written as I_C/V_T , the CE voltage gain can be written as the ratio of V_{RC} and V_T .
- V_{RC} is the potential difference between V_{CC} and V_{CE} , and V_{CE} cannot go below V_{BE} in order for the transistor to be in active region.

I/O Impedances of CE Stage

- When measuring output impedance, the input port has to be grounded so that $V_{in} = 0$.



$$R_{in} = \frac{v_X}{i_X} = r_\pi$$



$$R_{out} = \frac{v_X}{i_X} = R_C$$

Intrinsic Gain

$$A_v = -g_m(R_C \parallel r_o)$$

$$R_{out} = R_C \parallel r_o$$

$$A_v = -g_m r_o$$

$$|A_v| = \frac{V_A}{V_T}$$

- As R_C goes to infinity, the voltage gain reaches the product of g_m and r_o , which represents the maximum voltage gain the amplifier can have.
- The intrinsic gain is independent of the bias current.

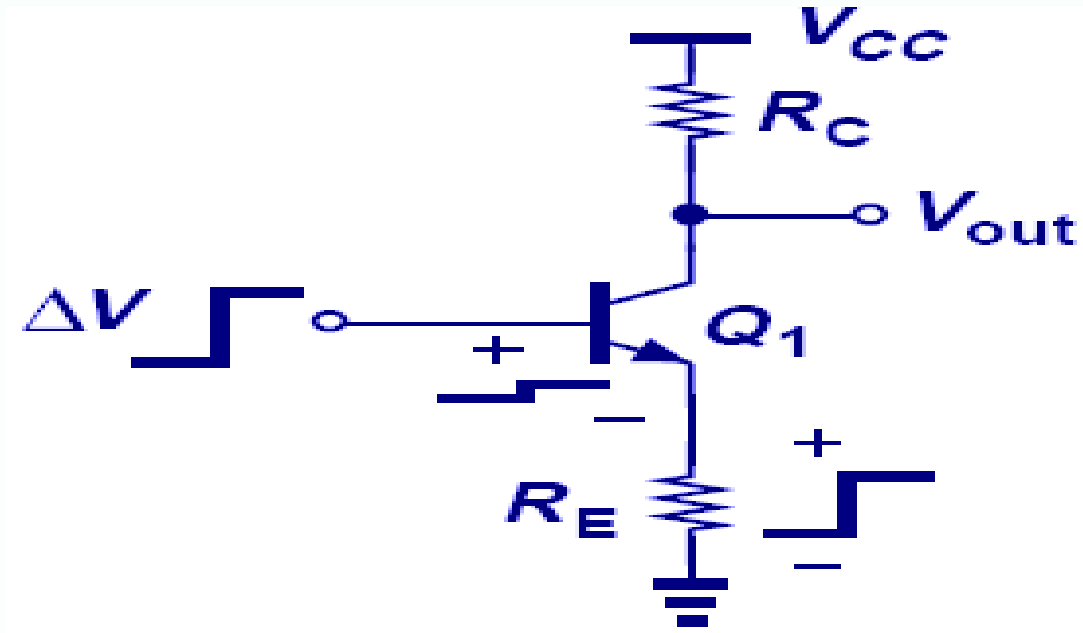
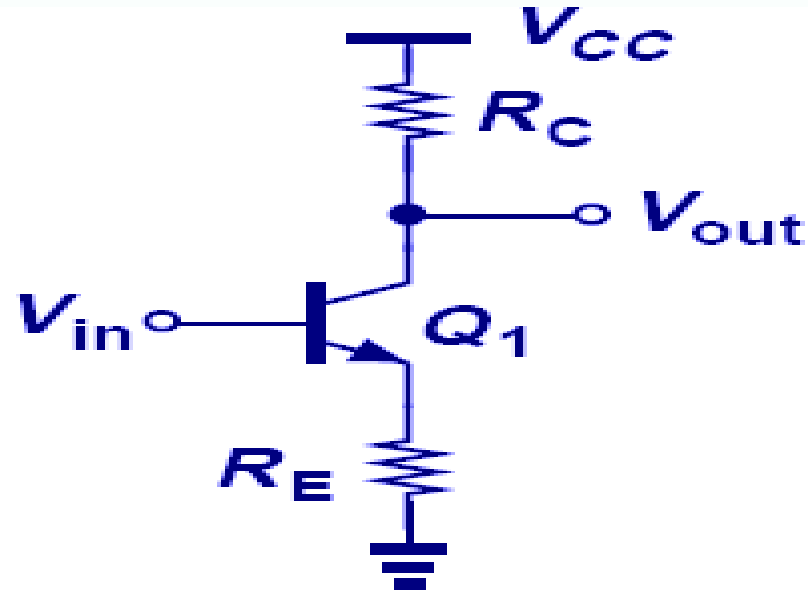
Current Gain

$$A_I = \frac{i_{out}}{i_{in}}$$

$$A_I|_{CE} = \beta$$

- Another parameter of the amplifier is the current gain, which is defined as the ratio of current delivered to the load to the current flowing into the input.
- For a CE stage, it is equal to β .

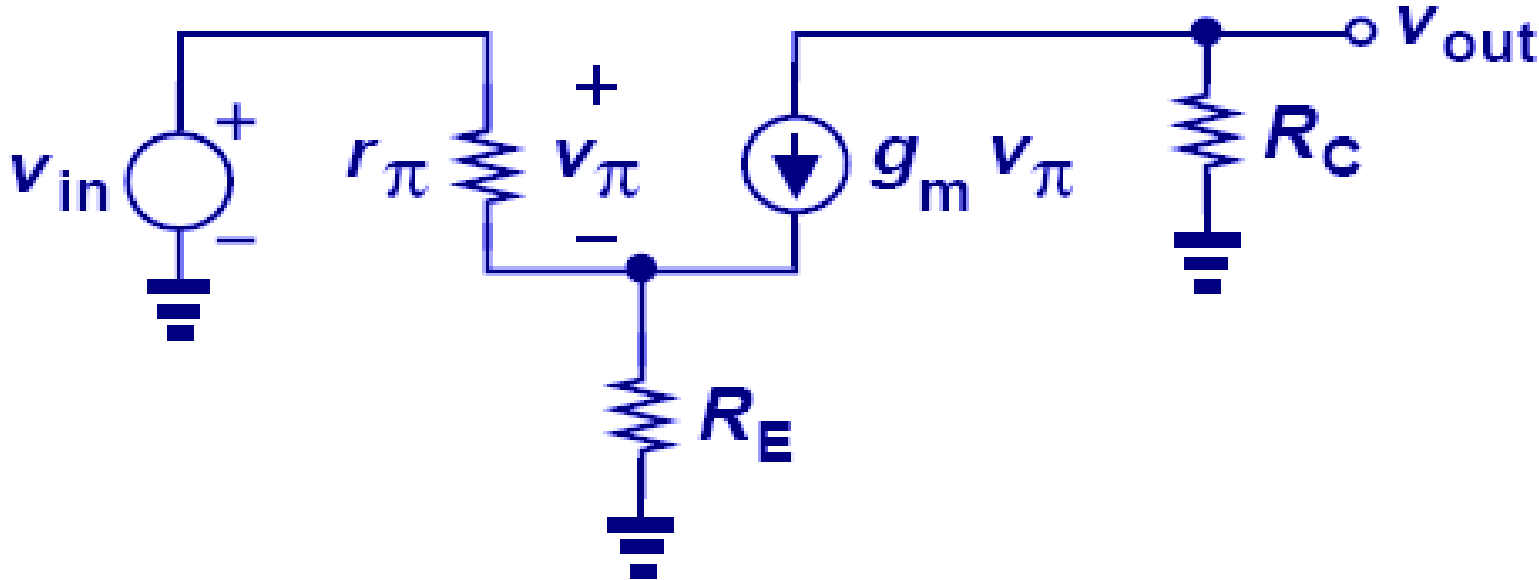
Emitter Degeneration



- By inserting a resistor in series with the emitter, we “degenerate” the CE stage.

This topology will decrease the gain of the amplifier but improve other aspects, such as linearity, and input impedance.

Small-Signal Model

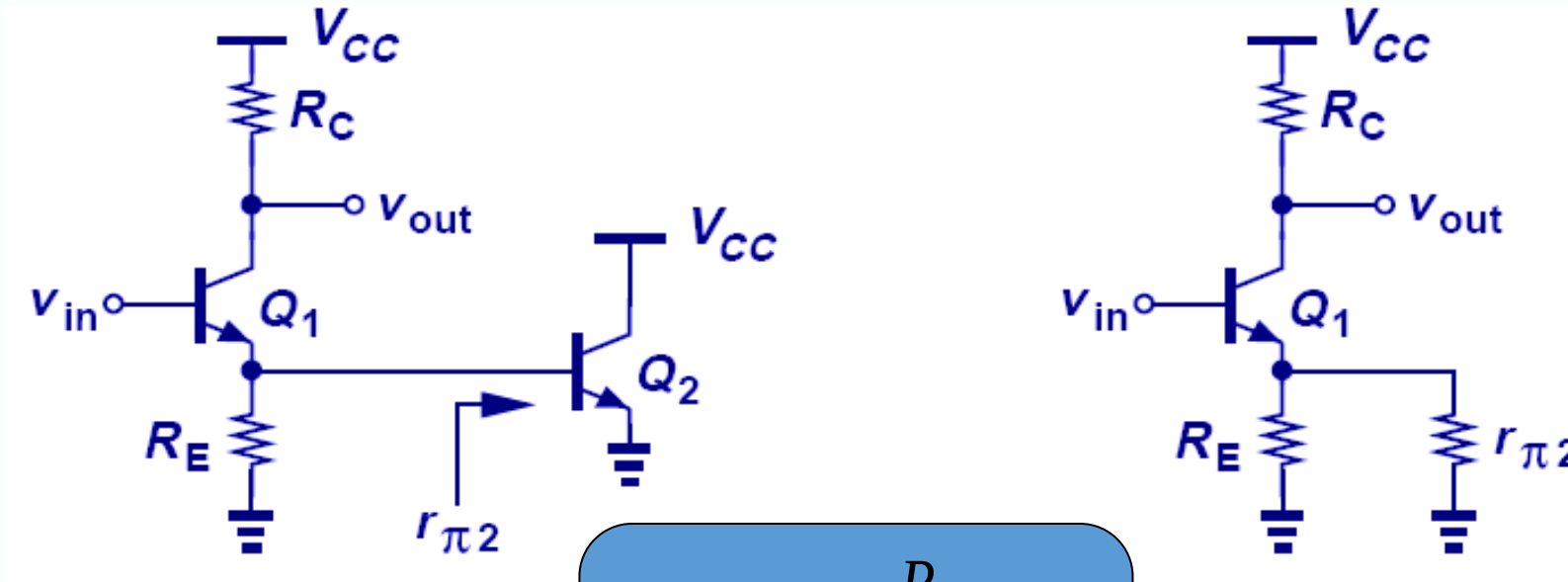


$$A_v = -\frac{g_m R_C}{1 + g_m R_E}$$

$$A_v = -\frac{R_C}{\frac{1}{g_m} + R_E}$$

- Interestingly, this gain is equal to the total load resistance to ground divided by $1/g_m$ plus the total resistance placed in series with the emitter.

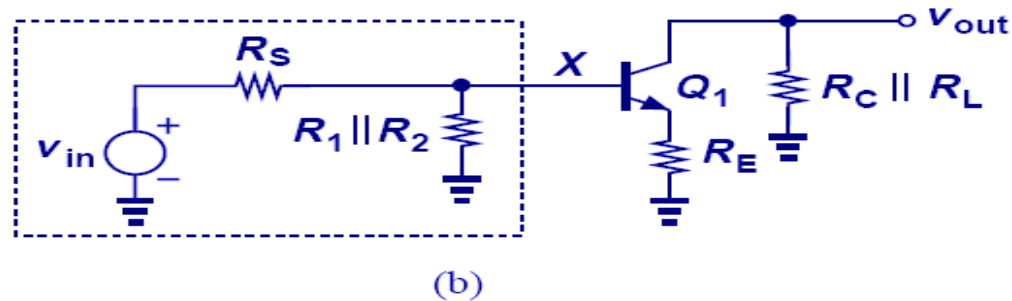
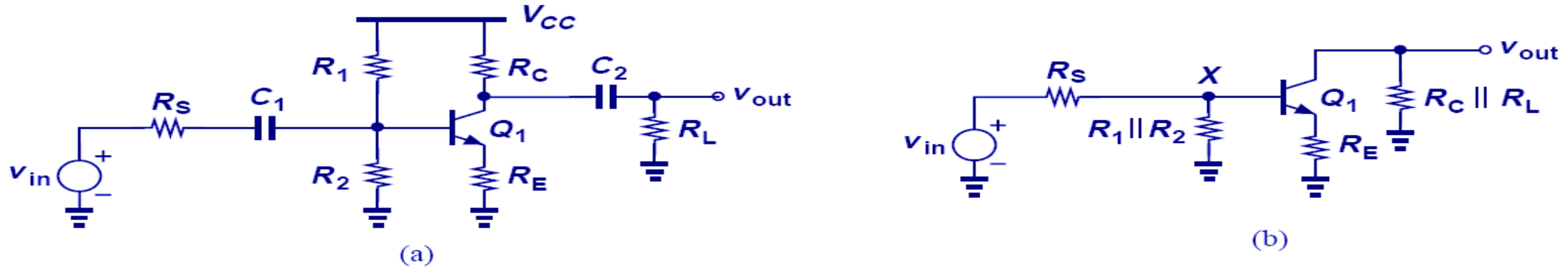
Emitter Degeneration Example



$$A_v = -\frac{R_C}{\frac{1}{g_{m1}} + R_E \parallel r_{\pi 2}}$$

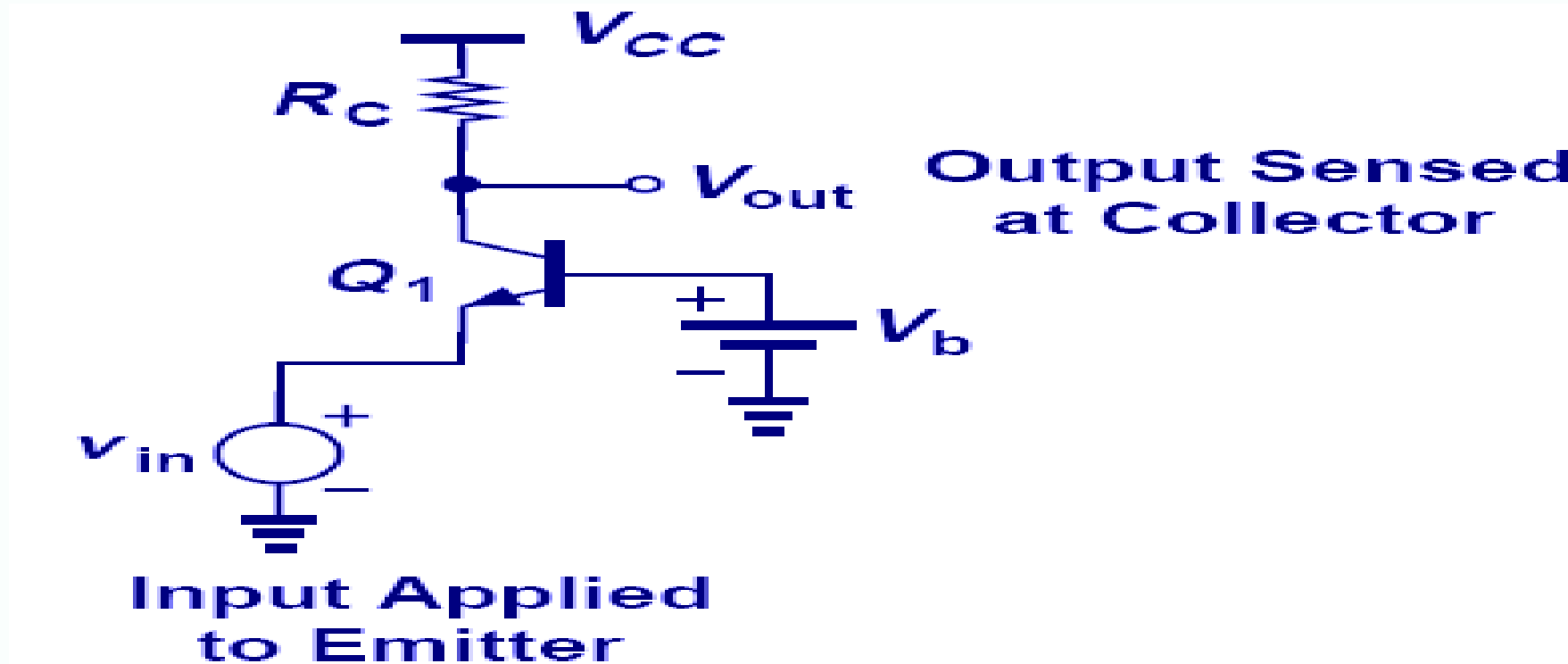
- The input impedance of Q_2 can be combined in parallel with R_E to yield an equivalent impedance that degenerates Q_1 .

Complete CE Stage



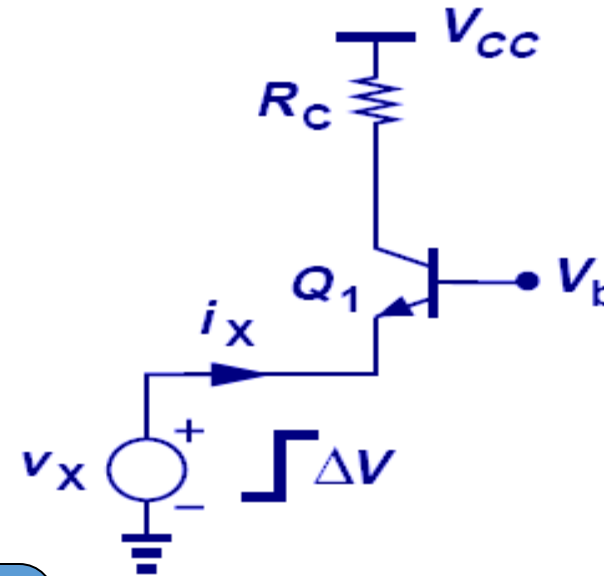
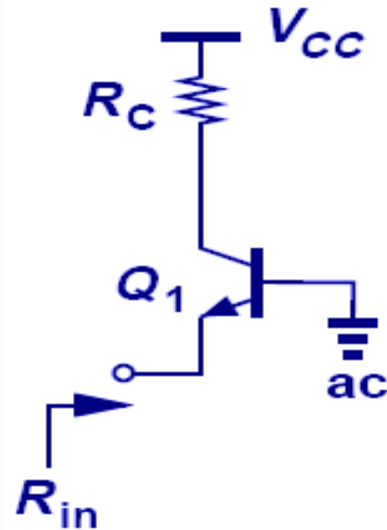
$$A_v = \frac{-R_C \parallel R_L}{\frac{1}{g_m} + R_E + \frac{R_s \parallel R_1 \parallel R_2}{\beta + 1}}$$

Common Base (CB) Amplifier



- In common base topology, where the base terminal is biased with a fixed voltage, emitter is fed with a signal, and collector is the output.

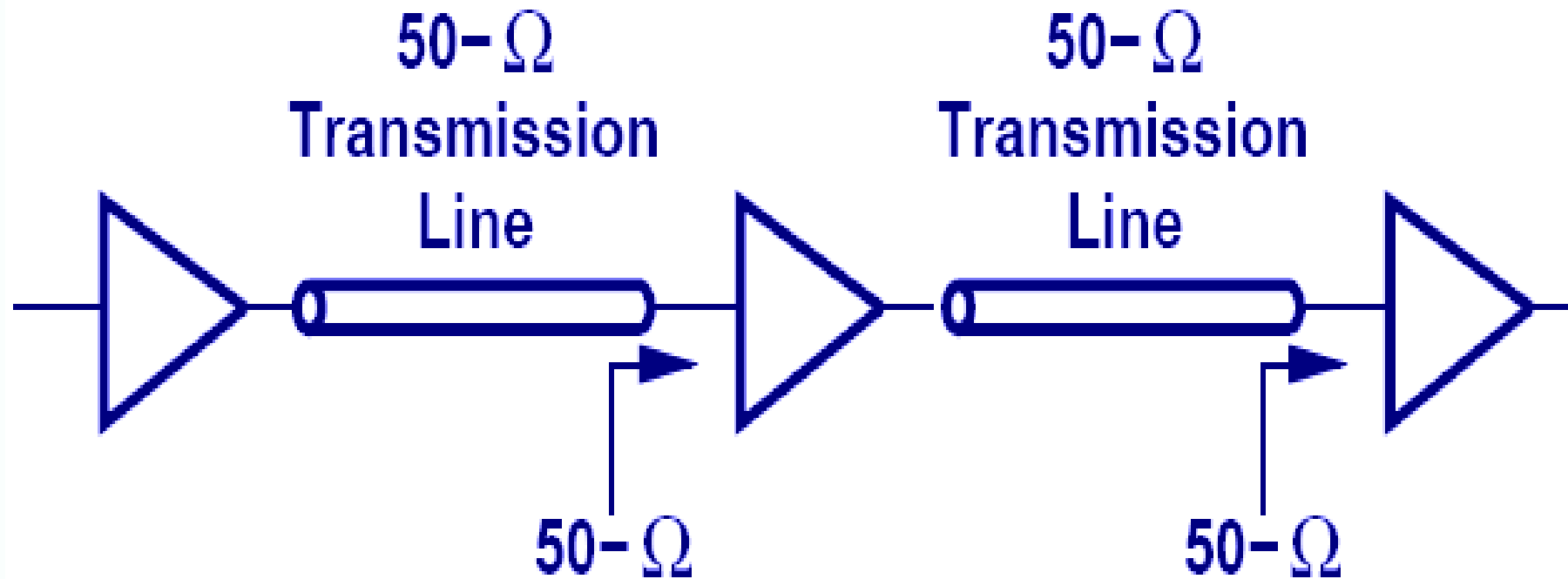
Input Impedance of CB



$$R_{in} = \frac{1}{g_m}$$

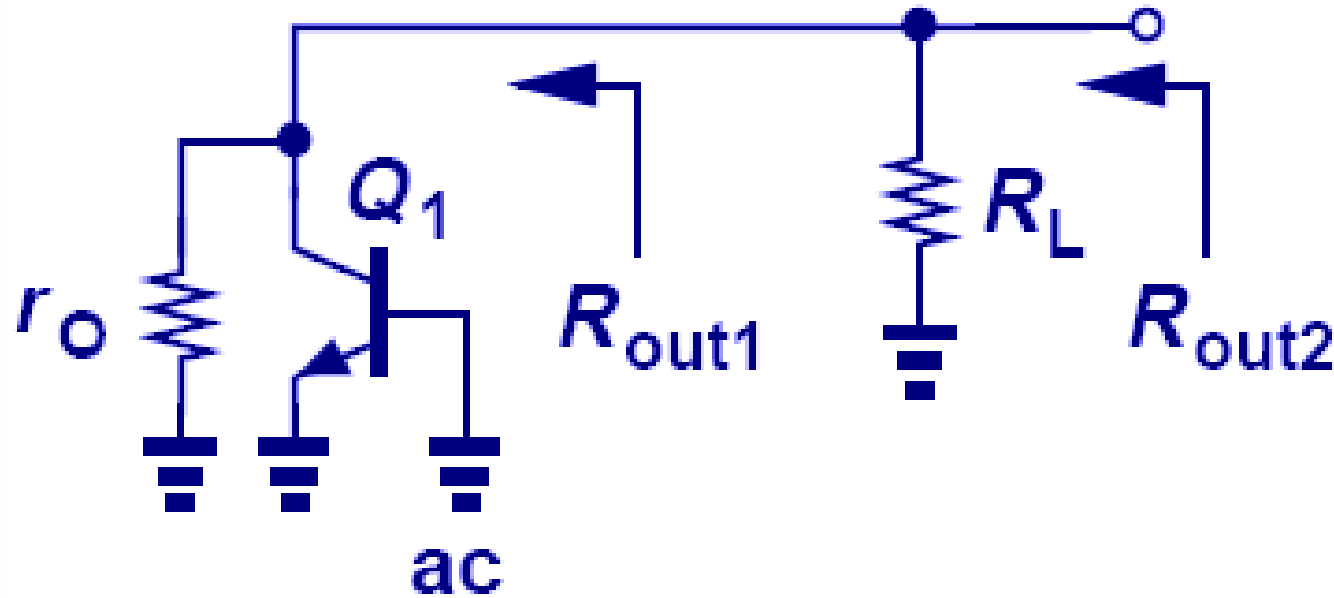
- The input impedance of CB stage is much smaller than that of the CE stage.

Practical Application of CB Stage



- To avoid “reflections”, need impedance matching.
- CB stage’s low input impedance can be used to create a match with $50\text{ }\Omega$.

Output Impedance of CB Stage

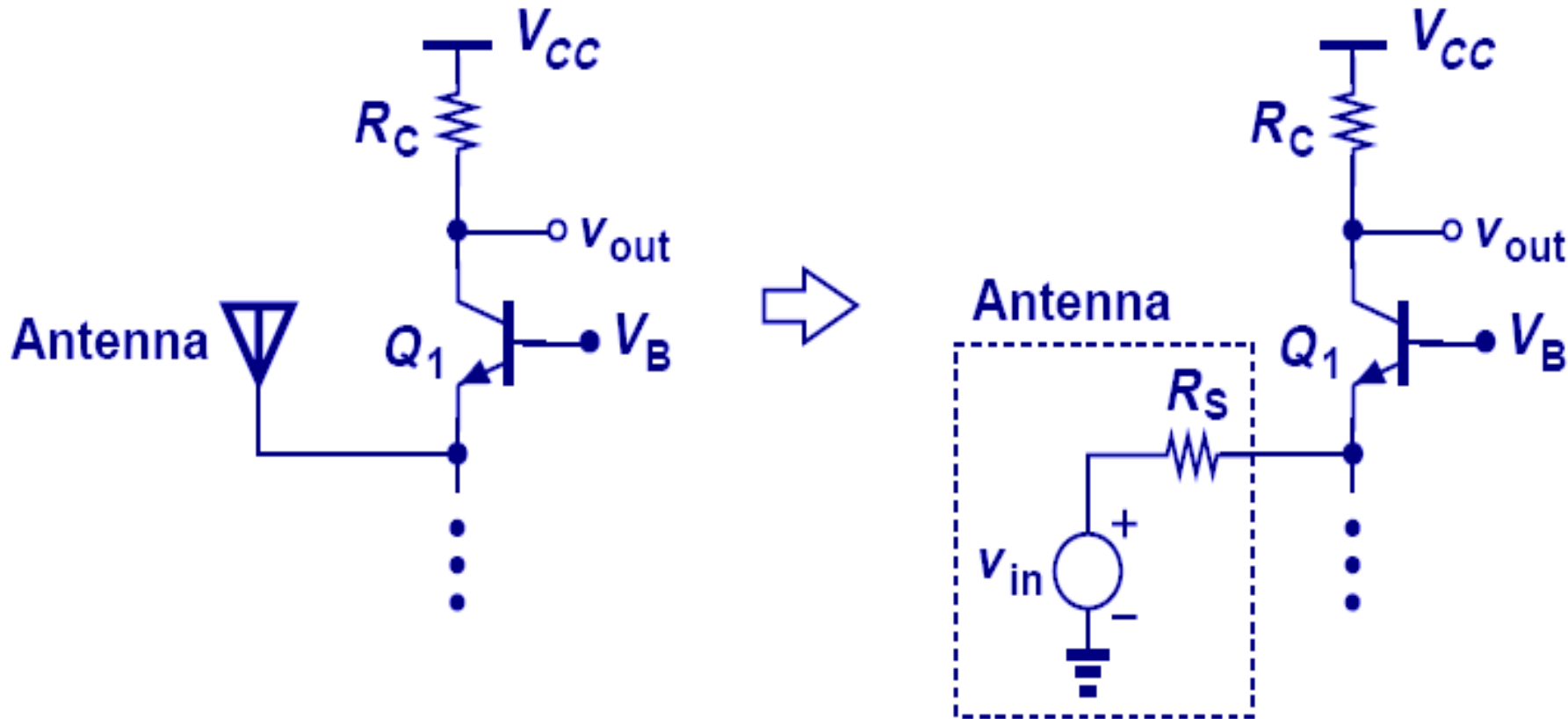


$$R_{out} = r_O \parallel R_C$$

- The output impedance of CB stage is similar to that of CE stage.

Practical Example of CB Stage

- An antenna usually has low output impedance; therefore, a common base (CB) stage is a good choice for the





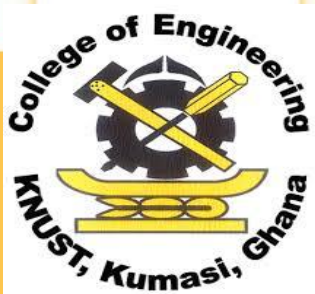
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Thank You

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COE 271 Semiconductor Devices

Lecture 6

Field Effect Transistors - JFET

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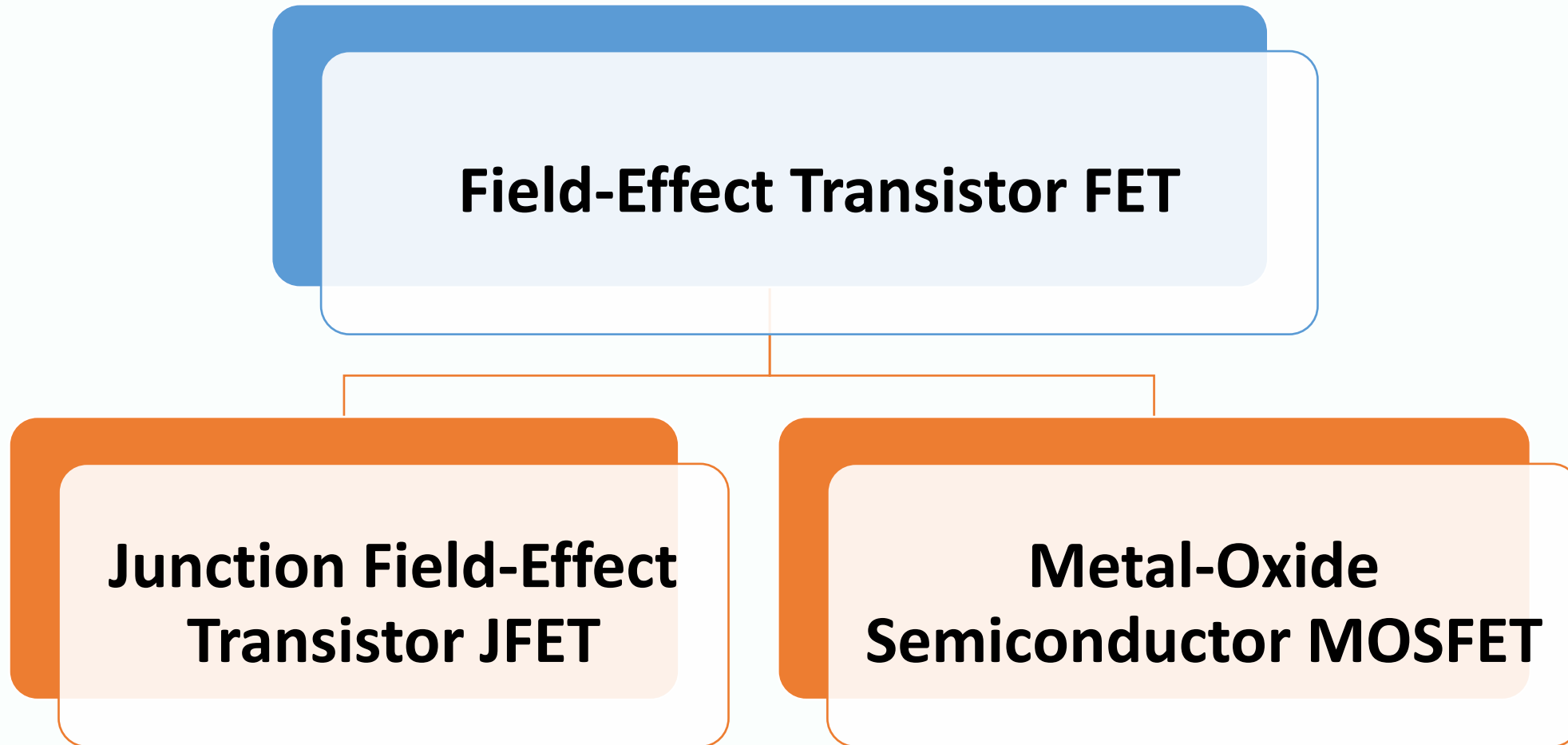
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“Technological change is like an axe in the hands of a pathological criminal”

Albert Einstein

Overview



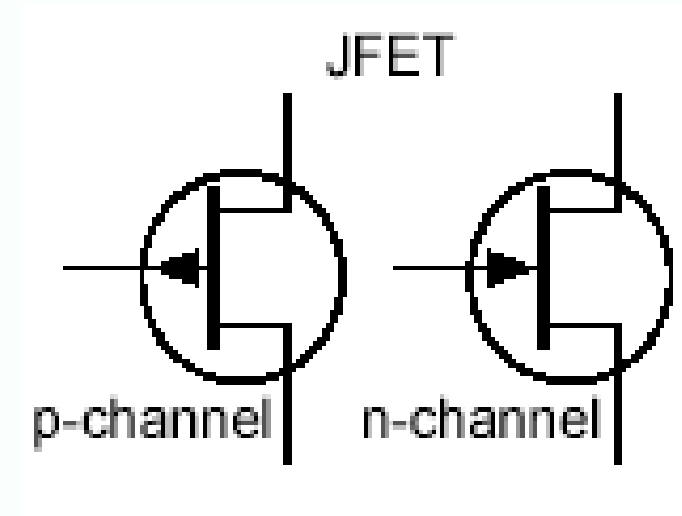
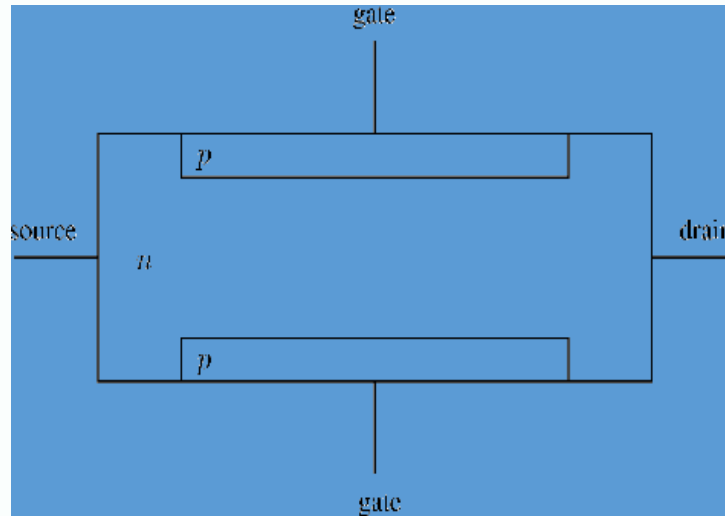
Field-Effect Transistors FET

- The field-effect transistor (FET) controls the current between two points but does so differently than the bipolar transistor.
- The FET operates by the effects of an electric field on the flow of electrons through a single type of semiconductor material.
- This is why the FET is sometimes called a unipolar transistor.

Field-Effect Transistors FET

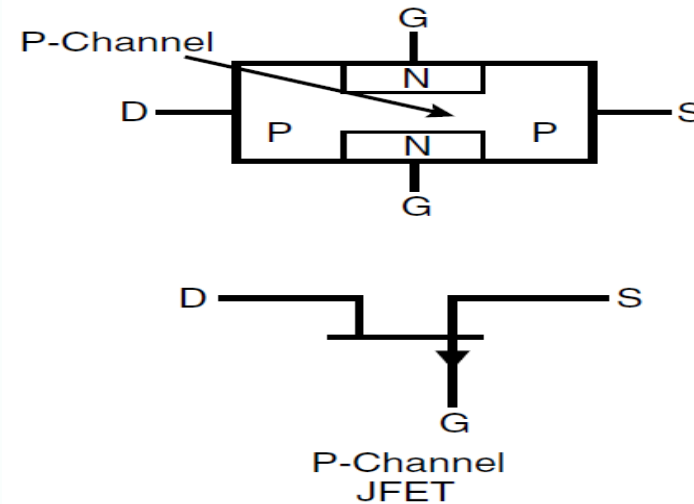
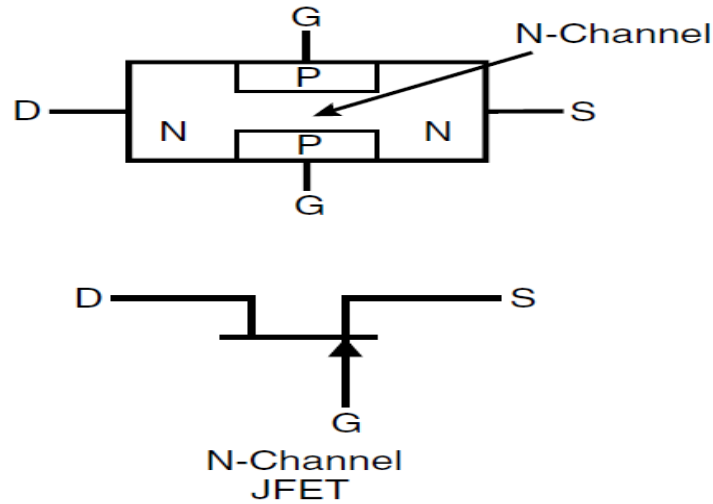
- Current moves within the FET in a channel, from the source (S) connection to the drain (D) connection.
- A gate (G) terminal generates an electric field that controls the current.
- The channel is made of either N-type or P-type semiconductor material
- FET is specified as either an N-channel or P-channel device
- Majority carriers flow from source to drain.

Field-Effect Transistors FET



- -In N-channel devices, electrons flow so the drain potential must be higher than that of the Source ($V_{DS} > 0$)
- - In P-channel devices, the flow of holes requires that $V_{DS} < 0$

Field-Effect Transistors FET



- Like bipolar junction transistors, field-effect transistors (FETs) are three terminal semiconductor devices capable of power gain.
- Qualitatively, they operate much like junction transistors, but they have much higher input impedance and lower transconductance and voltage gain.
- Also, they have a larger variation in their ' V_{BE} ' equivalent (called V_{GS}) than bipolar transistors

Field-Effect Transistors FET

- a conducting *channel* between the *drain* and *source* terminals is controlled by a voltage applied to the *gate* terminal.
- The channel can be made of either N-type or P-type material.
- N-channel is more common since the conductivity of N-type semiconductor (in which electrons carry the current) is higher than that of P-type (in which holes do)
- The gate region of a JFET consists of material of opposite type to that of the channel; thus, the gate and channel form a diode

Field-Effect Transistors FET

- The transconductance of a JFET is proportional to $\sqrt{I_D}$

$$g_m = \Delta I_D / \Delta V_{GS} .$$

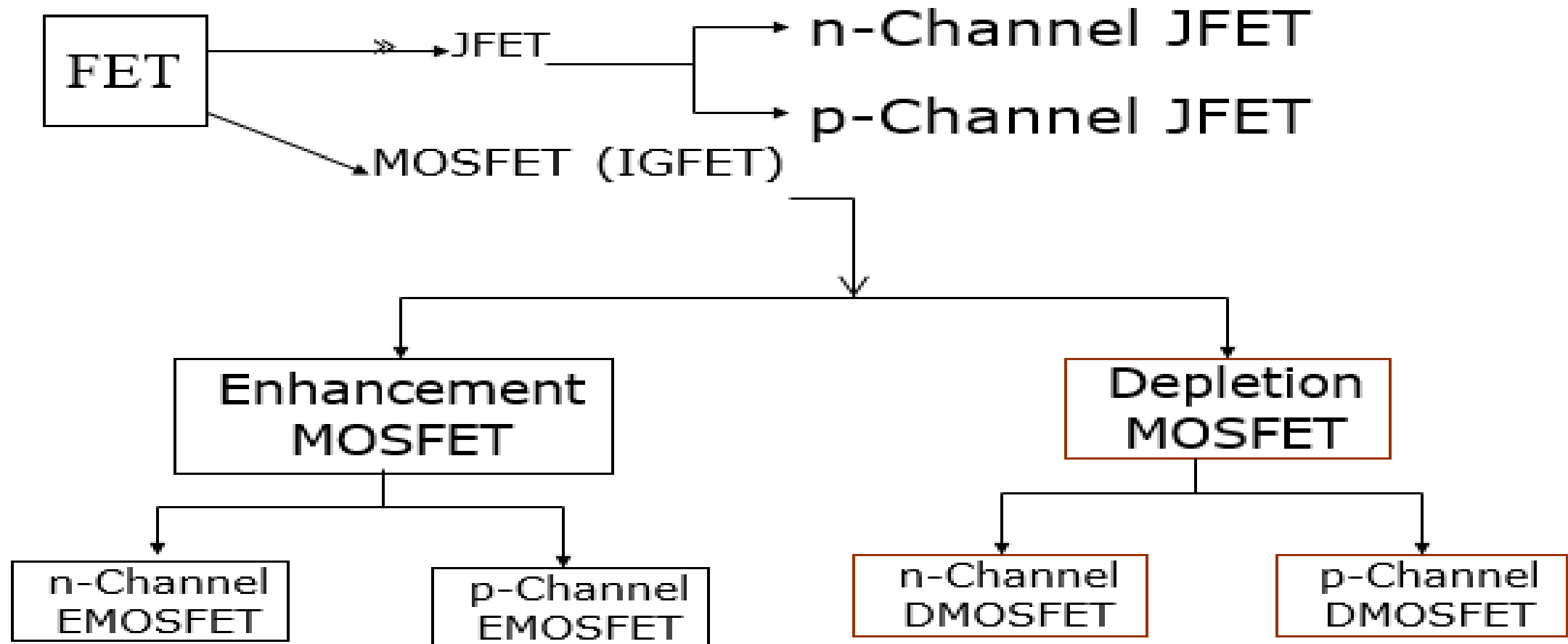
FET Advantages over BJT

- Unipolar device i. e. operation depends on only one type of charge carriers (h or e)
- Voltage controlled Device (gate voltage controls drain current)
- Very high input impedance ($\approx 10^9 - 10^{12} \Omega$)
- Source and drain are interchangeable in most Low-frequency applications
- Low Voltage Low Current Operation is possible (Low-power consumption)

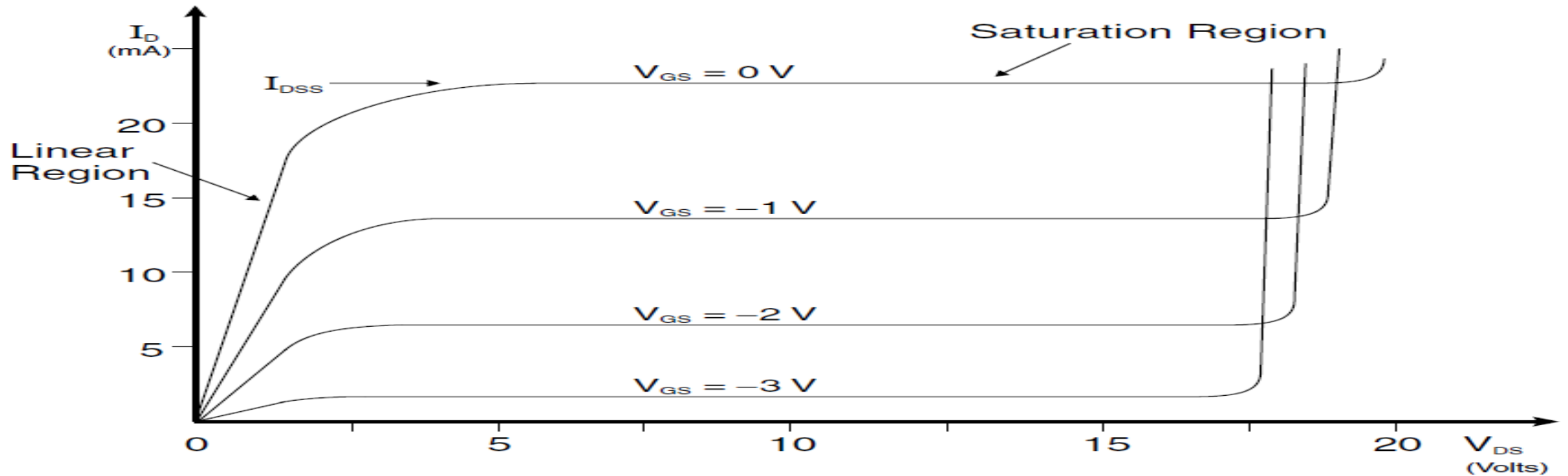
FET Advantages over BJT

- Less Noisy as Compared to BJT
- No minority carrier storage (Turn off is faster)
- Self limiting device
- Very small in size, occupies very small space in ICs
- Low voltage low current operation is possible in MOSFETS
- Zero temperature drift of output is possible

Types of FET



Characteristics of JFET (transfer curve)



- The voltage-controlled current-source behavior occurs as long as the drain-source voltage V_{DS} is sufficiently high.
- This is called the *saturation* region of the FET characteristic

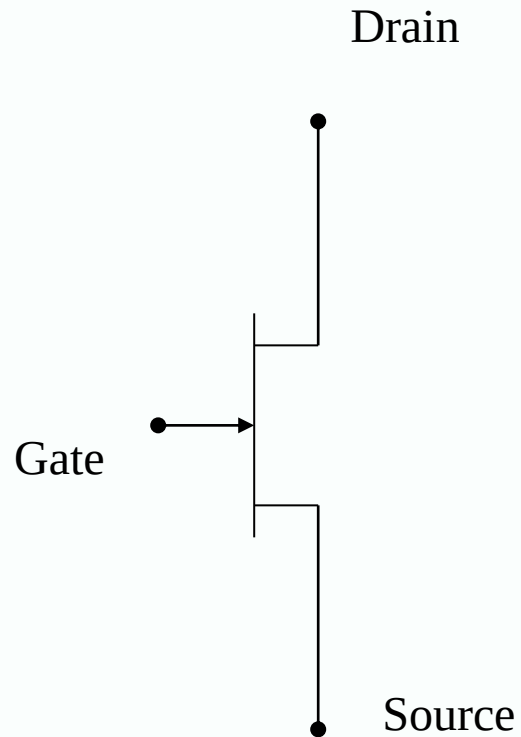
Characteristic of JFET

- For V_{DS} smaller than a volt or two, a JFET behaves like a voltage controlled resistor rather than a current source,
- i.e. the *slope* of the $I-V$ characteristic is controlled by the gate–source voltage.
- This is the *linear* region of the FET characteristic, useful for automatic gain control (AGC) and modulation applications
- For FETs operated in the saturation region, the relationship is quadratic (shockley's equation):

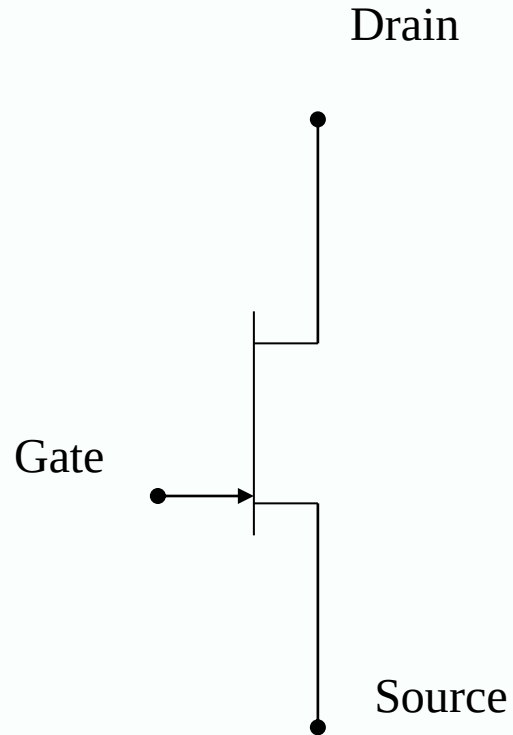
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

where V_P is the pinch-off voltage and I_{DSS} is the saturation drain current for $V_{GS} = 0$ (i.e. gate shorted to source).

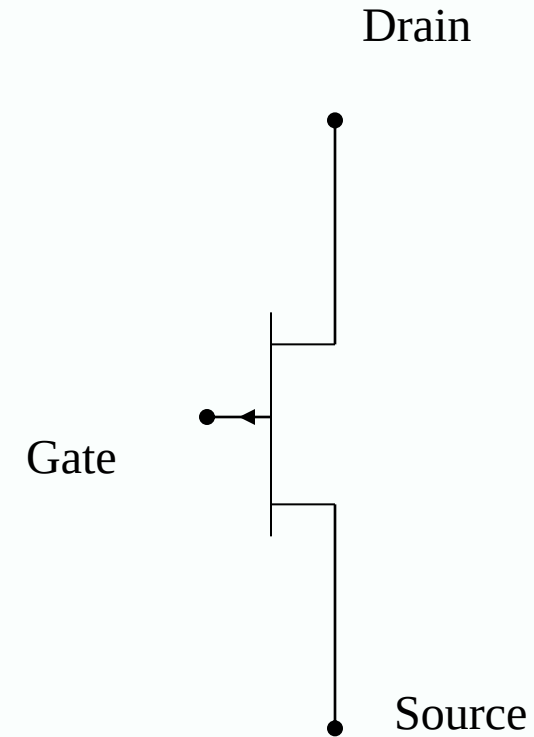
JFET Symbols



n-channel JFET



n-channel JFET
Offset-gate symbol



p-channel JFET

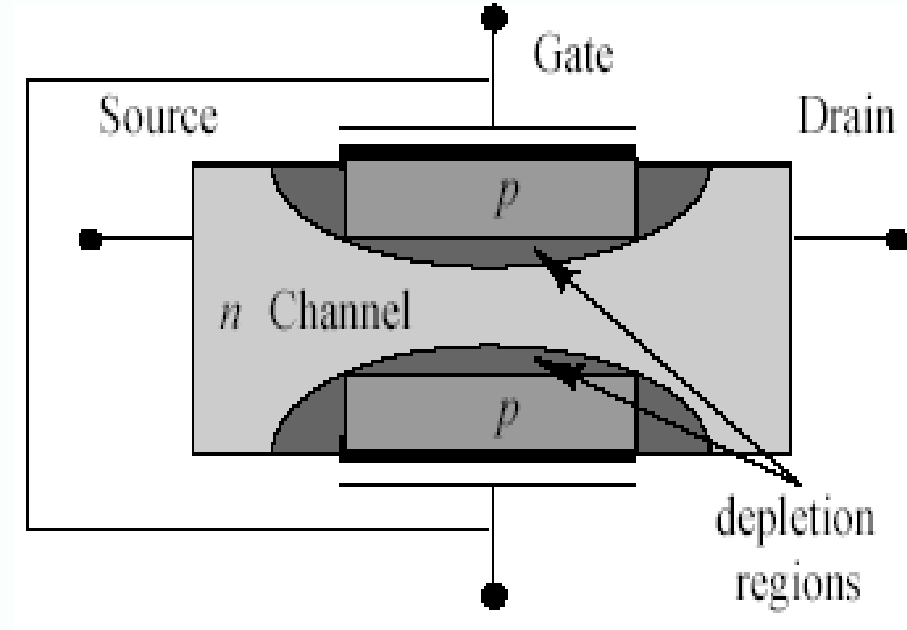
Operation Regions of JFET

Ohmic Region

Saturation Region

Breakdown Region

JFET- Ohmic Region



- Ohmic Region – the depletion regions that form around the gate section (for small drain-source voltages)
- These regions widen as gate voltage increases and thus the channel narrows

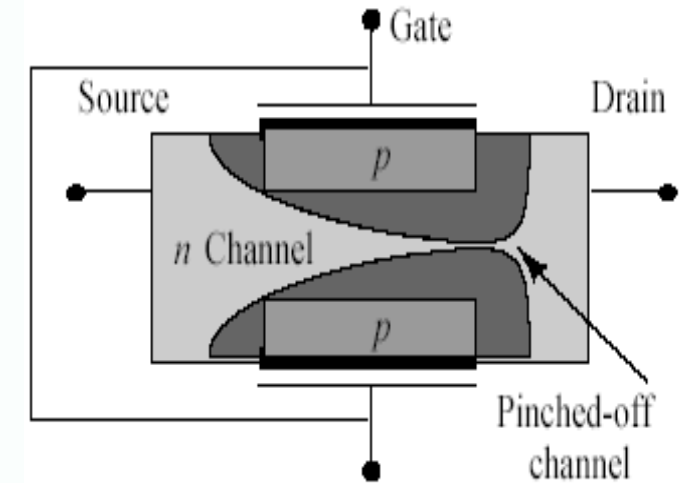
JFET- Ohmic Region

As the gate-source voltage is increased above $-V_p$ for small values of the drain-source voltage, the pn-junction between gate and channel becomes more reverse biased and the width of the depletion region increases, narrowing the channel

This increases the channel resistance, giving the voltage-controlled resistor behaviour of the ohmic region

JFET – Saturation Region

As the drain-source voltage is increased, the depletion regions further widen near the drain end and the channel becomes narrower, because the reverse bias of the pn junction is larger near the drain



For sufficiently large drain-source voltage, the channel will eventually be **pinched off** and any further increase in V_{DS} will not change the drain current.

This is the saturation region

JFET – Breakdown Region

- When the drain-source voltage increase above a breakdown voltage, V_B , the drain current will increase very quickly due to avalanche condition
- This causes excessive generation of heat and the destruction of the JFET
- This is the **breakdown** region

JFET Equations

The following equations describe the operation of n-channel JFETs. They also apply for p-channel JFETs by substituting V_{SG} for V_{GS} and V_{SD} for V_{DS}

Cut — off region : $V_{GS} < -V_P$

Breakdown region : $V_{DS} > V_B$

JFET Equations

Ohmic region : $v_{DS} < 0.25(v_{GS} + V_P)$, $v_{GS} > -V_P$

$$R_{DS} = \frac{V_P^2}{2I_{DSS}(v_{GS} + V_P)} \quad (\text{equivalent drain - to - source resistance})$$

$$i_D \approx \frac{v_{DS}}{R_{DS}}$$

Saturation region : $v_{DS} \geq v_{GS} + V_P$, $v_{GS} > -V_P$

$$i_D = \frac{I_{DSS}}{V_P^2} (v_{GS} + V_P)^2$$

Applications of JFET

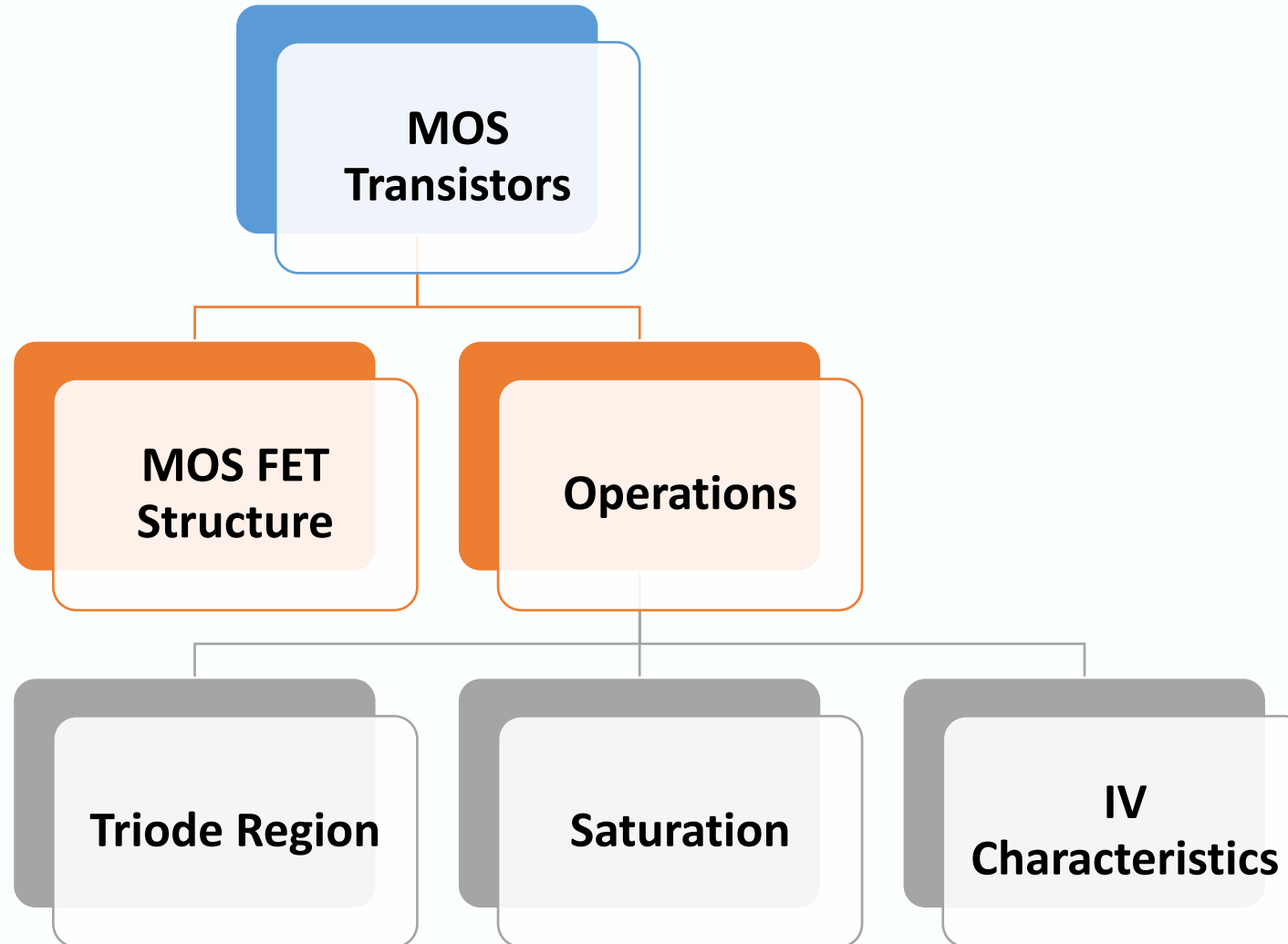
- MOSFET

- Primary component in high-density VLSI chips such as memories and microprocessors

JFET

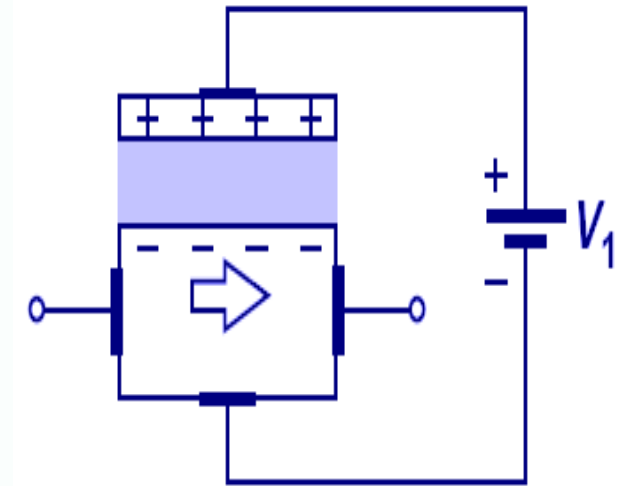
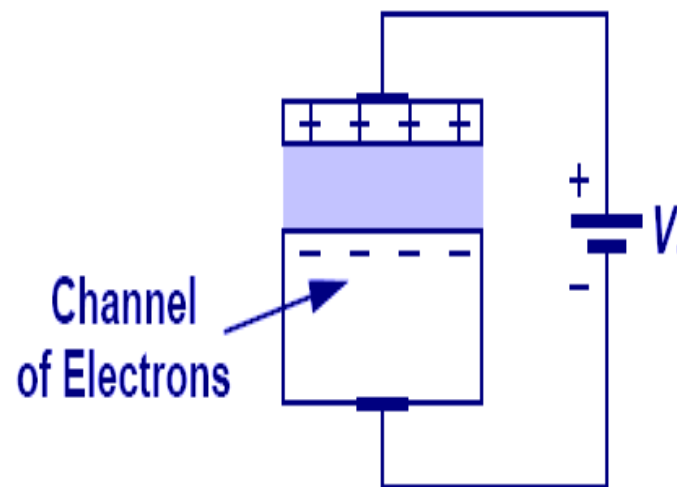
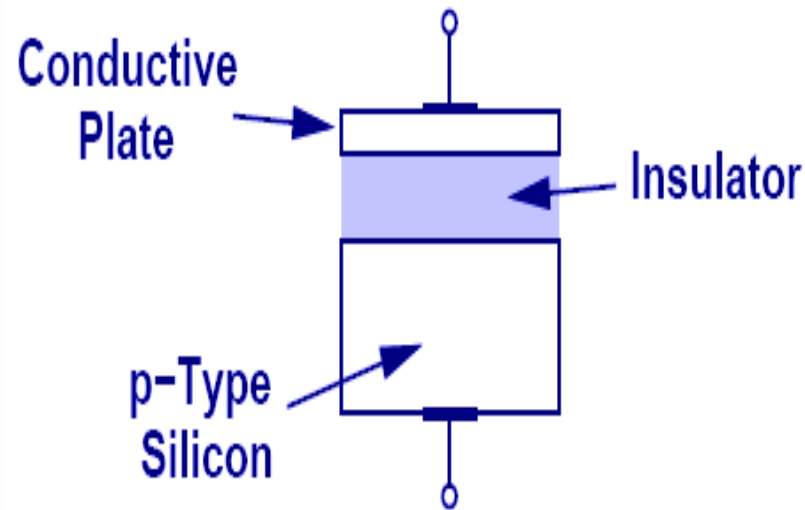
- Finds application especially in analog and RF circuit design

Overview

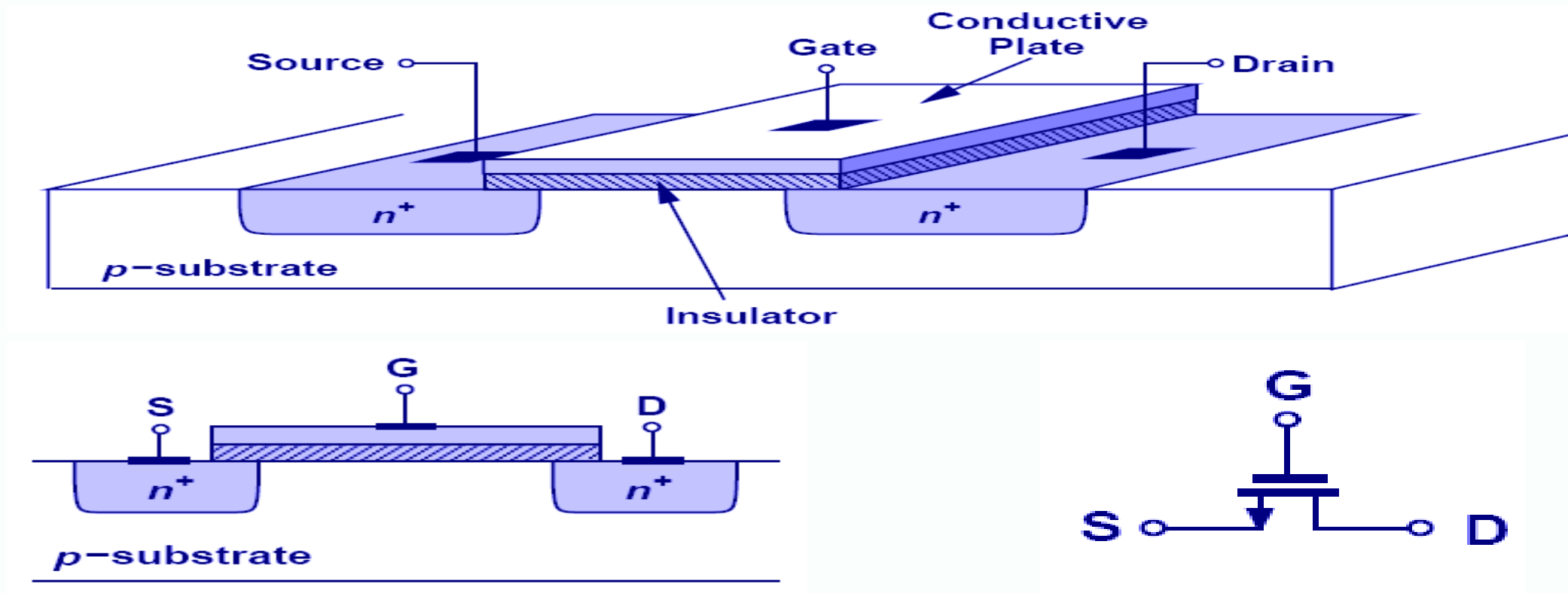


Metal-Oxide-Semiconductor (MOS) Capacitor

- The MOS structure can be thought of as a parallel-plate capacitor, with the top plate being the positive plate, oxide being the dielectric, and Si substrate being the negative plate.
- (We are assuming P-substrate.)



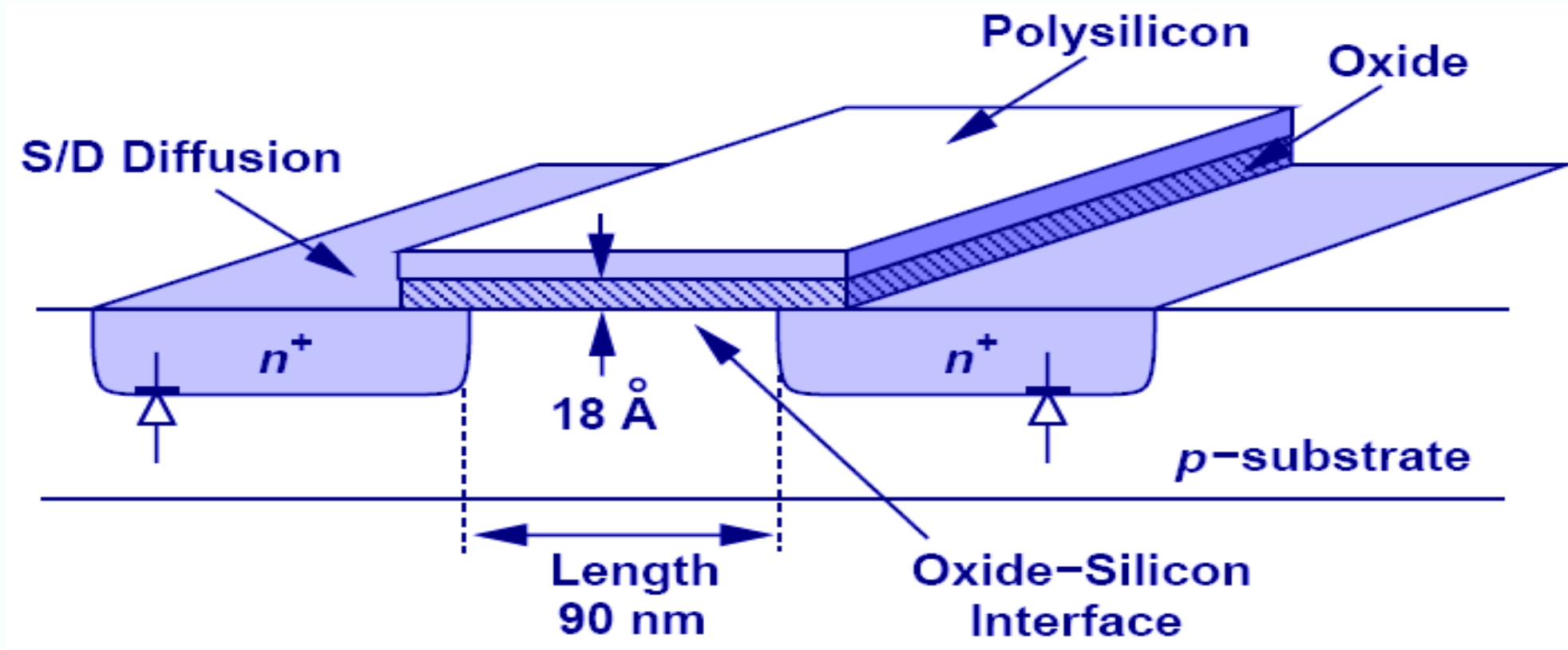
Structure and Symbol of MOSFET



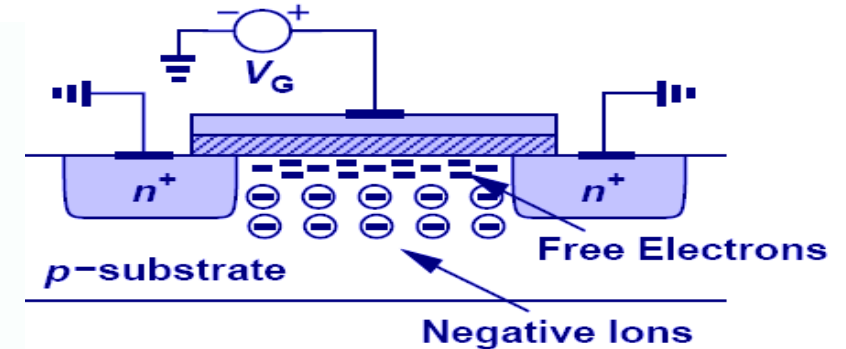
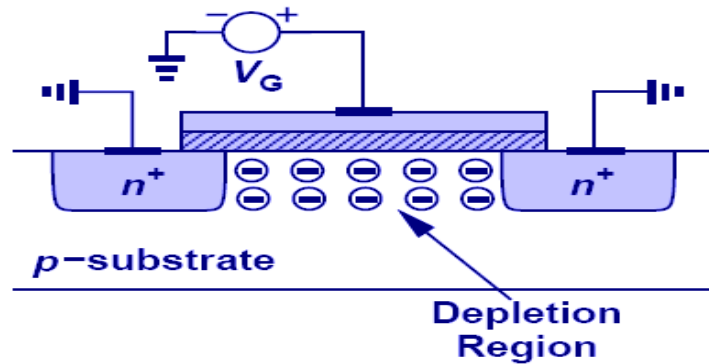
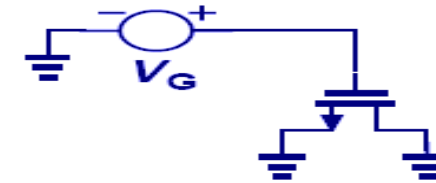
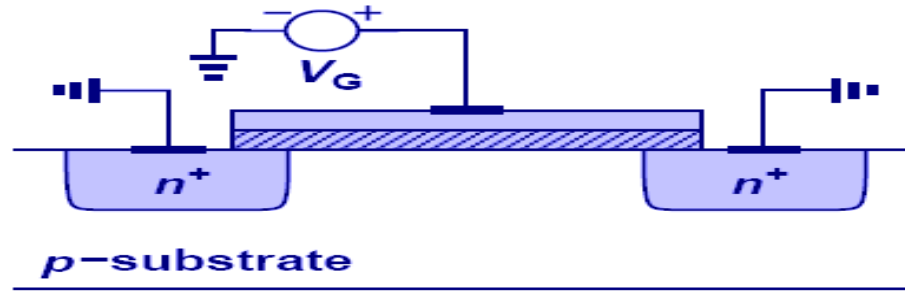
- This device is symmetric,
- so either of the n^+ regions can be source or drain.

State of the Art MOSFET Structure

- The gate is formed by poly-silicon, and the insulator by Silicon dioxide.



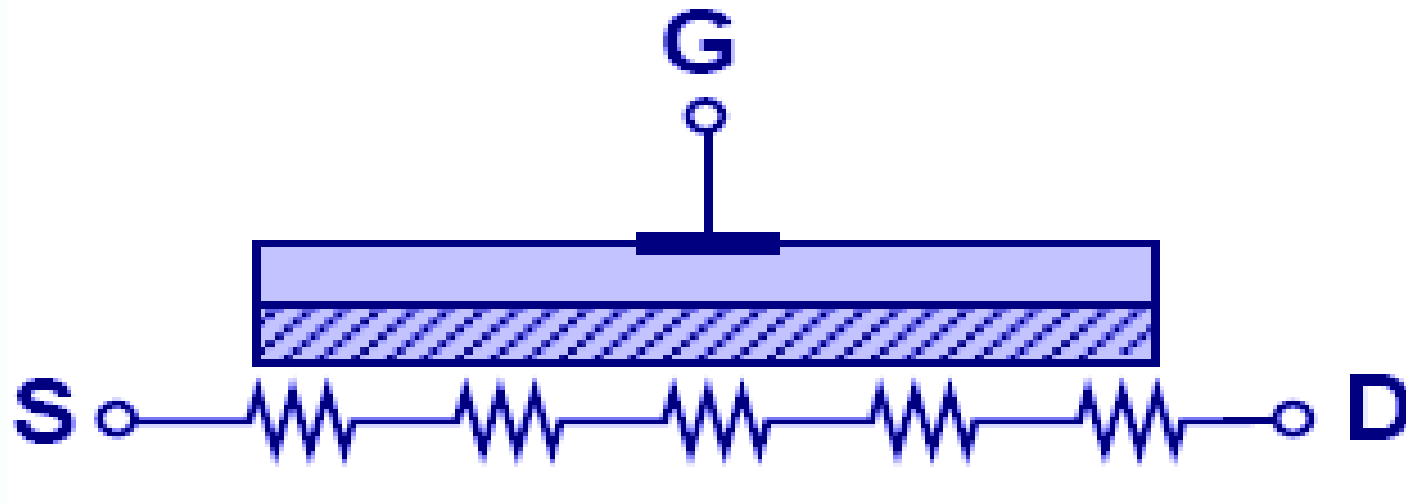
Formation of Channel



- First, the holes are repelled by the positive gate voltage, leaving behind negative ions and forming a depletion region.

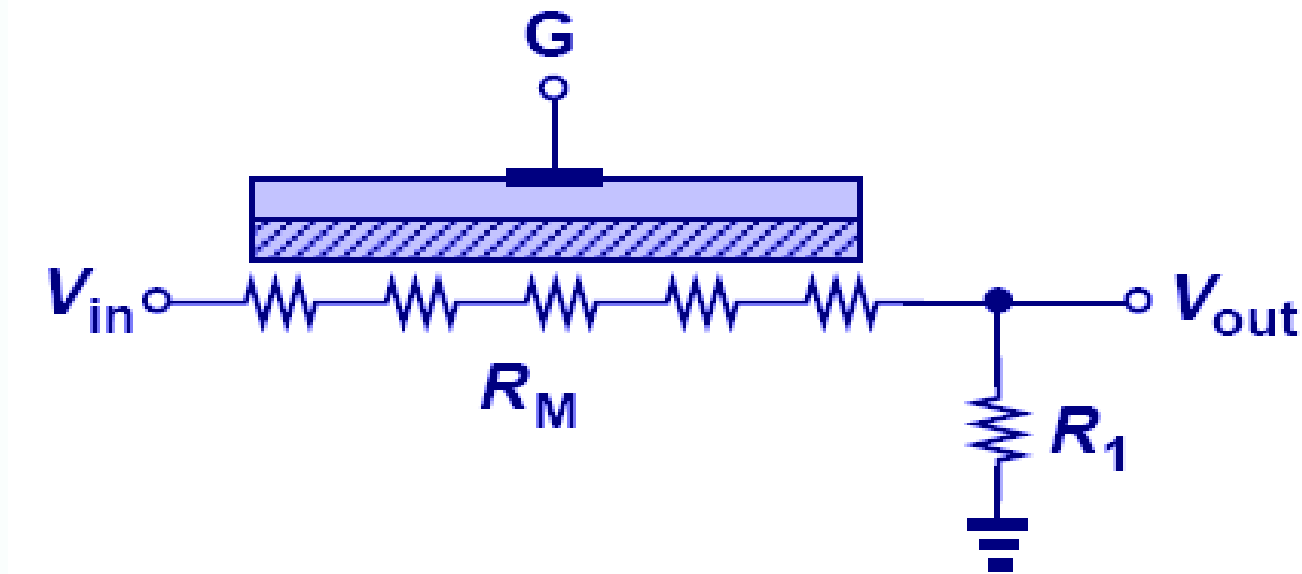
Next, electrons are attracted to the interface, creating a channel ("inversion layer").

Voltage-Dependent Resistor



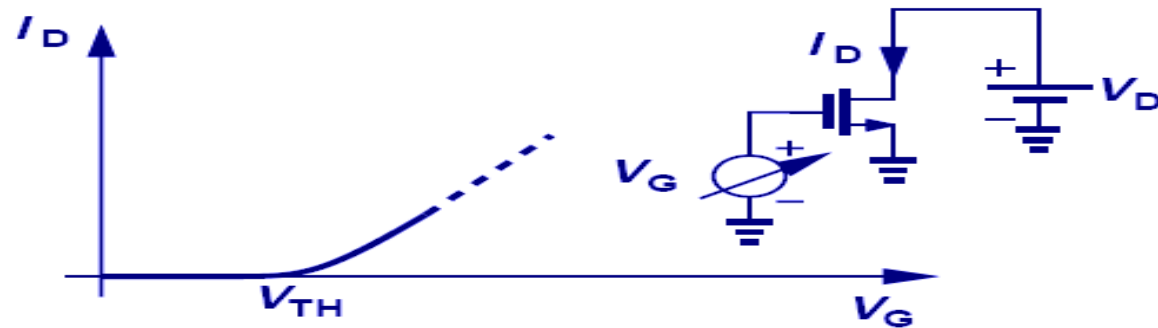
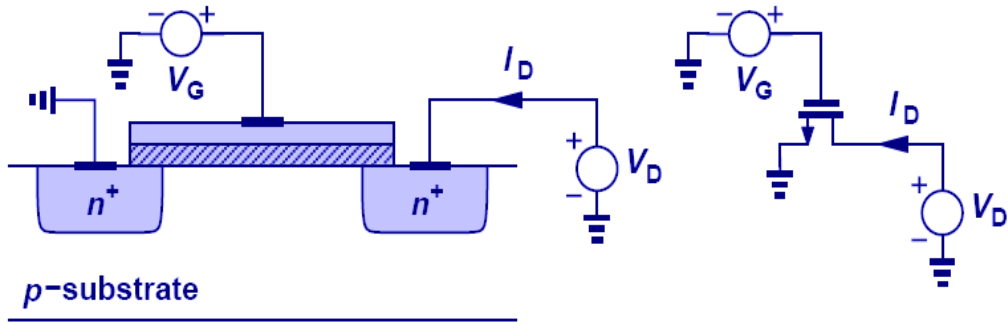
- The inversion channel of a MOSFET can be seen as a resistor.
- Since the charge density inside the channel depends on the gate voltage, this resistance is also voltage-dependent.

Voltage-Controlled Attenuator



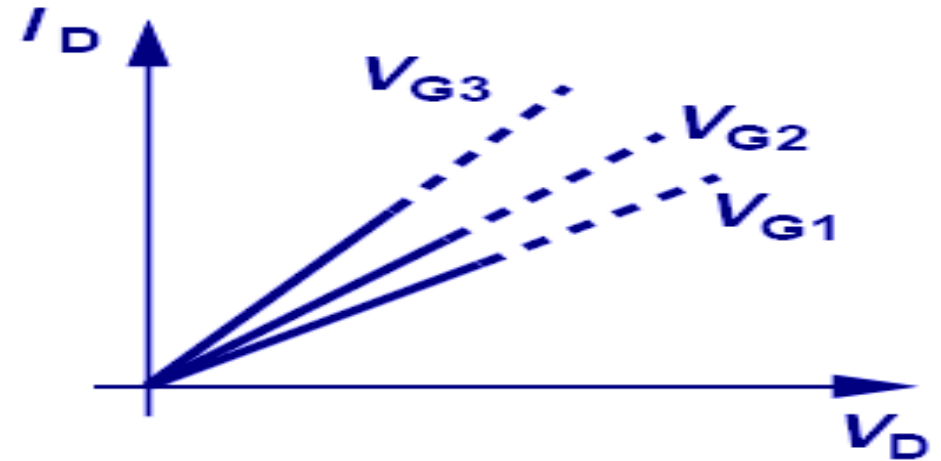
- As the gate voltage decreases, the output drops because the channel resistance increases.
- This type of gain control finds application in cell phones to avoid saturation near base stations.

MOSFET Characteristics

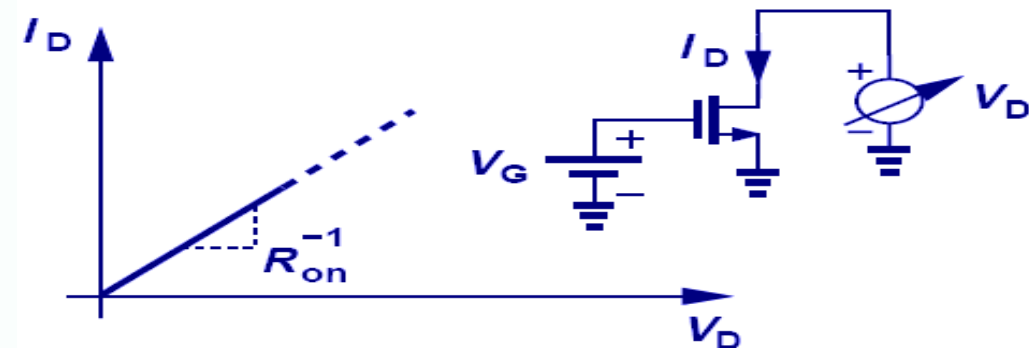


- The MOS characteristics are measured
- by varying V_G while keeping V_D constant,

and varying V_D while keeping V_G constant.

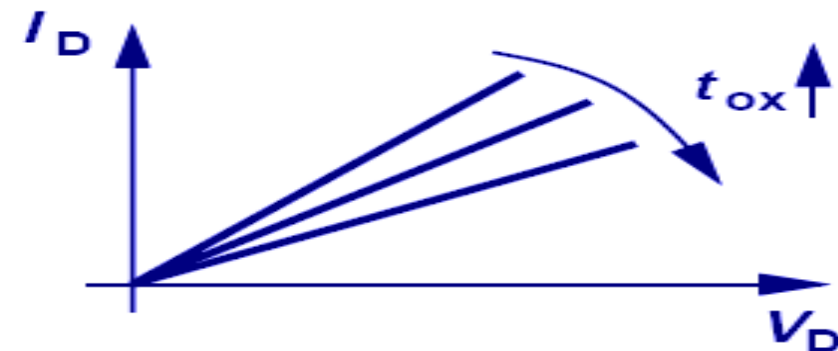
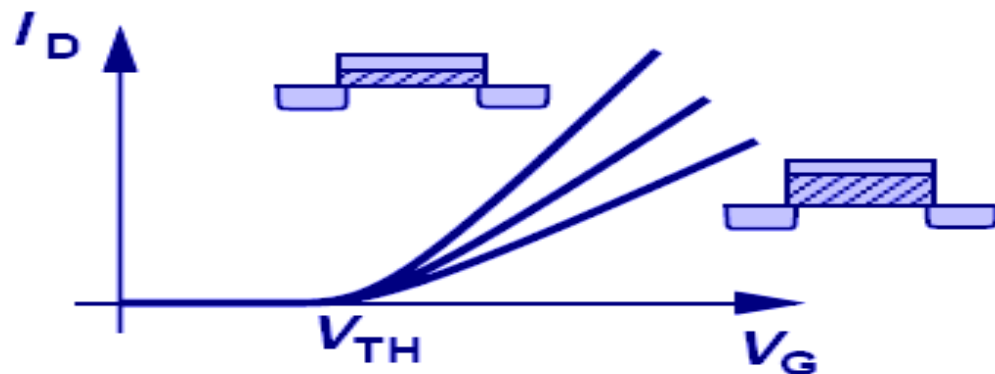
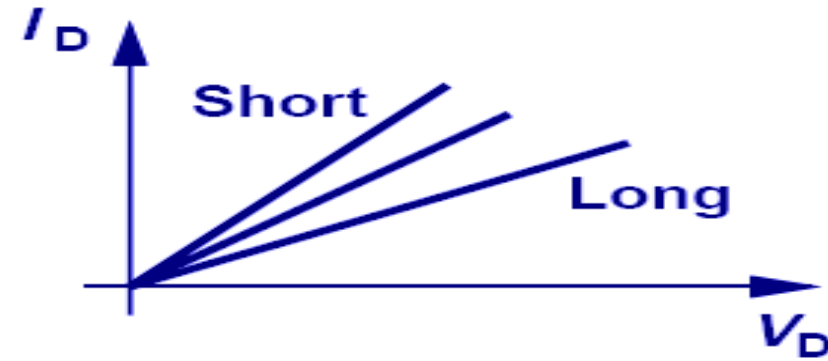
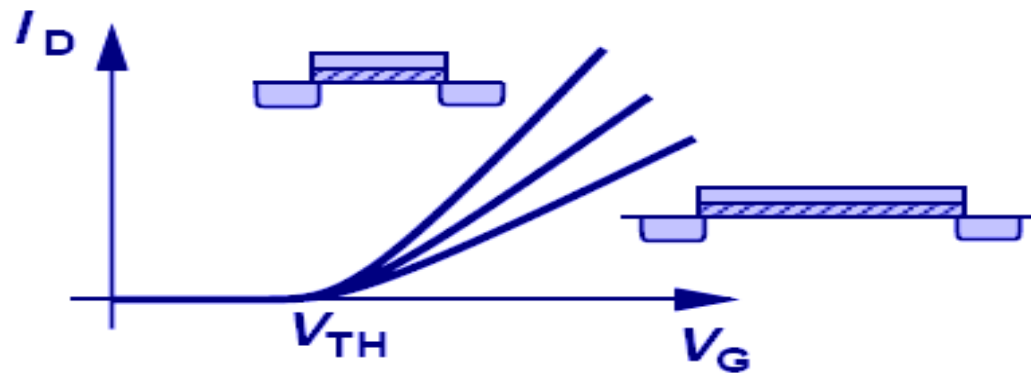


the voltage dependence of channel resistance

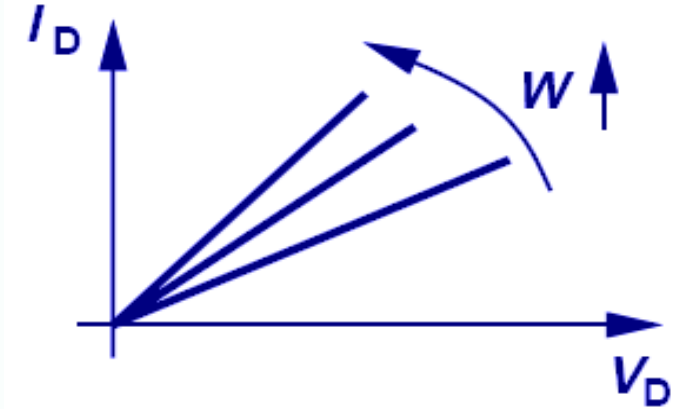
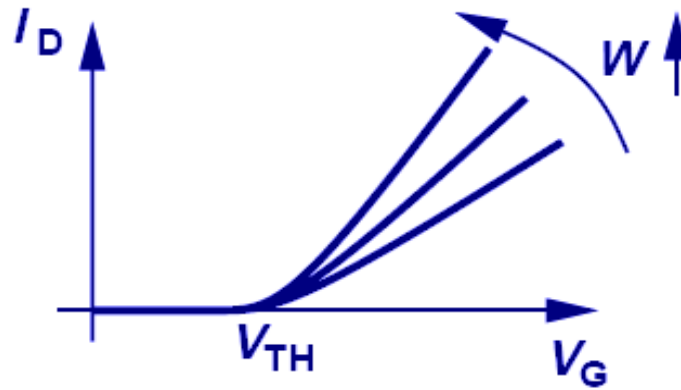
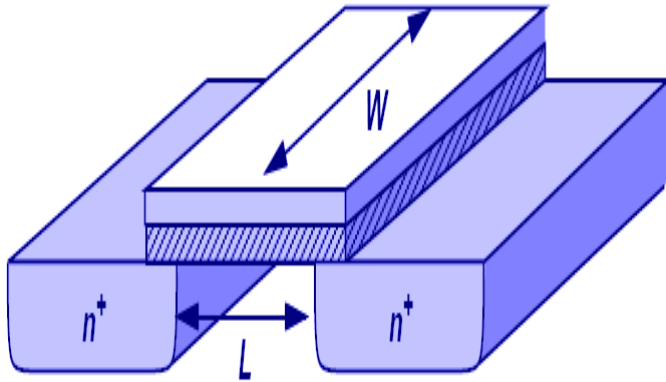


Gate Length L and Oxide Thickness t_{ox} Dependence

- Small gate length L and oxide thickness t_{ox} yield low channel resistance, which will increase the drain current.



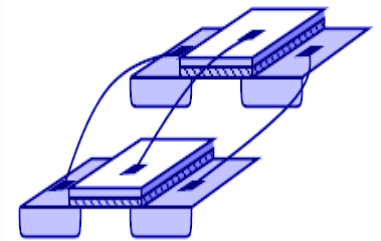
Effect of Width (W)



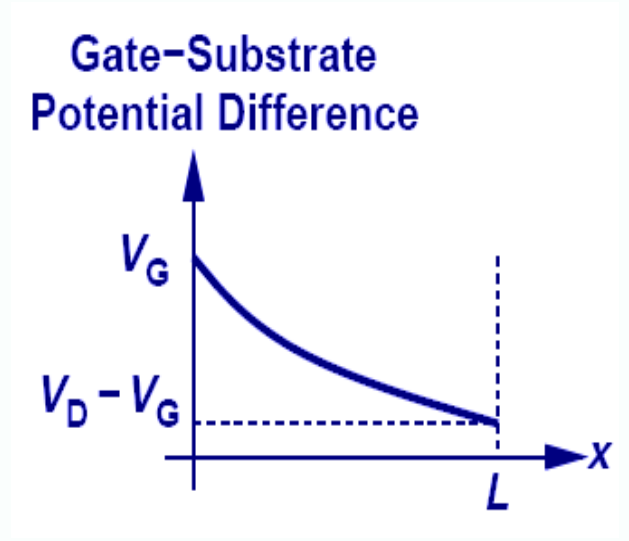
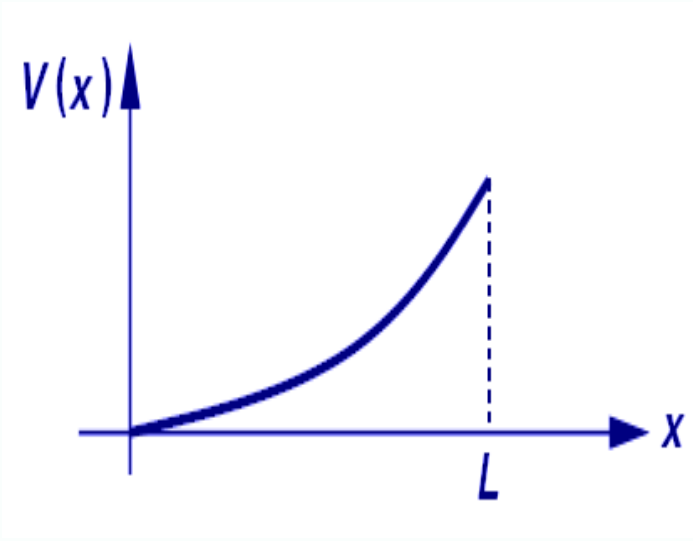
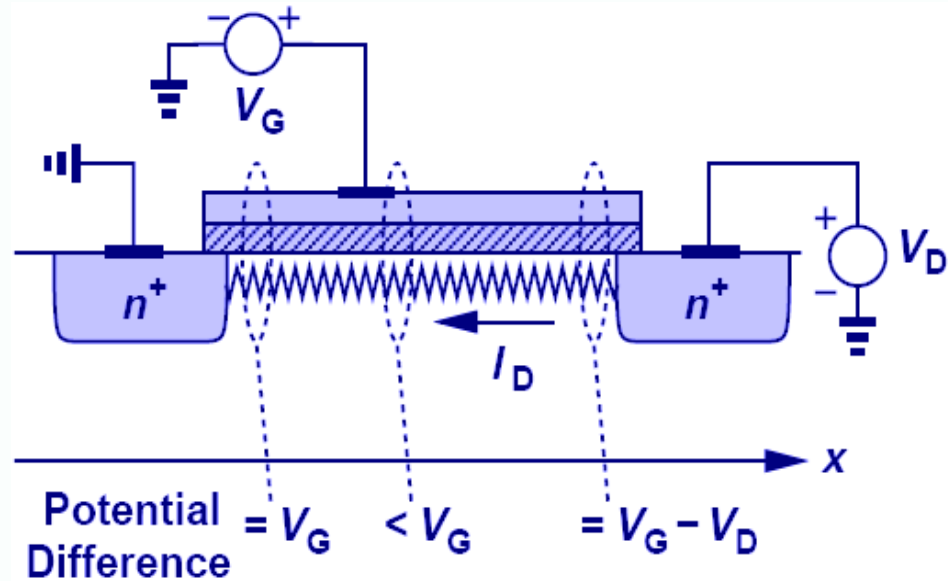
- As the gate width W increases, the current increases due to a decrease in resistance.

However, gate capacitance also increases thus, limiting the speed of the circuit.

An increase in W can be seen as two devices in parallel.

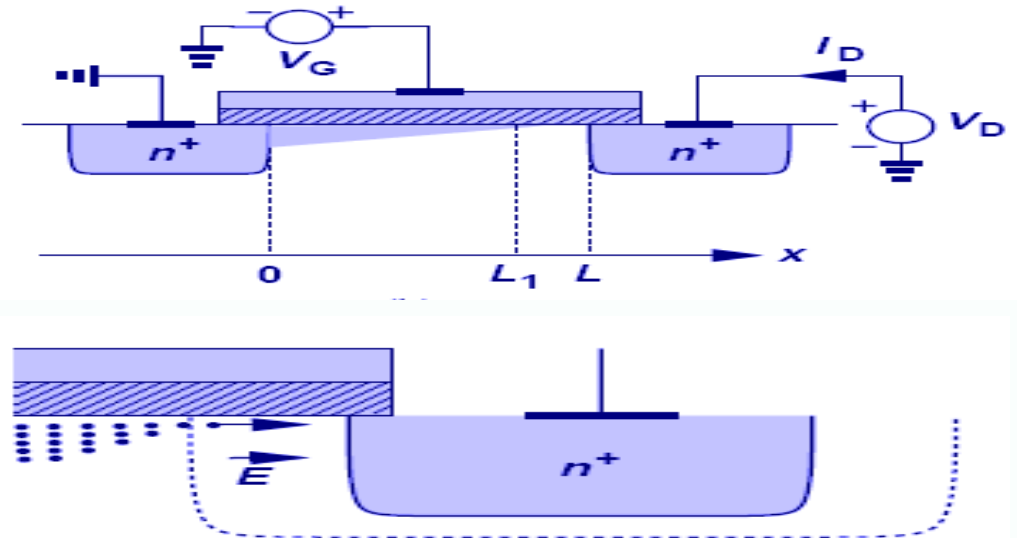
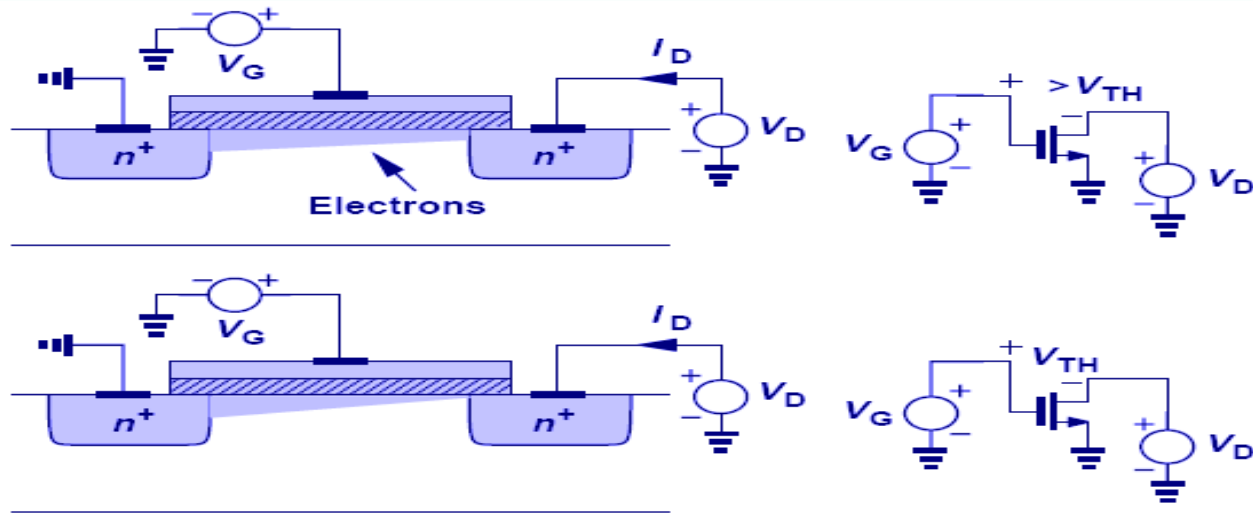


Channel Potential Variation



- Since there's a channel resistance between drain and source, and if drain is biased higher than the source,
- channel potential increases from source to drain,
- and the potential between gate and channel will decrease from source to drain.

Channel Pinch-Off



- As the potential difference between drain and gate becomes more positive, the inversion layer beneath the interface starts to pinch off around drain.

When $V_D - V_G = V_{th}$, the channel at drain totally pinches off, and when $V_D - V_G > V_{th}$, the channel length starts to decrease.



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Thank You

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COE 271 Semiconductor Devices

Lecture 7

FET - MOSFET

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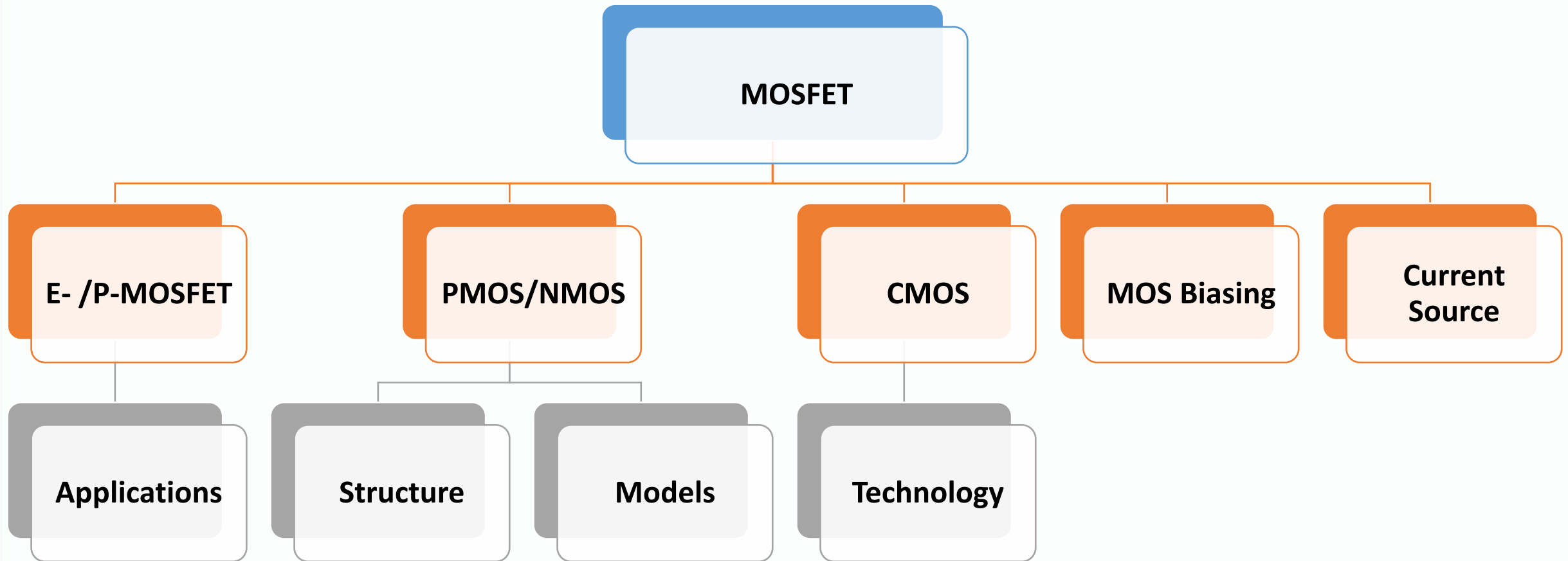
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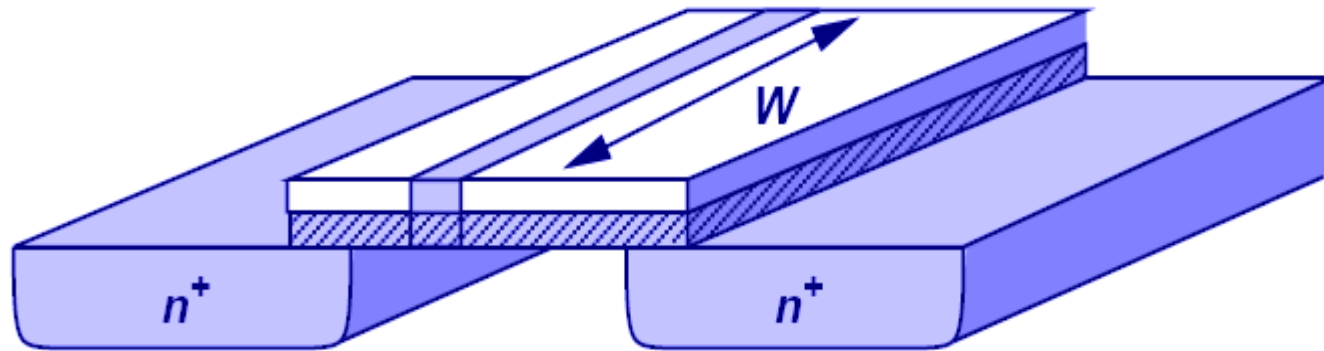
“A ship is always safe at the shore- but that is
NOT what it is built for”

Albert Einstein

Overview



Channel Charge Density

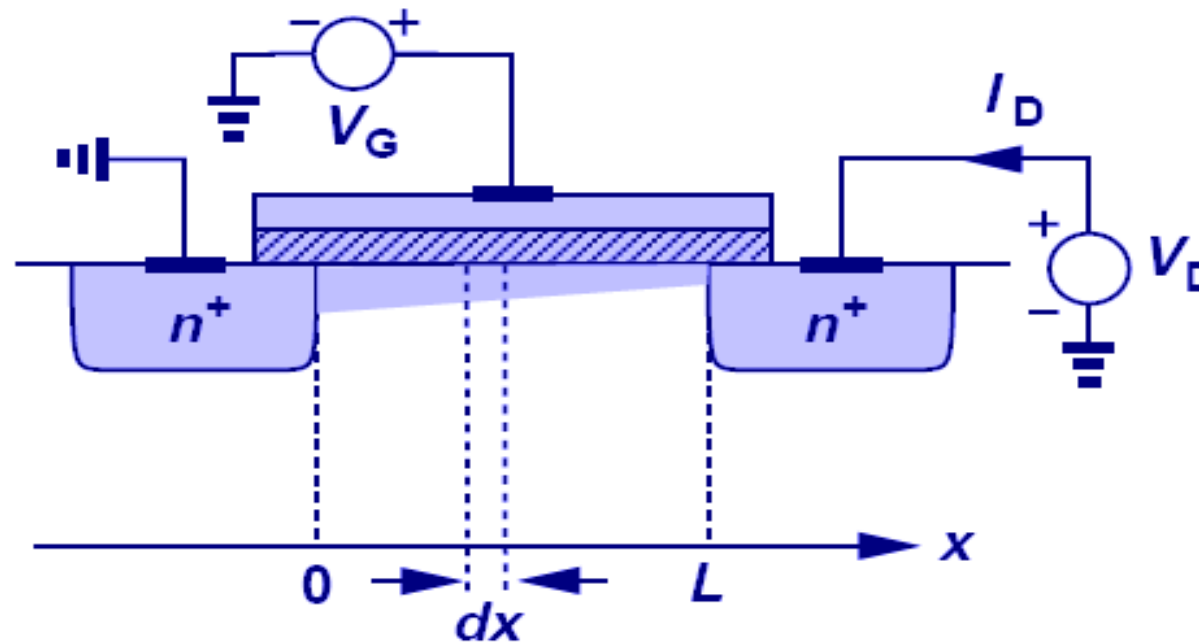


$$Q = WC_{ox}(V_{GS} - V_{TH})$$

- The channel charge density is equal to the gate capacitance times the gate voltage in excess of the threshold voltage.

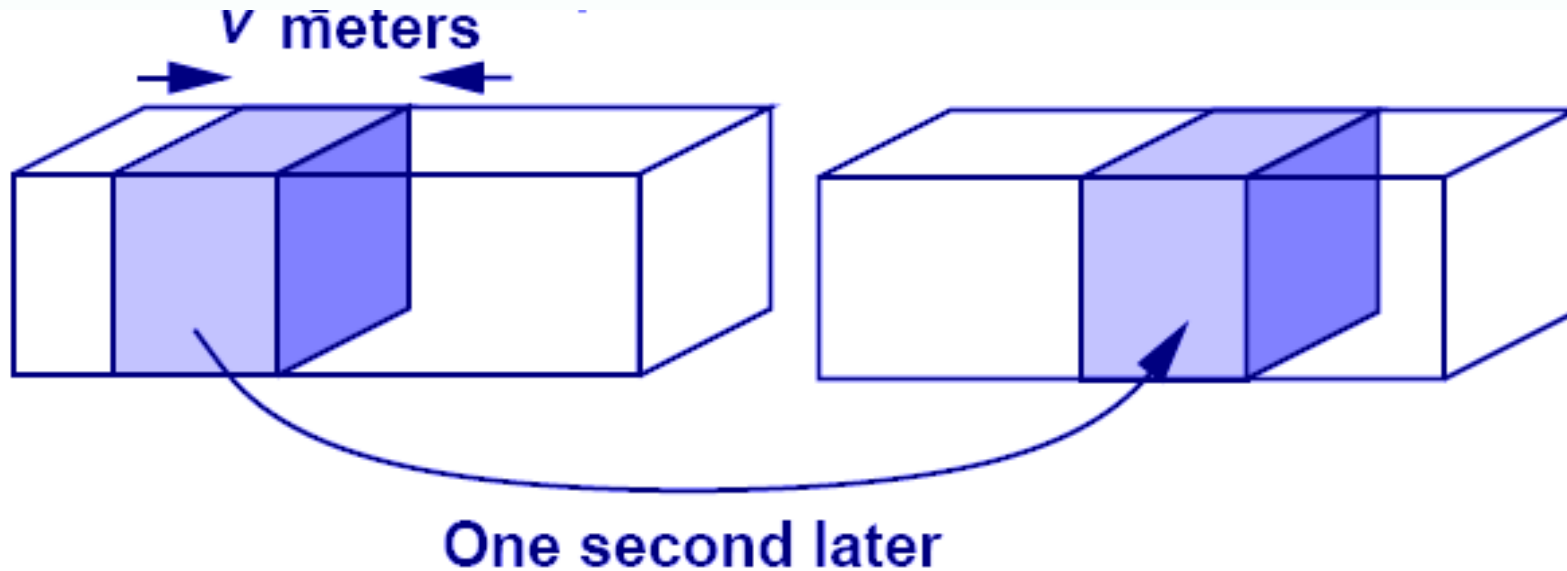
Charge Density at a Point

- Let x be a point along the channel from source to drain, and $V(x)$ its potential; the expression above gives the charge density (per unit length).



$$Q(x) = WC_{ox} [V_{GS} - V(x) - V_{TH}]$$

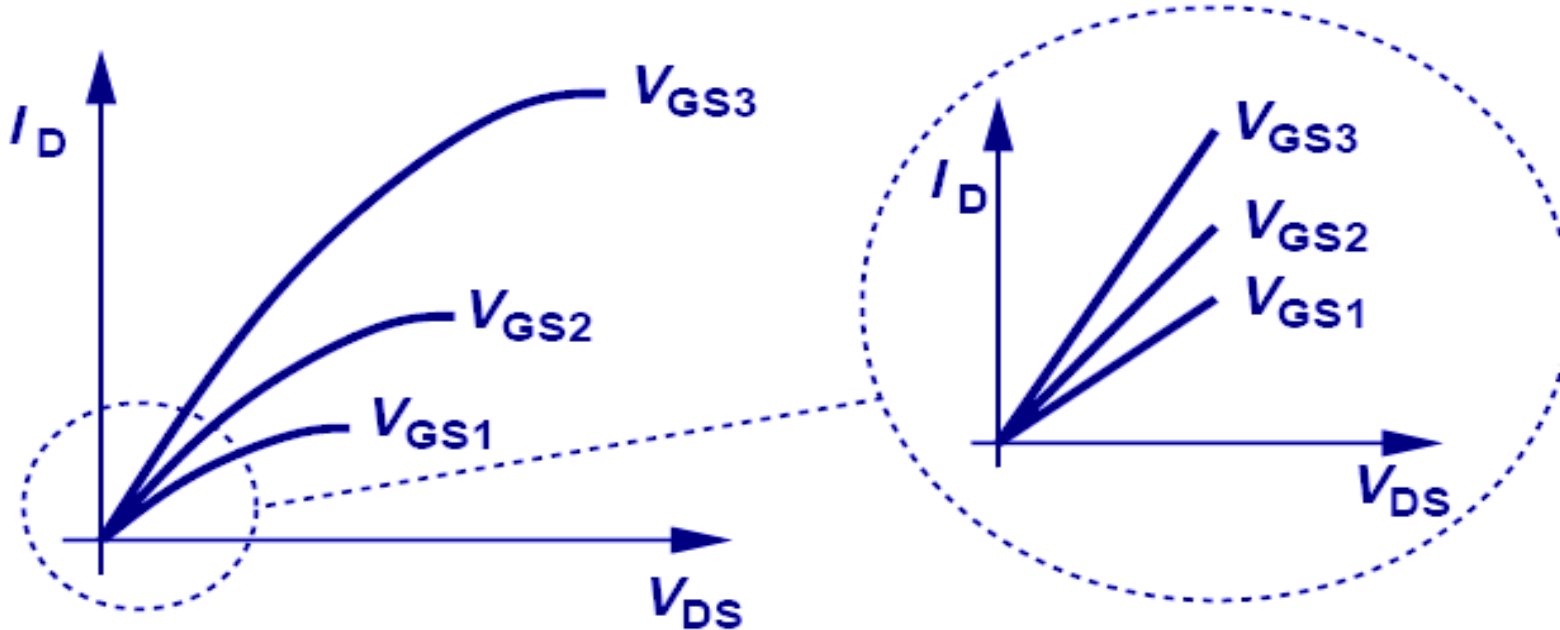
Charge Density and Current



$$I = Q \cdot v$$

- The current that flows from source to drain (electrons) is related to the charge density in the channel by the charge velocity.

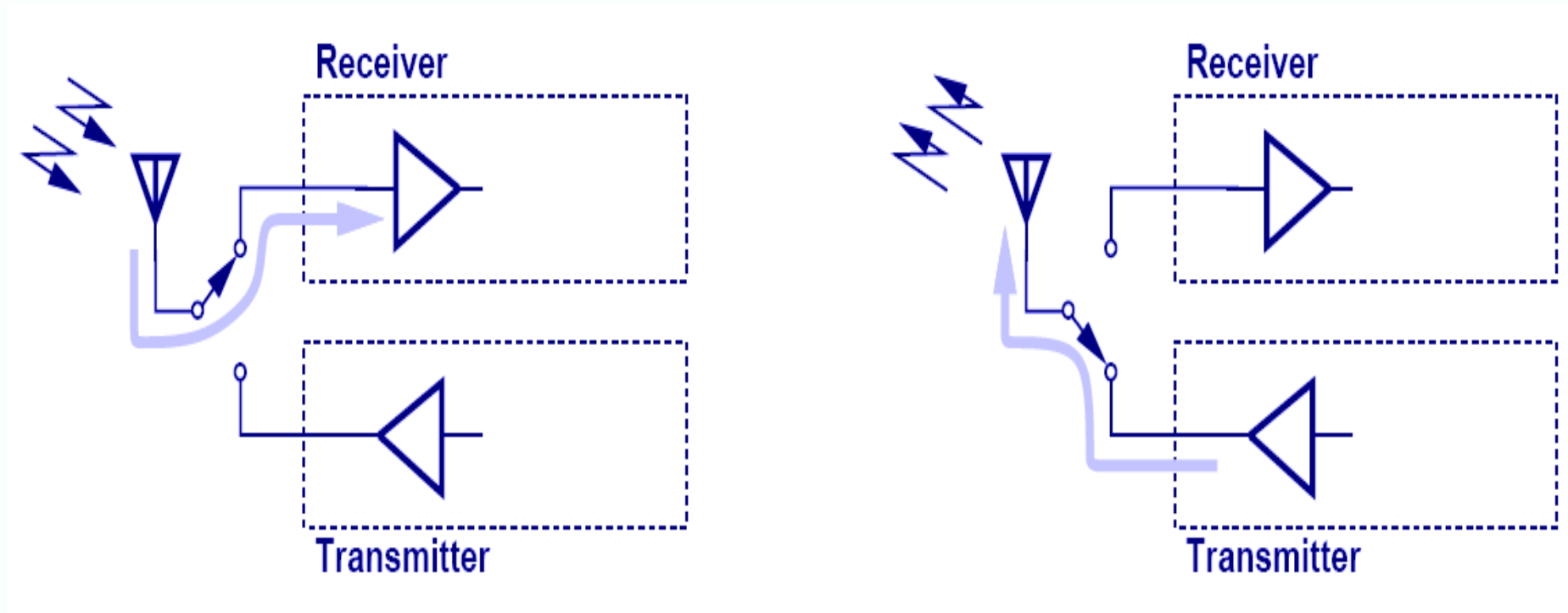
Linear Resistance



$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})}$$

- At small V_{DS} , the transistor can be viewed as a resistor, with the resistance depending on the gate voltage.
- It finds application as an electronic switch.

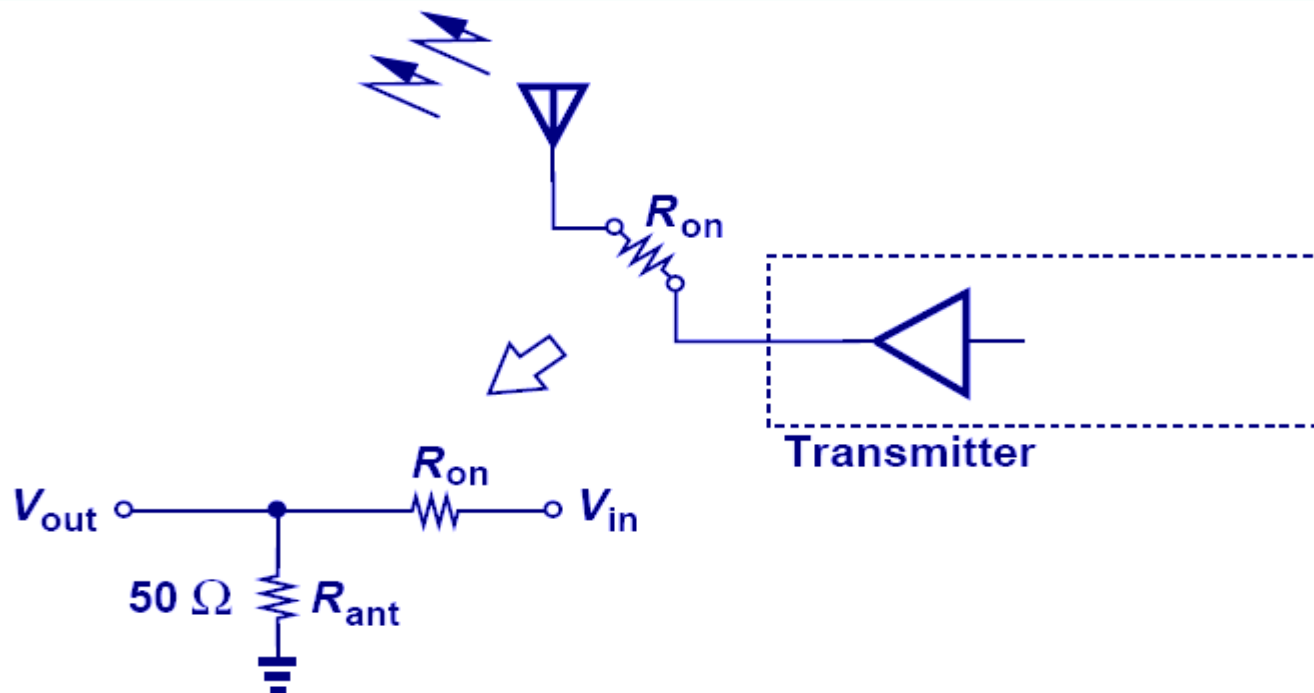
Application of Electronic Switches



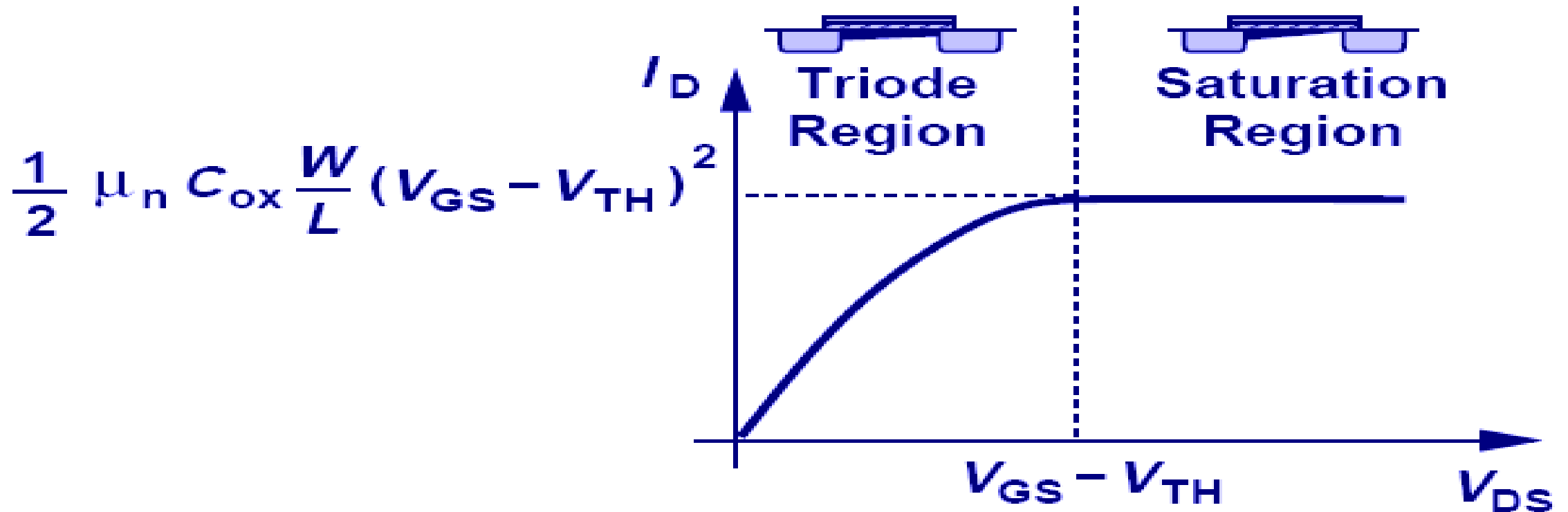
- In a cordless telephone system in which a single antenna is used for both transmission and reception, a switch is used to connect either the receiver or transmitter to the antenna.

Effects of On-Resistance

- To minimize signal attenuation, R_{on} of the switch has to be as small as possible. This means larger W/L aspect ratio and greater V_{GS} .



Different Regions of Operation



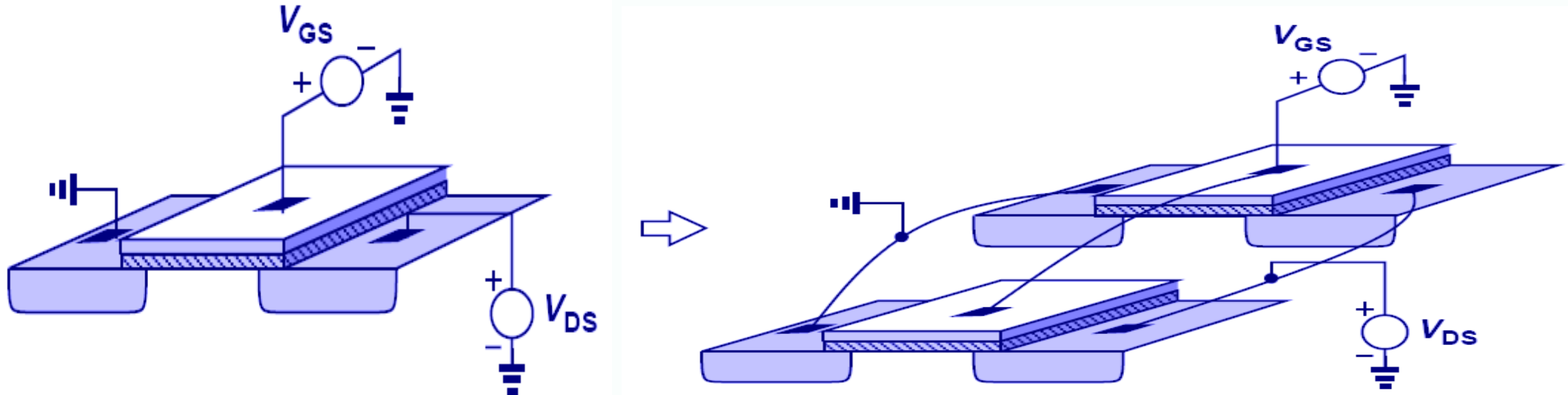
Transconductance

$\frac{W}{L}$ Constant $V_{GS} - V_{TH}$ Variable	$\frac{W}{L}$ Variable $V_{GS} - V_{TH}$ Constant	$\frac{W}{L}$ Variable $V_{GS} - V_{TH}$ Constant
$g_m \propto \sqrt{I_D}$ $g_m \propto V_{GS} - V_{TH}$	$g_m \propto I_D$ $g_m \propto \frac{W}{L}$	$g_m \propto \sqrt{\frac{W}{L}}$ $g_m \propto \frac{1}{V_{GS} - V_{TH}}$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \quad g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \quad g_m = \frac{2I_D}{V_{GS} - V_{TH}}$$

- Transconductance is a measure of how strong the drain current changes when the gate voltage changes.
- It has three different expressions.

Doubling of g_m Due to Doubling W/L



- If W/L is doubled, effectively two equivalent transistors are added in parallel, thus doubling the current (if $V_{GS} - V_{TH}$ is constant) and hence g_m .

Comparison of Bipolar and MOS Transistors

Bipolar Transistor	MOSFET
Exponential Characteristic Active: $V_{CB} > 0$ Saturation: $V_{CB} < 0$ Finite Base Current Early Effect Diffusion Current –	Quadratic Characteristic Saturation: $V_{DS} > V_{GS} - V_{TH}$ Triode: $V_{DS} < V_{GS} - V_{TH}$ Zero Gate Current Channel-Length Modulation Drift Current Voltage-Dependent Resistor

- Bipolar devices have a higher g_m than MOSFETs for a given bias current due to its exponential IV characteristics.

MOSFET

MOSFETs have characteristics similar to JFETs and additional characteristics that make them very useful

There are 2 types of MOSFET's:

- Depletion mode MOSFET (D-MOSFET)
 - Operates in Depletion mode the same way as a JFET when $V_{GS} \leq 0$
 - Operates in Enhancement mode like E-MOSFET when $V_{GS} > 0$
- Enhancement Mode MOSFET (E-MOSFET)
 - Operates in Enhancement mode
 - $I_{DSS} = 0$ until $V_{GS} > V_T$ (threshold voltage)

MOSFET Handling

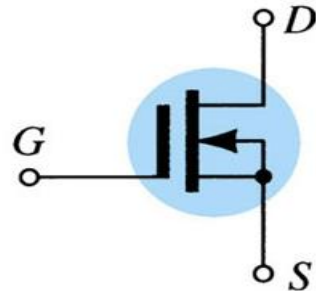
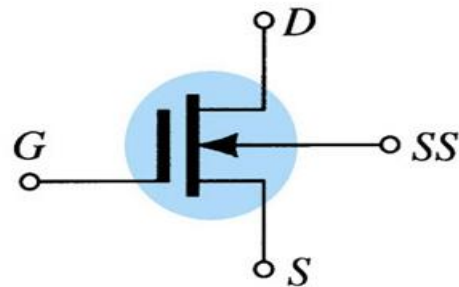
MOSFETs are very static sensitive. Because of the very thin SiO_2 layer between the external terminals and the layers of the device, any small electrical discharge can establish an unwanted conduction.

Protection:

- Always transport in a static sensitive bag
- Always wear a static strap when handling MOSFETs
- Apply voltage limiting devices between the Gate and Source, such as back-to-back Zeners to limit any transient voltage

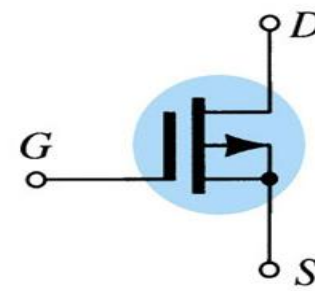
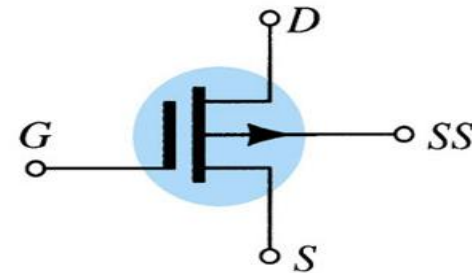
D-MOSFET Symbols

n-channel



(a)

p-channel



(b)

D-MOSFET Data Sheet

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	20	Vdc
Gate-Source Voltage	V_{GS}	± 10	Vdc
Drain Current	I_D	20	mA
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	200 1.14	mW mW/°C
Junction Temperature Range	T_J	+175	°C
Storage Channel Temperature Range	T_{stg}	-65 to +200	°C

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Drain-Source Breakdown Voltage ($V_{GS} = -7.0\text{ V}$, $I_D = 5.0\text{ }\mu\text{A}$)	$V_{(BR)DSX}$	20	25	—	Vdc
Gate Reverse Current (1) ($V_{DS} = -10\text{ V}$, $V_{GS} = 0$) ($V_{GS} = -10\text{ V}$, $V_{DS} = 0$, $T_A = 150^\circ\text{C}$)	I_{GSS}	—	—	1.0 200	pA
Gate-Source Cutoff Voltage ($I_D = 2.0\text{ }\mu\text{A}$, $V_{DS} = 10\text{ V}$)	$V_{GS(off)}$	—	-5.0	-7.0	Vdc
Drain-Gate Reverse Current (1) ($V_{DG} = 10\text{ V}$, $I_S = 0$)	I_{DGO}	—	—	1.0	pA
ON CHARACTERISTICS					
Zero-Gate-Voltage Drain Current ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$)	I_{DSS}	2.0	2.9	6.0	mA
On-State Drain Current ($V_{DS} = 10\text{ V}$, $V_{GS} = +3.5\text{ V}$)	$I_{D(on)}$	9.0	14	18	mA

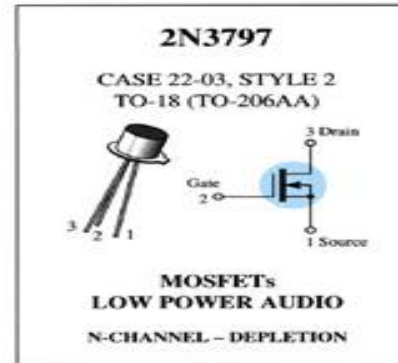
SMALL-SIGNAL CHARACTERISTICS

Forward Transfer Admittance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$)	$ Y_{fs} $	1500	2300	3000	μmhos
($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)		1500	—	—	
Output Admittance ($I_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$)	$ Y_{os} $	—	27	60	μmhos
Input Capacitance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)	C_{iss}	—	6.0	8.0	pF
Reverse Transfer Capacitance ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ MHz}$)	C_{rss}	—	0.5	0.8	pF

FUNCTIONAL CHARACTERISTICS

Noise Figure ($V_{DS} = 10\text{ V}$, $V_{GS} = 0$, $f = 1.0\text{ kHz}$, $R_N = 3\text{ megohms}$)	NF	—	3.8	—	dB
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(1) This value of current includes both the FET leakage current as well as the leakage current associated with the test socket and fixture when measured under best attainable conditions.

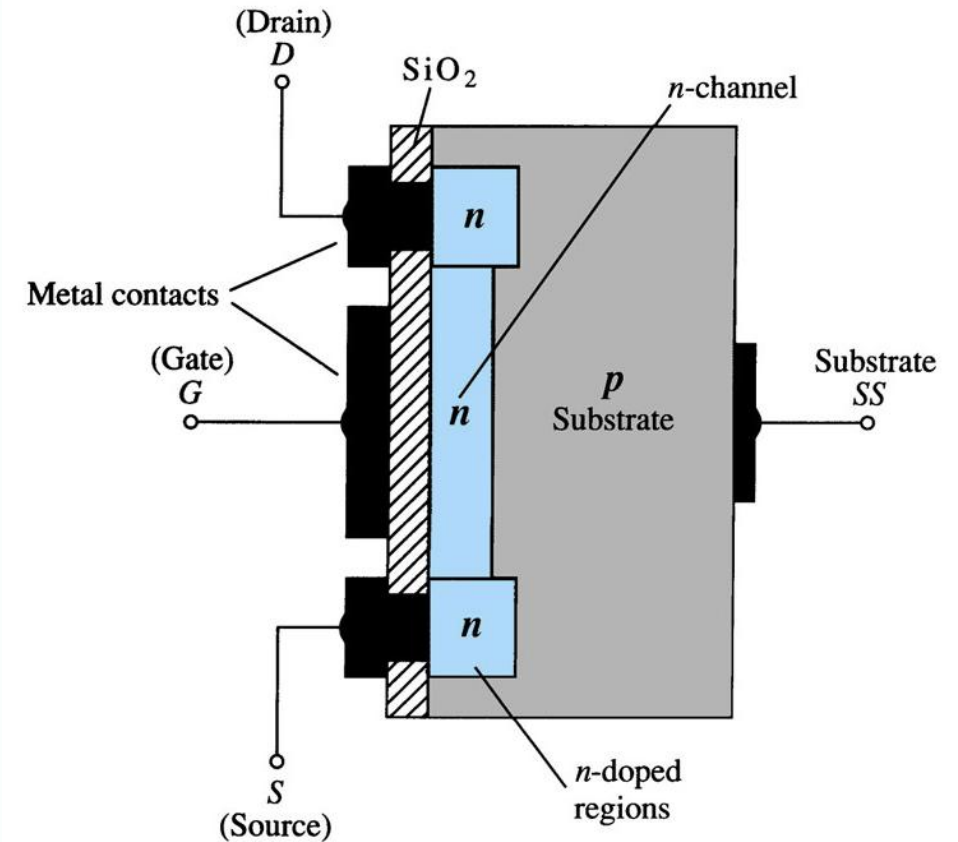


D-MOSFET Construction

The Drain (D) and Source (S) leads connect to the n-doped regions. These N-doped regions are connected via an n-channel.

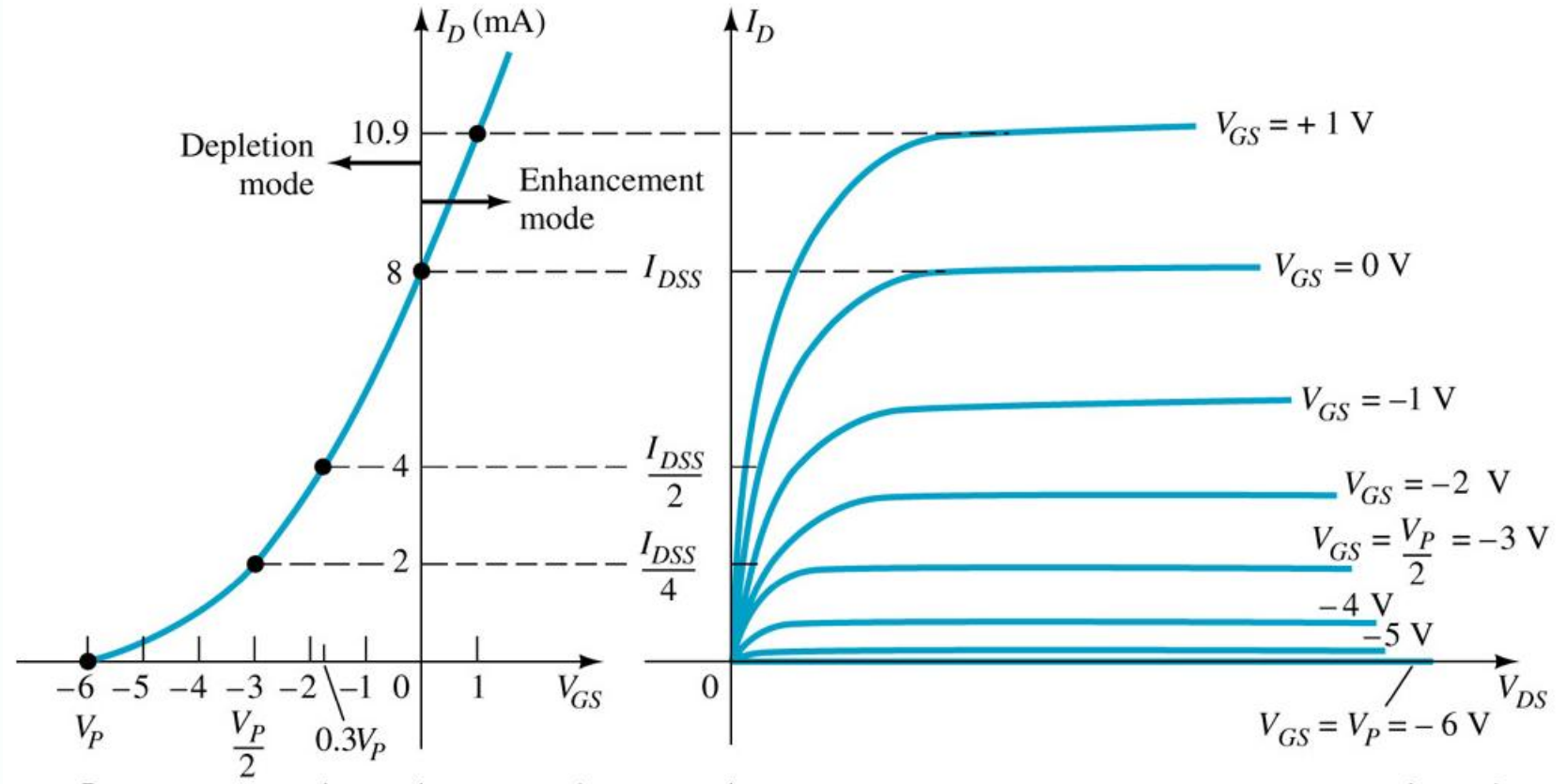
This n-channel is connected to the Gate (G) via a thin insulating layer of SiO_2 .

The n-doped material lies on a p-doped substrate that may have an additional terminal connection called SS.



D-MOSFET Basic Operation

A D-MOSFET may be biased to operate in two modes:
the **Depletion** mode
or the **Enhancement** mode



D-MOSFET Depletion Mode Operation

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

The transfer characteristics are similar to the JFET

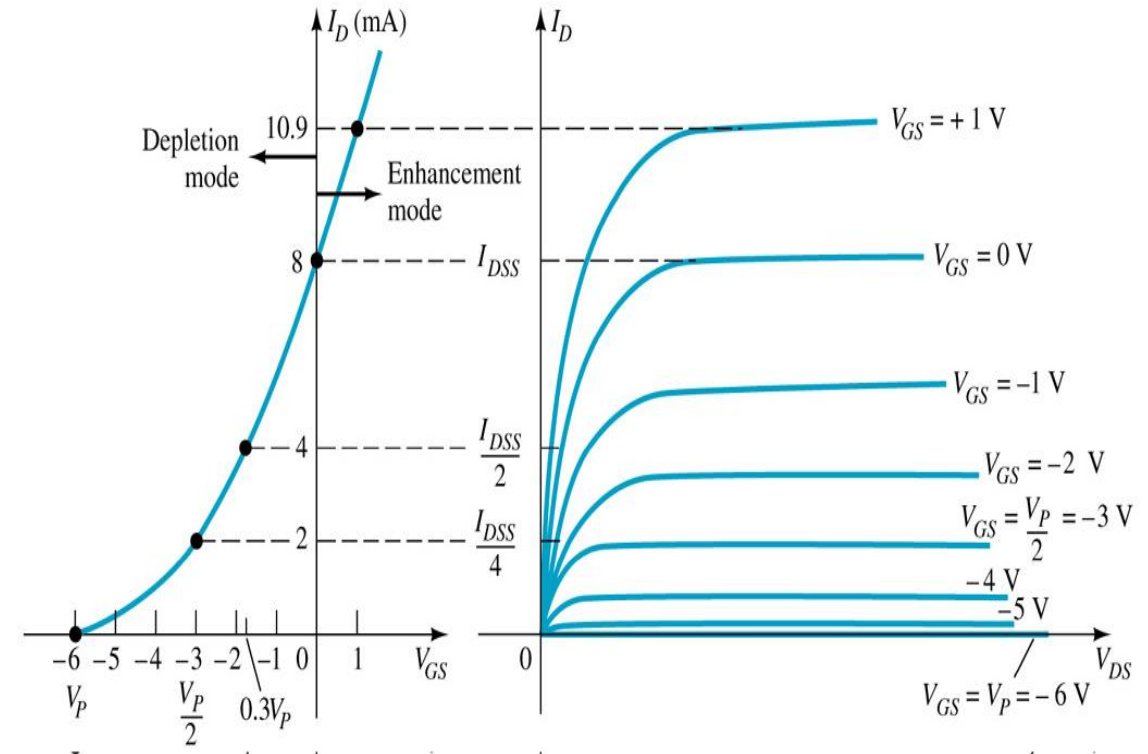
In Depletion Mode operation:

When $V_{GS} = 0V$, $I_D = I_{DSS}$

When $V_{GS} < 0V$, $I_D < I_{DSS}$

When $V_{GS} > 0V$, $I_D > I_{DSS}$

The formula used to plot the Transfer Curve, is:

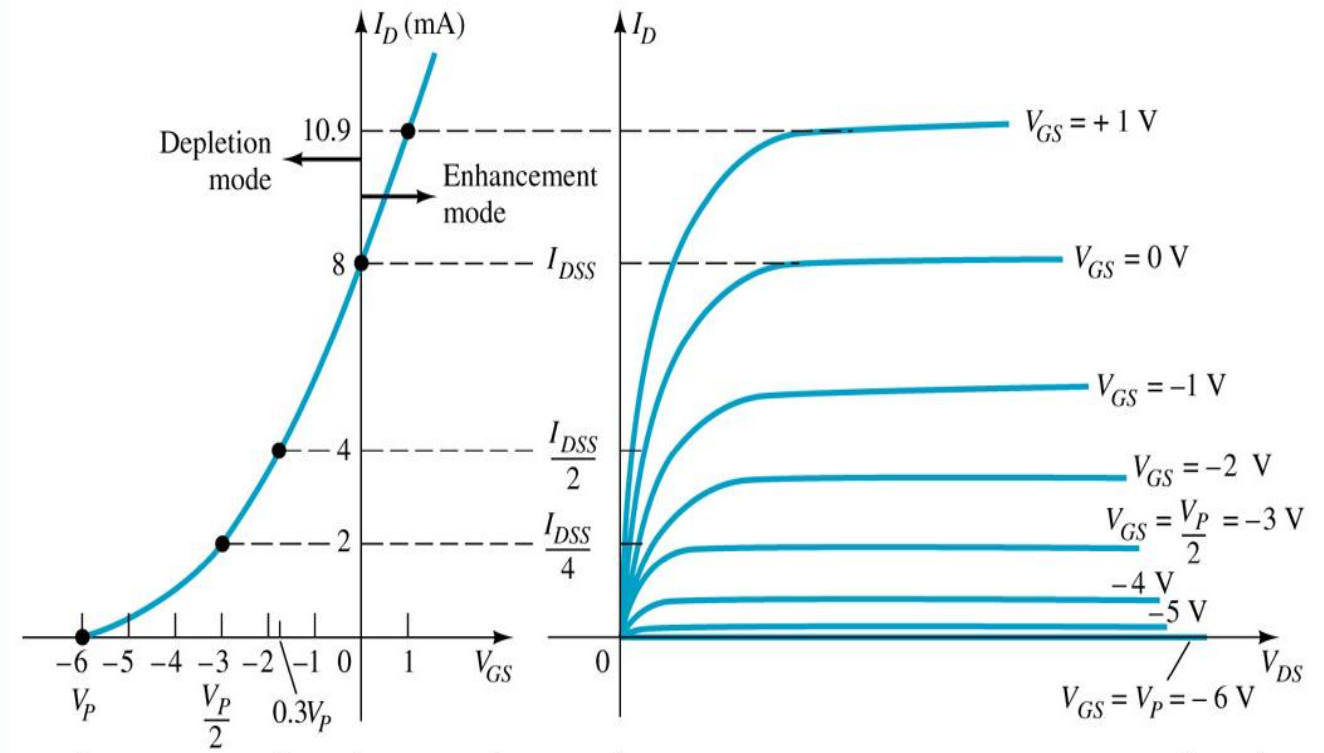


D-MOSFET Enhancement Mode Operation

Enhancement Mode operation

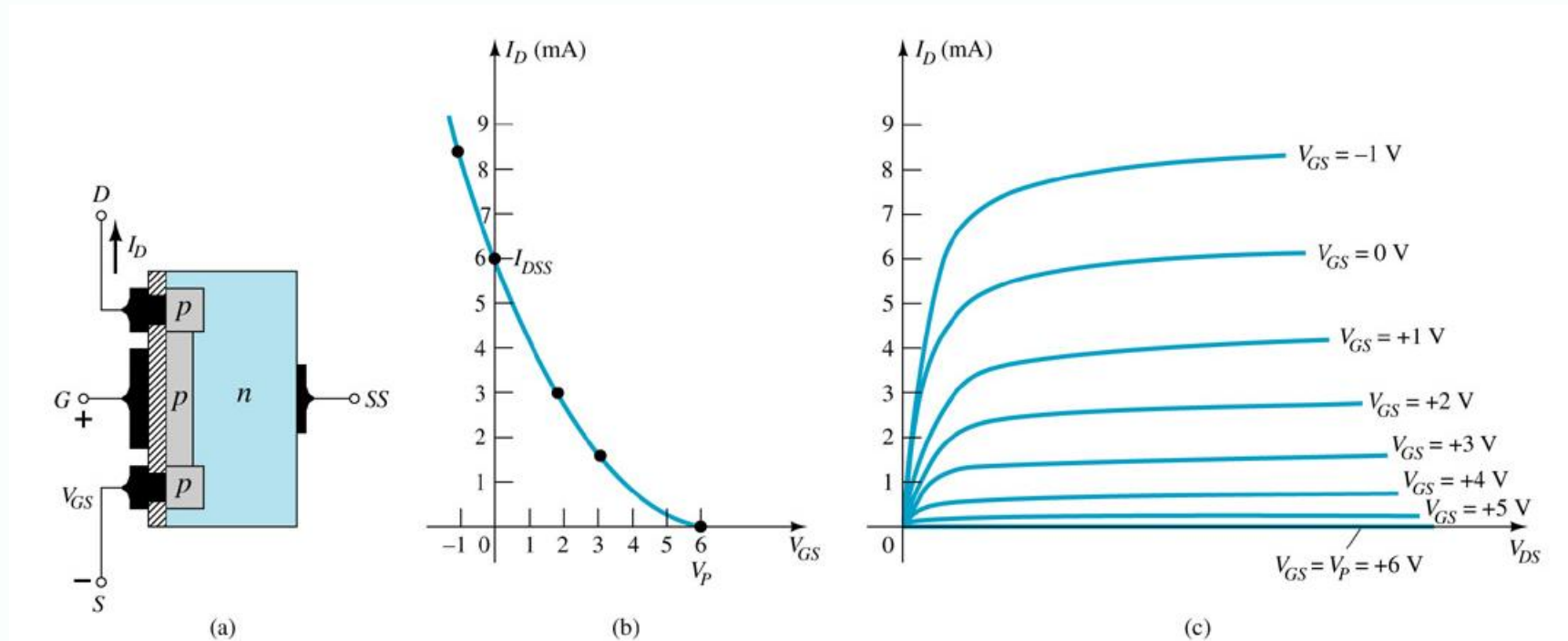
In this mode, the transistor operates with $V_{GS} > 0V$, and I_D increases above I_{DSS}

Shockley's equation, the formula used to plot the Transfer Curve, still applies but V_{GS} is positive:



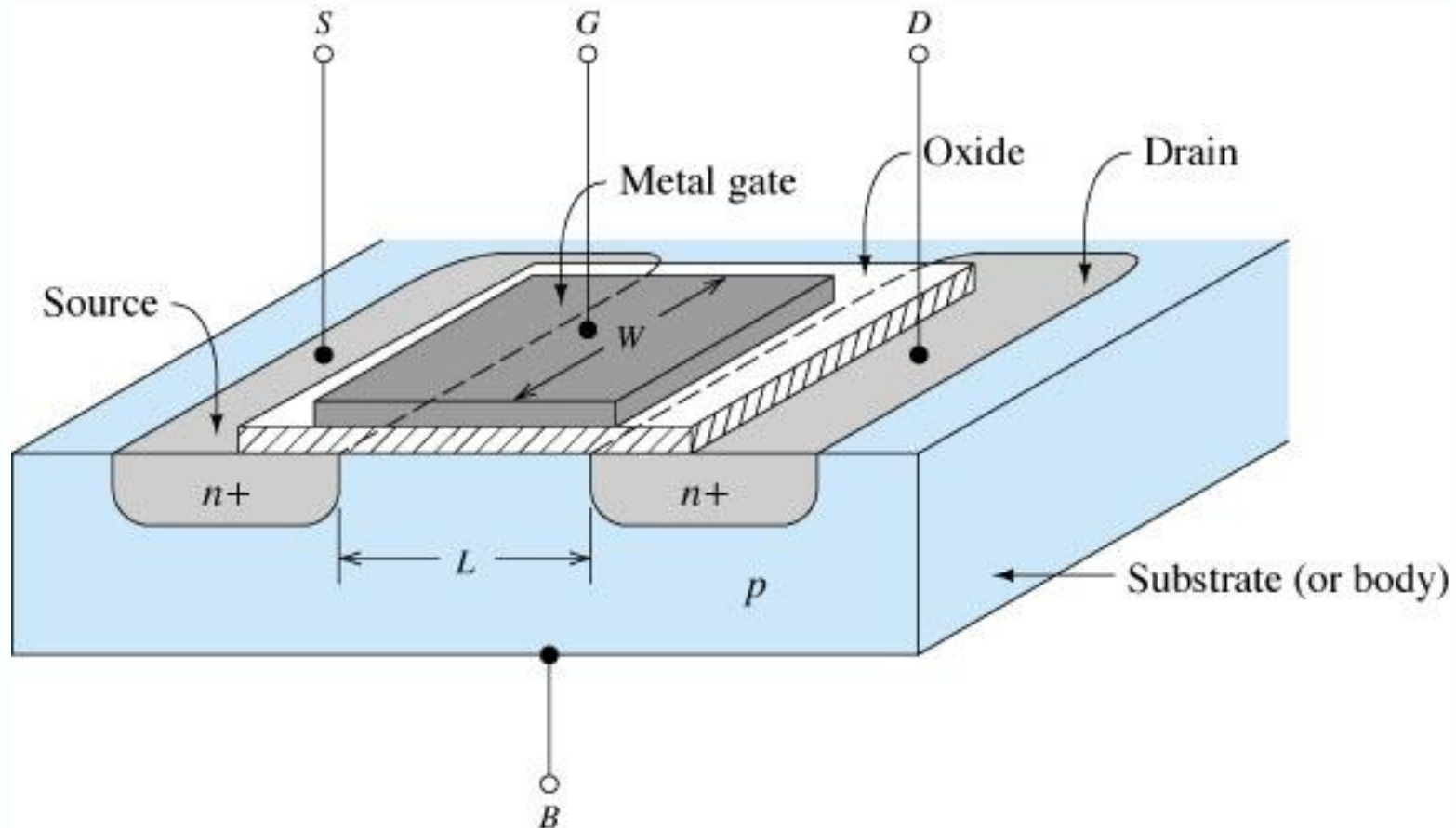
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

P-Channel Depletion Mode MOSFET



The p-channel Depletion mode MOSFET is similar to the n-channel except that the voltage polarities and current directions are reversed

Enhancement Mode MOSFET's



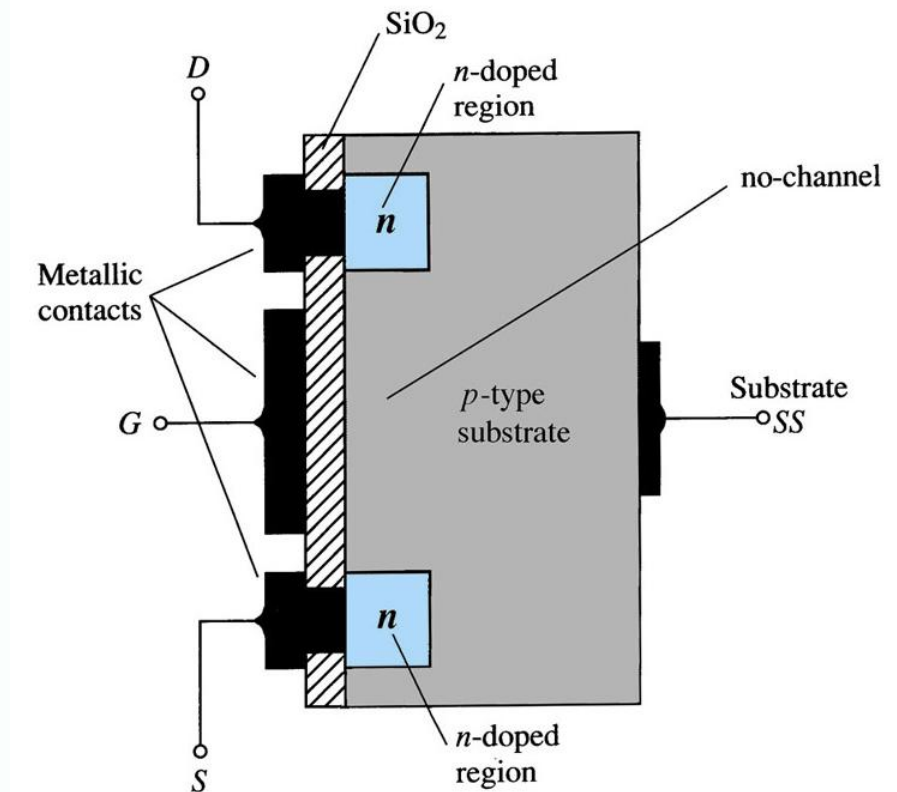
Enhancement Mode MOSFET Construction

The Drain (D) and Source (S) connect to the n-doped regions

These n-doped regions are not connected via an n-channel without an external voltage

The Gate (G) connects to the p-doped substrate via a thin insulating layer of SiO_2

The n-doped material lies on a p-doped substrate that may have an additional terminal connection called SS



E-MOSFET Data Sheet

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	25	Vdc
Drain-Gate Voltage	V_{DG}	30	Vdc
Gate-Source Voltage*	V_{GS}	30	Vdc
Drain Current	I_D	30	mA dc
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	300 1.7	mW mW/°C
Junction Temperature Range	T_J	175	°C
Storage Temperature Range	T_{stg}	-65 to +175	°C

* Transient potentials of ± 75 Volts will not cause gate-oxide failure.

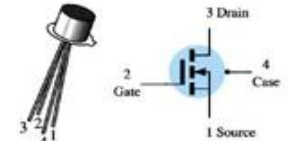
ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Drain-Source Breakdown Voltage ($I_D = 10 \mu\text{A}$, $V_{GS} = 0$)	$V_{(BR)DSX}$	25	—	Vdc
Zero-Gate-Voltage Drain Current ($V_{DS} = 10 \text{ V}$, $V_{GS} = 0$) $T_A = 25^\circ\text{C}$ $T_A = 150^\circ\text{C}$	I_{DSS}	—	10 10	nA dc $\mu\text{A dc}$
Gate Reverse Current ($V_{GS} = \pm 15 \text{ Vdc}$, $V_{DS} = 0$)	I_{GSS}	—	± 10	pA dc
ON CHARACTERISTICS				
Gate Threshold Voltage ($V_{DS} = 10 \text{ V}$, $I_D = 10 \mu\text{A}$)	$V_{GS(TH)}$	1.0	5	Vdc
Drain-Source On-Voltage ($I_D = 2.0 \text{ mA}$, $V_{GS} = 10 \text{ V}$)	$V_{DS(on)}$	—	1.0	V
On-State Drain Current ($V_{GS} = 10 \text{ V}$, $V_{DS} = 10 \text{ V}$)	$I_{D(on)}$	3.0	—	mA dc
SMALL-SIGNAL CHARACTERISTICS				
Forward Transfer Admittance ($V_{DS} = 10 \text{ V}$, $I_D = 2.0 \text{ mA}$, $f = 1.0 \text{ kHz}$)	$ y_{fs} $	1000	—	μmho
Input Capacitance ($V_{DS} = 10 \text{ V}$, $V_{GS} = 0$, $f = 140 \text{ kHz}$)	C_{iss}	—	5.0	pF
Reverse Transfer Capacitance ($V_{DS} = 0$, $V_{GS} = 0$, $f = 140 \text{ kHz}$)	C_{rs}	—	1.3	pF
Drain-Substrate Capacitance ($V_{DSUB} = 10 \text{ V}$, $f = 140 \text{ kHz}$)	$C_{d(sub)}$	—	5.0	pF
Drain-Source Resistance ($V_{GS} = 10 \text{ V}$, $I_D = 0$, $f = 1.0 \text{ kHz}$)	$r_{ds(on)}$	—	300	ohms
SWITCHING CHARACTERISTICS				
Turn-On Delay (Fig. 5)	t_{d1}	—	45	ns
Rise Time (Fig. 6)	t_r	—	65	ns
Turn-Off Delay (Fig. 7)	t_{d2}	—	60	ns
Fall Time (Fig. 8)	t_f	—	100	ns

$I_D = 2.0 \text{ mA dc}$, $V_{DS} = 10 \text{ Vdc}$,
($V_{GS} = 10 \text{ Vdc}$)
(See Figure 9; Times Circuit Determined)

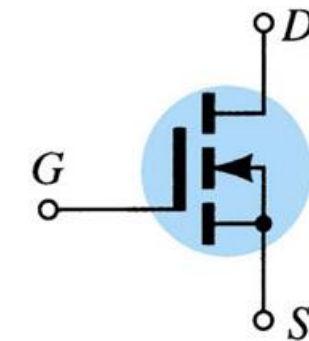
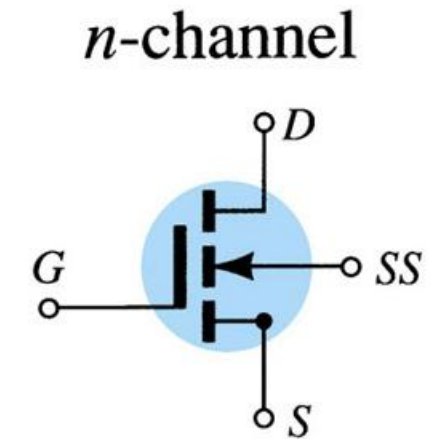
2N4351

CASE 20-03, STYLE 2
TO-72 (TO-206AF)



**MOSFET
SWITCHING**
N-CHANNEL – ENHANCEMENT

E-MOSFET Symbols



(a)

E-MOSFET Basic Operation

The Enhancement mode MOSFET only operates in the enhancement mode.

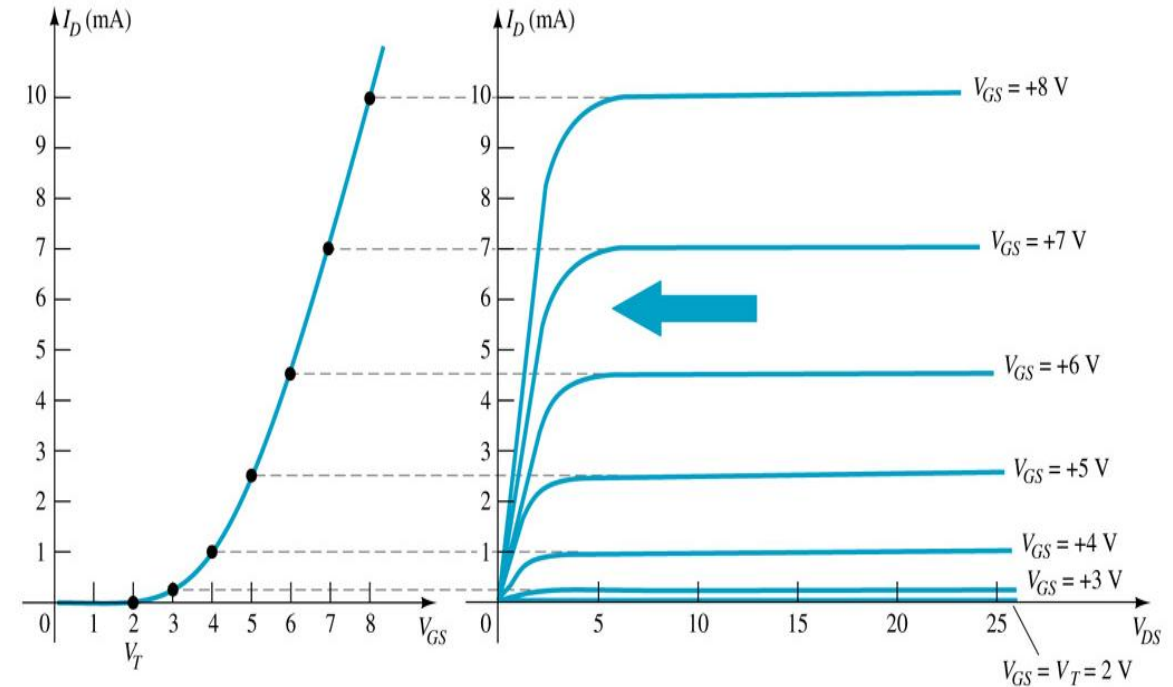
V_{GS} is always positive

$I_{DSS} = 0$ when $V_{GS} < V_T$

As V_{GS} increases above V_T , I_D increases

If V_{GS} is kept constant and V_{DS} is increased, then I_D saturates (I_{DSS})

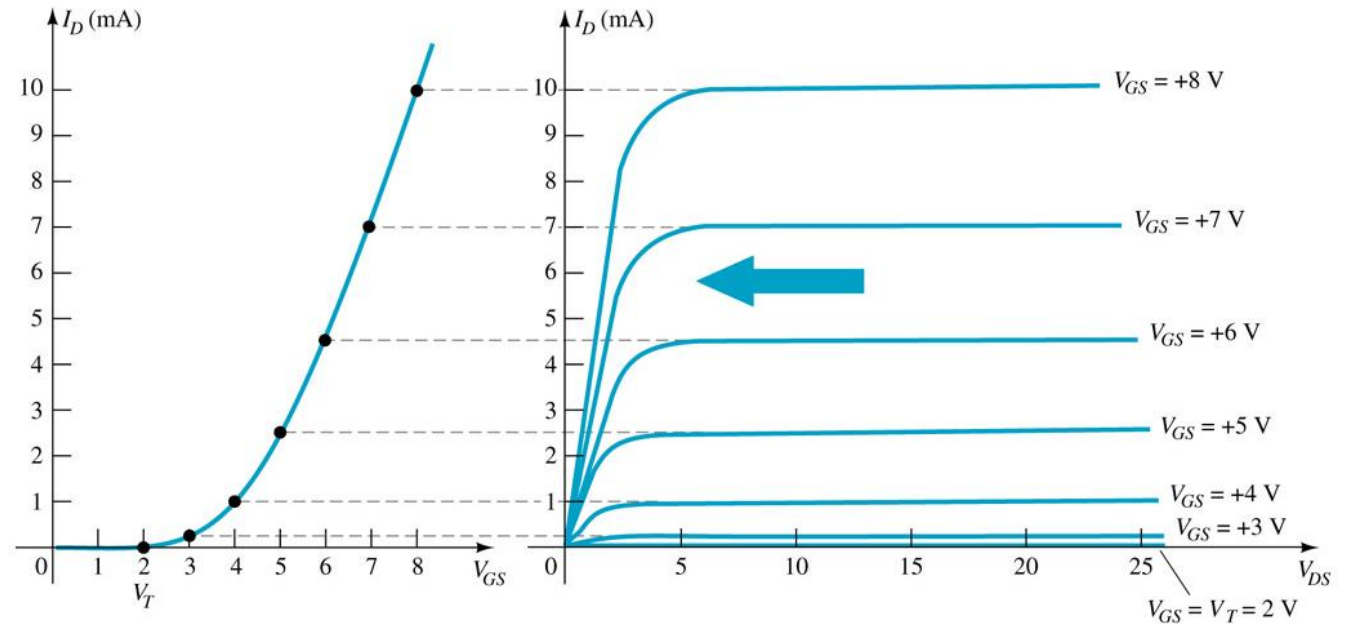
The saturation level, V_{DSsat} is reached.



E-MOSFET Transfer Curve

$$I_D = k (V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2}$$



To determine I_D given V_{GS} :

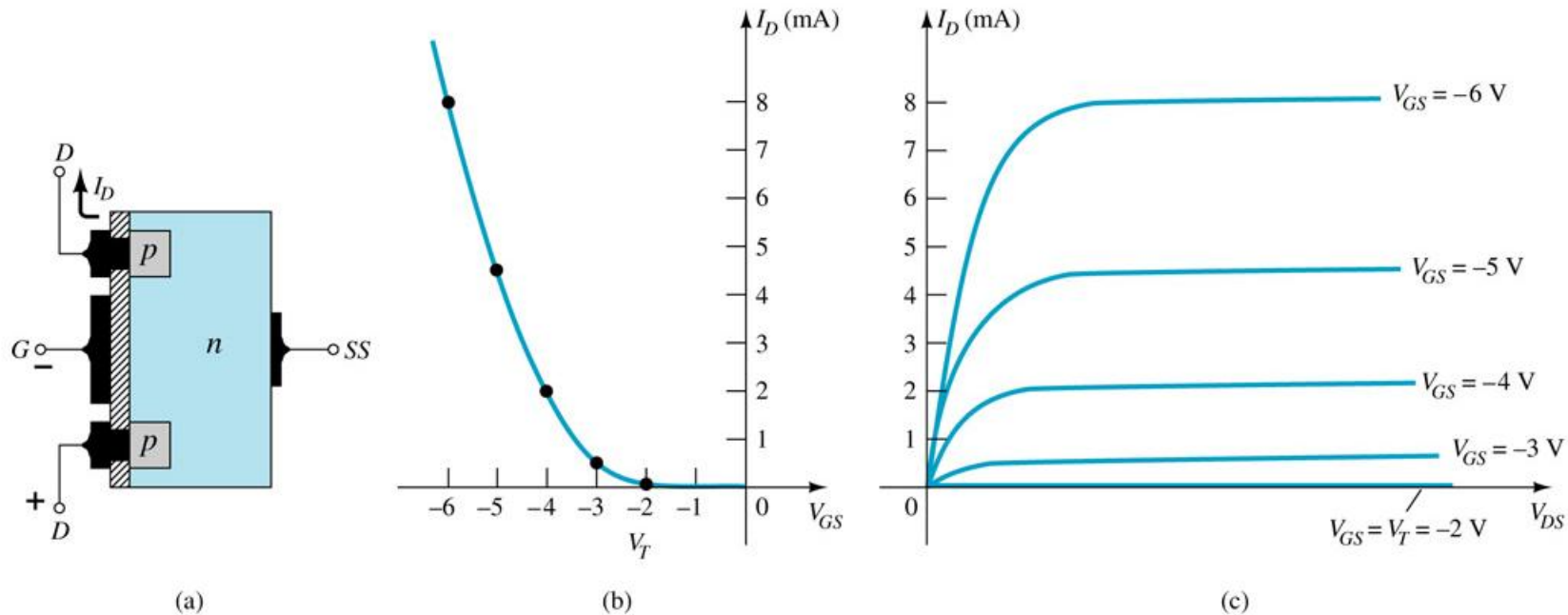
where V_T = threshold voltage or voltage at which the MOSFET turns on.

k = constant found in the specification sheet

The PSpice determination of k is based on the geometry of the device:

$$k = \left(\frac{W}{L} \right) \left(\frac{KP}{2} \right) \quad \text{where } KP = \mu_n C_{ox}$$

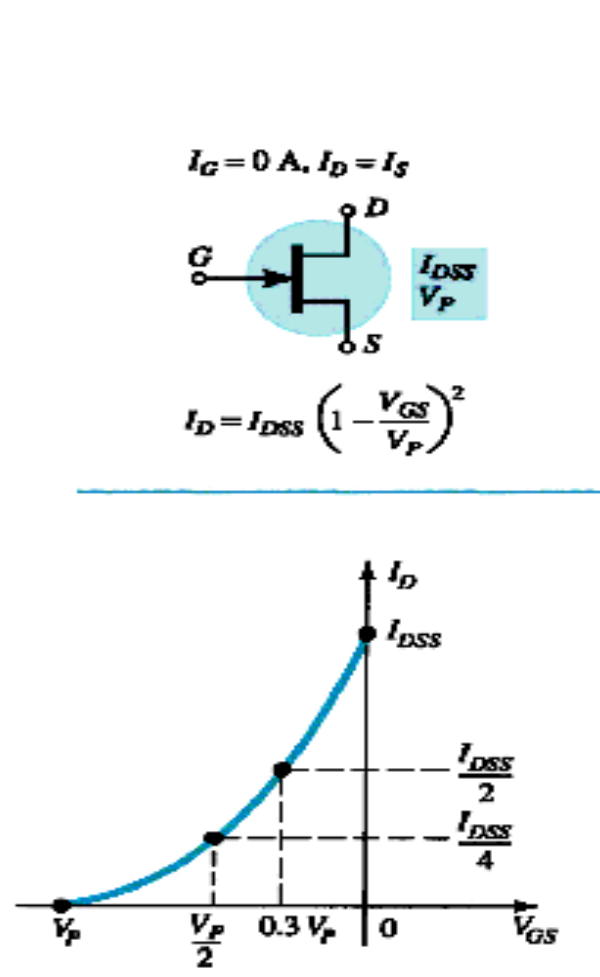
P-Channel Enhancement Mode MOSFET



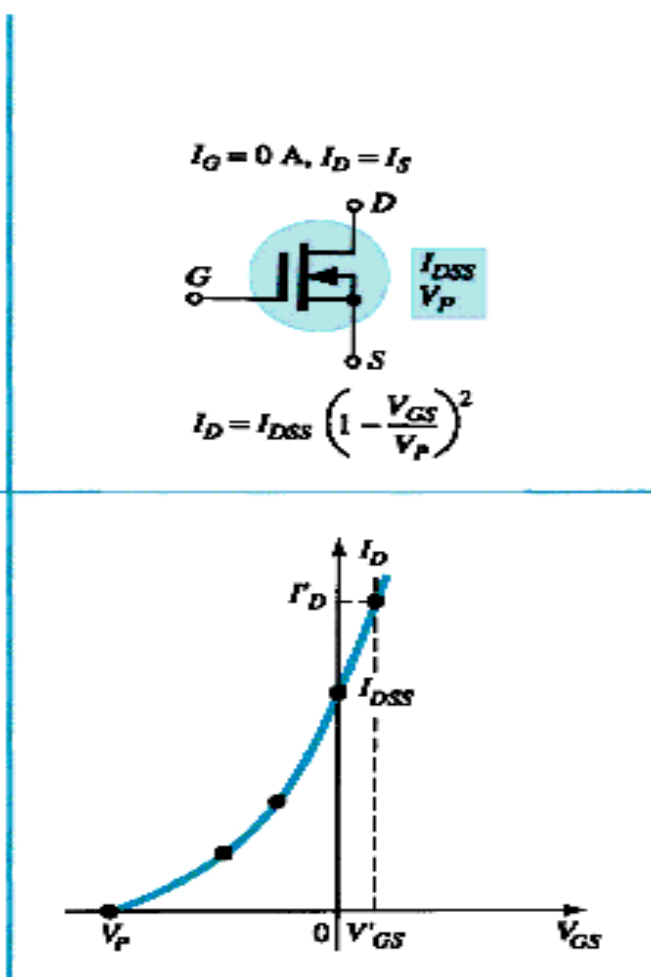
The p-channel Enhancement mode MOSFET is similar to the n-channel except that the voltage polarities and current directions are reversed.

FET Summary

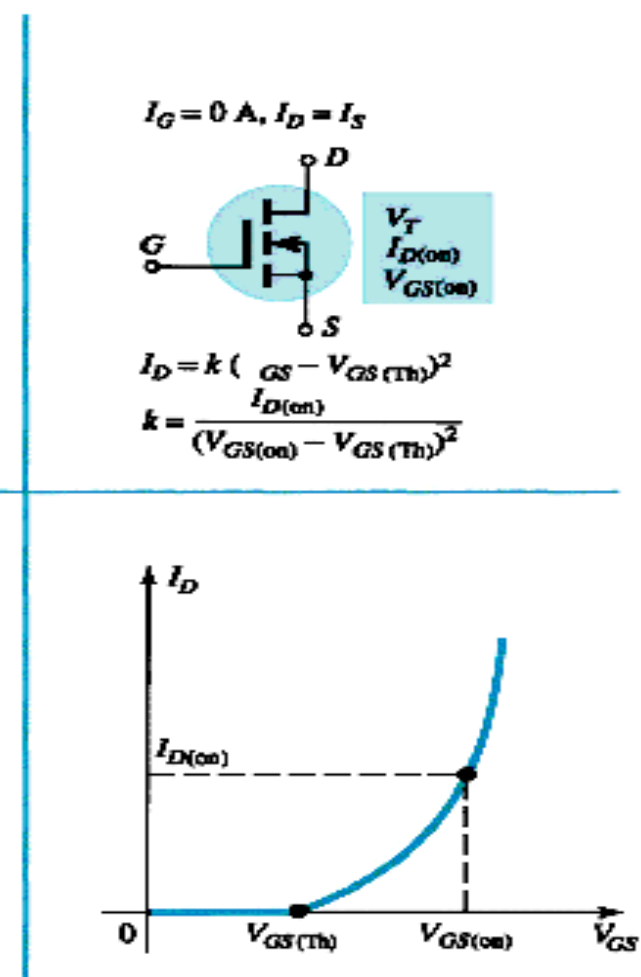
JFET



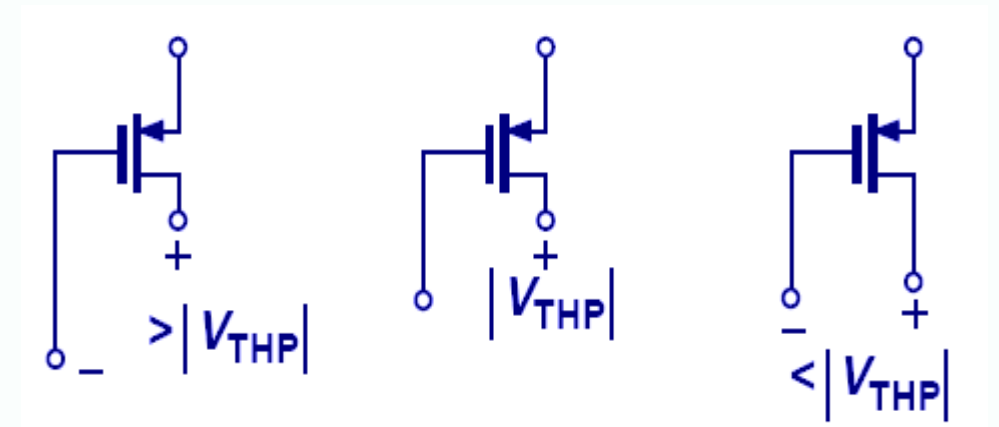
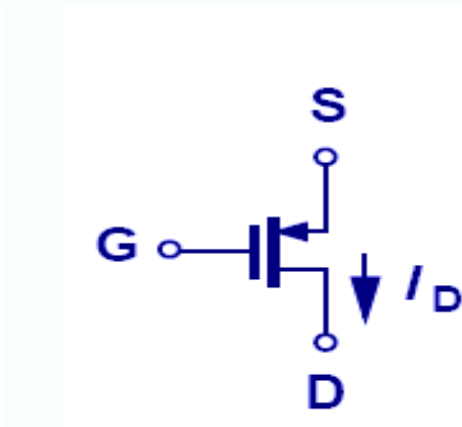
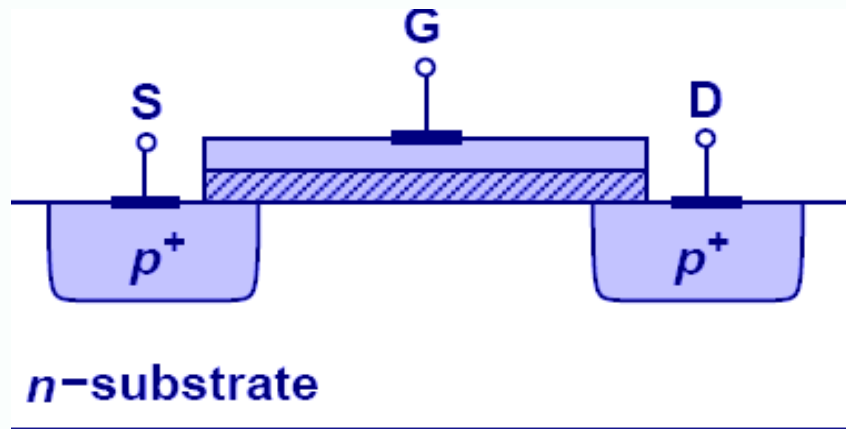
D-MOSFET



E-MOSFET

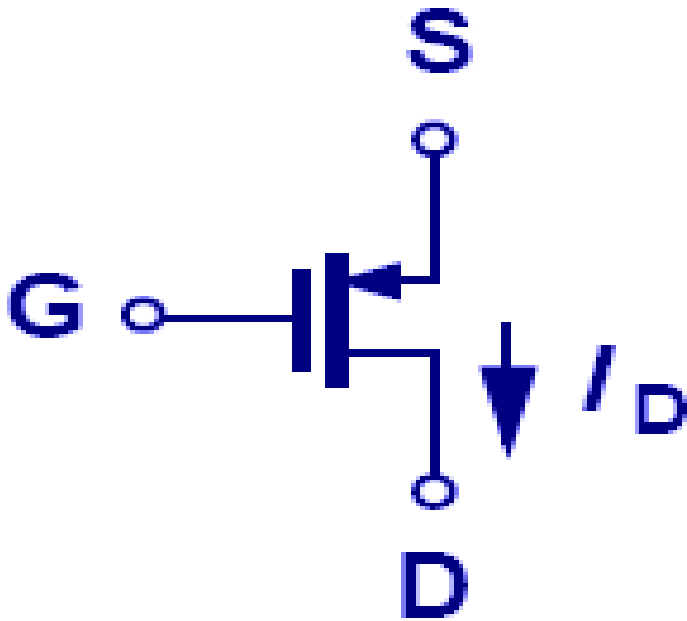


PMOS Transistor



- Just like the PNP transistor in bipolar technology, it is possible to create a MOS device where holes are the dominant carriers.
- It is called the PMOS (p-channel) transistor.
- It behaves like an NMOS (n-channel) device with all the polarities reversed.

PMOS Equations



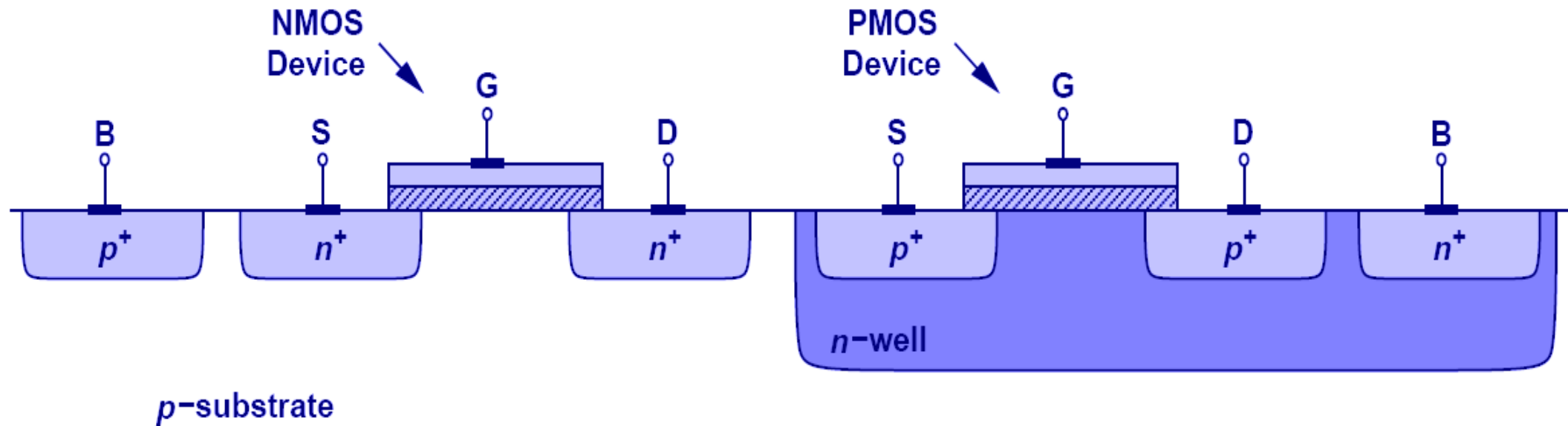
$$I_{D,sat} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 - \lambda V_{DS})$$

$$I_{D,tri} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} [2(V_{GS} - V_{TH})V_{DS} - V_{DS}^2]$$

$$I_{D,sat} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (|V_{GS}| - |V_{TH}|)^2 (1 + \lambda |V_{DS}|)$$

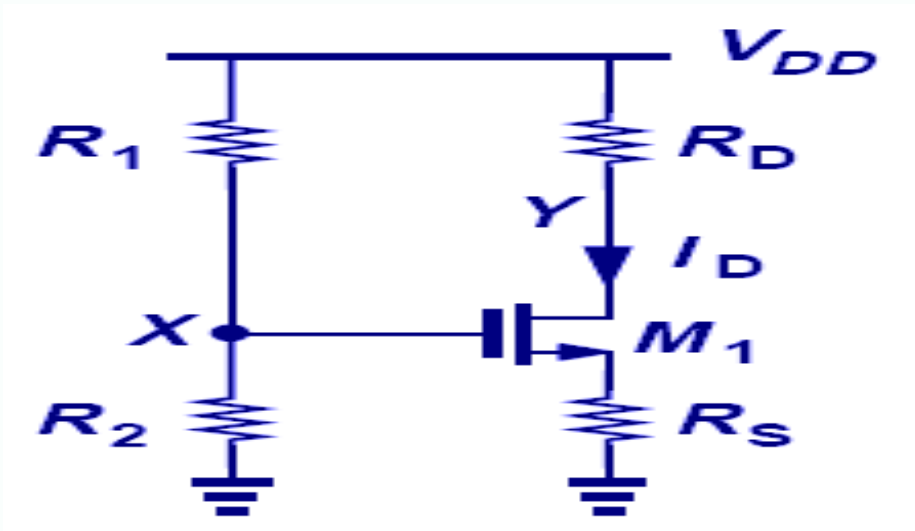
$$I_{D,tri} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} [2(|V_{GS}| - |V_{TH}|)|V_{DS}| - V_{DS}^2]$$

CMOS Technology



- It possible to grow an n-well inside a p-substrate to create a technology where both NMOS and PMOS can coexist.
- It is known as CMOS, or “Complementary MOS”.

MOS Biasing



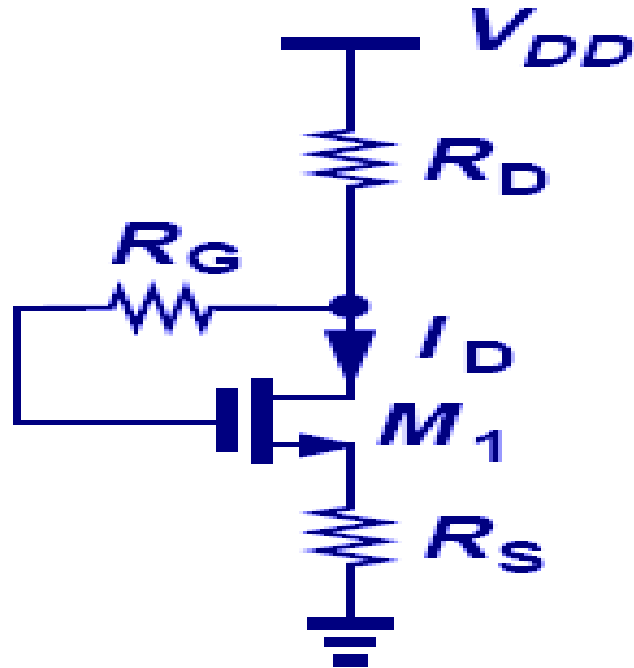
$$V_{GS} = -(V_1 - V_{TH}) + \sqrt{V_1^2 + 2V_1 \left(\frac{R_2 V_{DD}}{R_1 + R_2} - V_{TH} \right)}$$

$$V_1 = \frac{1}{\mu_n C_{ox} \frac{W}{L} R_S}$$

- Voltage at X is determined by V_{DD} , R_1 , and R_2 .

V_{GS} can be found using the equation above, and I_D can be found by using the NMOS current equation.

Self-Biased MOS



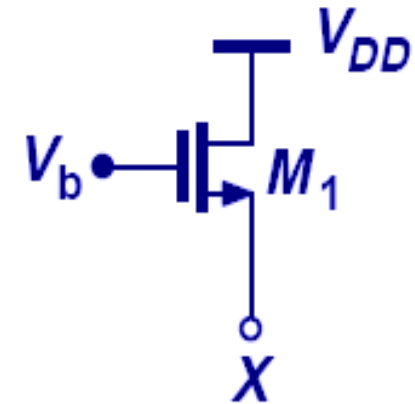
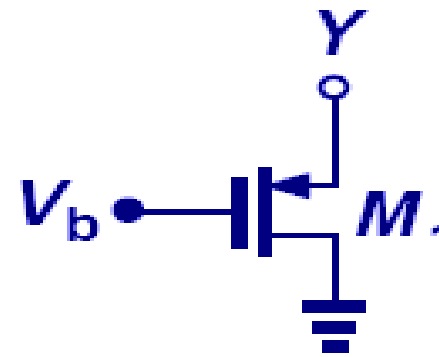
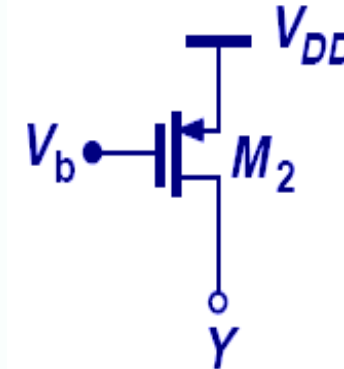
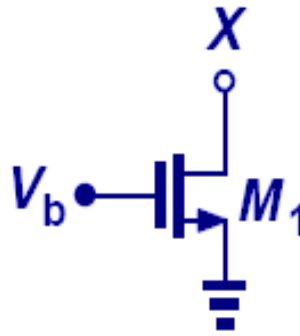
$$I_D R_D + V_{GS} + R_S I_D = V_{DD}$$

- if M_1 is in saturation and no potential drop appears across R_G .

Current Sources

- When in saturation region, a MOSFET behaves as a current source.

NMOS draws current from a point to ground (sinks current), whereas PMOS draws current from V_{DD} to a point (sources current).





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COE 271 Semiconductor Devices

Lecture 8

Operational Amplifiers OPAMP

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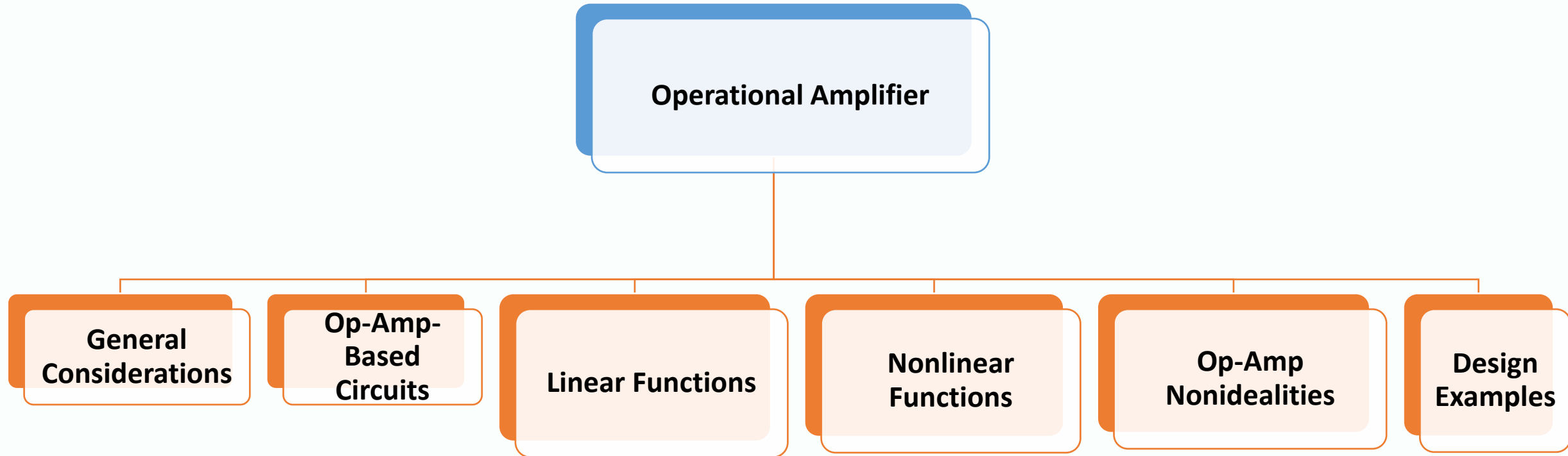
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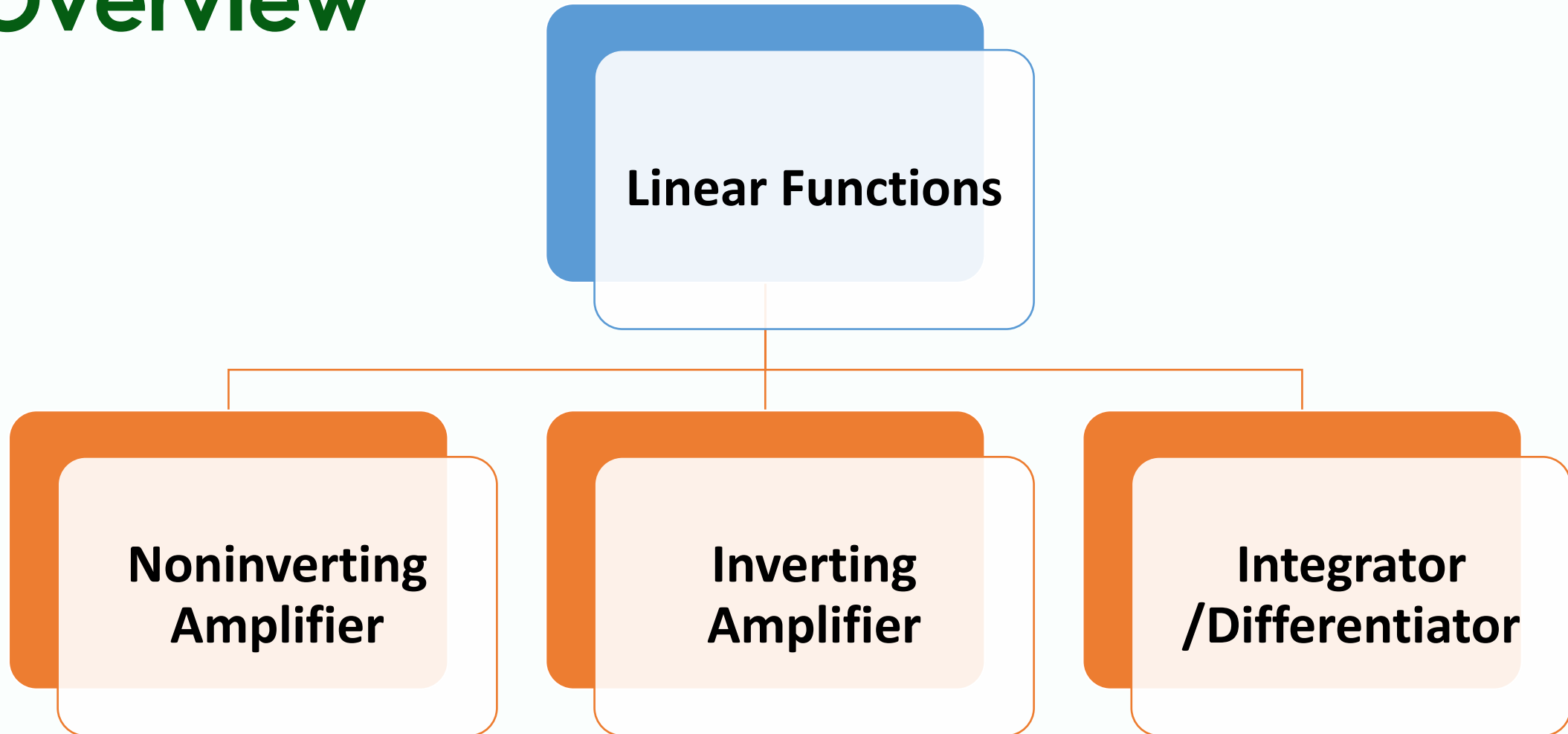
“Learn from yesterday, live for today, hope for tomorrow.
The important thing is to not stop questioning.”

Albert Einstein

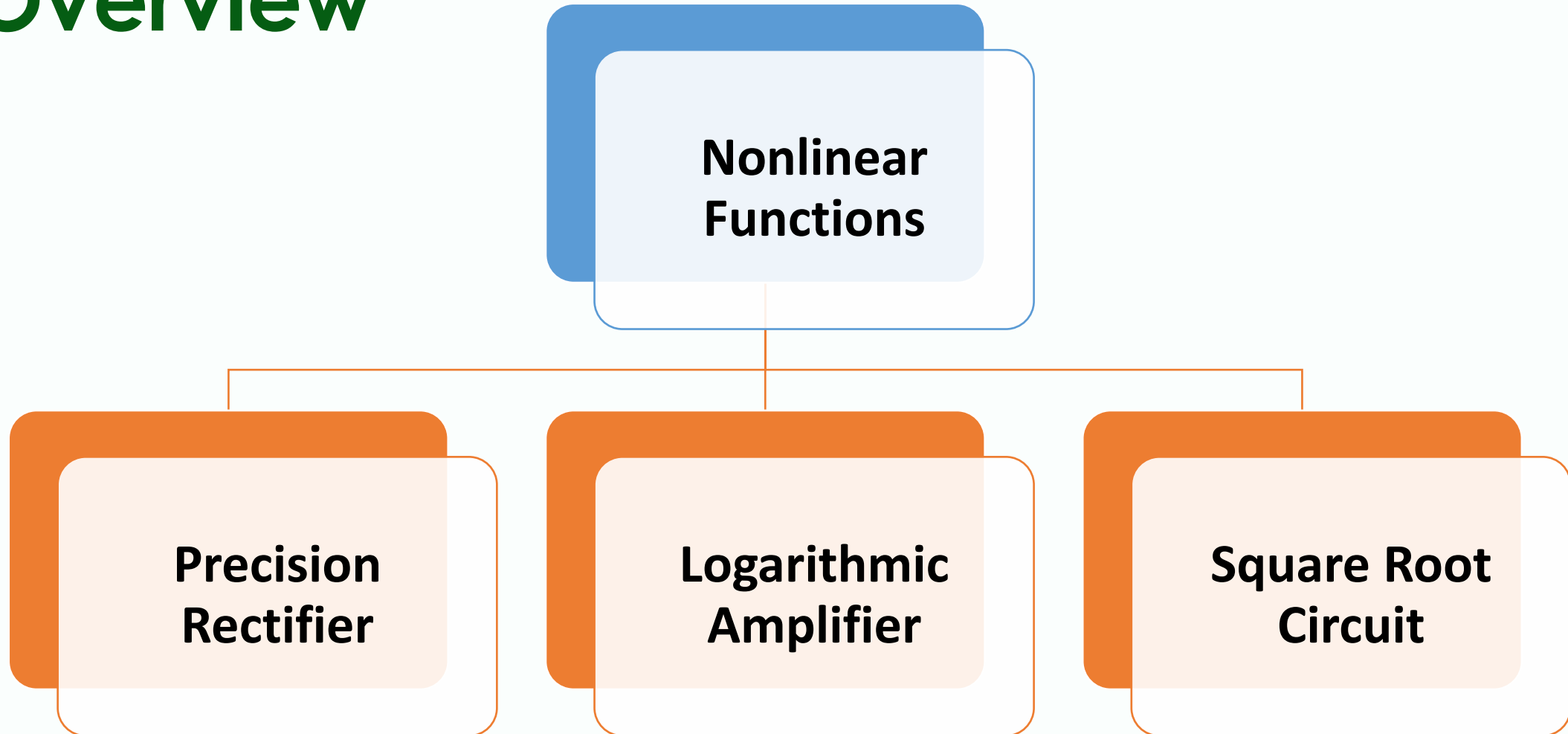
Overview



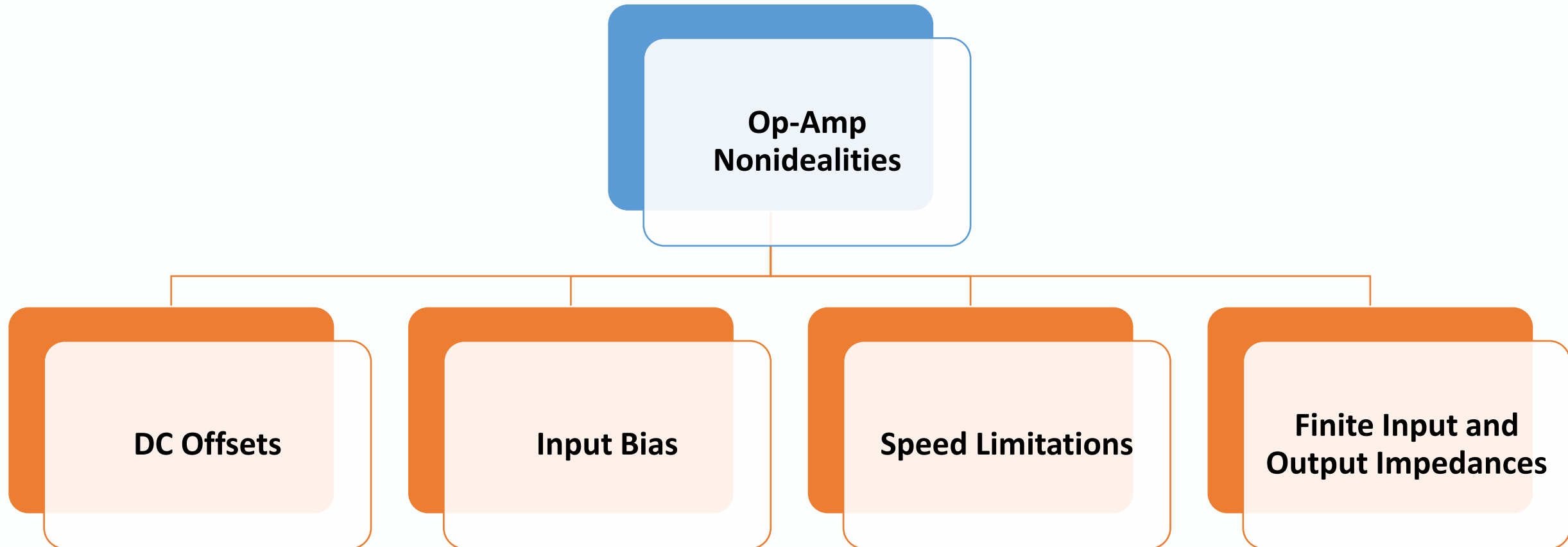
Overview



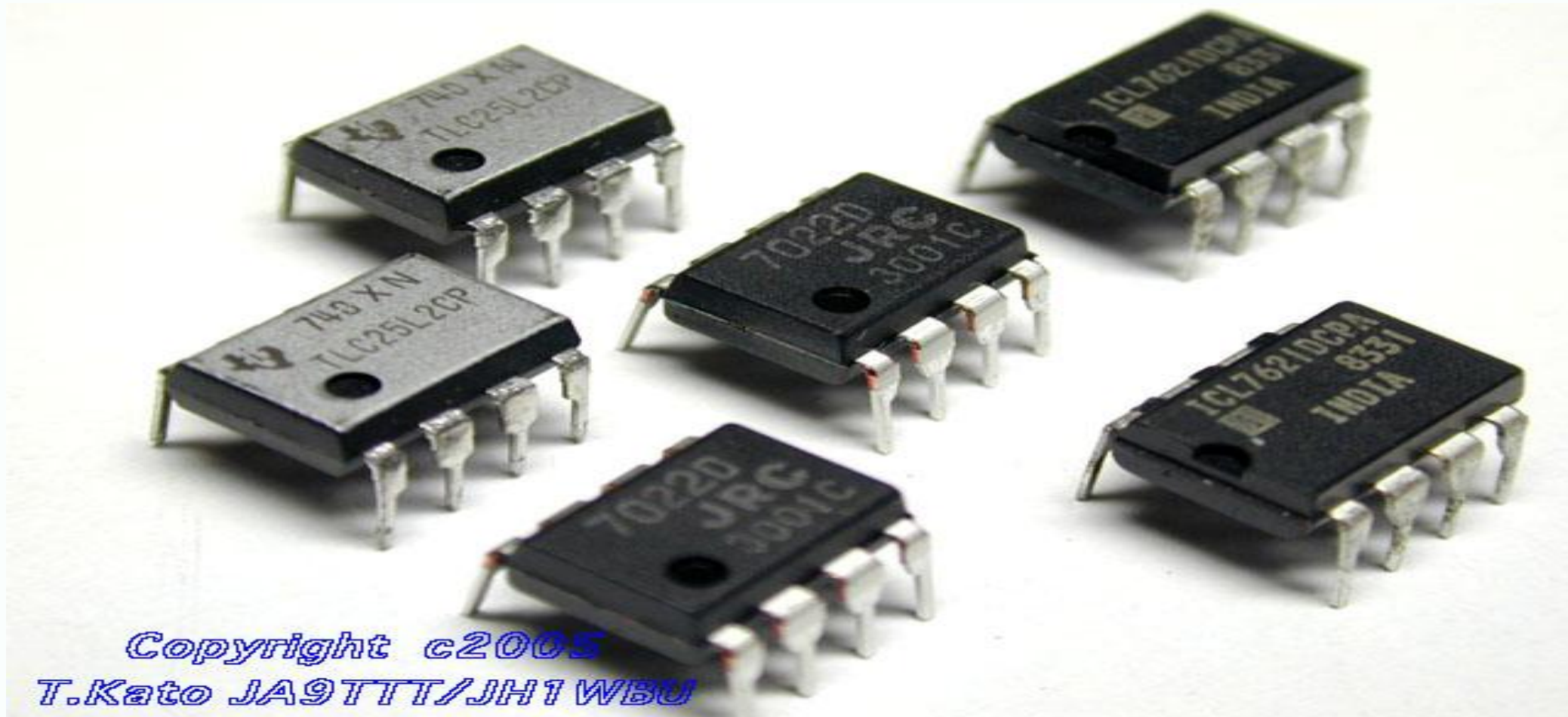
Overview



Overview



Introduction to OP



Introduction to OP

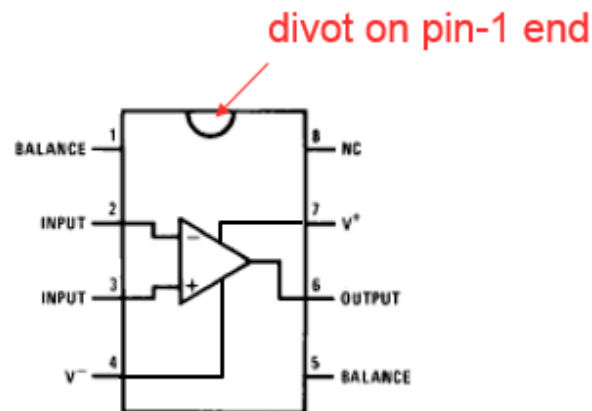
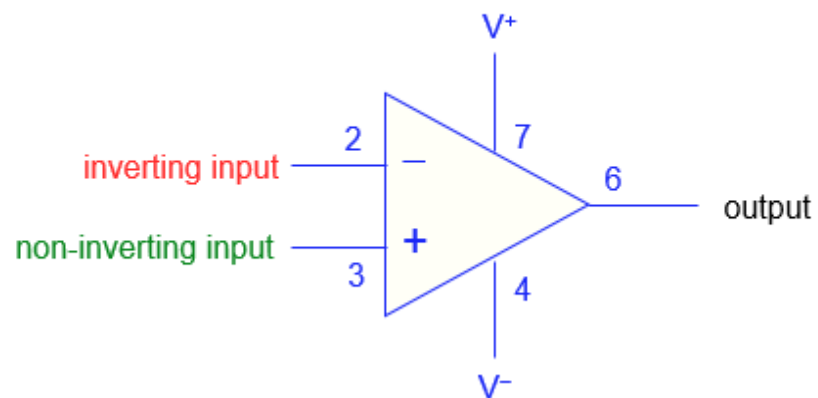
OP (-also called amplifiers or buffers in general)

Op is drawn as a triangle in a schematic

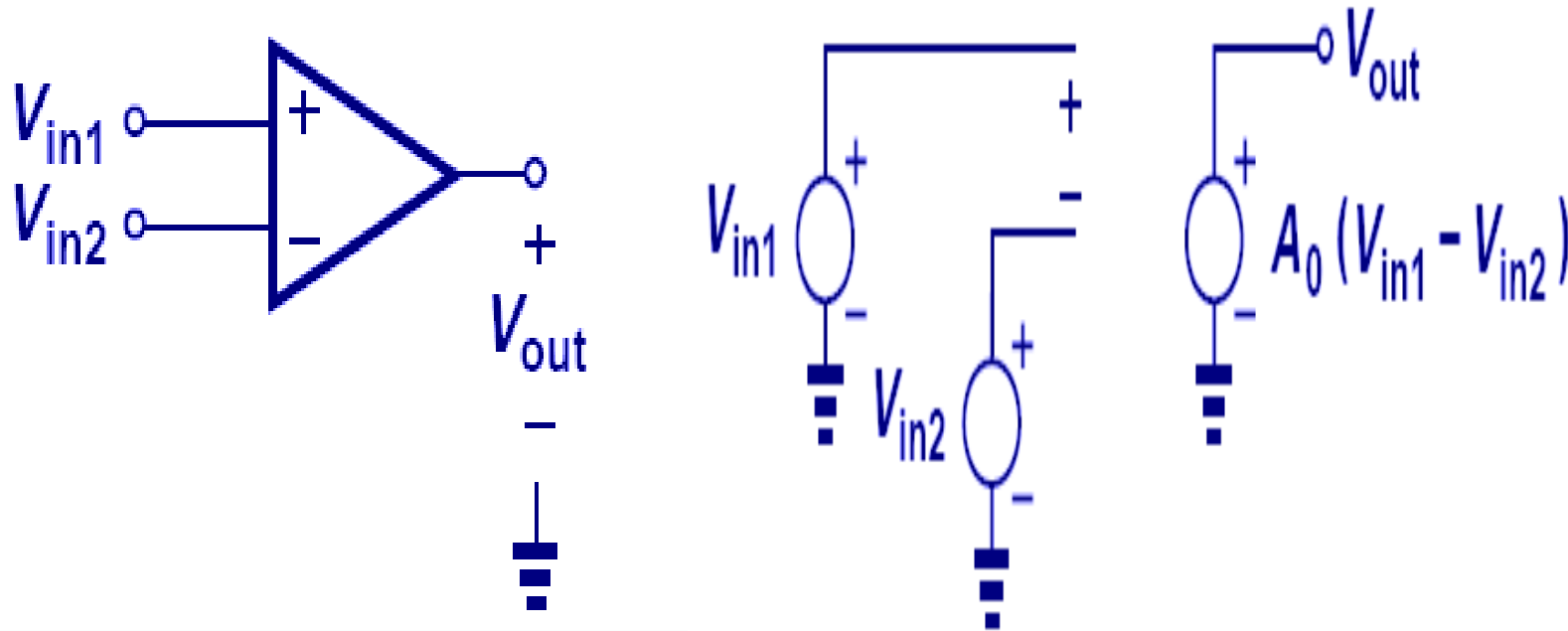
Op has two inputs (inverting and non-inverting)

Op has one output

Op also has power connections (NB: no explicit ground)



Basic Operational Amplifier

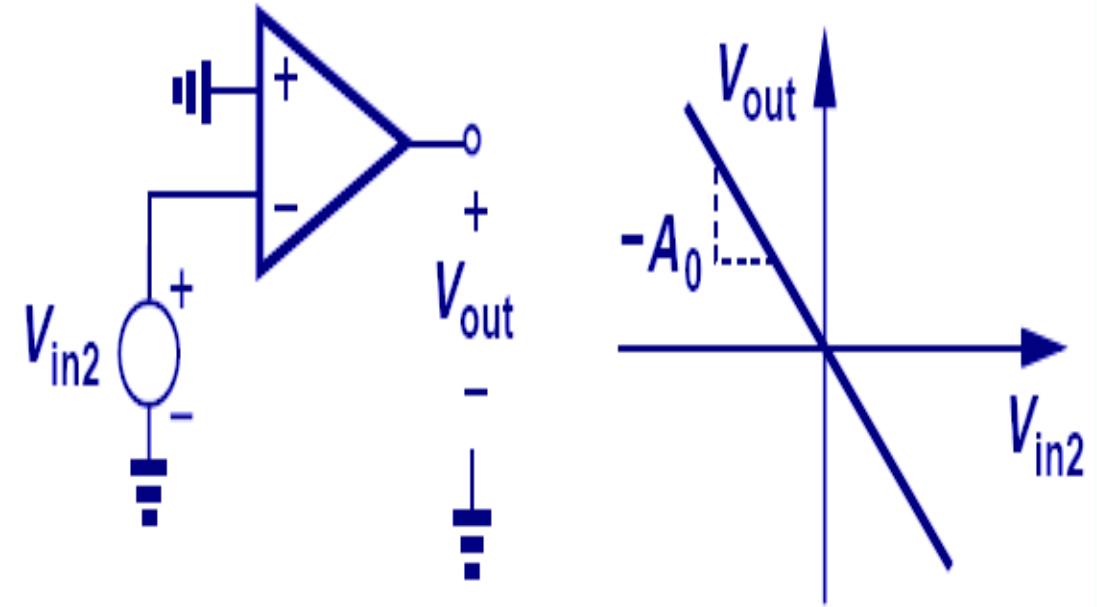
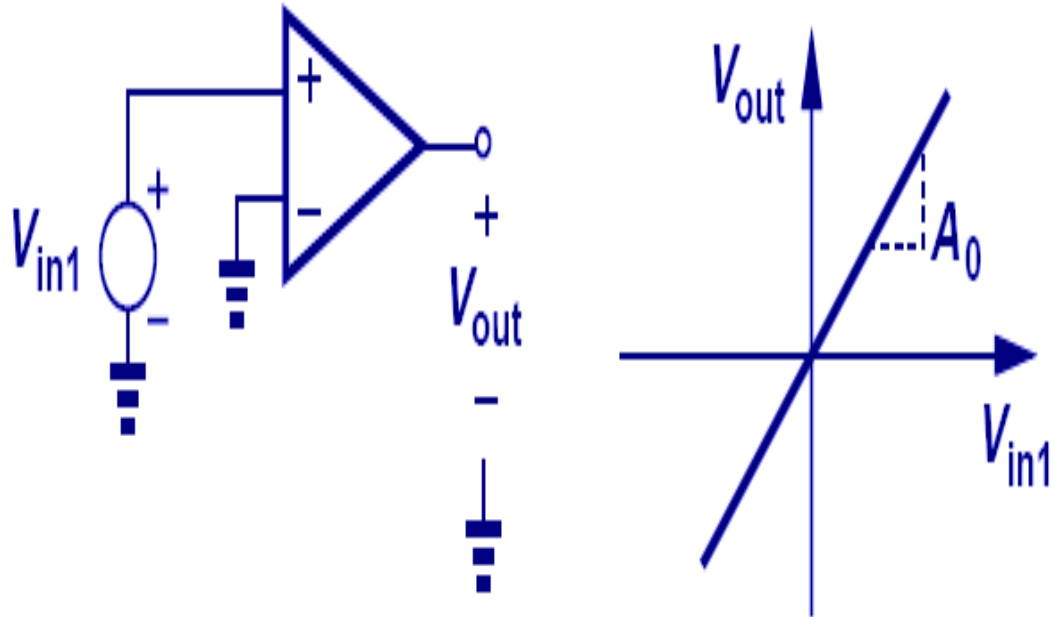


Internal Op-amplifier formula

$$V_{out} = A_0 (V_{in1} - V_{in2})$$

- Op amp is a circuit that has two inputs and one output.
- It amplifies the difference between the two inputs.

Inverting and Non-inverting Op Amp



- If the negative input is grounded, the gain is positive.
If the positive input is grounded, the gain is negative.

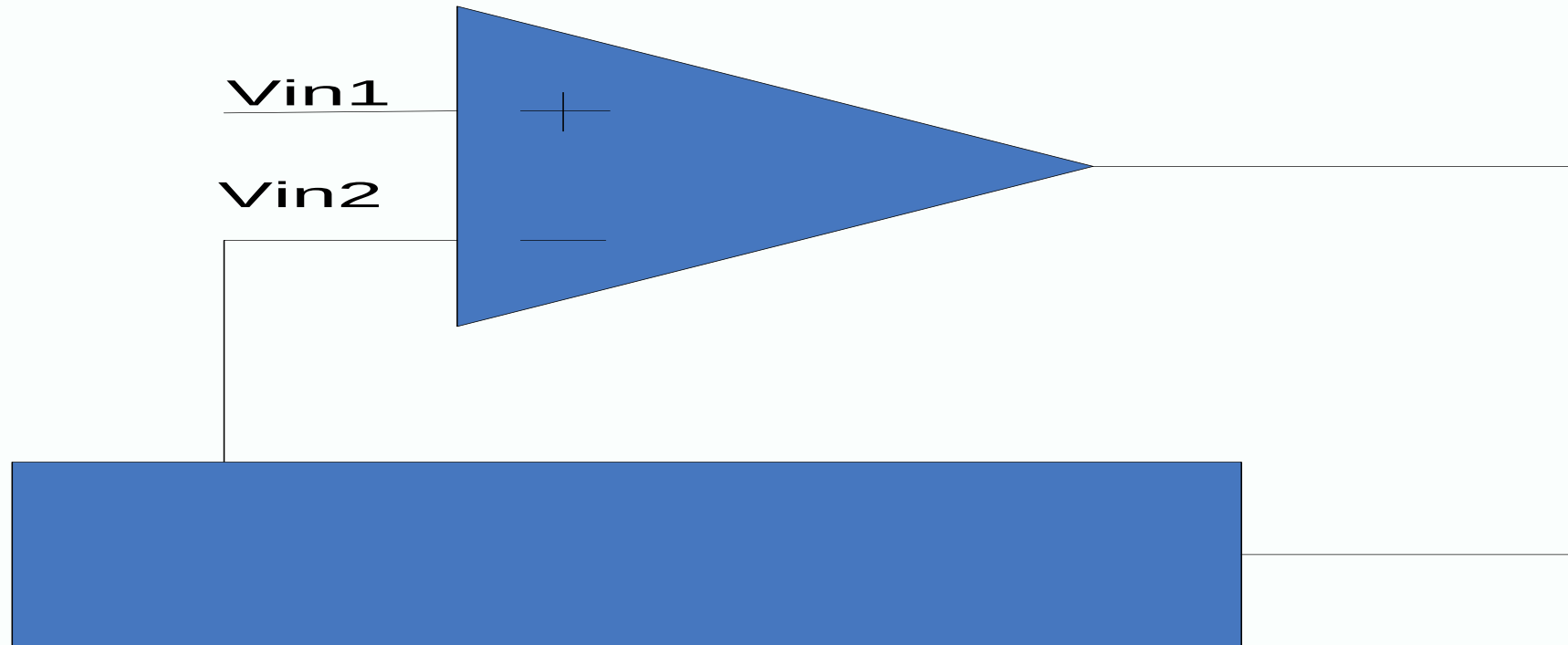
Ideal Operational Amplifier

- Infinite gain
- (- a voltage difference at the two inputs is magnified infinitely)
- (-meaning that the difference between + and – terminal is amplified by say 200000)

Ideal Operational Amplifier

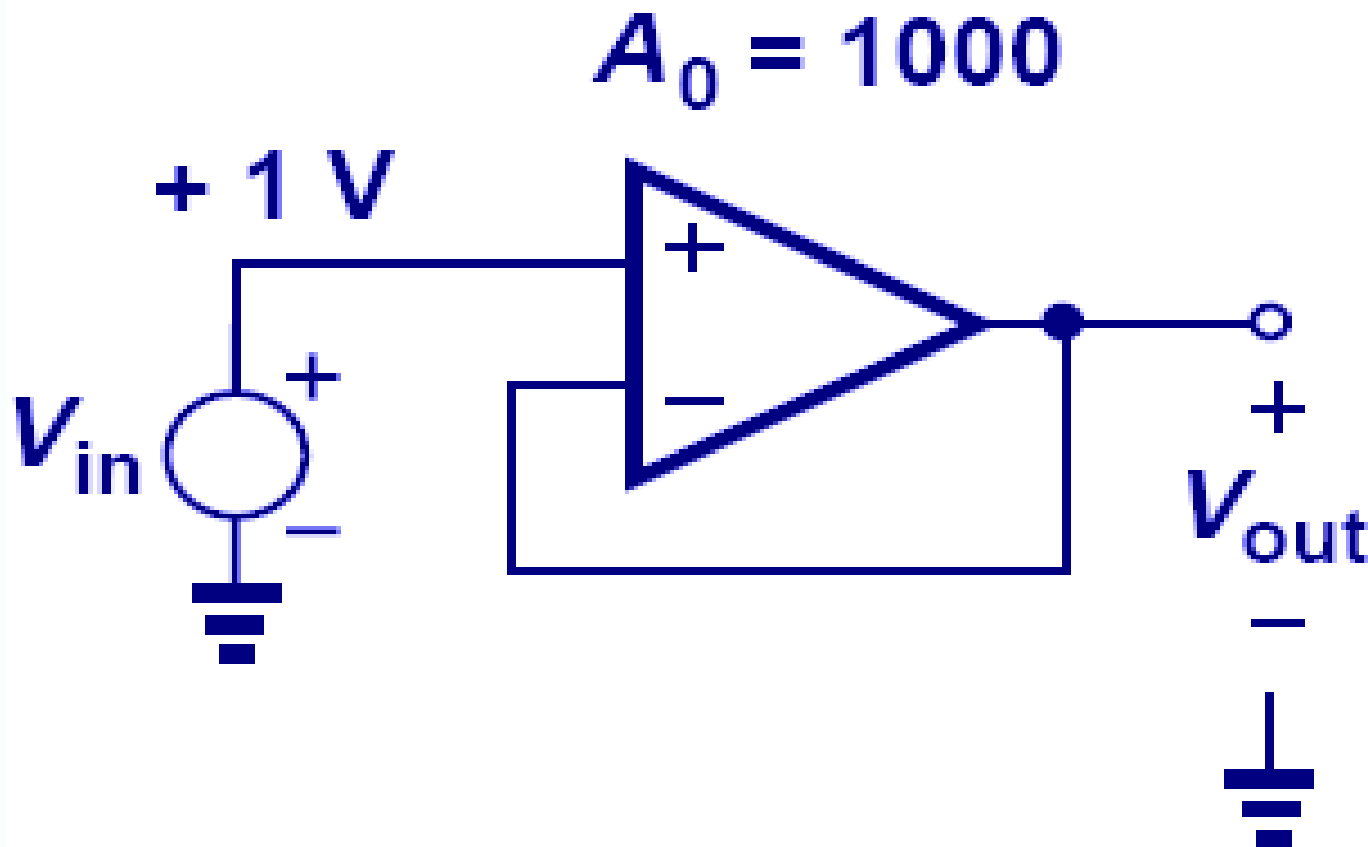
- Infinite input impedance
- (-no current flows into the inputs)
- (-impedance is about $10^{12}\Omega$ for FET)
- Zero output impedance
- (-rock-solid independent of load)
- Infinite speed
- (-limited to few MHz range)

Virtual Short



- Due to infinite gain of op amp, the circuit forces V_{in2} to be close to V_{in1} , thus creating a virtual short.

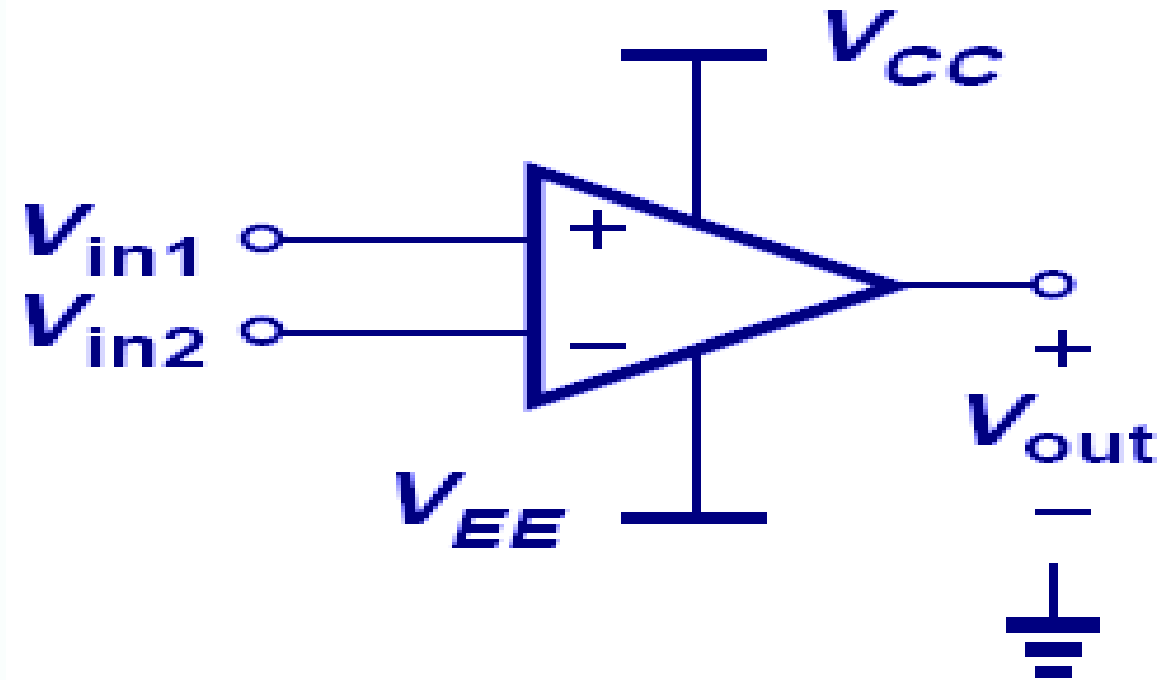
Unity Gain Amplifier



$$V_{out} = A_0 (V_{in} - V_{out})$$

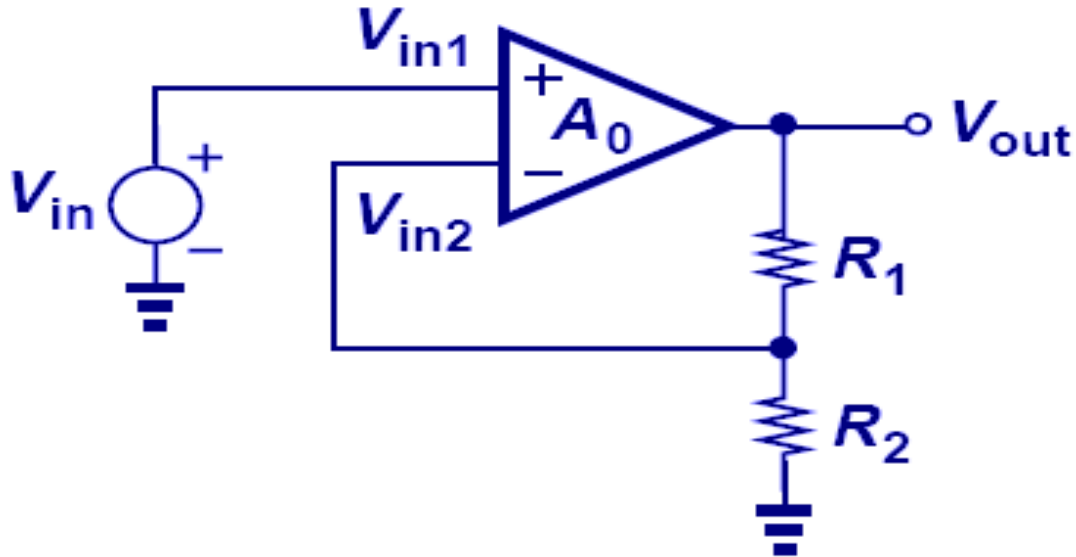
$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + A_0}$$

Op Amp with Supply Rails



- To explicitly show the supply voltages, V_{CC} and V_{EE} are shown.
- In some cases, V_{EE} is zero.

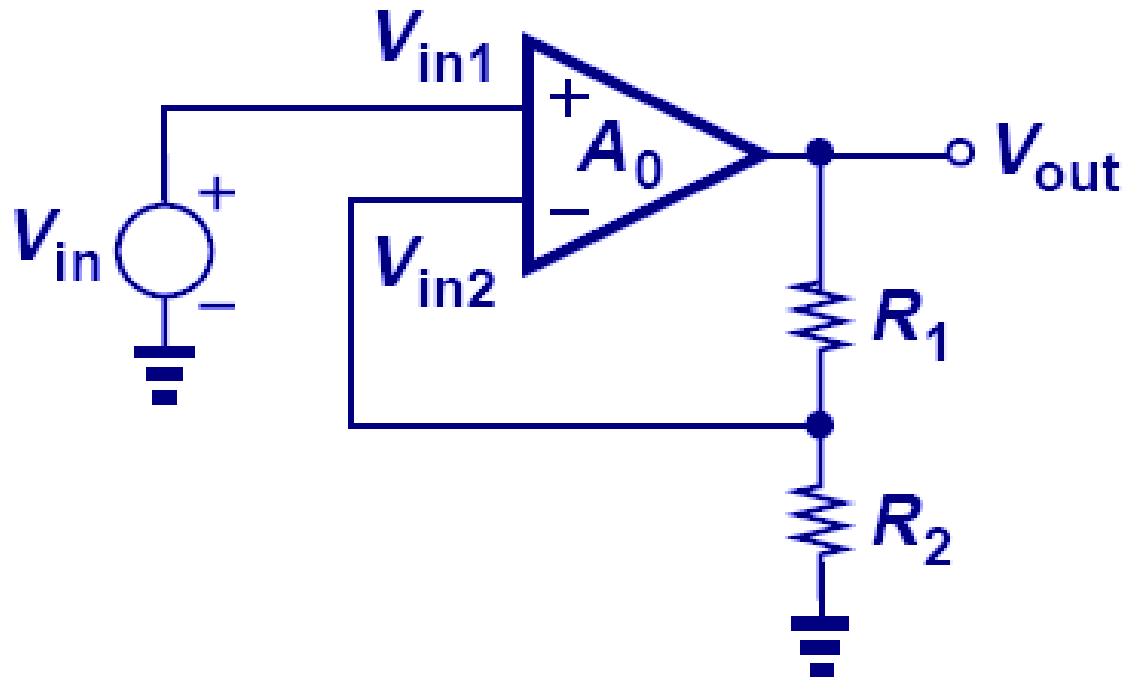
Noninverting Amplifier (Infinite A_0)



$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_1}{R_2}$$

- A noninverting amplifier returns a fraction of output signal thru a resistor divider to the negative input.
- With a high A_0 , V_{out}/V_{in} depends only on ratio of resistors, which is very precise.

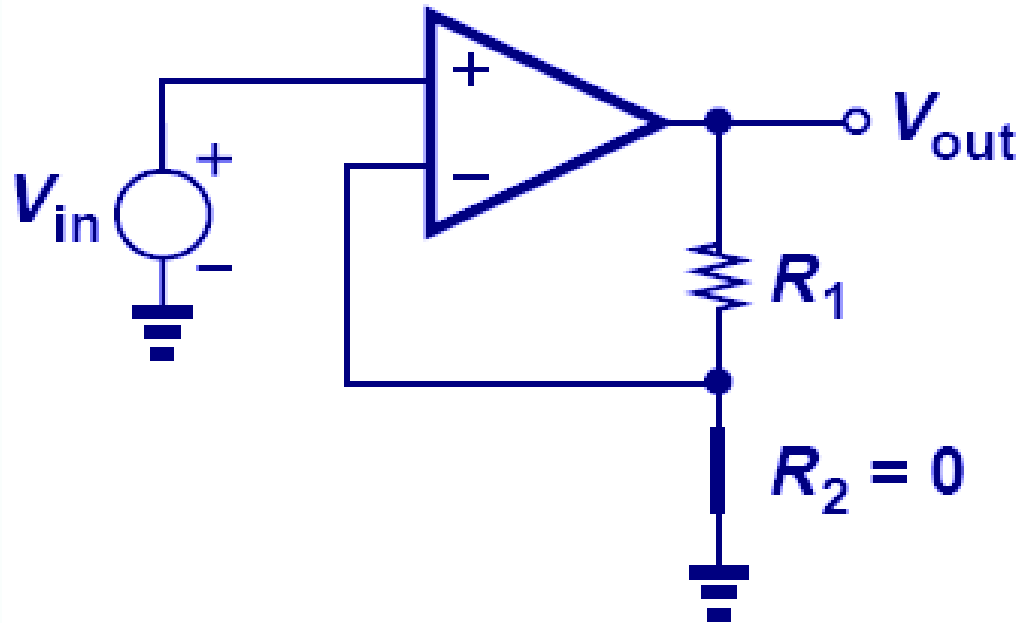
Noninverting Amplifier (Finite A_0)



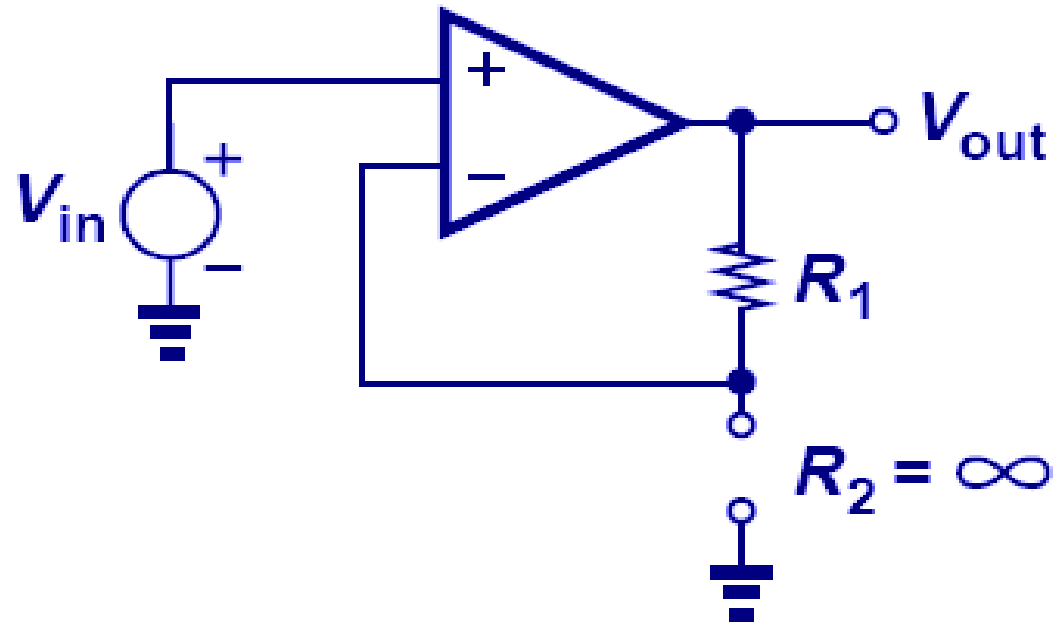
$$\frac{V_{out}}{V_{in}} \approx \left(1 + \frac{R_1}{R_2}\right) \left[1 - \left(1 + \frac{R_1}{R_2}\right) \frac{1}{A_0}\right]$$

- The error term indicates that the larger the closed-loop gain, the less accurate the circuit becomes.

Extreme Cases of R_2 (Infinite A_0)

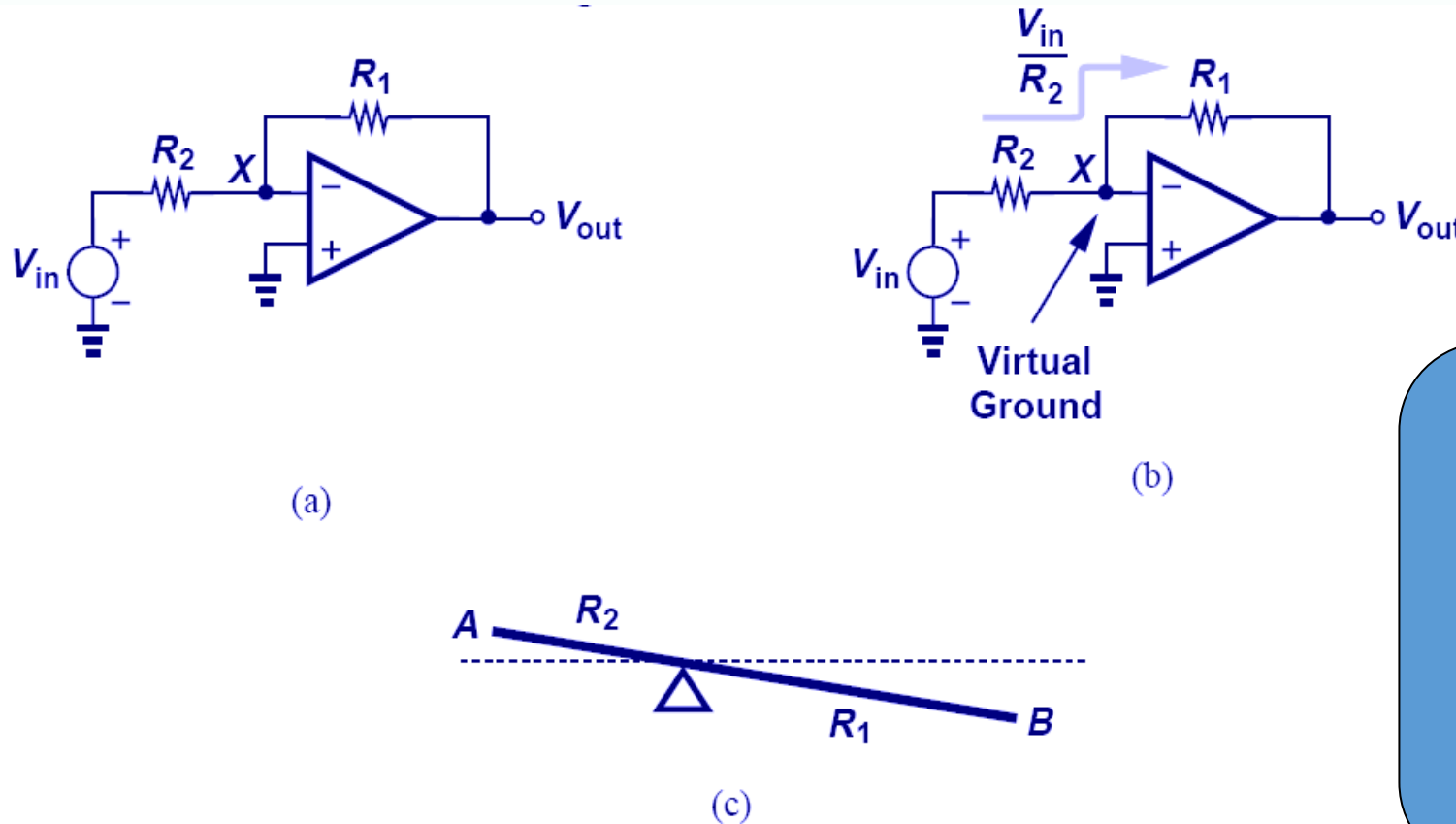


- If R_2 is zero, the loop is open and V_{out}/V_{in} is equal to the intrinsic gain of the op amp.



If R_2 is infinite, the circuit becomes a unity-gain amplifier and V_{out}/V_{in} becomes equal to one.

Inverting Amplifier

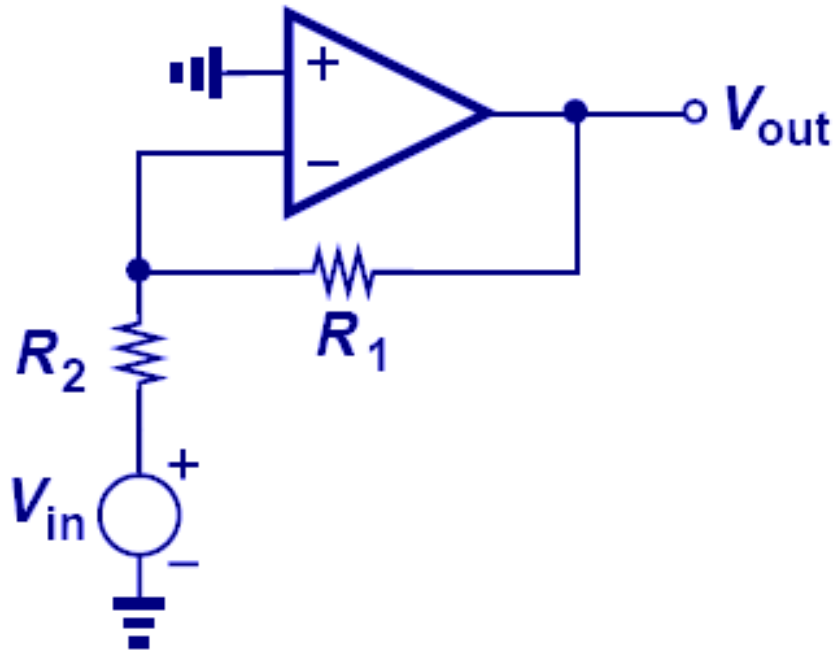


$$\frac{0 - V_{out}}{R_1} = \frac{V_{in}}{R_2}$$

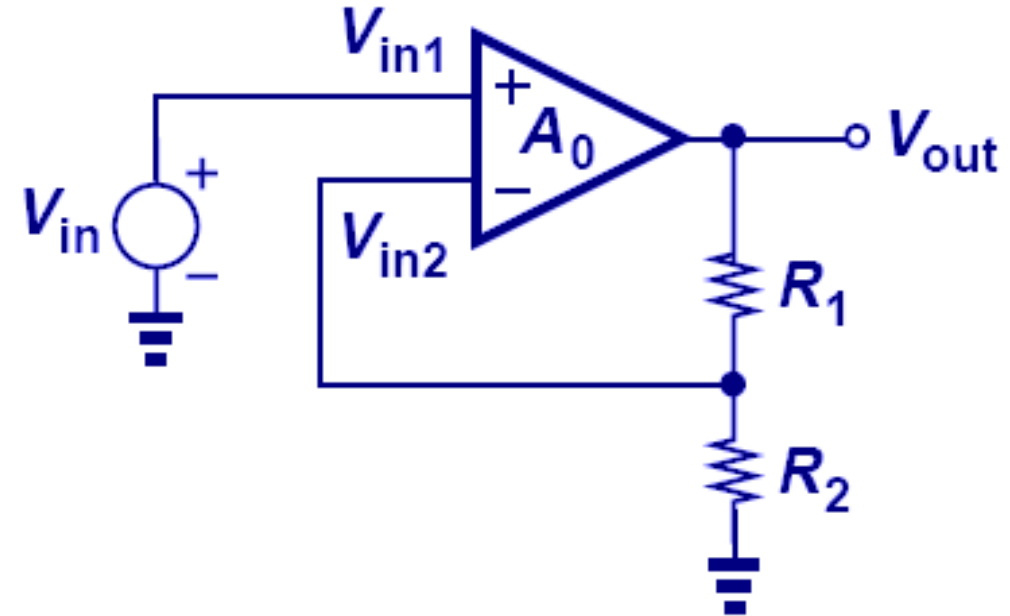
$$\frac{V_{out}}{V_{in}} = \frac{-R_1}{R_2}$$

- Infinite A_0 forces the negative input to be a virtual ground.

Another View of Inverting Amplifier

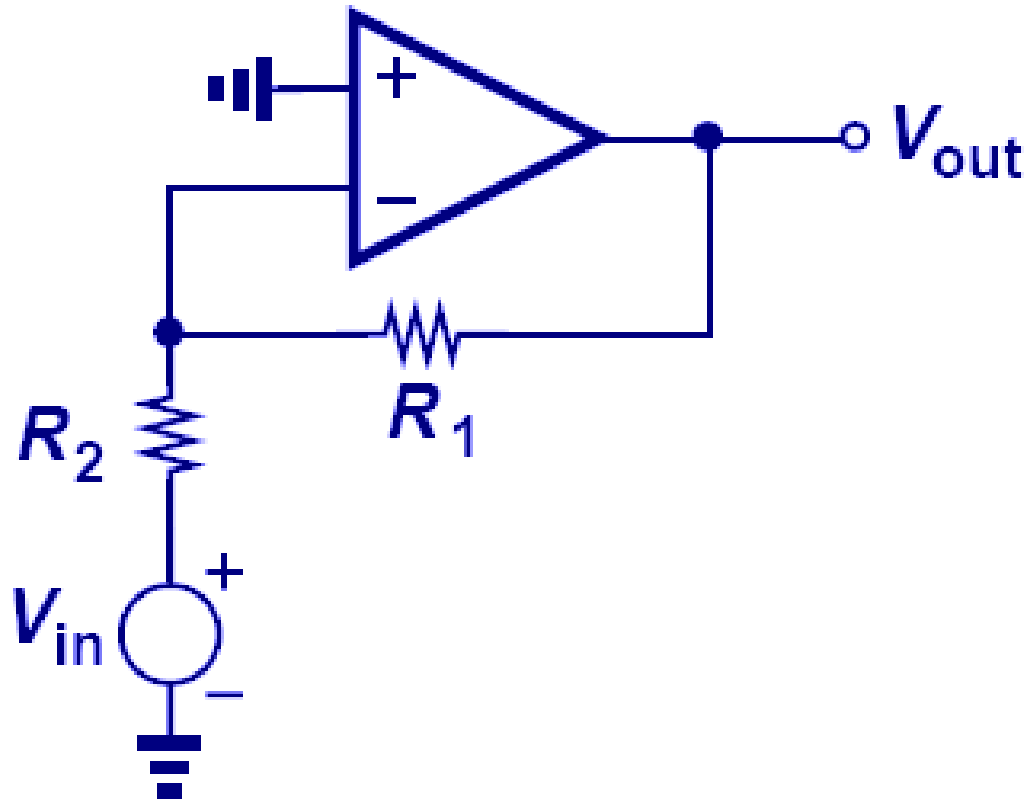


Inverting



Noninverting

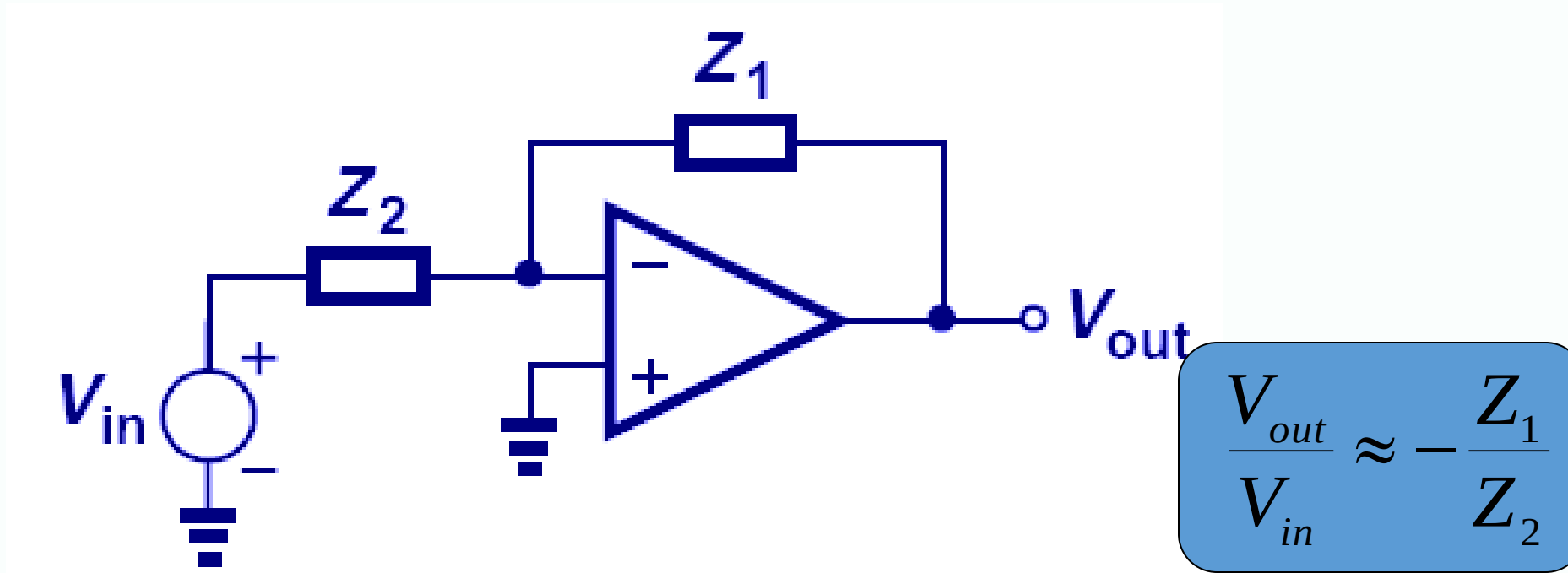
Gain Error Due to Finite A_0



$$\frac{V_{out}}{V_{in}} \approx -\frac{R_1}{R_2} \left[1 - \frac{1}{A_0} \left(1 + \frac{R_1}{R_2} \right) \right]$$

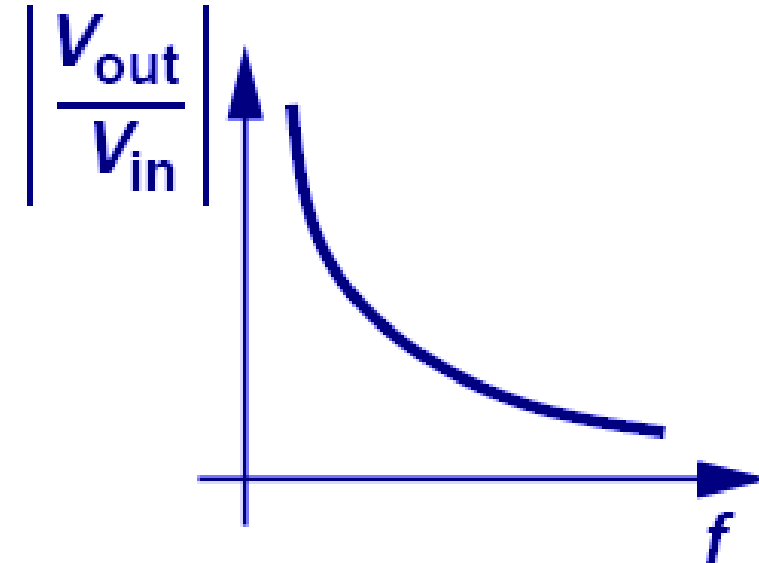
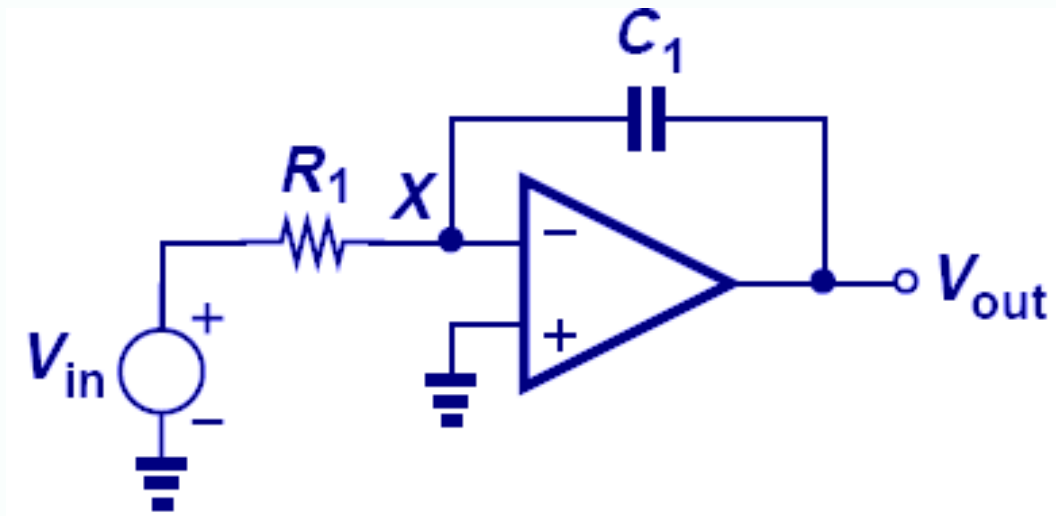
The larger the closed loop gain, the more inaccurate the circuit is.

Complex Impedances Around the Op Amp



- The closed-loop gain is still equal to the ratio of two impedances.

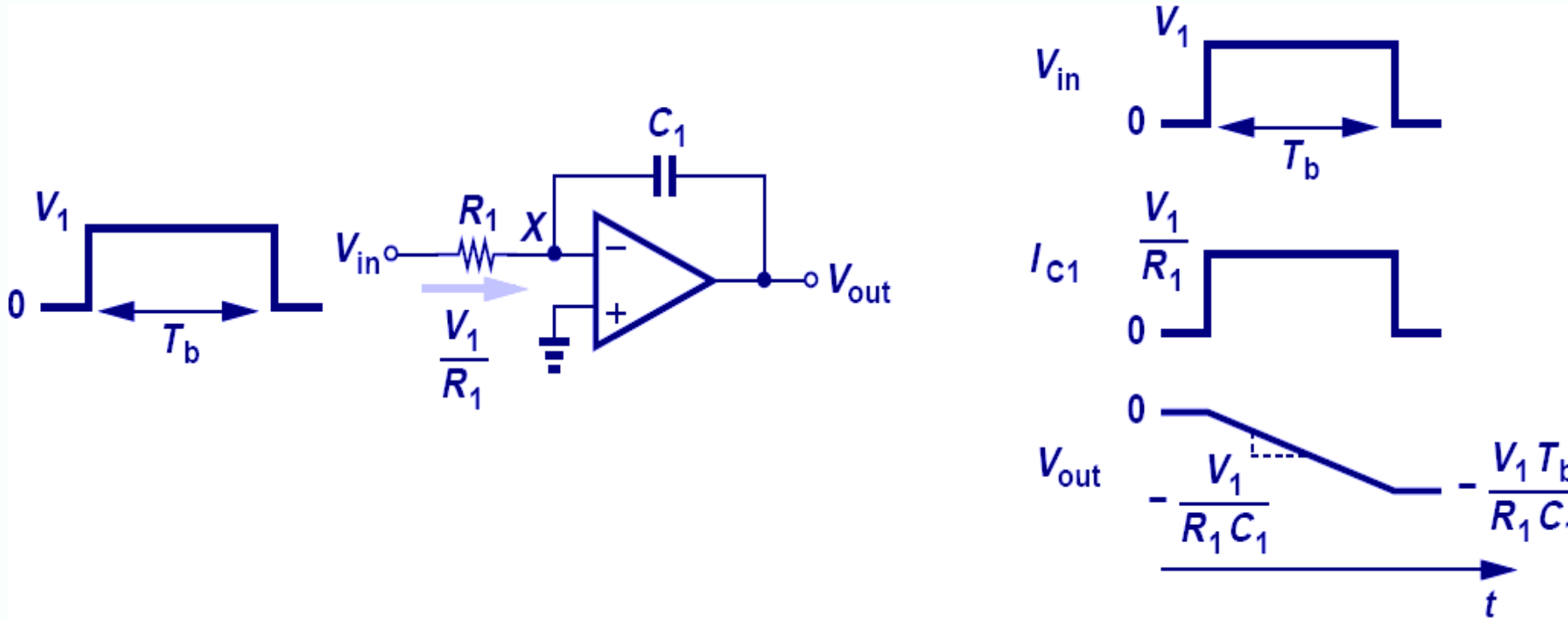
Integrator



$$\frac{V_{out}}{V_{in}} = -\frac{1}{R_1 C_1 s}$$

$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in} dt$$

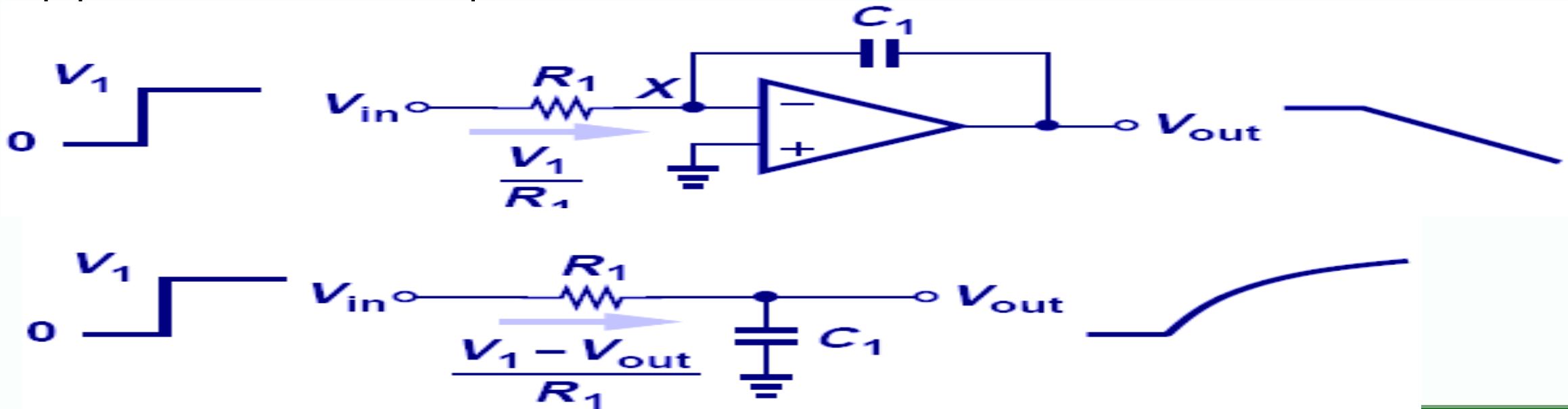
Integrator with Pulse Input



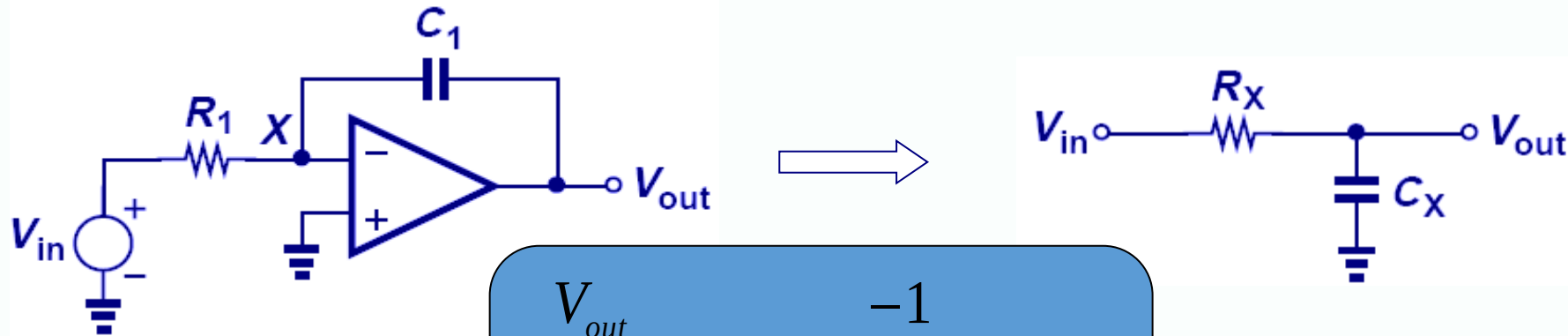
$$V_{out} = -\frac{1}{R_1 C_1} \int V_{in} dt = -\frac{V_1}{R_1 C_1} t \quad 0 < t < T_b$$

Comparison of Integrator and RC Lowpass Filter

- The RC low-pass filter is actually a “passive” approximation to an integrator.
- With the RC time constant large enough, the RC filter output approaches a ramp.



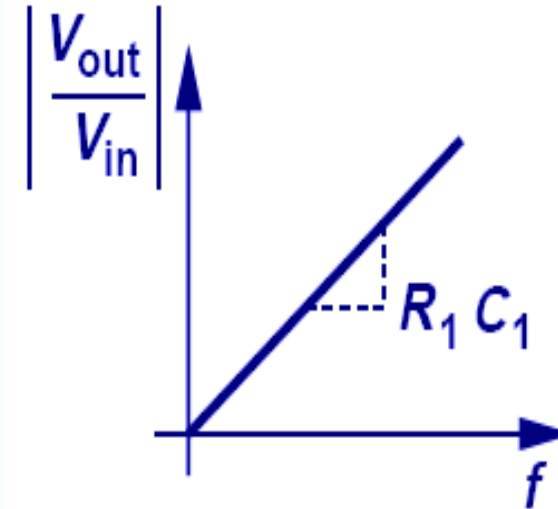
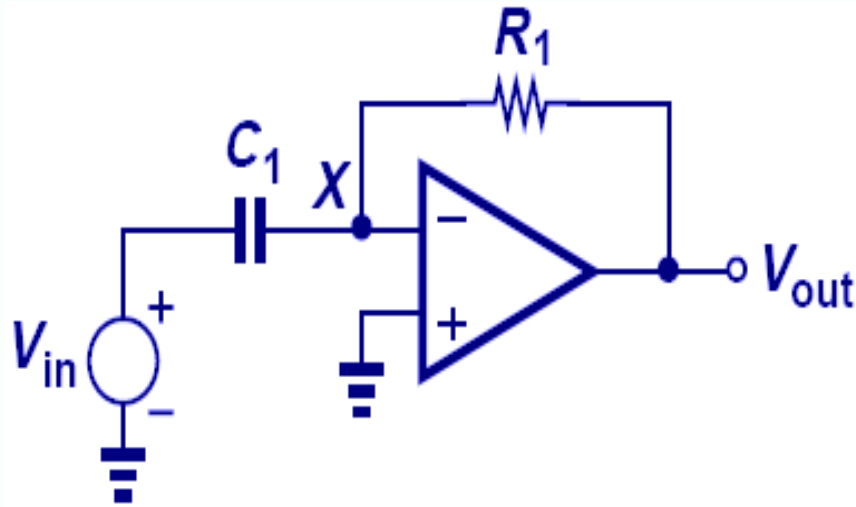
Lossy Integrator



$$\frac{V_{out}}{V_{in}} = \frac{-1}{\frac{1}{A_0} + \left(1 + \frac{1}{A_0}\right) R_1 C_1 s}$$

- When finite op amp gain is considered, the integrator becomes lossy as the pole moves from the origin to $-1/[(1+A_0)R_1C_1]$.
- It can be approximated as an RC circuit with C boosted by a factor of A_0+1 .

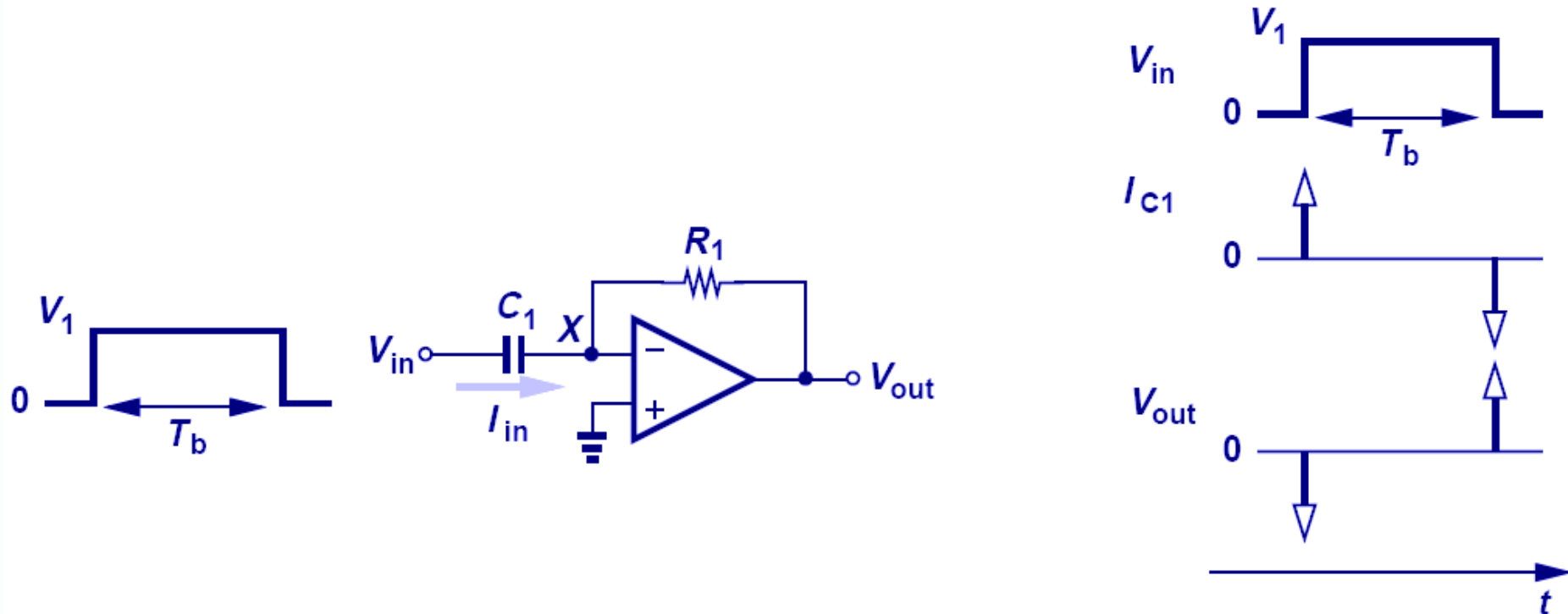
Differentiator



$$V_{out} = -R_1 C_1 \frac{dV_{in}}{dt}$$

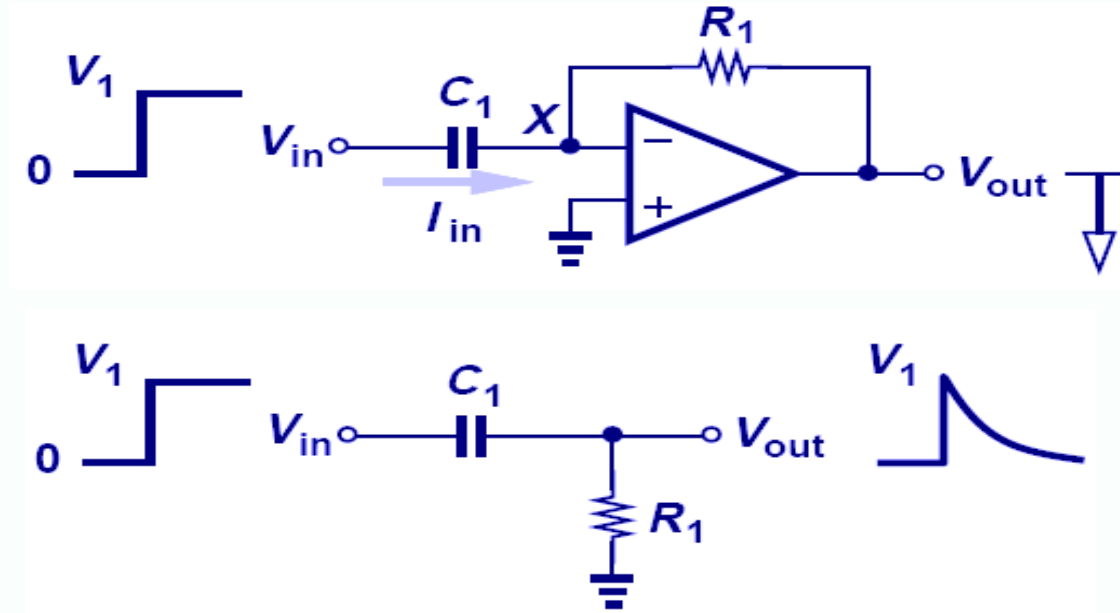
$$\frac{V_{out}}{V_{in}} = -\frac{R_1}{\frac{1}{C_1 s}} = -R_1 C_1 s$$

Differentiator with Pulse Input



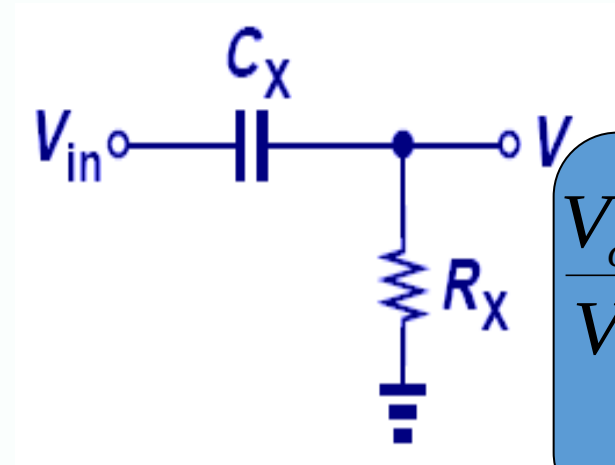
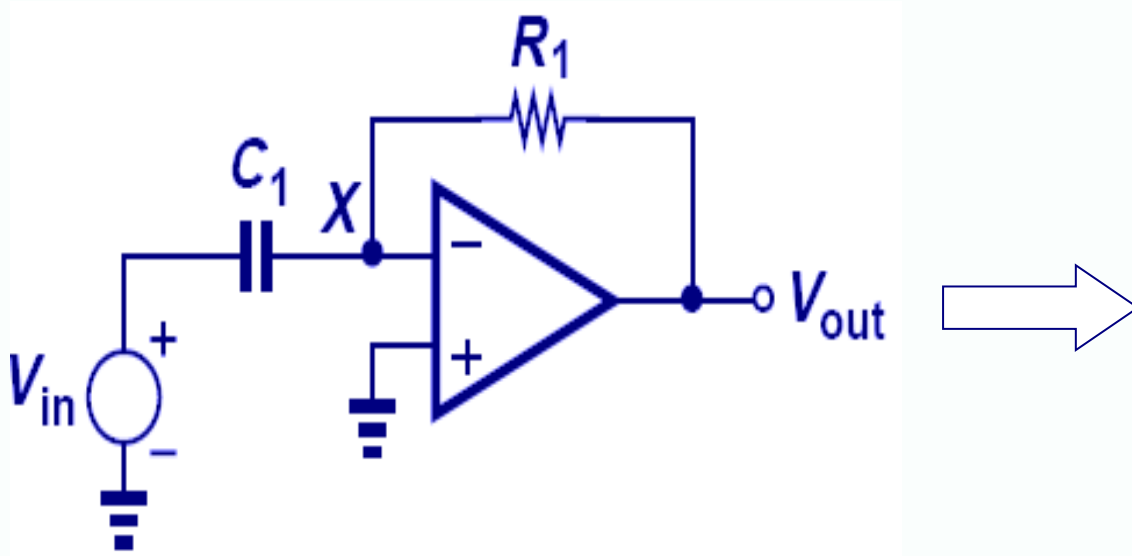
$$V_{out} = \mp R_1 C_1 V_1 \delta(t)$$

Comparison of Differentiator and High-Pass Filter



- The RC high-pass filter is actually a passive approximation to the differentiator.
- When the RC time constant is small enough, the RC filter approximates a differentiator.

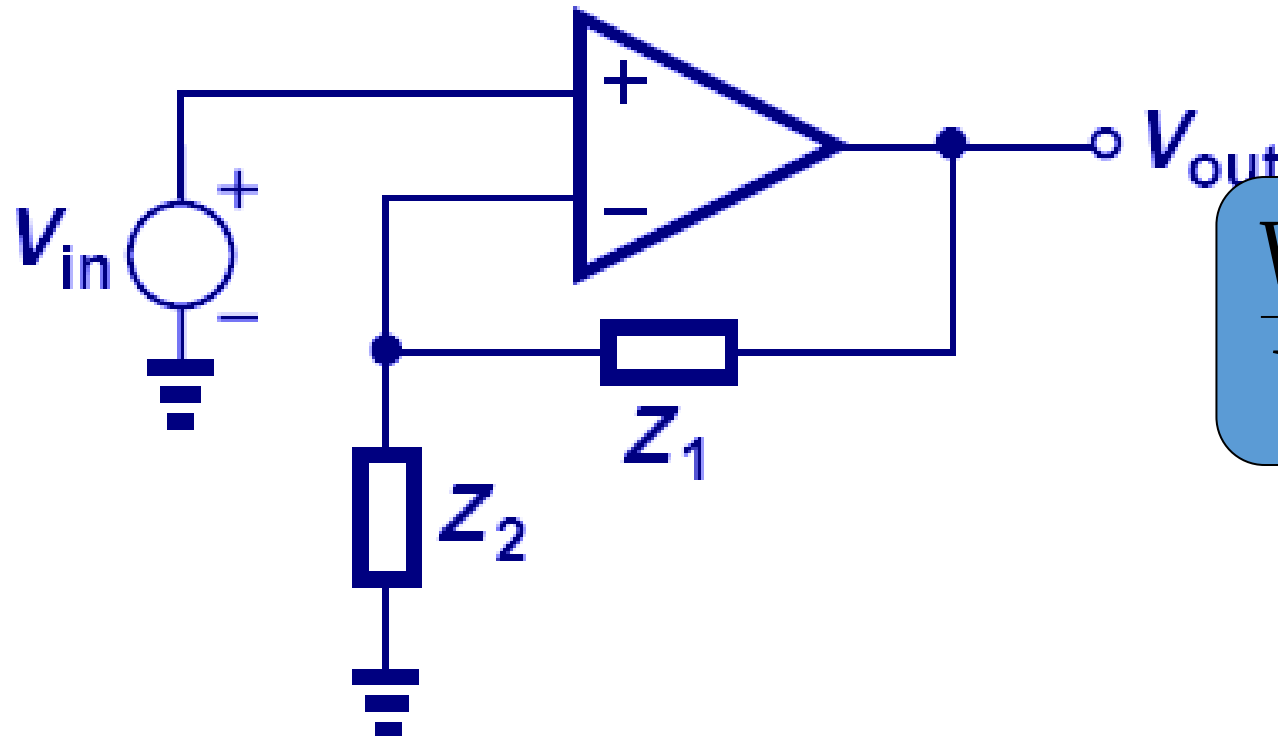
Lossy Differentiator



$$\frac{V_{out}}{V_{in}} = \frac{-R_1 C_1 s}{1 + \frac{1}{A_0} + \frac{R_1 C_1 s}{A_0}}$$

- When finite op amp gain is considered, the differentiator becomes lossy as the zero moves from the origin to $-(A_0+1)/R_1 C_1$.
- It can be approximated as an RC circuit with R reduced by a factor of (A_0+1) .

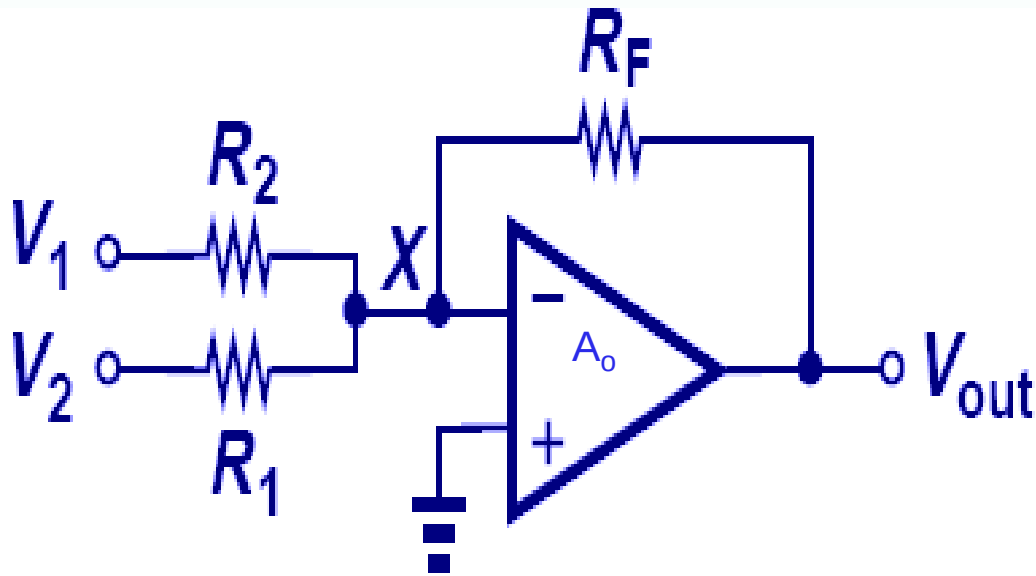
Op Amp with General Impedances



$$\frac{V_{out}}{V_{in}} = 1 + \frac{Z_1}{Z_2}$$

- This circuit cannot operate as ideal integrator or differentiator.

Voltage Adder



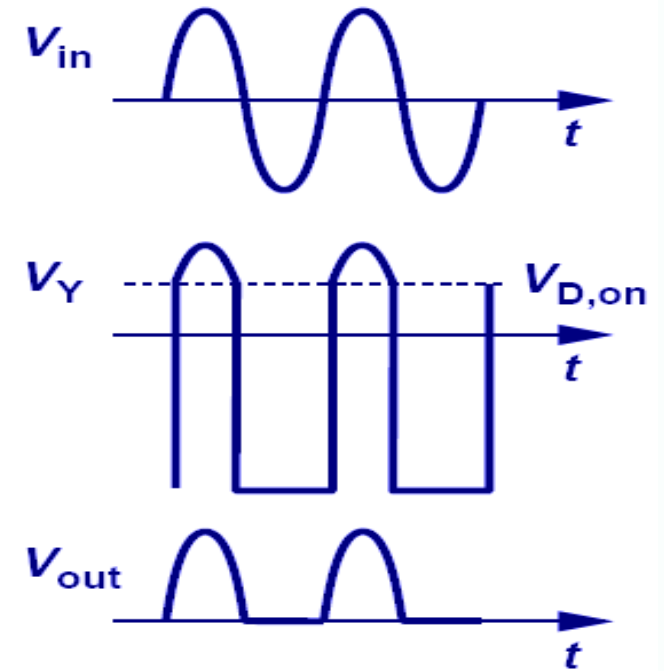
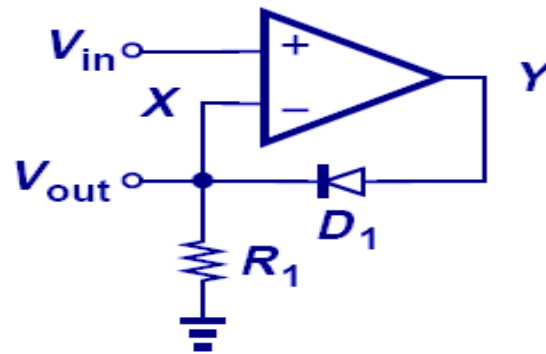
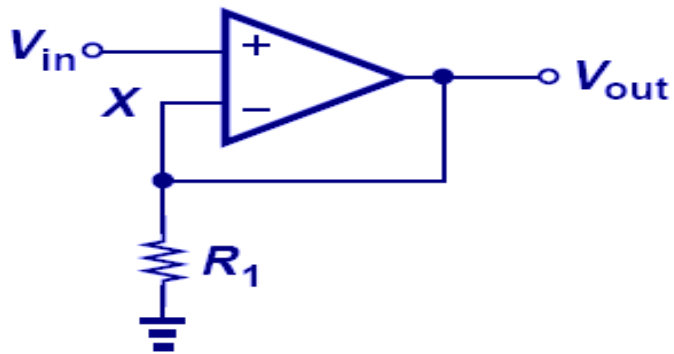
$$V_{out} = -R_F \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

$$V_{out} = \frac{-R_F}{R} (V_1 + V_2)$$

$$\text{If } R_1 = R_2 = R$$

- If A_o is infinite, X is pinned at ground, currents proportional to V_1 and V_2 will flow to X and then across R_F to produce an output proportional to the sum of two voltages.

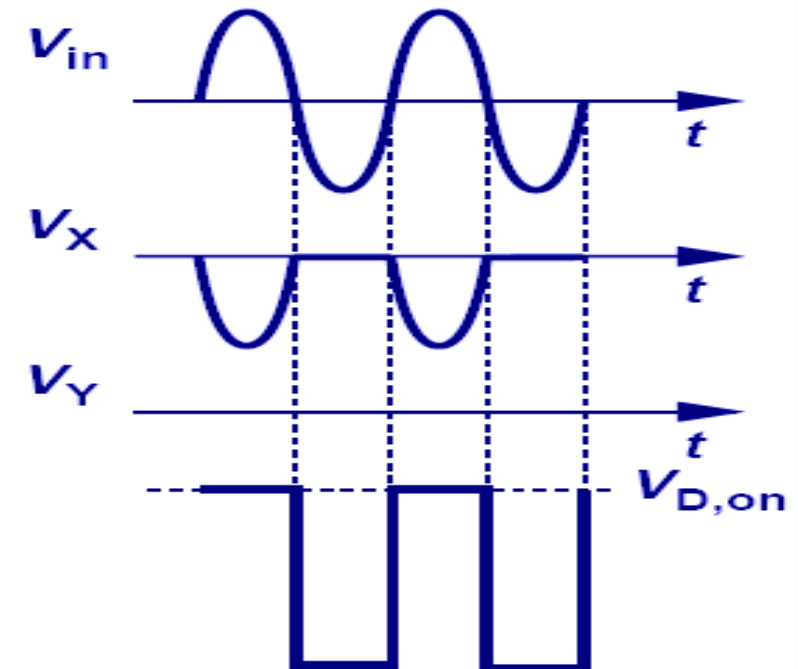
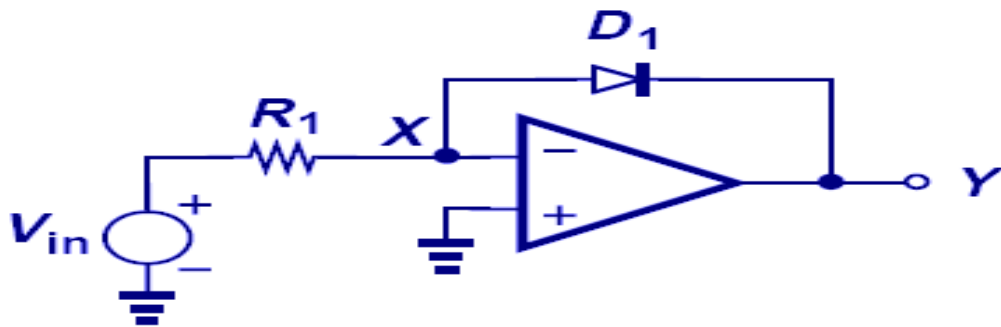
Precision Rectifier



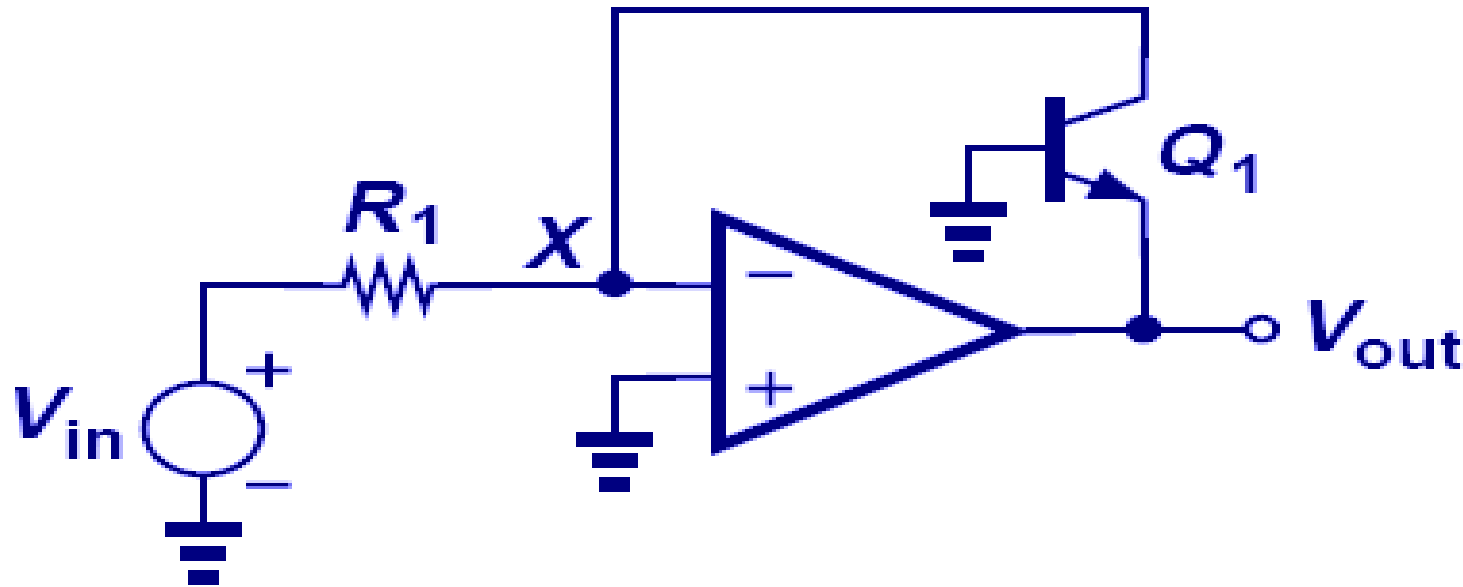
- When V_{in} is positive, the circuit in b) behaves like that in a), so the output follows input.
- When V_{in} is negative, the diode opens, and the output drops to zero. Thus performing rectification.

Inverting Precision Rectifier

- When V_{in} is positive, the diode is on, V_y is pinned around $V_{D,on}$, and V_x at virtual ground.
- When V_{in} is negative, the diode is off, V_y goes extremely negative, and V_x becomes equal to V_{in} .



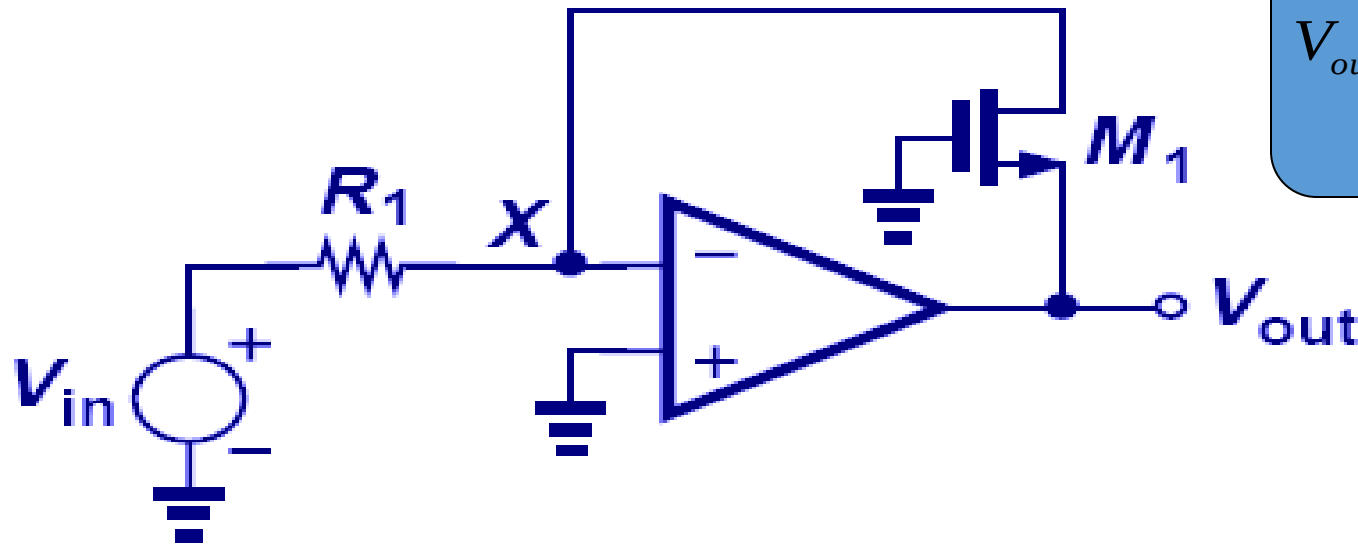
Logarithmic Amplifier



$$V_{out} = -V_T \ln \frac{V_{in}}{R_1 I_S}$$

- By inserting a bipolar transistor in the loop, an amplifier with logarithmic characteristic can be constructed.
- This is because the current to voltage conversion of a bipolar transistor is a natural logarithm.

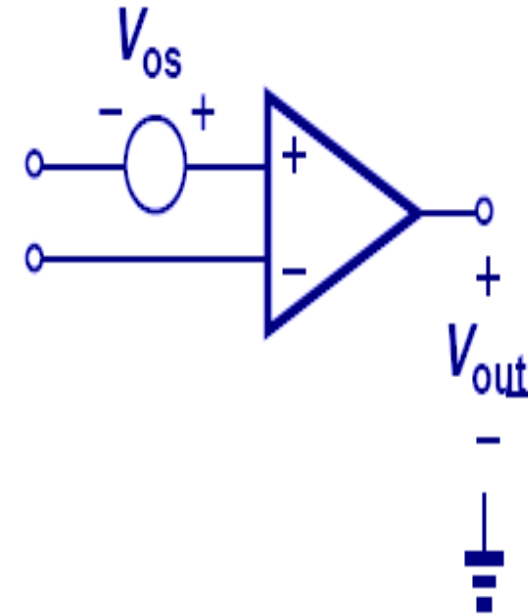
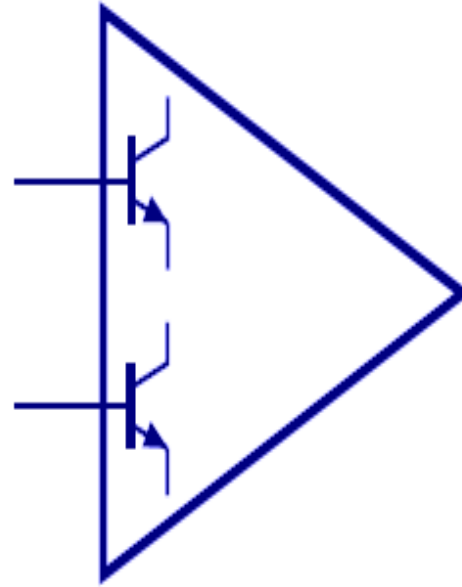
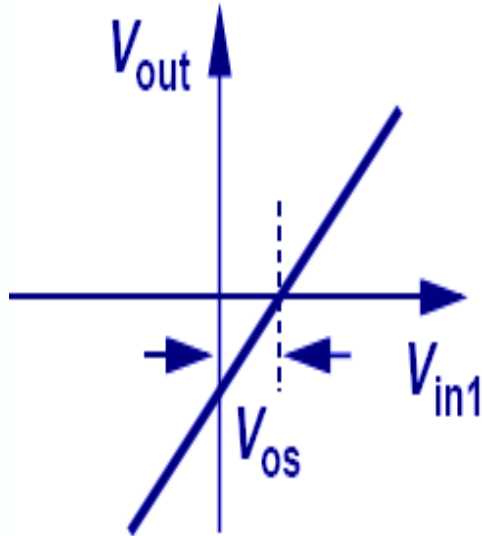
Square-Root Amplifier



$$V_{out} = - \sqrt{\frac{2V_{in}}{\mu_n C_{ox} \frac{W}{L} R_1}} - V_{TH}$$

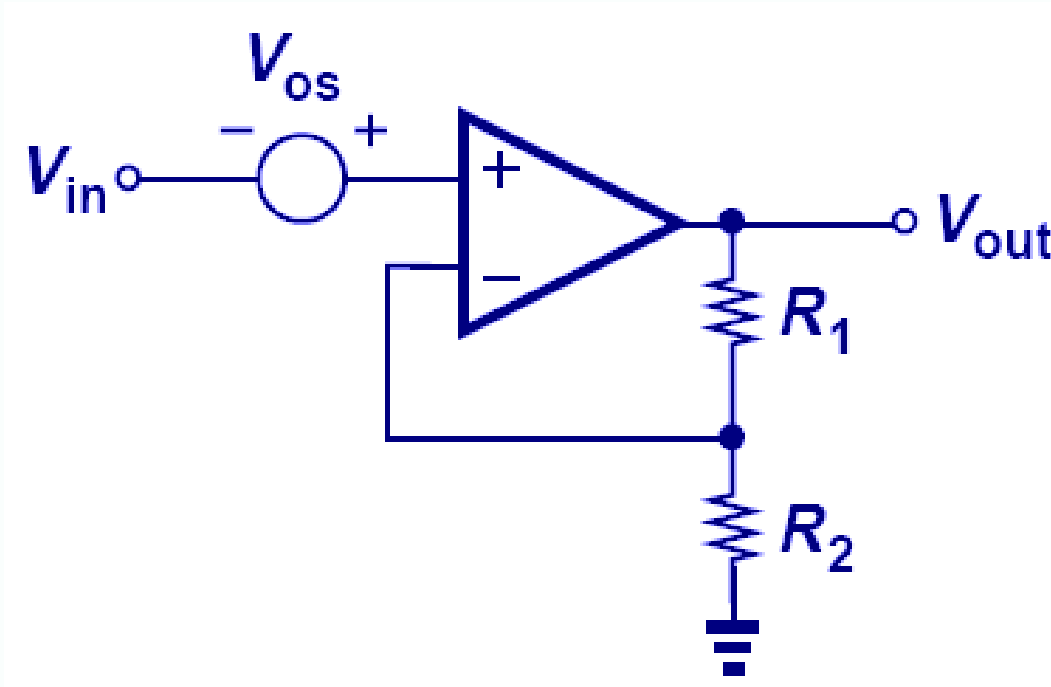
- By replacing the bipolar transistor with a MOSFET, an amplifier with a square-root characteristic can be built.
- This is because the current to voltage conversion of a MOSFET is square-root.

Op Amp Nonidealities: DC Offsets



- Offsets in an op amp that arise from input stage mismatch cause the input-output characteristic to shift in either the positive or negative direction (the plot displays positive direction).

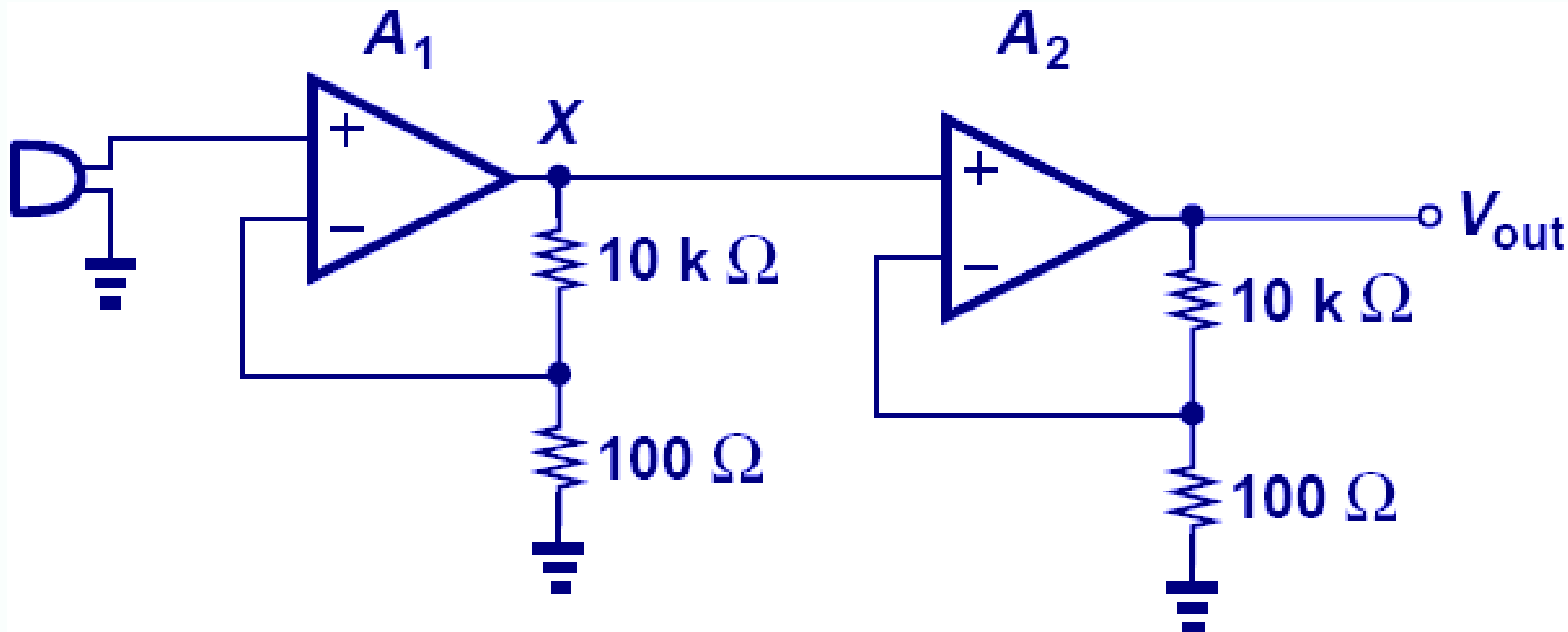
Effects of DC Offsets



$$V_{out} = \left(1 + \frac{R_1}{R_2}\right)(V_{in} + V_{os})$$

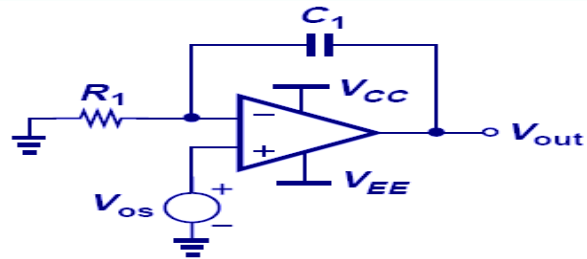
- As it can be seen, the op amp amplifies the input as well as the offset, thus creating errors.

Saturation Due to DC Offsets

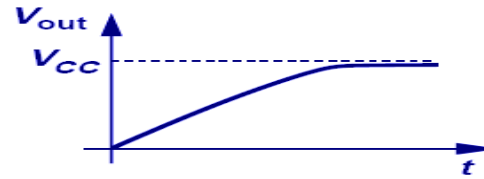


- Since the offset will be amplified just like the input signal, output of the first stage may drive the second stage into saturation.

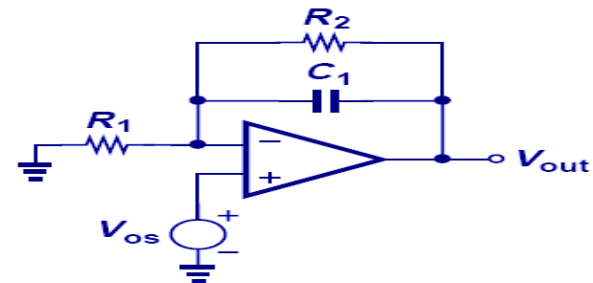
Offset in Integrator



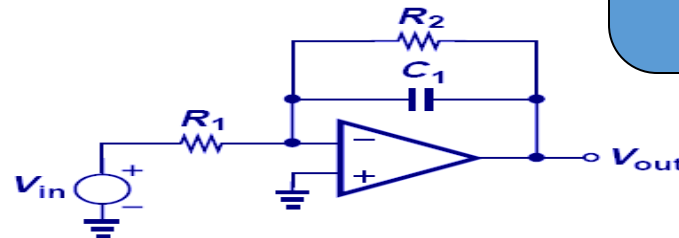
(a)



(b)



(c)



(d)

$$\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1} \frac{1}{R_2 C_1 s + 1}$$

- A resistor can be placed in parallel with the capacitor to “absorb” the offset. However, this means the closed-loop transfer function no longer has a pole at origin.



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COE 271 Semiconductor Devices

Lecture 9

Other Two Terminal Devices

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“Logic will get you from A to Z; imagination will get you everywhere”

Albert Einstein

Overview

Other Two Terminal Devices

Schottky Barrier
Diodes

Solar Cells

Photodetectors

Thermistors

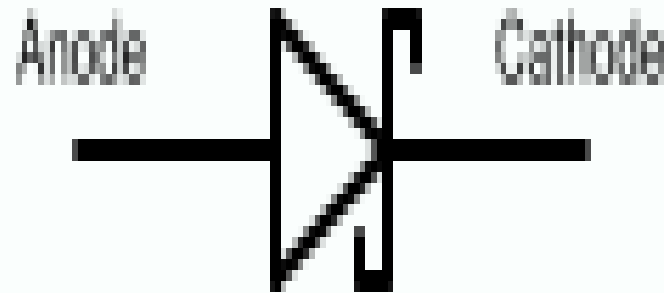
Other Two Terminal Devices

- Like the semiconductor diode,
- there are other two terminal devices having a single p-n junction,
- but with different mode of operation,
- terminal characteristics and areas of application

Other Two Terminal Devices

- These include:
- Schottky barrier diode
- Photodiode
- Photoconductive cell
- Solar cell
- IR emitter
- Tunnel diodes
- Thermistors

Schottky Barrier Diode



- The Schottky diode (named after German physicist Walter H. Schottky; also known as hot carrier diode) is a semiconductor diode with a low forward voltage drop and a very fast switching action.
- When current flows through a diode there is a small voltage drop across the diode terminals.

Schottky Barrier Diode

- A normal silicon diode has a voltage drop between 0.6–1.7 volts, while a Schottky diode voltage drop is between approximately 0.15–0.45 volts.
- This lower voltage drop can provide higher switching speed and better system efficiency.

Construction of Schottky Barrier Diode

- A metal–semiconductor junction is formed between a metal and a semiconductor, creating a Schottky barrier (instead of a semiconductor–semiconductor junction as in conventional diodes).
- Typical metals used are **molybdenum, platinum, chromium or tungsten**; and the semiconductor would typically be N-type silicon.
- The metal side acts as the anode and N-type semiconductor acts as the cathode of the diode.
- This Schottky barrier results in both very fast switching and low forward voltage drop.

Reverse Recovery Time

- The most important difference between the p-n and Schottky diode is reverse recovery time, when the diode switches from conducting to non-conducting state.
- Where in a p-n diode the reverse recovery time can be in the order of hundreds of nanoseconds and less than 100 ns for fast diodes, Schottky diodes do not have a recovery time, as there is nothing to recover from (i.e. no charge carrier depletion region at the junction).

Reverse Recovery Time

- The switching time is **~100 ps** for the small signal diodes, and up to tens of nanoseconds for special high-capacity power diodes. With p-n junction switching, there is also a reverse recovery current, which in high-power semiconductors brings **increased EMI (electromagnetic interference) noise**.
- With Schottky diodes switching essentially instantly with only **slight capacitive loading**, this is much less of a concern.

Schottky Barrier Diode

- Schottky diode is a "majority carrier" semiconductor device.
- This means that if the semiconductor body is doped n-type, only the n-type carriers (mobile electrons) play a significant role in normal operation of the device.
- The majority carriers are quickly injected into the conduction band of the metal contact on the other side of the diode to become free moving electrons.

Schottky Barrier Diode

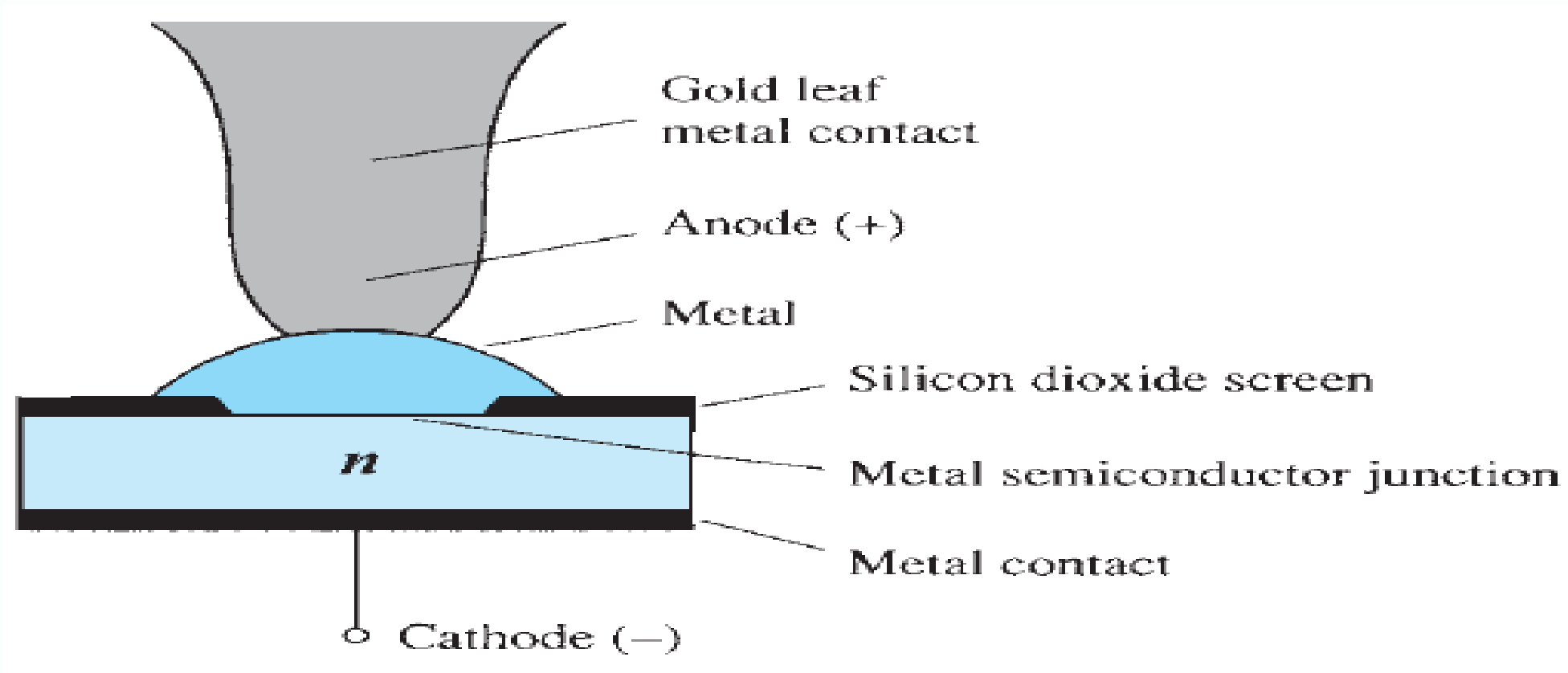
- Therefore no slow, random recombination of n- and p-type carriers is involved, so that this diode can cease conduction faster than an ordinary p-n rectifier diode.
- This property in turn allows a smaller device area, which also makes for a faster transition.

Schottky Barrier Diodes

- Schottky barrier or hot-carrier diodes are used mostly in
radar systems,
Schottky TTL (transistor-transistor logic) for computers,
Mixers and detectors in communication equipment,
Instrumentation and Analog-digital converters

Schottky Barrier Diodes

- Schottky barrier is created by joining a n- or p-type silicon to a host of different metal



Schottky Barrier Diodes

- Electron is the majority carrier in both materials
- When the materials are joined, the electrons in the n-type silicon material flow immediately into the adjoining metal, establishing a heavy flow of majority carriers
- Schottky barrier diode are also called hot-carrier because the injected carriers have a very high kinetic energy level compared to the electrons of the metal
- Electrons are injected into a region of the same electron plurality.
- Schottky diodes are therefore unique in that conduction is entirely by majority carriers.

Schottky Barrier Diodes

- The heavy flow of electrons into the metal creates a region near the junction surface depleted of carriers in the silicon material—much like the depletion region in the $p - n$ junction diode.
- The additional carriers in the metal establish a “negative wall” in the metal at the boundary between the two materials.
- The net result is a “surface barrier” between the two materials, preventing any further current. That is, any electrons (negatively charged) in the silicon material face a carrier-free region and a “negative wall” at the surface of the metal

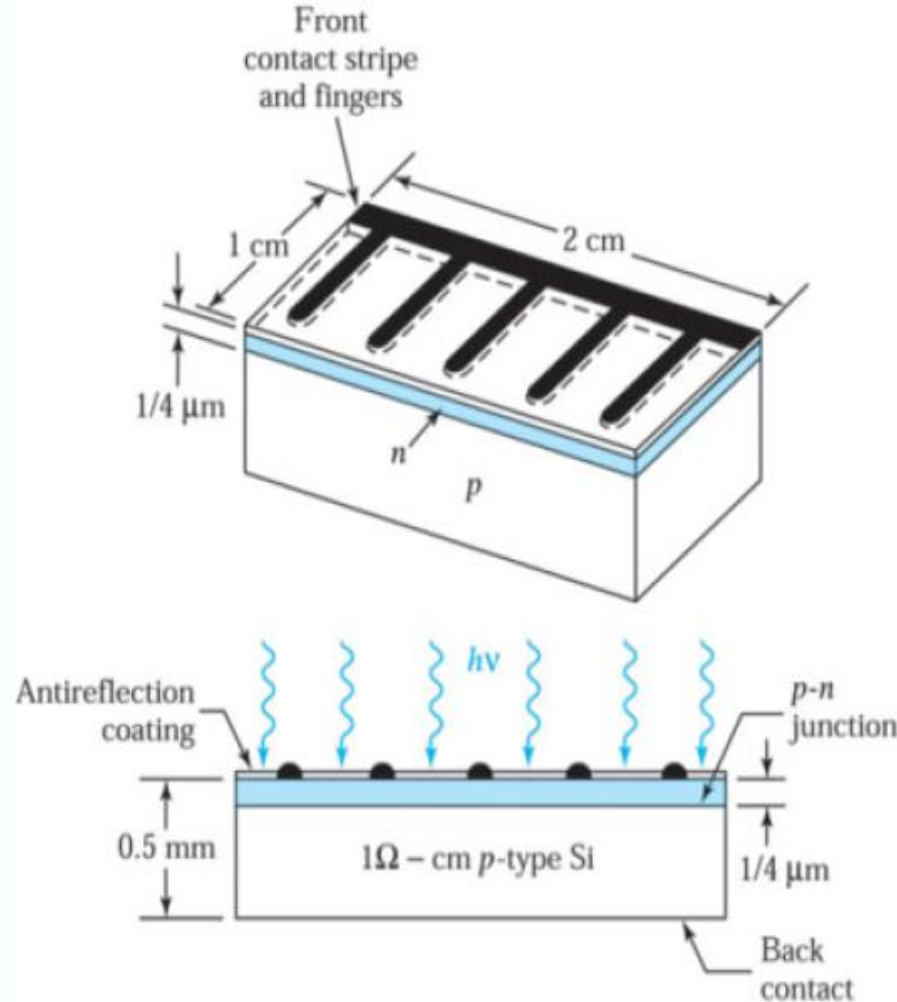
Solar Cells

- Solar cells are useful both for space and terrestrial applications
- Solar cell provide the long-duration power supply for satellites
- Solar cell is an important candidate for an alternative terrestrial energy because it converts sunlight directly to electricity with good conversion efficiency
- Solar cell provides nearly permanent power at low operating cost
- Solar cell is virtually non-polluting

Solar Cells

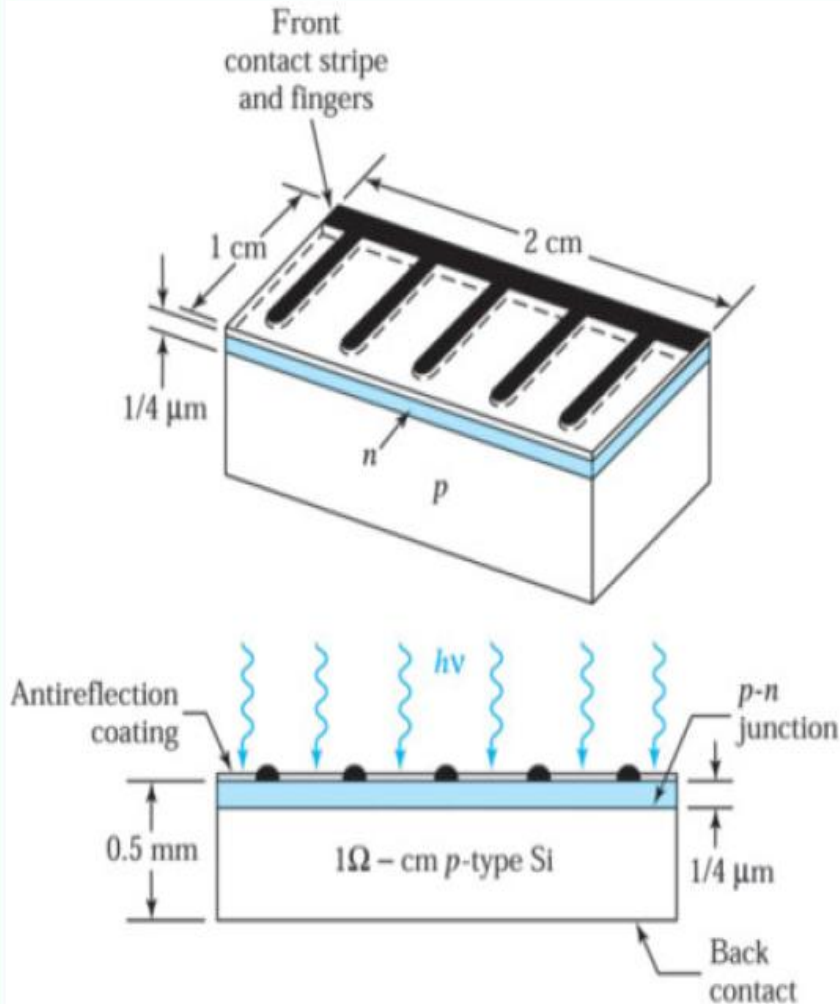
- Solar constant is defined as the intensity of solar radiation outside the earth's atmosphere, at the average distance of its orbit around the sun
- Solar constant = 1367 W/m^2
- Terrestrially, the sunlight is attenuated by clouds and by atmospheric scattering and absorption.
- The attenuation depends primarily on the length of the light's path through the atmosphere, or the mass of air through which it passes.
- This "air mass" is defined as $1/\cos \phi$, where ϕ is the angle between the vertical and the sun's position.

p-n Junction Solar Cells



- A *p-n* junction solar cell consists of :
 - a shallow *p-n* junction formed on the surface,
 - a front ohmic contact stripe and fingers,
 - a back ohmic contact that covers the entire back surface, and an
 - antireflection coating on the front surface.

p-n Junction Solar Cells

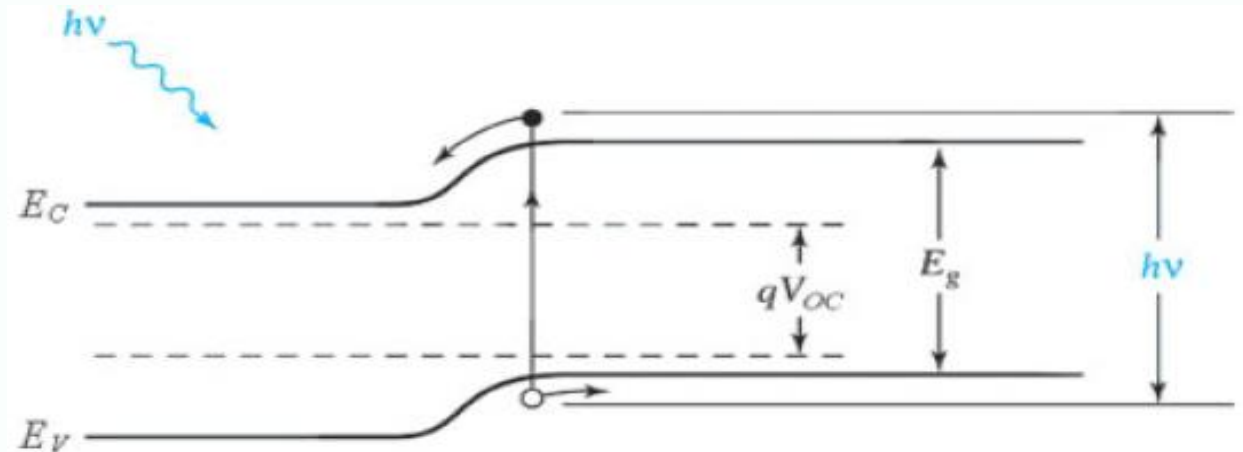


- The surface reflection of the incident light from air ($n = 1$) into semiconductor silicon ($n = 3.5$) is about 0.31.
- This means that 31% of incident light is reflected and is not available for conversion to electrical energy in a silicon solar cell.

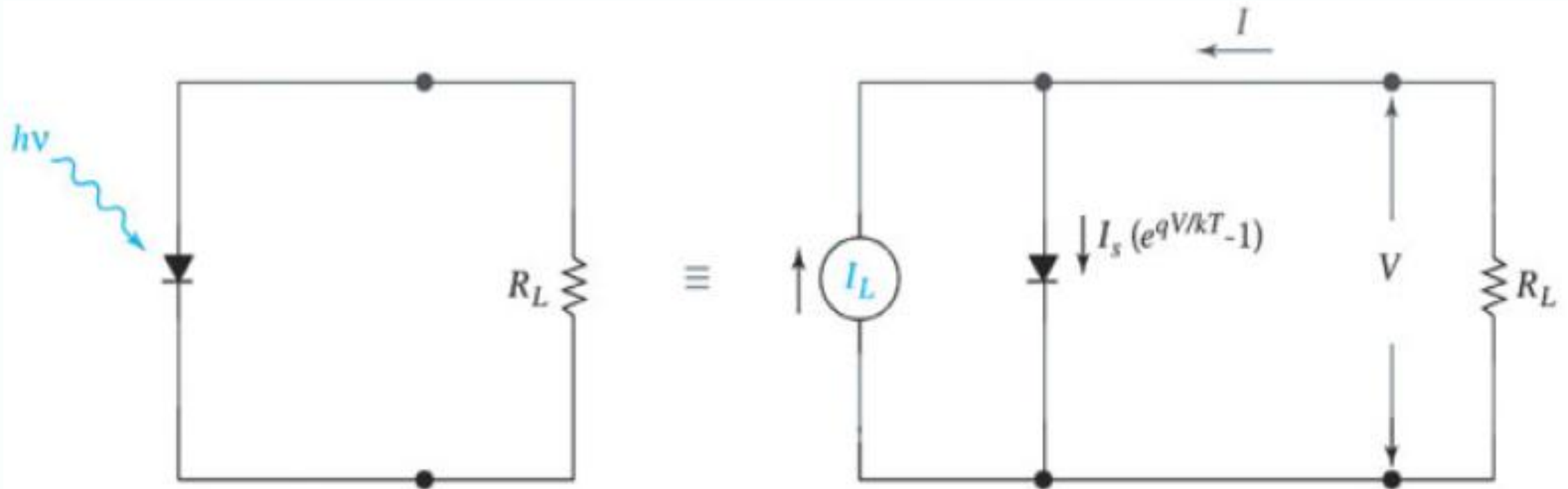
Solar Cells

- When the cell is exposed to the solar spectrum, a photon that has an energy less than the bandgap E_g makes no contribution to the cell output.
- A photon that has energy greater than E_g contributes an energy E_g to the cell output.
- Energy greater than E_g is wasted as heat

Energy band diagram of a p-n junction solar cell under solar radiation



Conversion Efficiency



Idealized equivalent circuit of a solar cell

Conversion Efficiency

- From the equivalent circuit above the constant-current source is in parallel with the junction.
- The source current I_L results from the excitation of excess carriers by solar radiation
- I_s is the diode saturation current and R_L is the load resistance
- The ideal I-V characteristics of device is given by

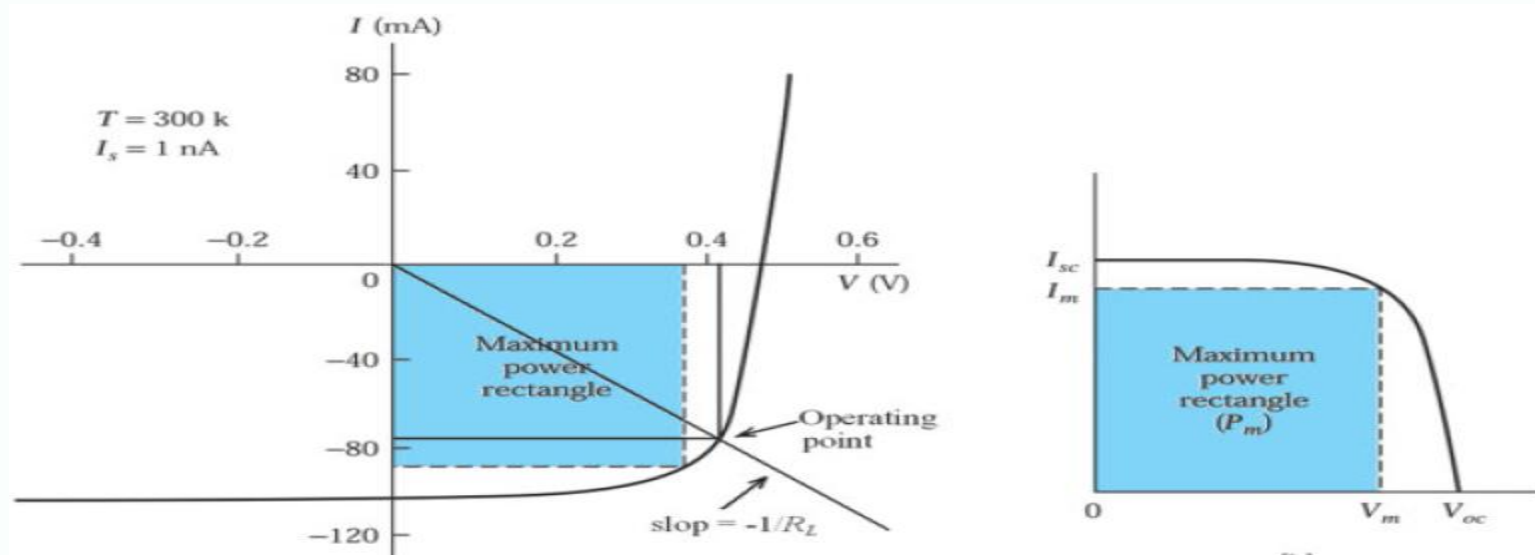
$$I = I_s(e^{qV/kT} - 1) + I_L,$$

$$J_s = \frac{I_s}{A} = qN_C N_V \left(\frac{1}{N_A} \sqrt{\frac{D_n}{\tau_n}} + \frac{1}{N_D} \sqrt{\frac{D_p}{\tau_p}} \right) \cdot e^{-E_g/kT},$$

where A is the device area

$$V_{OC} = \frac{kT}{q} \ln \left(\frac{I_L}{I_s} + 1 \right) \cong \frac{kT}{q} \ln \left(\frac{I_L}{I_s} \right)$$

I-V Characteristics of Solar Cell



- For a given I_L , V_{oc} increases logarithmically with decreasing saturation current I_s
- The maximum output power

$$P_m = I_m V_m \cong I_L \left[V_{oc} - \frac{kT}{q} \ln \left(1 + \frac{qV_m}{kT} \right) - \frac{kT}{q} \right]$$

Solar Cell Types

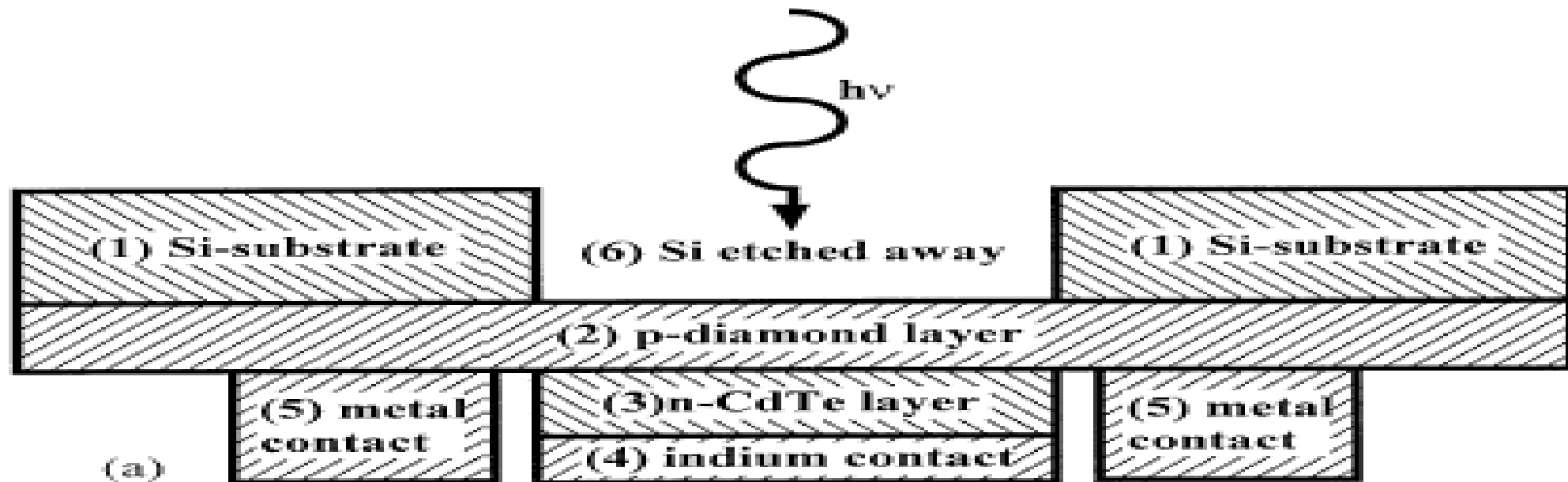
- Thin film solar cells
- Inverted thin film solar cells
- Tandem cells
- Multiband cells
- Multi quantum well
- Thermophotonic cells
- Thermophotovoltaic cell

Thin Film Solar Cells

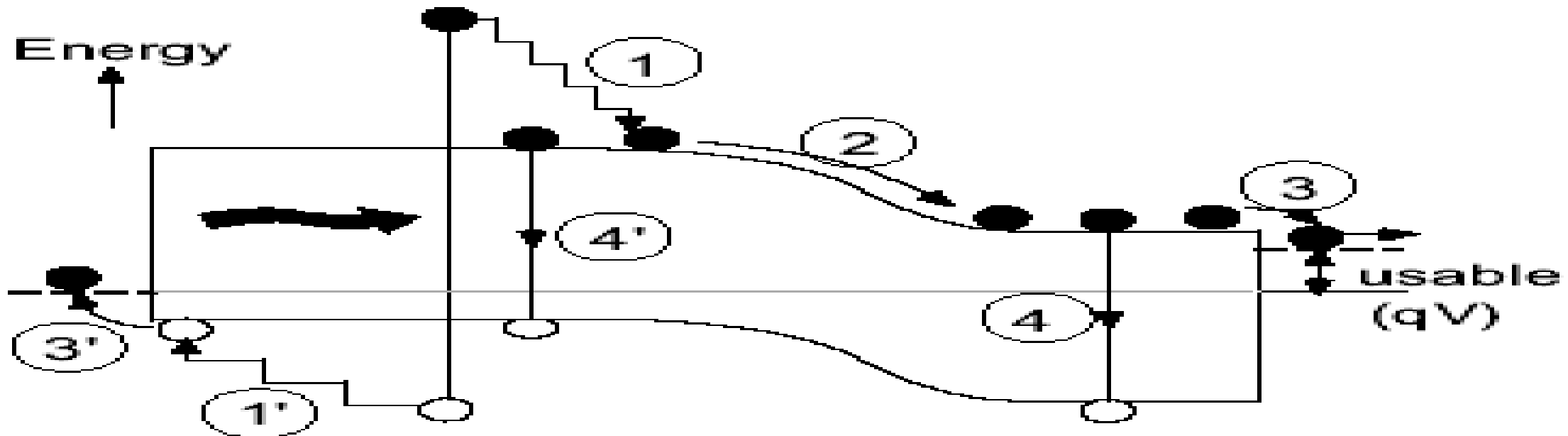
- Produced from cheaper polycrystalline materials and glass
- High optical absorption coefficients
- Bandgap suited to solar spectrum

Inverted Thin Film Cell

- p-diamond (Bandgap 5.5 eV) as a window layer
- n-CdTe layer as an absorption layer



Efficiency Losses in Solar Cell



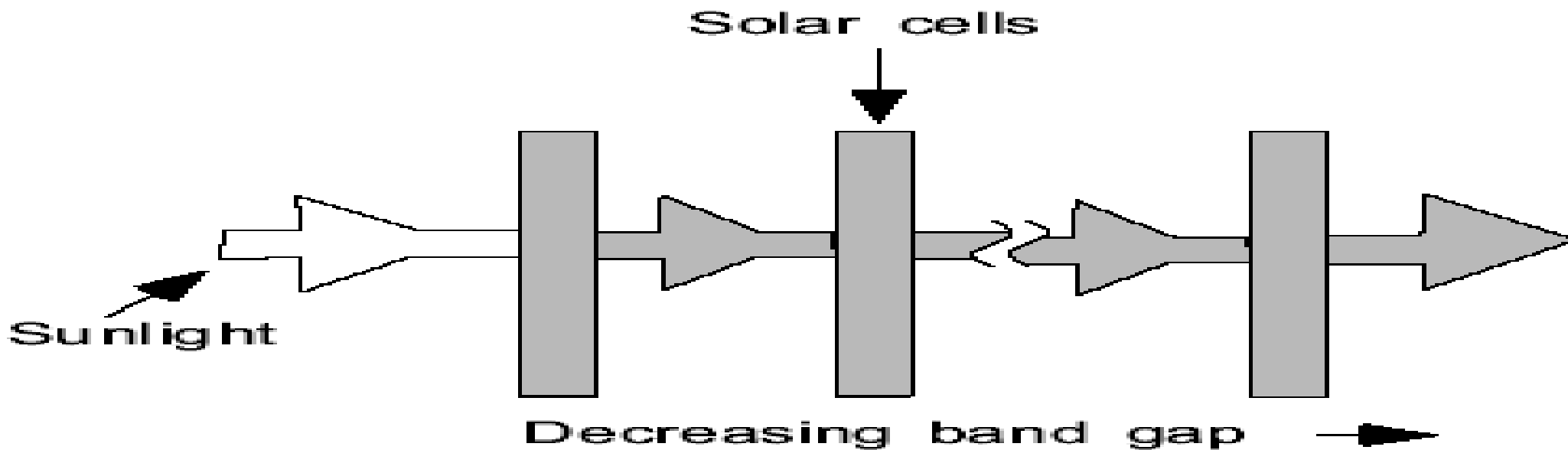
1 = Thermalization loss

2 and 3 = Junction and contact voltage loss

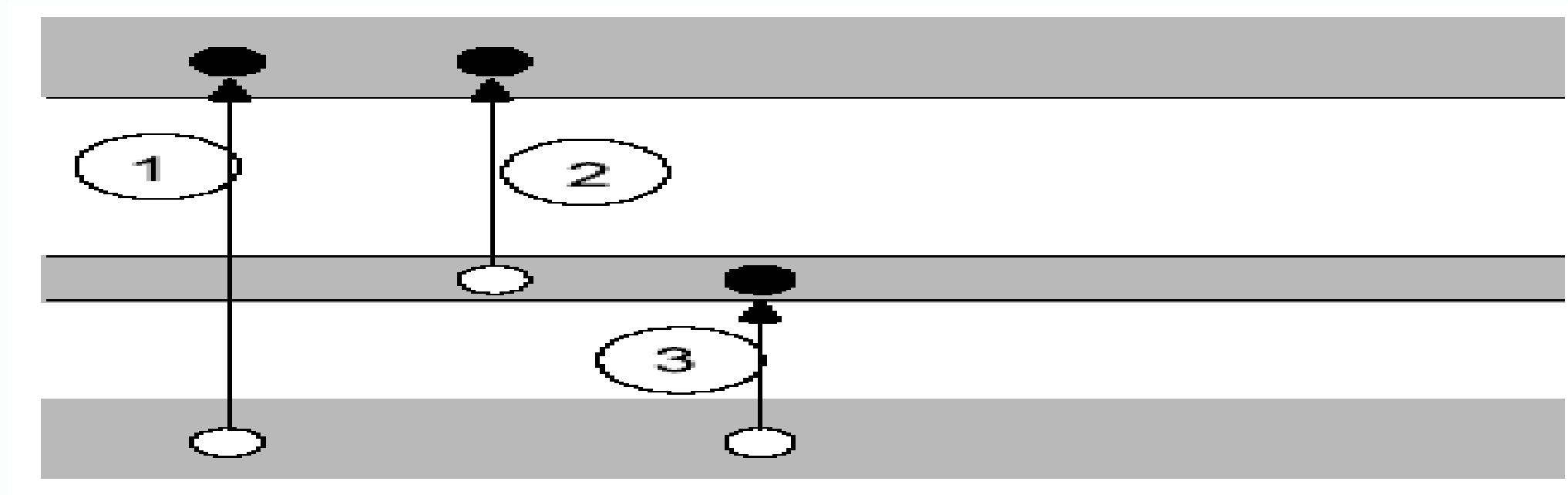
4 = Recombination loss

Tandem Cells

- Current output matched for individual cells
- Ideal efficiency for infinite stack is 86.8%
- GaInP/GaAs/Ge tandem cells (efficiency 40%)

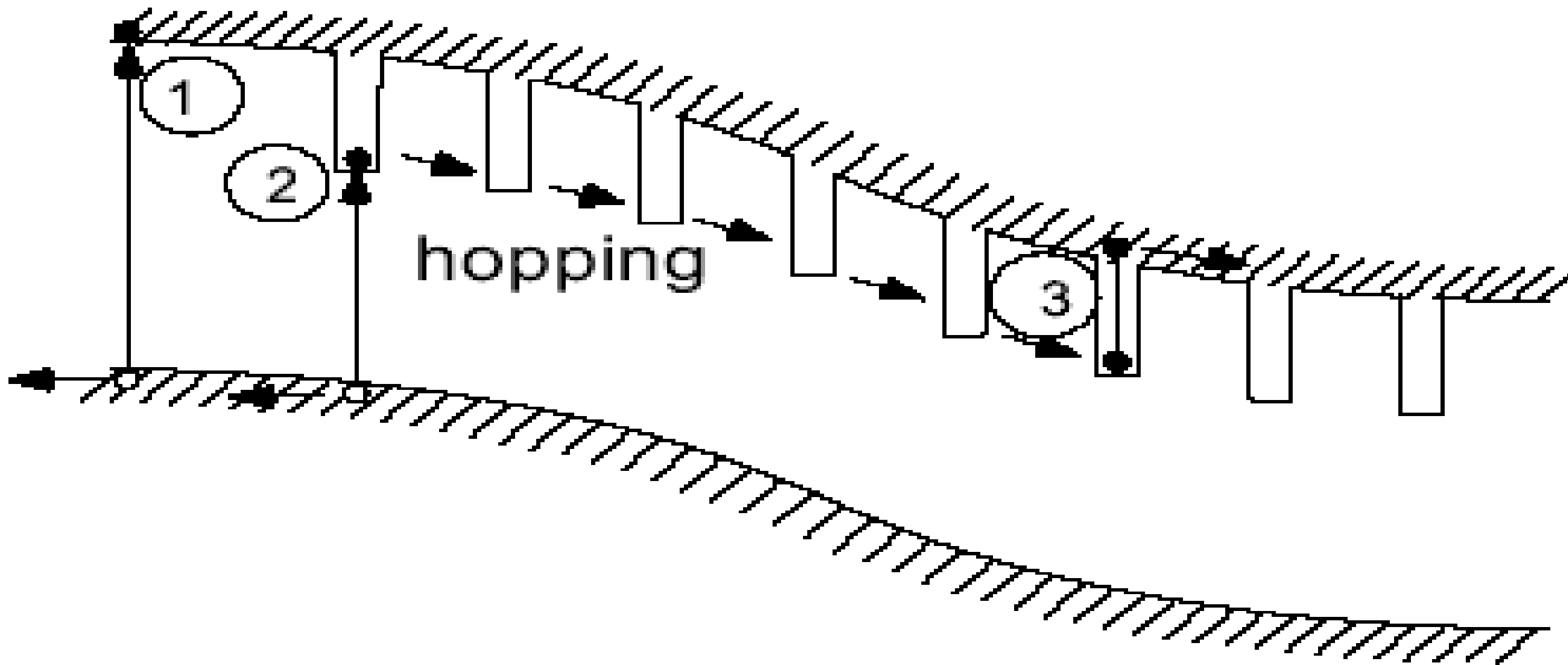


Multiband Cells



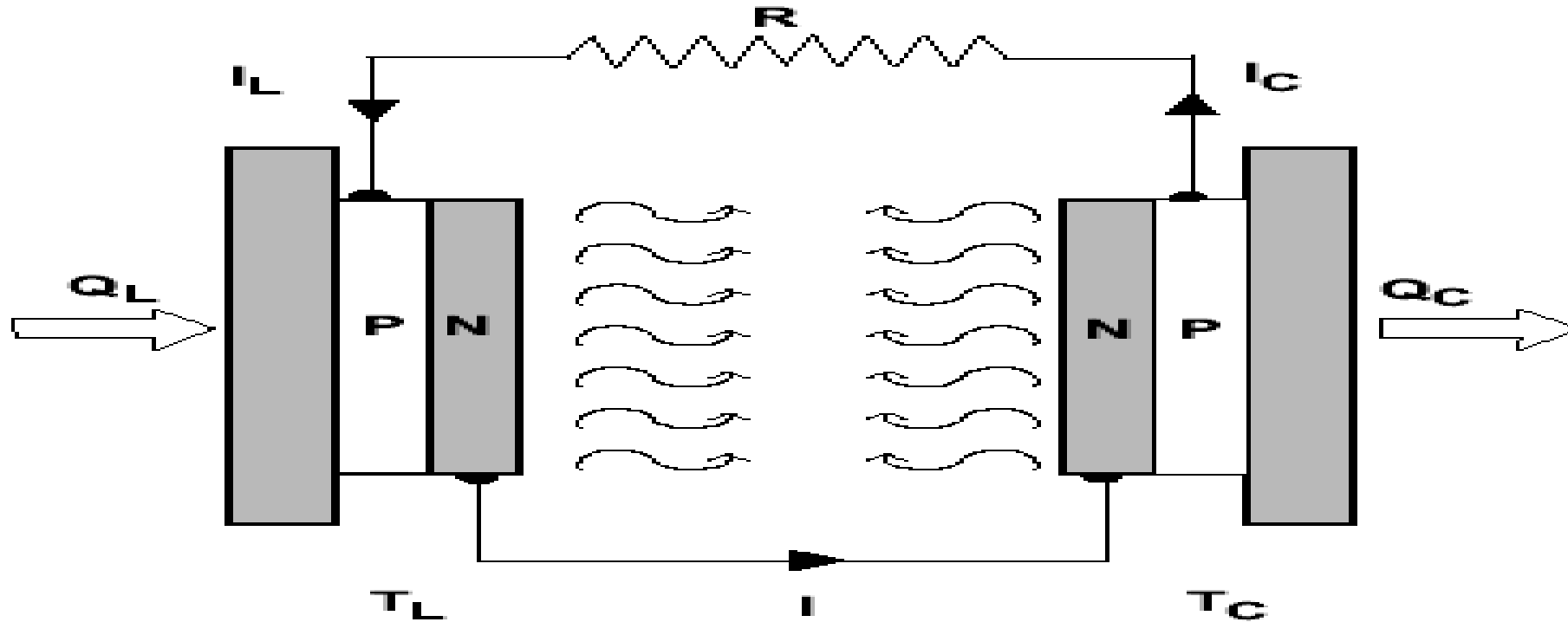
- Intermediate band formed by impurity levels.
- Process 3 also assisted by phonons
- Limiting efficiency is 86.8%

Multiple Quantum Well



- Principle of operation similar to multiband cells

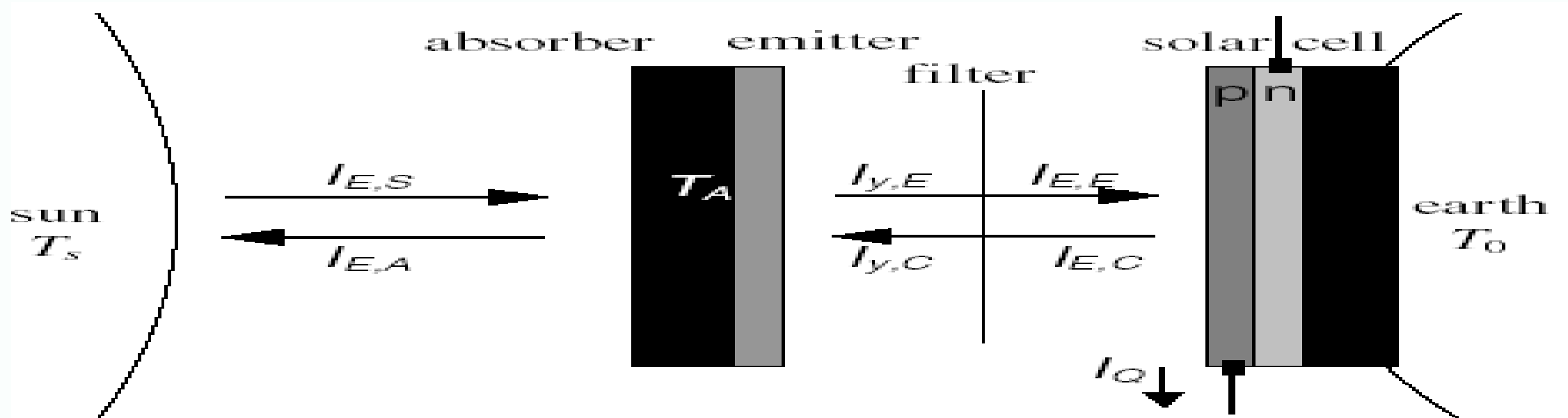
Thermophotonic Cells



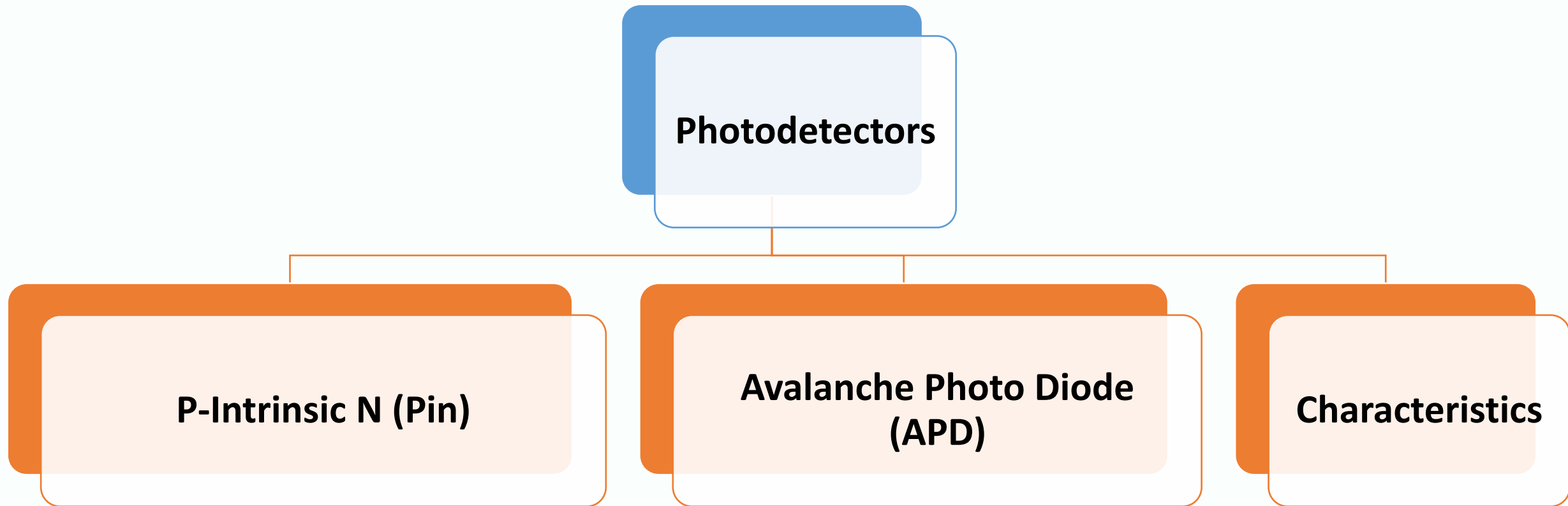
- Heated semiconductor emits narrow bandwidth radiations
- Diode with higher temperature has lower voltage

Thermophotovoltaic Cell

- Filter passes radiations of energy equal to bandgap of solar cell material
- Emitter radiation matched with spectral sensitivity of cell
- High Illumination Intensity ($\sim 10 \text{ kW/m}^2$)



Photodetectors - Overview



Photodetectors

These are **Opto-electric devices** i.e. to convert the optical signal back into electrical impulses.

The light detectors are commonly made up of semiconductor material.

When the light strikes the light detector a current is produced in the external circuit proportional to the intensity of the incident light.

Photodetectors

- ☐ A **high sensitivity** to the emission wavelength range of the received light signal
- ☐ A **minimum** addition of **noise** to the signal
- ☐ A **fast response** speed to handle the desired data rate
- ☐ Be **insensitive** to **temperature** variations
- ☐ Be **compatible** with the physical dimensions of the **fiber**
- ☐ Have a **Reasonable cost** compared to other system components
- ☐ Have a long **operating lifetime**

Type of Photodetectors

PIN Photodiode

Avalanche Photodiode



PIN photodiode

InGaAs avalanche photodiode

Ing. B. Kommey, Semiconductor Devices

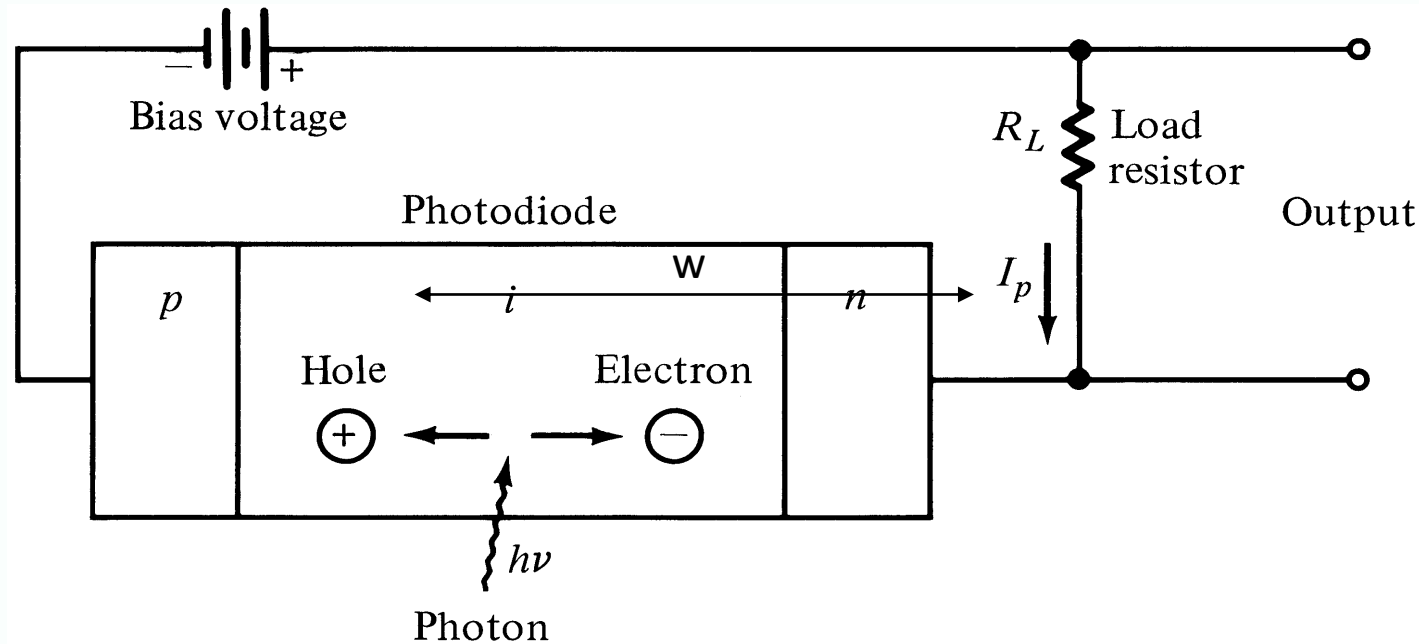
p-intrinsic-n pin Photodetector

The device structure consists of **p** and **n** semiconductor regions separated by a very **lightly n-doped intrinsic (i)** region.

In normal operation a reverse-bias voltage is applied across the device so that **no free electrons or holes** exist in the **intrinsic region**.

Incident photon having energy **greater than or equal** to the **bandgap energy** of the semiconductor material, **give up its energy** and **excite an electron** from the valence band to the conduction band

p-intrinsic-n pin Photodetector



The high electric field present in the depletion region causes photo-generated carriers to separate and be collected across the reverse –biased junction. This give rise to a current flow in an external circuit, known as **photocurrent**.

p-intrinsic-n pin Photodetector

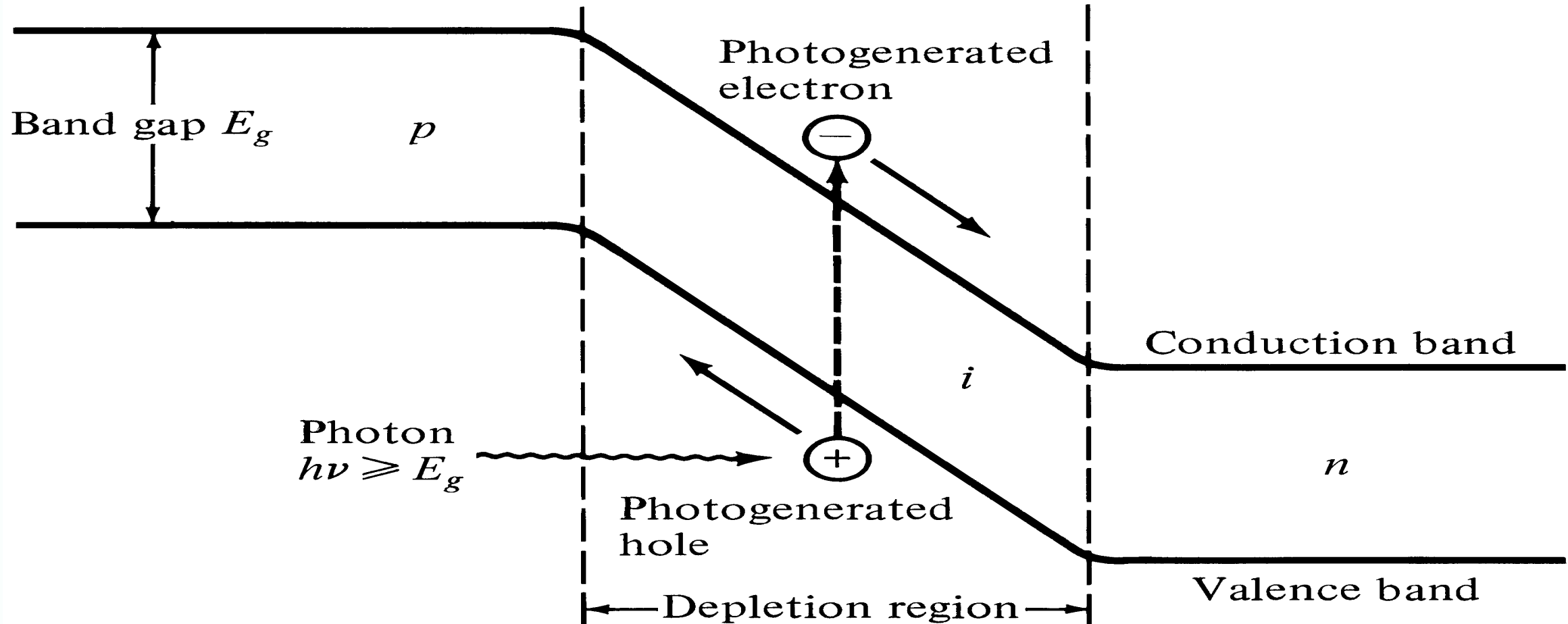
Photocarriers:

Incident photon, generates free (mobile) **electron-hole pairs in the intrinsic region**. These charge carriers are known as photocarriers, since they are generated by a photon.

Photocurrent:

The electric field across the device causes the **photocarriers to be swept out of the intrinsic region**, thereby giving rise to a **current flow in an external circuit**. This current flow is known as the photocurrent.

Energy-Band Diagram for a *pin* Photodiode



Avalanche Photodiode (APD)

APD has an internal gain M , which is obtained by having a high electric field that energizes photo-generated electrons.

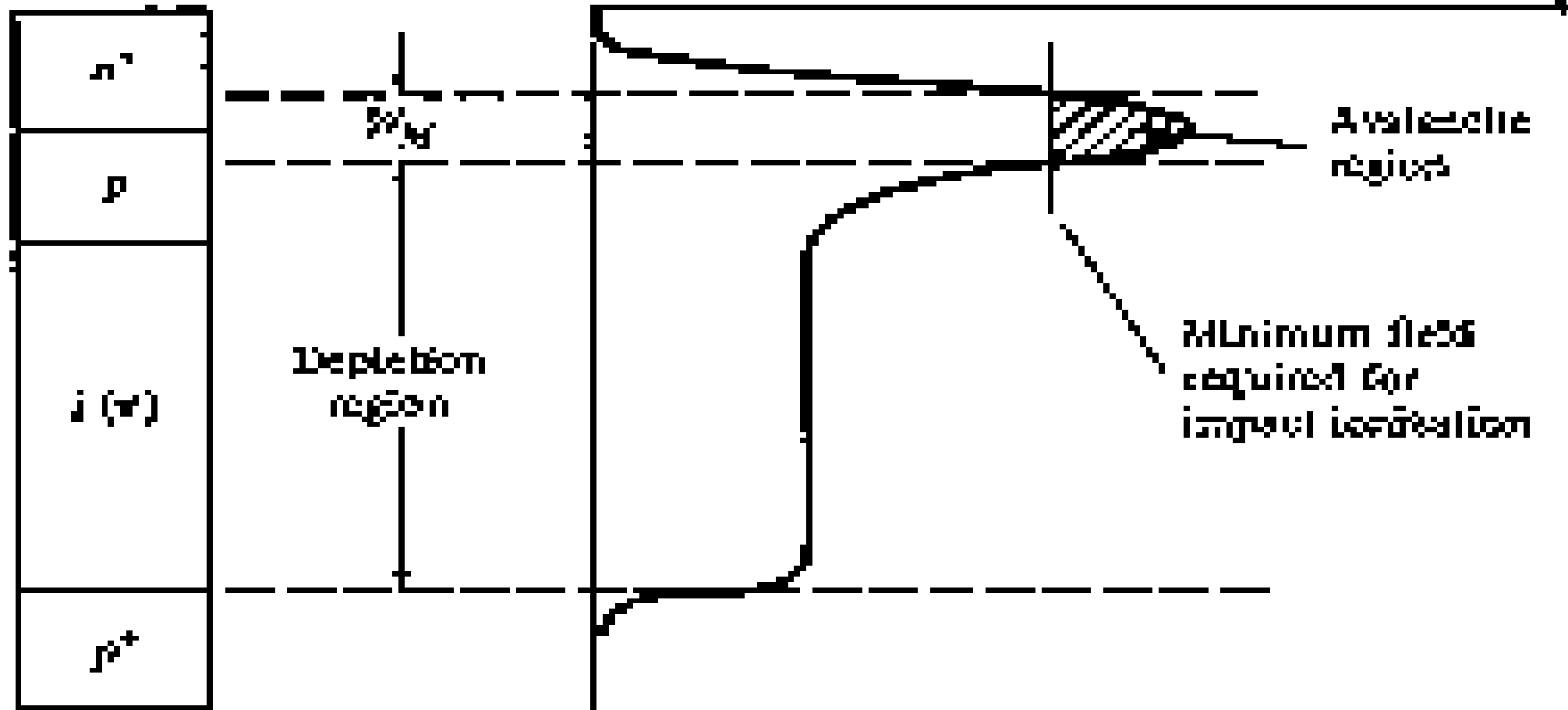
These electrons ionize bond electrons in the valence band upon colliding with them which is known as *impact ionization*

The newly generated electrons and holes are also accelerated by the high electric field and gain energy to cause further impact ionization

This phenomena is the avalanche effect

Avalanche Photodiode (APD)

Electric field



Photocurrent

• Optical power absorbed, $P(x)$

in the depletion region P_0

can be written in terms of incident optical power, :

$$P(x) = P_0 (1 - e^{-\alpha_s(\lambda)x})$$

Photocurrent

- Absorption coefficient $\alpha_s(\lambda)$ strongly depends on wavelength. The upper wavelength cutoff for any semiconductor can be determined by its energy gap as follows:

$$\lambda_c (\mu\text{m}) = \frac{1.24}{E_g (\text{eV})}$$

Photocurrent

- Taking entrance face reflectivity into consideration, the absorbed power in the width of depletion region, w , becomes:

$$(1 - R_f)P(w) = P_0(1 - e^{-\alpha_s(\lambda)w})(1 - R_f)$$

Noise Sources in Photodetectors

- The principal noises associated with photodetectors are :

1- Quantum (Shot) noise: arises from statistical nature of the production and collection of photo-generated electrons upon optical illumination. It has been shown that the statistics follow a Poisson process.

2- Dark current noise: is the current that continues to flow through the bias circuit in the absence of the light. This is the combination of **bulk dark current**, which is due to thermally generated e and h in the pn junction, and the **surface dark current**, due to surface defects, bias voltage and surface area.

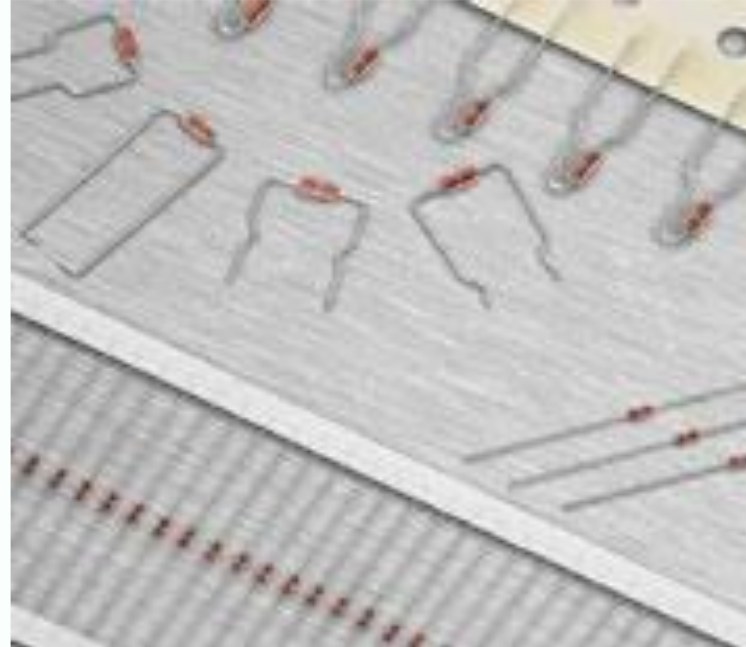
Noise Sources in Photodetectors

- In order to calculate the total noise presented in photodetector, we should sum up the root mean square of each noise current by assuming that those are uncorrelated.
- **Total photodetector noise current = quantum noise current + bulk dark current noise + surface current noise**

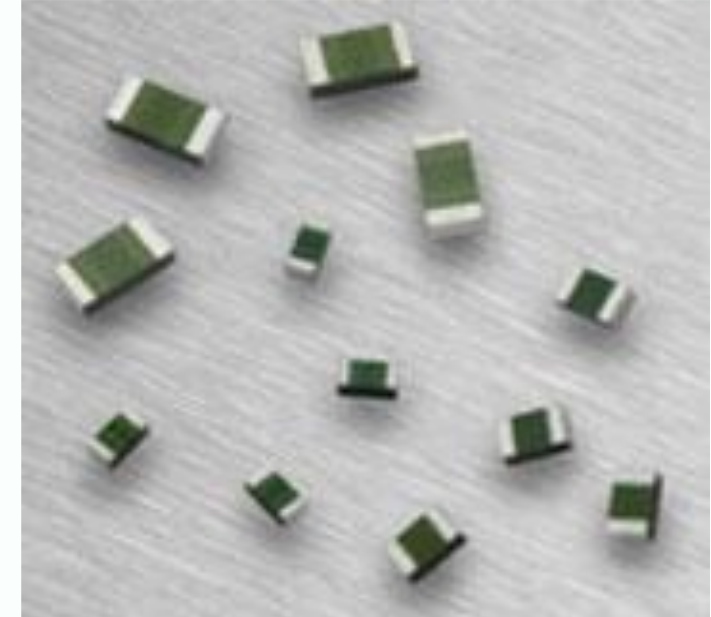
Thermistors



Leads, coated



Glass encased



Surface mount

Thermistors are made of semiconductor materials (metallic compounds including oxides such as manganese, copper, cobalt, and nickel, as well as single-crystal semiconductors silicon and germanium).

Thermistors

- *Thermistors* are sensors whose resistance changes as a function of temperature

Thermistors are classified as either
NTC (negative temperature coefficient) or
PTC (positive temperature coefficient)

Thermistors

- Resistance increases with temperature for PTCs;
- Resistance decreases with temperature for NTCs

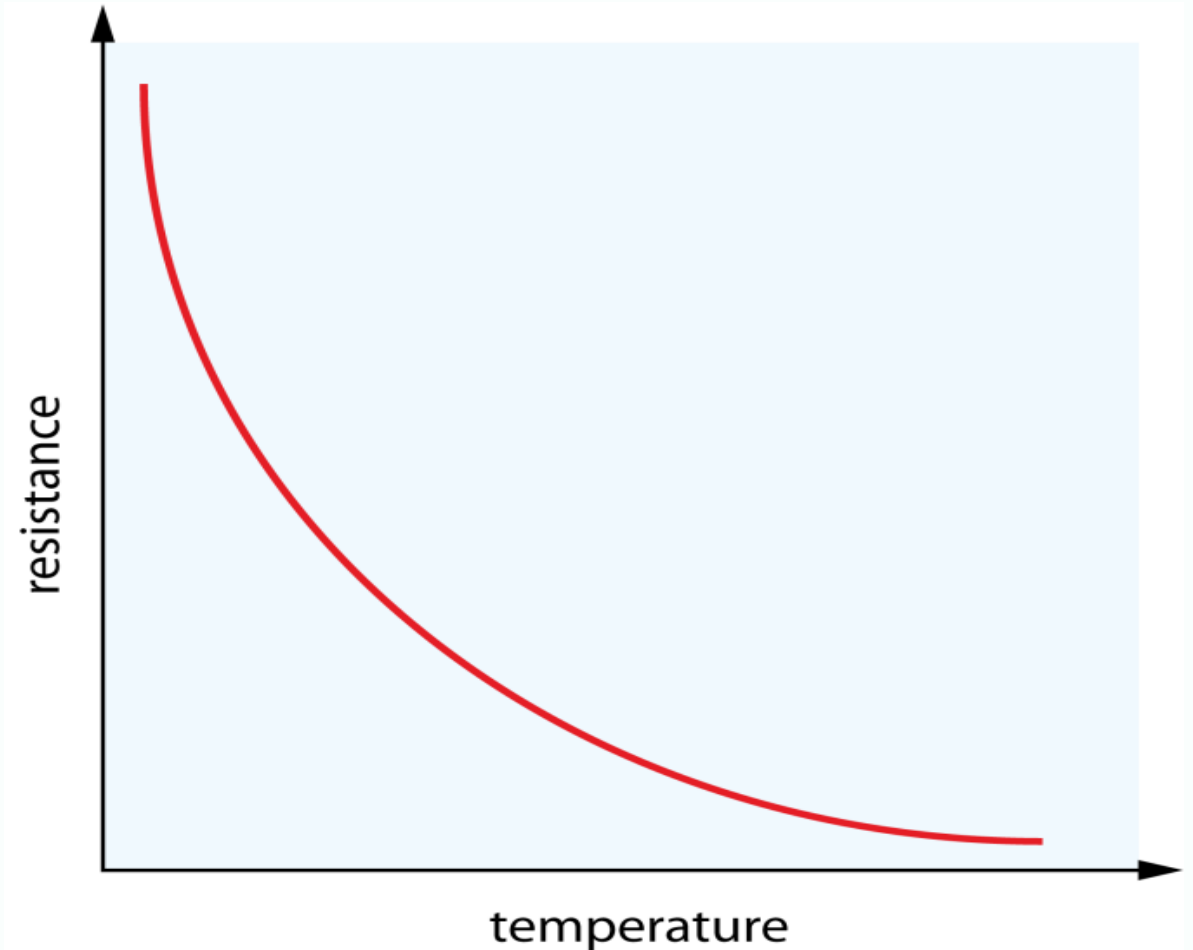
A resistance variation is generally not directly useful;
information is generally relayed with voltage

We need to convert the resistance change to a voltage change

Thermistors

This graph shows the thermistor's resistance against temperature.

Remember, the resistance of a thermistor decreases as the temperature increases.



Thermistors Applications

- Air conditioning and seat temperature controls.
- Electronic fuel injection, in which air-inlet, air/fuel mixture and cooling water temperatures are monitored to help determine the fuel concentration for optimum injection.
- Warning indicators such as oil and fluid temperatures, oil level and turbo-charger switch off.
- Fan motor control, based on cooling water temperature
- Frost sensors, for outside temperature measurement

Thermistors Advantages

High sensitivity to small temperature changes

Temperature measurements become more stable with use

Copper or nickel extension wires can be used

Thermistors Disadvantages

Limited temperature range

Fragile

Some initial accuracy “drift”

Decalibration if used beyond the sensor’s temperature ratings

Lack of standards for replacement



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Thank You

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COE 271 Semiconductor Devices

Lecture 10

PNPN Devices

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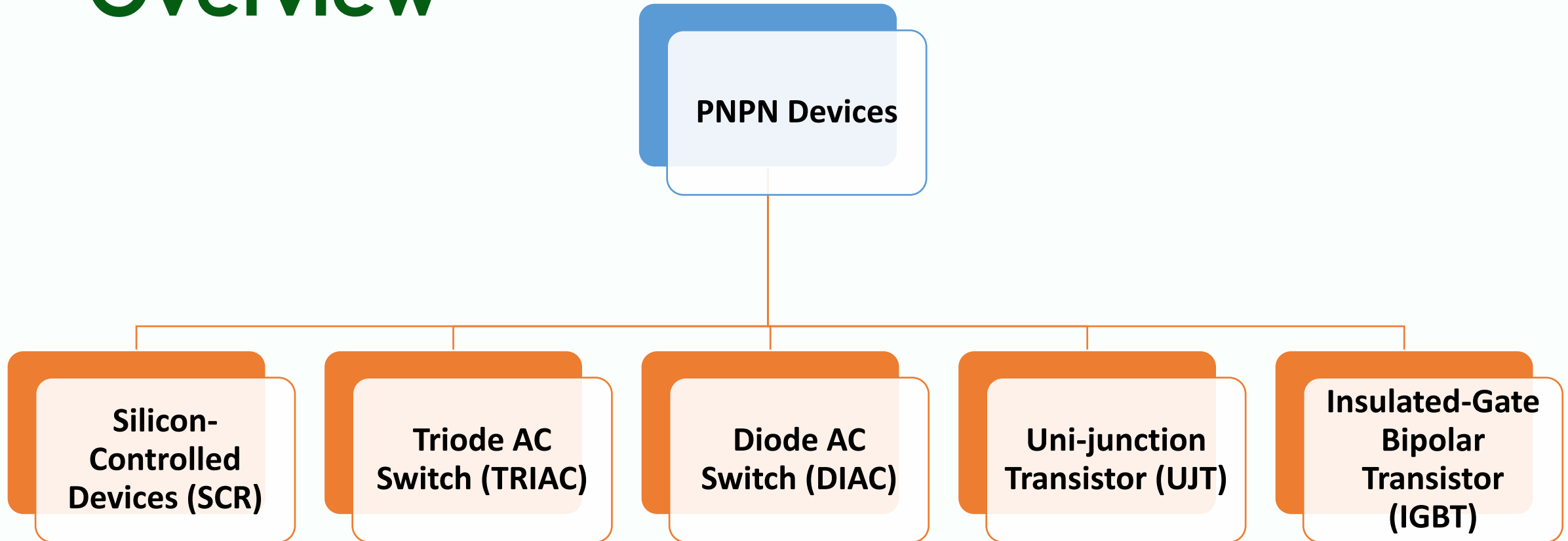
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“The person who reads too much and uses his brain too little will fall into lazy habits of thinking”

Albert Einstein

Overview



Silicon-Controlled Rectifier (SCR)

- The ***silicon-controlled rectifier*** or ***semiconductor controlled rectifier*** is a two-state device used for efficient power control.
- ***SCR*** is the parent member of the ***thyristor family*** and is used in ***high-power electronics***.

Silicon-Controlled Rectifier (SCR)

The SCR is the most important special semiconductor device. This device is popular for its ***Forward-Conducting*** and ***Reverse-blocking characteristics***.

SCR can be used in ***high-power devices***.

For example, in the central processing unit of the computer, the SCR is used in ***switch mode power supply (SMPS)***.

SCR Constructional Features

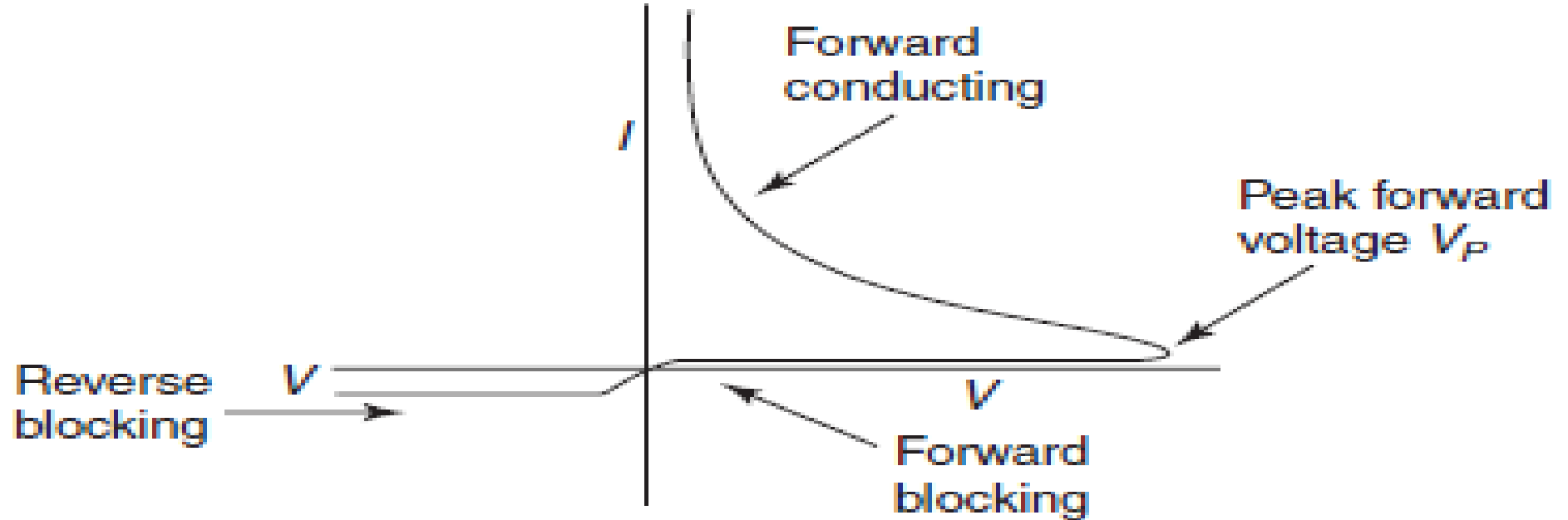
- The SCR is a **four-layer structure**,
- either ***p-n-p-n*** or ***n-p-n-p***,
- *that effectively blocks current through two terminals until it is turned **ON** by a small-signal at a third terminal.*
- SCR has **two states**:
- a **high-current low-impedance ON state** and
- a **low-current high-impedance OFF state.**
- The basic transistor action in a **four-layer p-n-p-n structure** is analysed first with only two terminals, and then the third control input is introduced.

Physical Operation and Characteristics:

- The physical operation of the SCR can be explained clearly with reference to the current–voltage characteristics.
- The forward-bias condition and reverse-bias condition illustrate the conducting state and the reverse blocking state respectively.

Physical Operation and Characteristics:

- Based on these two states a typical I –V characteristic of the SCR is as shown

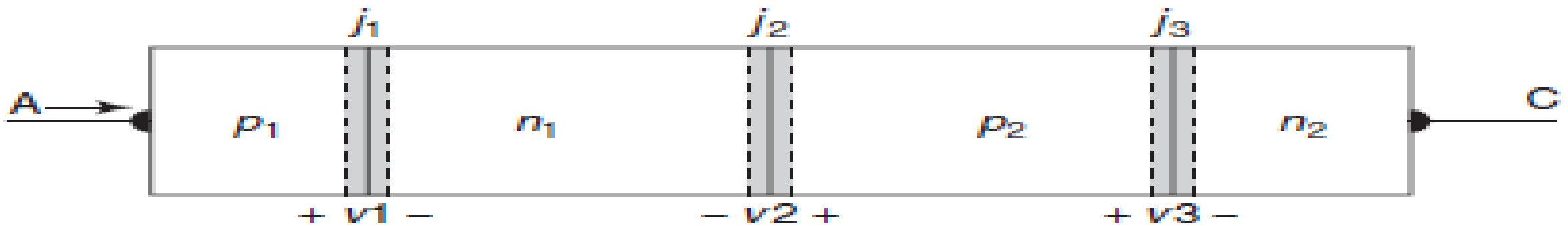


I–V characteristics of a two terminal $p-n-p-n$ device

SCR in Forward Bias

- There are two different states in which we can examine the SCR in the forward-biased condition:
 - (i) The high- impedance or forward-blocking state
 - (ii) The low-impedance or forward-conducting state

At a critical peak forward voltage V_p , the SCR switches from the blocking state to the conducting state, as shown in slide before.



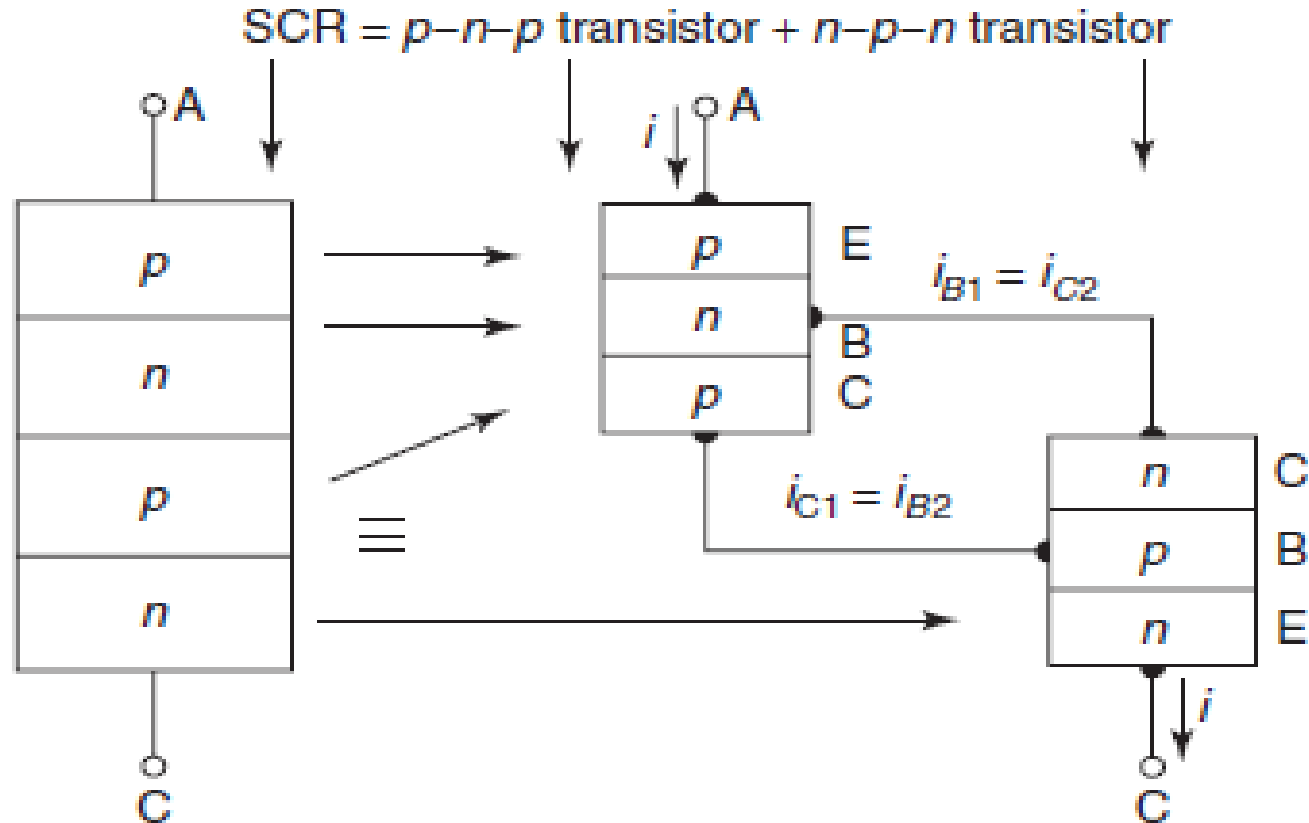
SCR in Forward Bias

- A positive voltage places junction $j1$ and $j3$ under forward-bias, and the centre junction $j2$ under reverse-bias.
- The forward voltage in the blocking state appears across the reverse-biased junction $j2$ as the applied voltage V is increased.
- The voltage from the anode A to cathode C , as shown in Figure, is very small after switching to the forward-conducting state, and all three junctions are forward-biased. The junction $j2$ switches from reverse-bias to forward-bias..

SCR in Reverse Bias:

- In the reverse-blocking state the junctions $j1$ and $j3$ are reverse-biased, and $j2$ is forward-biased.
- The supply of electrons and holes to junction $j2$ is restricted, and due to the thermal generation of electron-hole pairs near junctions $j1$ and $j2$ the device current is a small saturation current.
- In the reverse blocking condition the current remains small until avalanche breakdown occurs at a large reverse-bias of several thousand volts.
- An SCR $p-n-p-n$ structure is equivalent to one $p-n-p$ transistor and one $n-p-n$ transistor sharing some common terminals.

SCR in Reverse Bias



An SCR $p-n-p-n$: a combination of one $p-n-p$ transistor and one $n-p-n$ transistor

SCR in Reverse Bias:

- Collector current $I_{C1} = a_1 i + I_{CO1}$ having a transfer ratio a_1 for the p-n-p.
- Collector current $I_{C2} = a_2 i + I_{CO2}$ having a transfer ratio a_2 for the n-p-n.
- I_{CO1} and I_{CO2} stand for the respective collector-saturation currents.

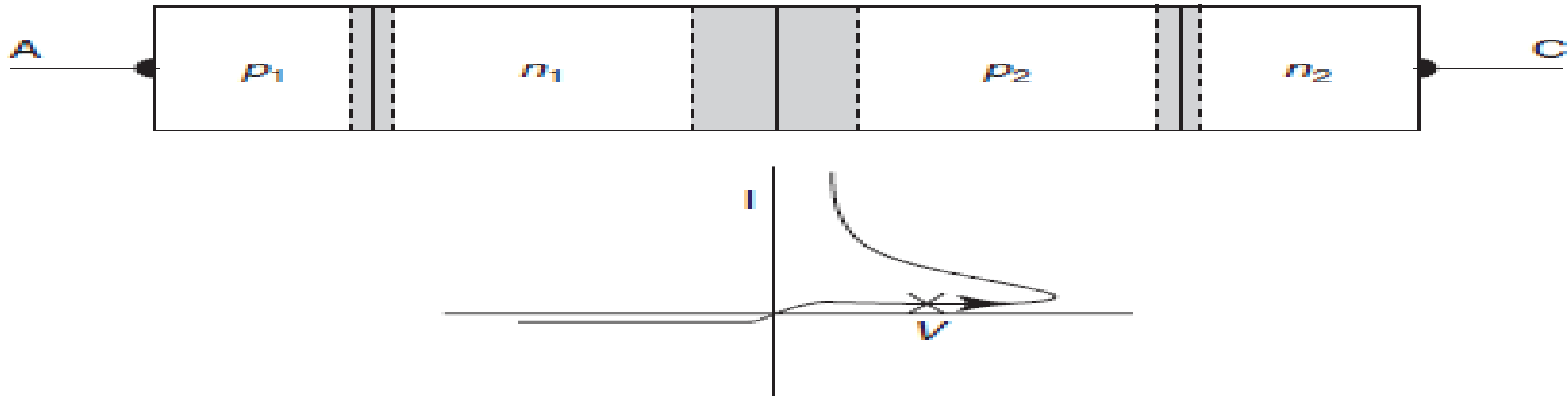
$$I_{C1} = a_1 i + I_{CO1} = I_{B2}$$

$$I_{C2} = a_2 i + I_{CO2} = I_{B1}$$

I-V Characteristics of the SCR

- **Forward-Blocking State:**

- When the device is biased in the forward-blocking state, as shown in the figure below, the applied voltage appears primarily across the reverse-biased junction j_2 .
- *Although the junctions j_1 and j_3 are forward-biased, the current is small.*



The forward-blocking state of the SCR

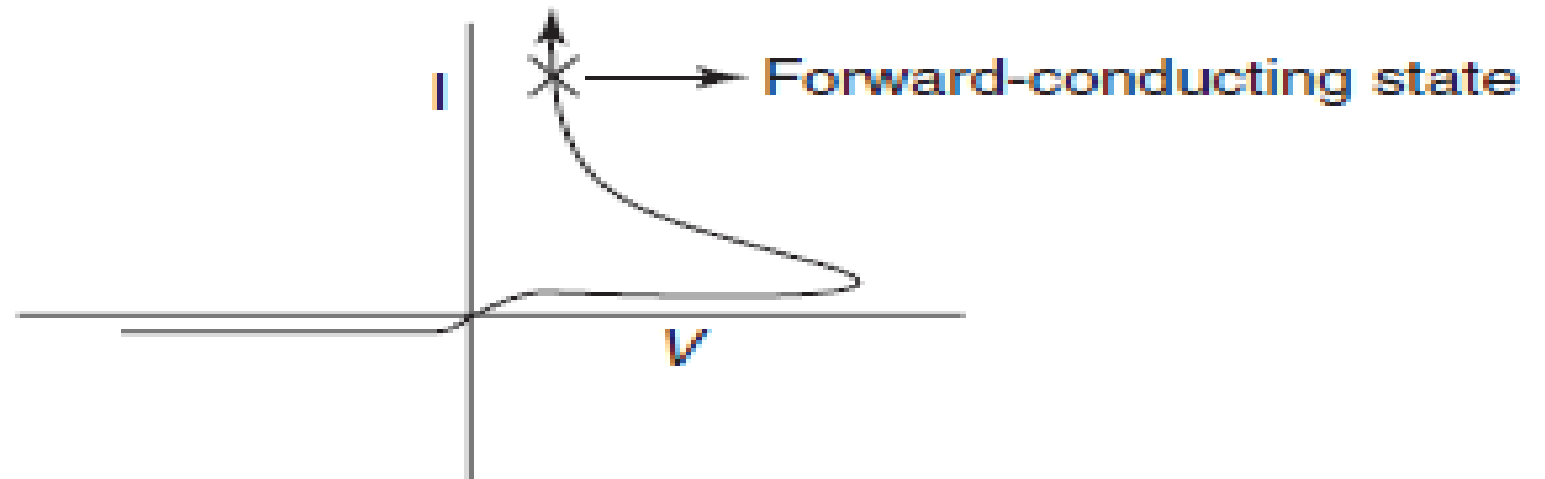
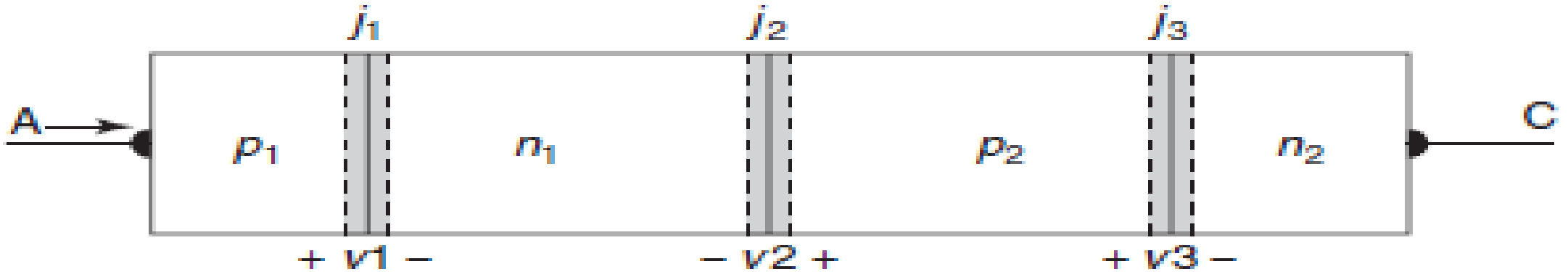
I-V Characteristics of the SCR

- **Forward-Conducting State of the SCR:**

As the value of $(\alpha_1 + \alpha_2)$ approaches unity through one of the mechanisms, many holes injected at j_1 survive to be swept across j_2 into p_2 .

- *This process helps feed the recombination in p_2 and support the injection of holes into n_2 . In a similar manner, the transistor action of electrons injected at j_3 and collected at j_2 supplies electrons for n_1 .*
- *The current through the device can be much **larger**.*

I-V Characteristics of the SCR

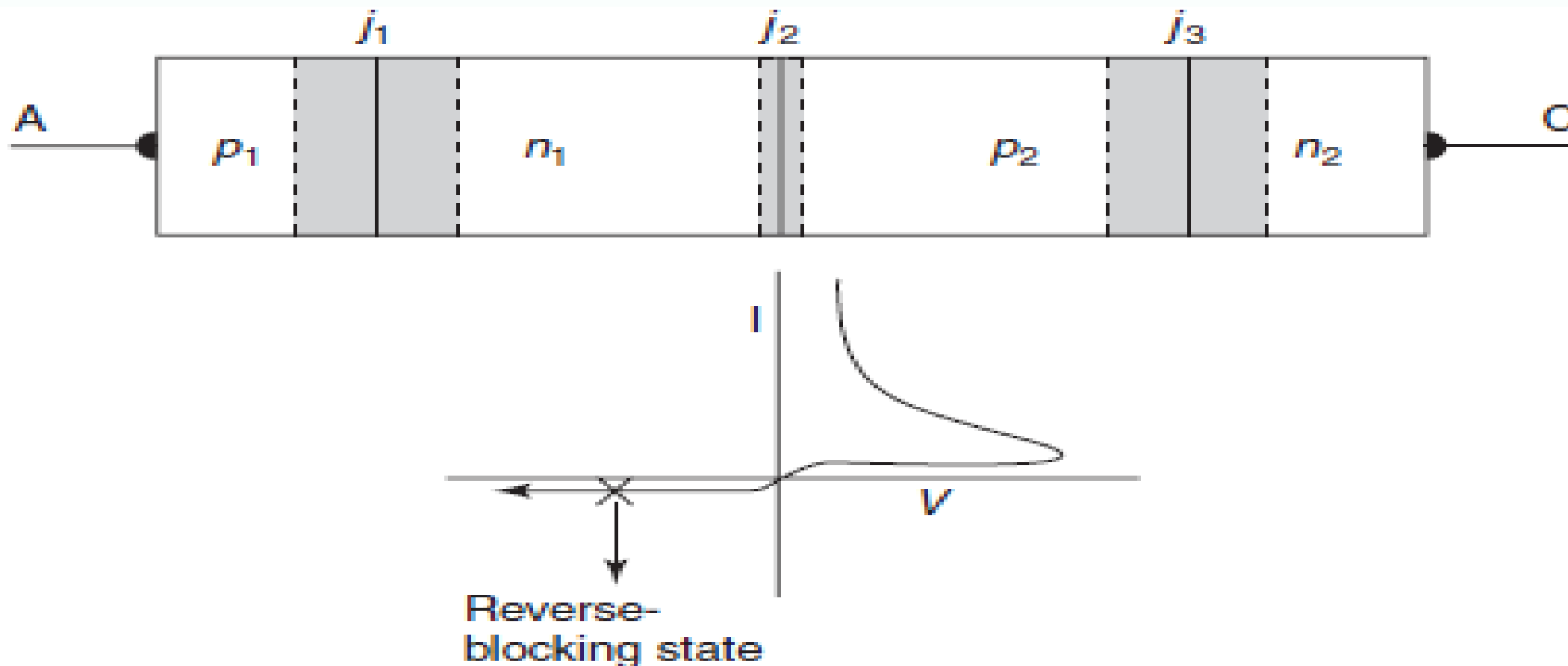


Forward-conducting state of the SCR

Reverse-Blocking State of the SCR

- The SCR in reverse-biased condition allows almost negligible current to flow through it.
- In the reverse-blocking state of the SCR, a small saturation current flows from anode to cathode.
- Holes will flow from the gate into p_2 , the base of the $n-p-n$ transistor, due to positive gate current.
- The required gate current for turn-on is only a few milli-amperes, therefore, the SCR can be turned on by a very small amount of power in the gate.

Reverse-Blocking State of the SCR

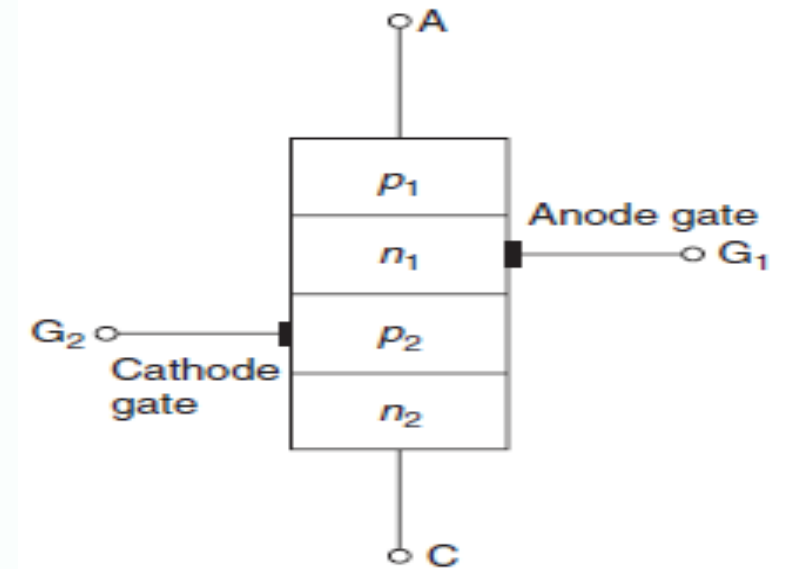


Reverse-blocking state of the SCR

Semiconductor-controlled switch (SCS)

- Few SCRs have two gate leads, G_2 attached to p_2 and G_1 attached to n_1 .

This configuration is called the semiconductor-controlled switch (SCS).

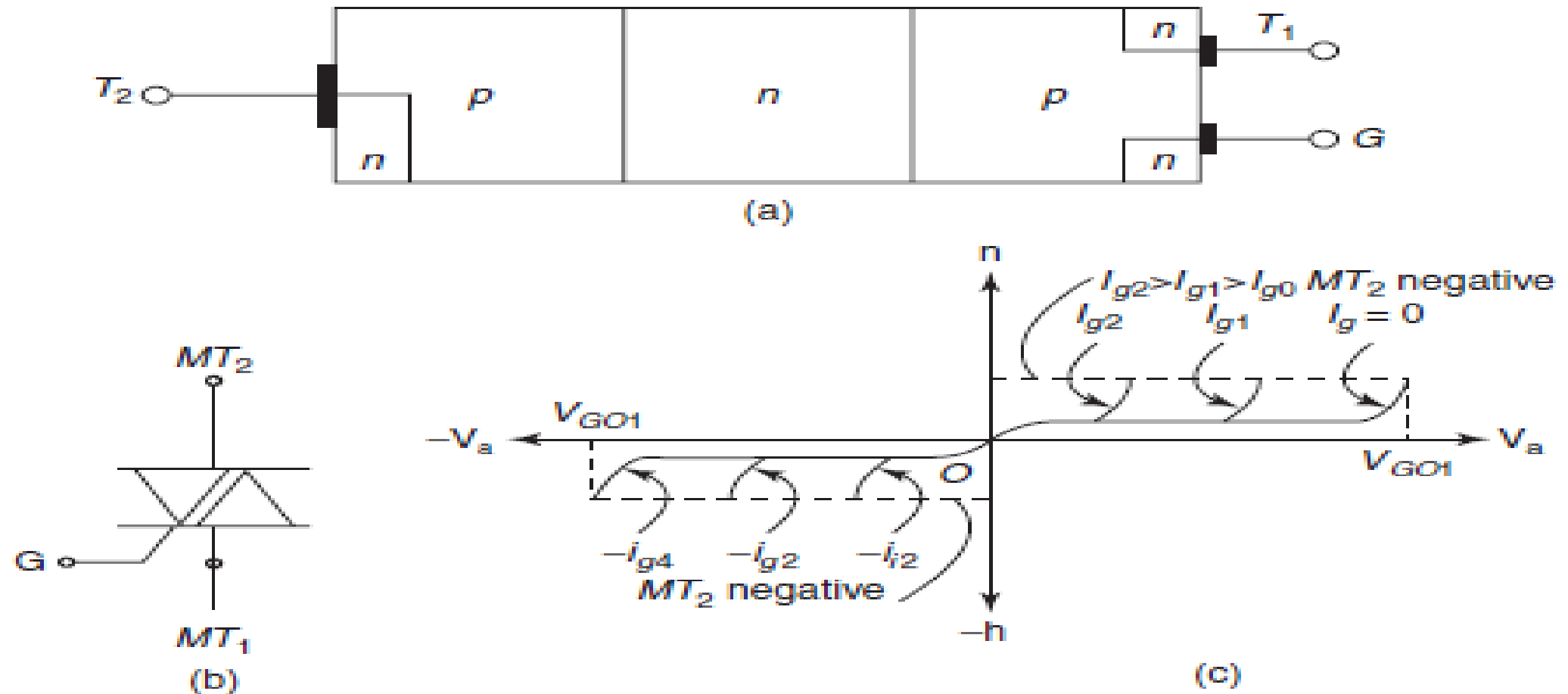


The SCS, biased in the forward-blocking state, can be switched to the conducting state by a negative pulse at the anode gate n_1 or by a positive current pulse applied to the cathode gate at p_2 .

Triode AC Switch (TRIAC)

- The term TRIAC is derived by combining the first three letters of the word “TRIODE” and the word “AC”.
- A TRIAC is capable of conducting in both the directions. The TRIAC, is thus, a bidirectional thyristor with three terminals. It is widely used for the control of power in ac circuits.

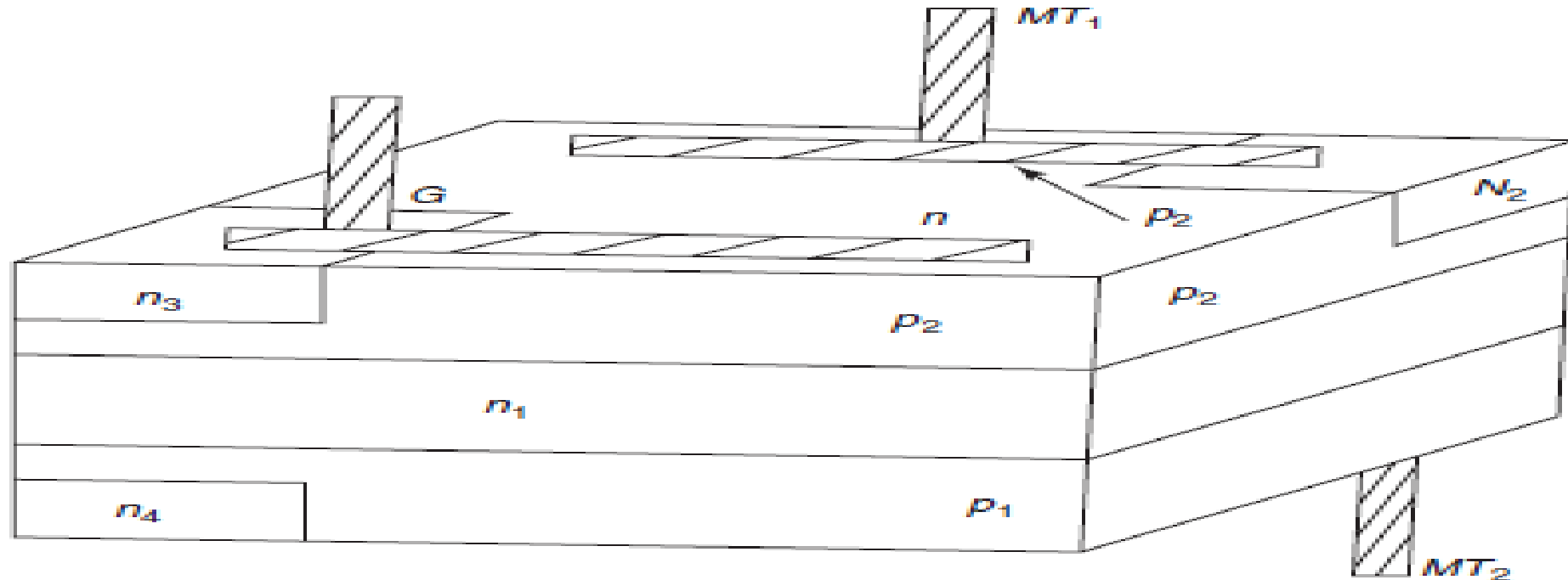
Triode AC Switch (TRIAC)



(a) Structure of TRIAC (b) Circuit symbol (c) Static I–V characteristics

Constructional Features

Depending upon the polarity of the gate pulse and the biasing conditions, the main four-layer structure that turns ON by a regenerative process could be one of $p_1 n_1$, $p_2 n_2$, $p_1 n_1 p_2 n_3$, or $p_2 n_1 p_1 n_4$



Cross-sectional view of the TRIAC

Advantages of the TRIAC

- The TRIAC has the following advantages
 - (i) They can be triggered with positive- or negative-polarity voltage.
 - (ii) They need a single heat sink of slightly larger size.
 - (iii) They need a single fuse for protection, which simplifies their construction.
 - (iv) In some dc applications, the SCR has to be connected with a parallel diode for protection against reverse voltage, whereas a TRIAC may work without a diode, as safe breakdown in either direction is possible.

Disadvantages of the TRIAC

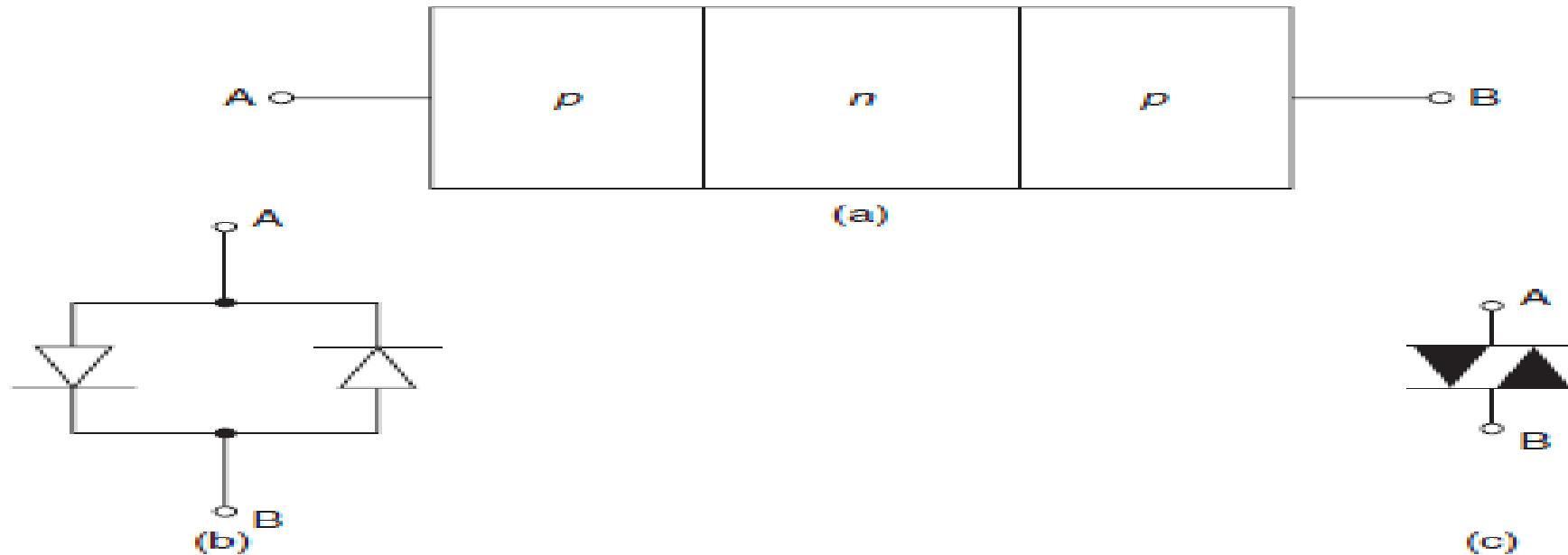
- The TRIAC has the following disadvantages
 - (i) TRIACs have low dv/dt ratings compared to SCRs.
 - (ii) Since TRIACs can be triggered in either direction, the trigger circuits with TRIACs needs careful consideration.
 - (iii) Reliability of TRIACs is less than that of SCRs.

Simple Applications of the TRIAC

- The TRIAC as a bidirectional thyristor has various applications.
- Some of the popular applications of the TRIAC are as follows:
 - (i) In speed control of single-phase ac series or universal motors.
 - (ii) In food mixers and portable drills.
 - (iii) In lamp dimming and heating control.
 - (iv) In zero-voltage switched ac relay.

Diode AC Switch (DIAC)

- The DIAC is a combination of two diodes. Diodes being unidirectional devices, conduct current only in one direction.
- If bidirectional (ac) operation is desired, two Shockley diodes may be joined in parallel facing different directions to form the DIAC.



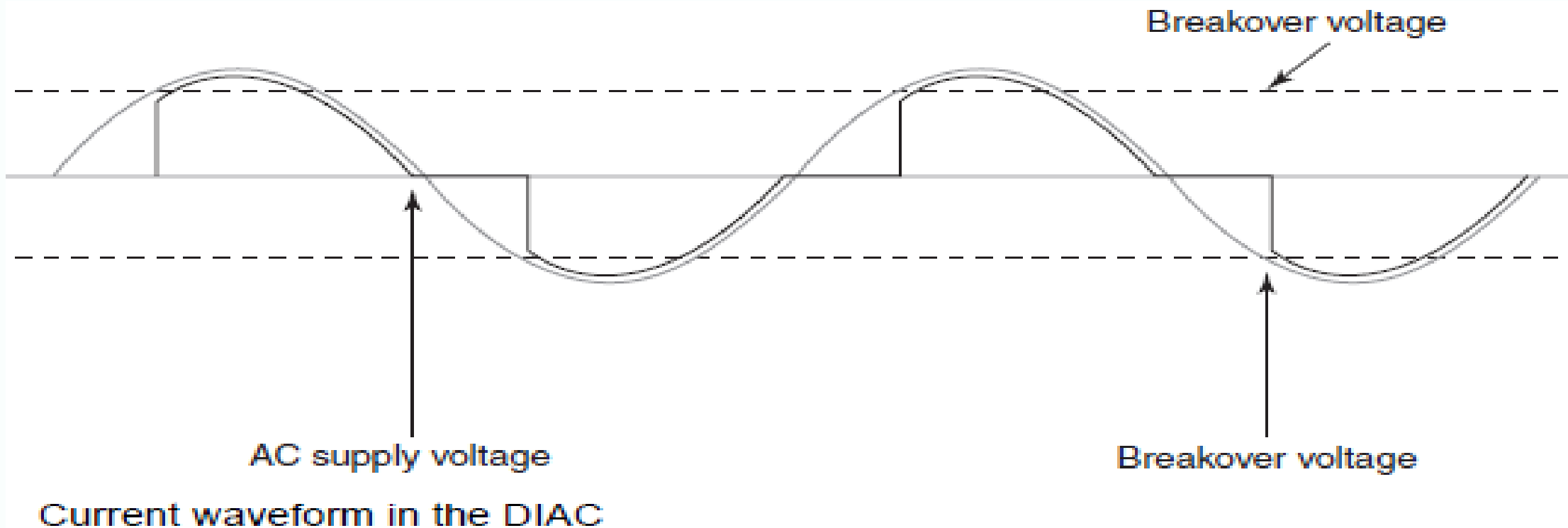
(a) Basic structure of the DIAC (b) Equivalent circuit of the DIAC (c) Symbol of the DIAC

DIAC Constructional Features

- The construction of DIAC looks like a transistor but there are major differences.
- **They are as follows**
- All the three layers, $p-n-p$ or $n-p-n$, are equally doped in the DIAC, whereas in the BJT there is a gradation of doping. The emitter is highly doped, the collector is lightly doped, and the base is moderately doped.

DIAC Constructional Features

- The DIAC is a two-terminal diode as opposed to the BJT, which is a three-terminal device.

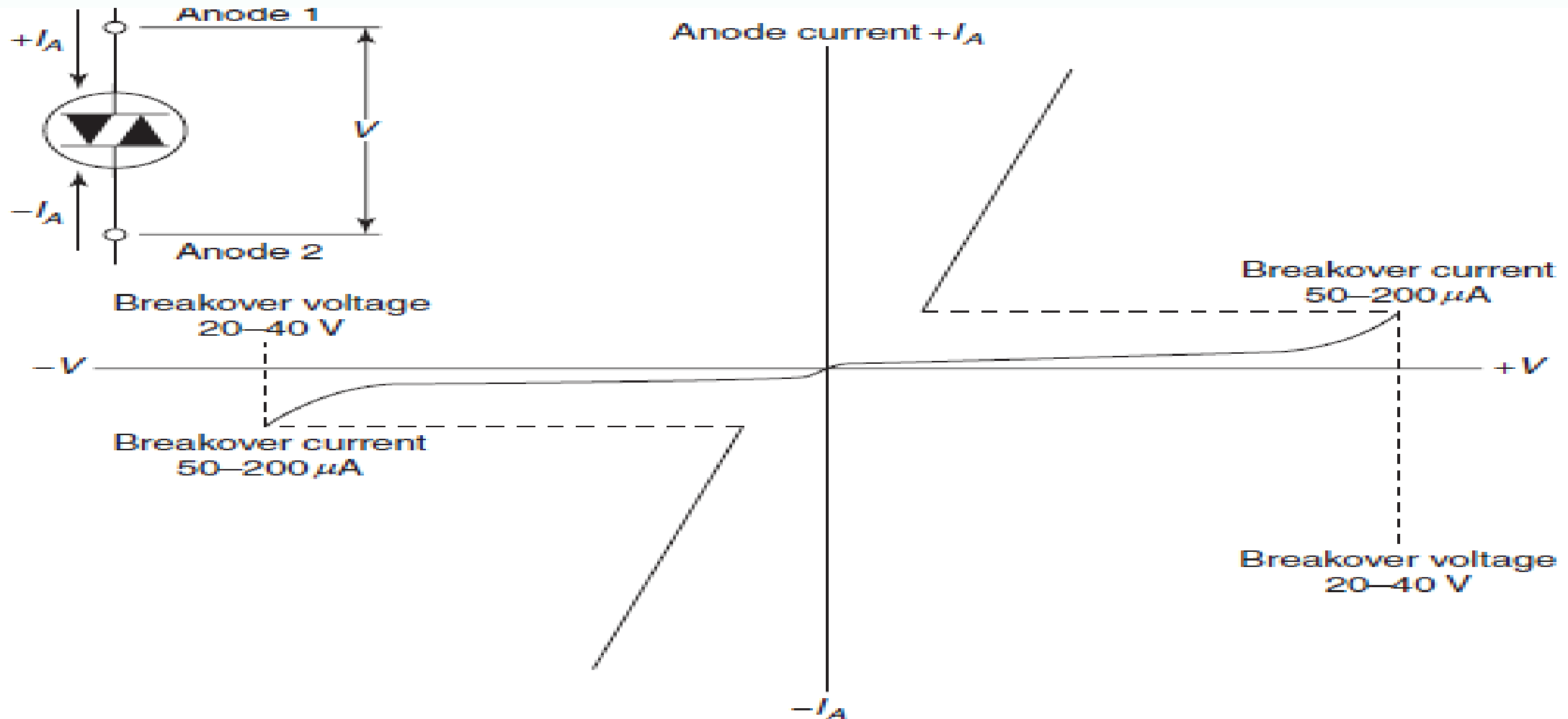


Physical Operation and Characteristics

- **The main characteristics of the DIAC are as follows**

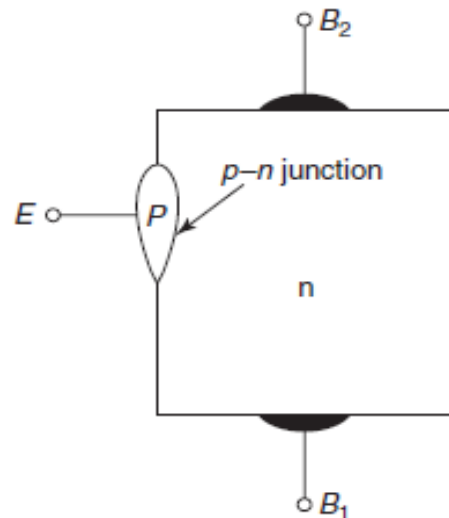
- (i) Break over voltage
 - (ii) Voltage symmetry
 - (iii) Break-back voltage
 - (iv) Break over current
 - (v) Lower power dissipation
- Although most DIACs have symmetric switching voltages, asymmetric DIACs are also available.
 - Typical DIACs have a power dissipation ranging from 1/2 to 1 watt.

I-V Characteristics of the DIAC

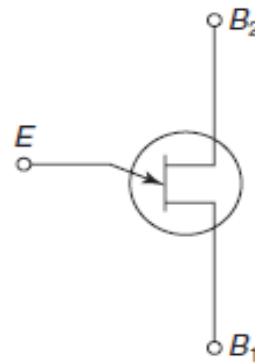


Unijunction Transistor (UJT)

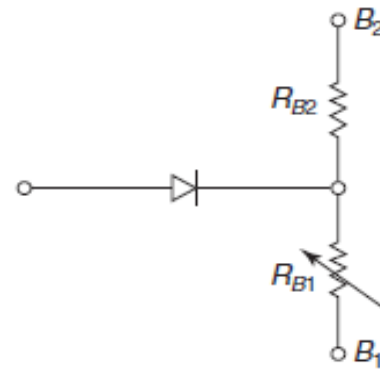
- The uni-junction transistor is a three-terminal single-junction device. The switching voltage of the UJT can be easily varied.
- The UJT is always operated as a switch in oscillators, timing circuits and in SCR/TRIAC trigger circuits.



(a)



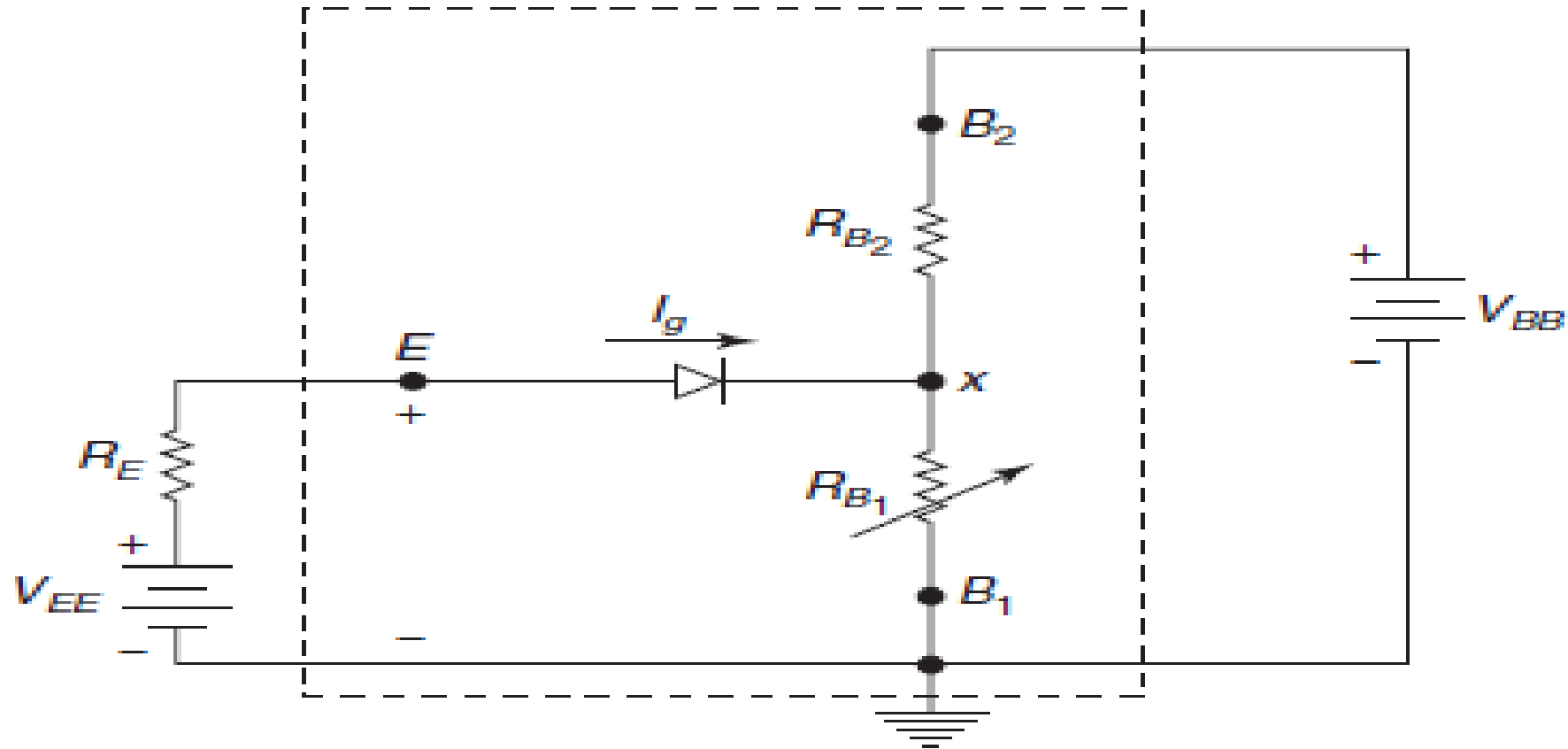
(b)



Constructional Features

- The UJT structure consists of a lightly doped *n-type silicon bar provided with ohmic contacts on either side.*
- The two end connections are called base *B1* and base *B2*. A *small heavily doped p-region is alloyed into one side of the bar.*
- *This p-region is the UJT emitter (E) that forms a p–n junction with the bar.*
- Between base *B1* and base *B2*, the resistance of the *n-type bar called inter-base resistance (R_B) and is in the order of a few kilo ohm.*

Equivalent circuit for UJT



Equivalent circuit for UJT analysis

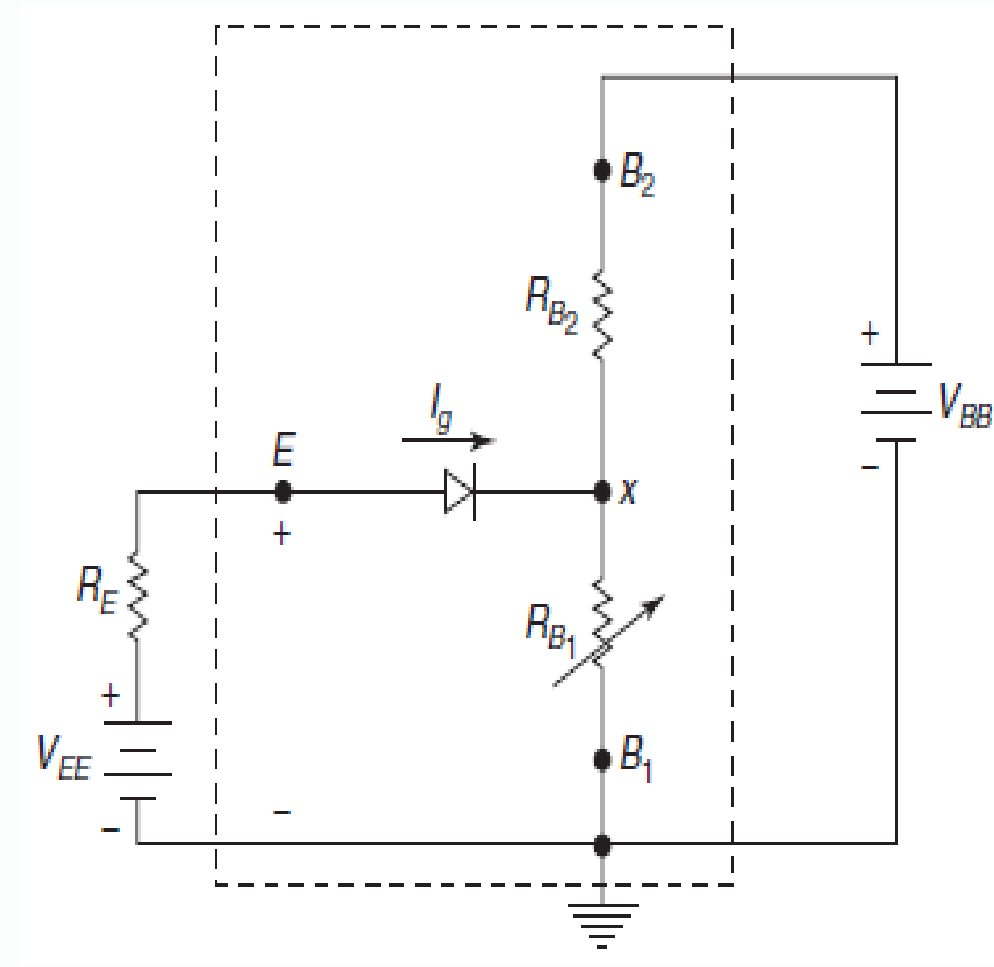
Constructional Features

- This inter-base resistance can be broken up into two resistances—the resistance from $B1$ to the emitter is $RB1$ and the resistance from $B2$ to the emitter is $RB2$.
- Since the emitter is closer to $B2$ the value of $RB1$ is greater than $RB2$.
- Total resistance is given by:

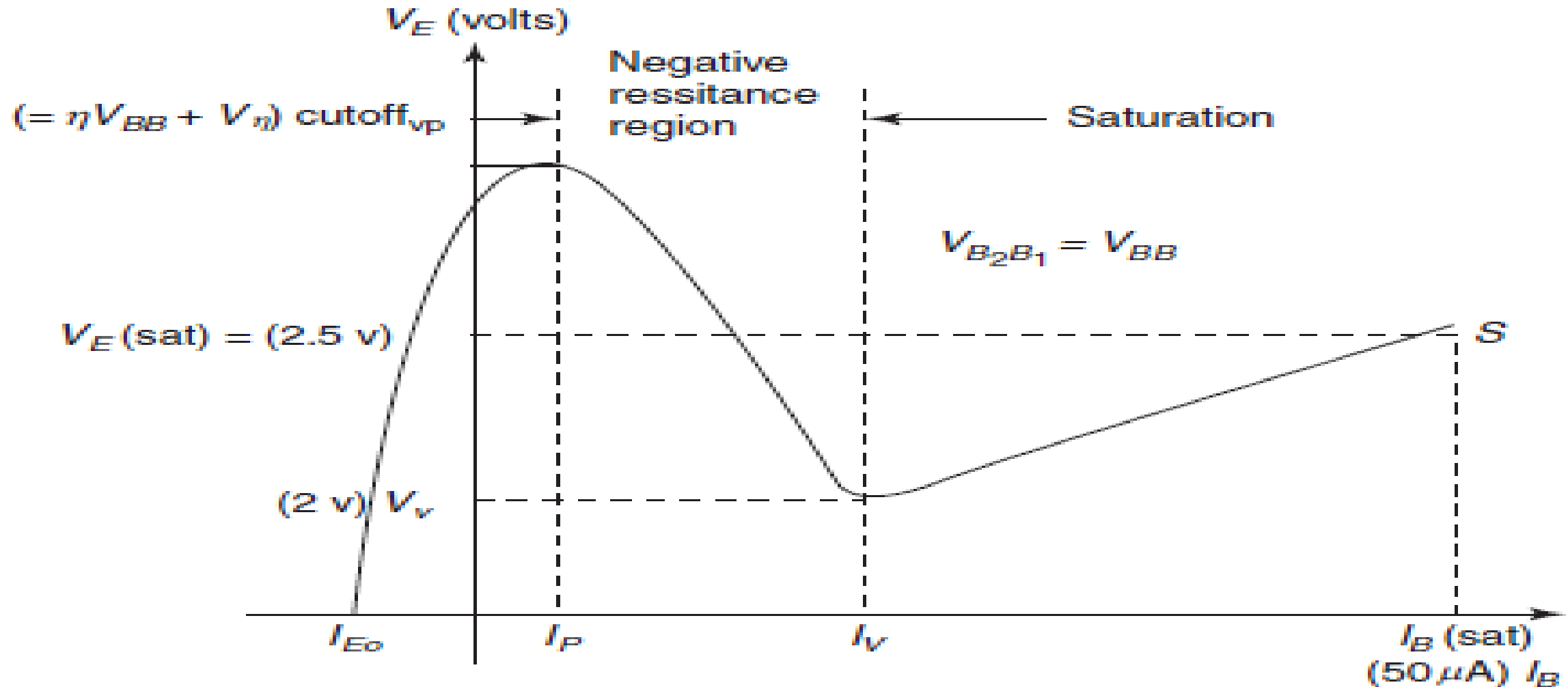
$$RB = RB1 + RB2$$

Equivalent circuit for UJT

- The V_{BB} source is generally fixed and provides a constant voltage from B_2 to B_1 .
- The UJT is normally operated with both B_2 and E positive biased relative to B_1 .
- B_1 is always the UJT reference terminal and all voltages are measured relative to B_1 . V_{EE} is a variable voltage source.



UJT V-I Characteristic Curves



ON State of the UJT Circuit

- As V_{EE} increases, the UJT stays in the OFF state until V_E approaches the peak point value V_P . As V_E approaches V_P the p-n junction becomes forward-biased and begins to conduct in the opposite direction.
- As a result I_E becomes positive near the peak point P on the $V_E - I_E$ curve. When V_E exactly equals V_P the emitter current equals I_P .

ON State of the UJT Circuit

- *At this point holes from the heavily doped emitter are injected into the n -type bar, especially into the $B1$ region. The bar, which is lightly doped, offers very little chance for these holes to recombine.*
- *The lower half of the bar becomes replete with additional current carriers (holes) and its resistance R_B is drastically reduced; the decrease in $BB1$ causes V_x to drop.*
- *This drop, in turn, causes the diode to become more forward-biased and I_E increases even further.*

OFF State of the UJT Circuit

- When a voltage V_{BB} is applied across the two base terminals $B1$ and $B2$, the potential of point p with respect to $B1$ is given by:

$$V_P = [V_{BB} / (R_{B1} + R_{B2})] * R_{B1} = \eta * R_{B1}$$

-
- η is called the intrinsic stand off ratio with its typical value lying between 0.5 and 0.8.

OFF State of the UJT Circuit

- The V_{EE} source is applied to the emitter which is the p-side. Thus, the emitter diode will be reverse-biased as long as V_{EE} is less than V_x . This is OFF state and is shown on the $V_E - I_E$ curve as being a very low current region.
- In the OFF the UJT has a very high resistance between E and B1, and I_E is usually a negligible reverse leakage current. With no I_E , the drop across R_E is zero and the emitter voltage equals the source voltage.

UJT Ratings

- **Maximum peak emitter current**
- This represents the maximum allowable value of a pulse of emitter current.
- **Maximum reverse emitter voltage**
- This is the maximum reverse-bias that the emitter base junction B_2 can tolerate before breakdown occurs.

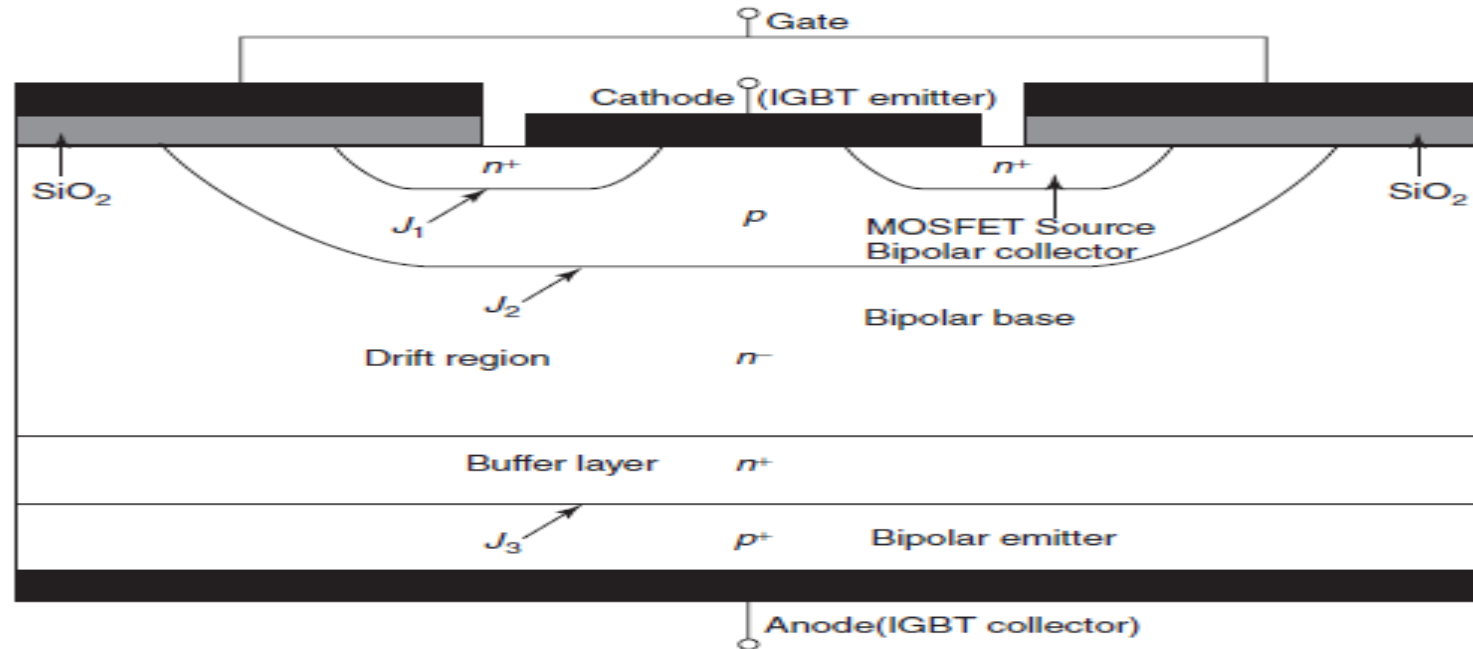
UJT Ratings

- **Maximum inter base voltage**
- This limit is caused by the maximum power that the *n-type base bar* can safely dissipate.
- **Emitter leakage current :**
- This is the emitter current which flows when *V_E is less than V_p and the UJT is in the OFF state.*

Applications

- The UJT is very popular today mainly due to its high switching speed.
- **A few select applications of the UJT are as follows**
 - (i) It is used to trigger SCRs and TRIACs
 - (ii) It is used in non-sinusoidal oscillators
 - (iii) It is used in phase control and timing circuits
 - (iv) It is used in saw tooth generators
 - (v) It is used in oscillator circuit design

Insulated-Gate Bipolar Transistor (IGBT)



The insulated-gate bipolar transistor is a recent model of a power-switching device that combines the advantages of a power BJT and a power MOSFET.

Both power MOSFET and IGBT are the continuously controllable voltage-controlled switch.

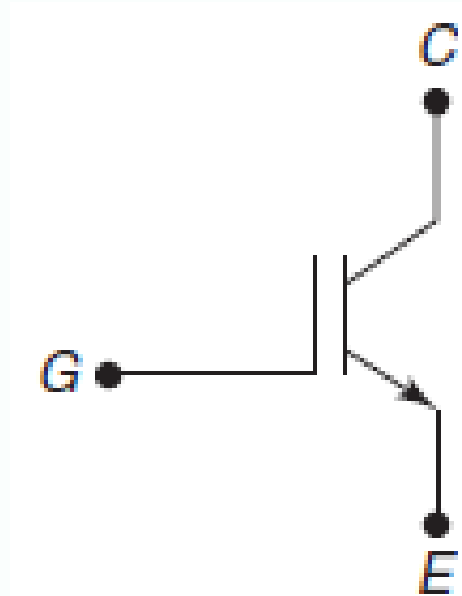
Insulated-Gated Bipolar Transistor (IGBT)

- **Constructional Features**

- The *p* region acts as a substrate which forms the anode region, i.e., the collector region of the IGBT. Then there is a buffer layer of *n* region and a bipolar-base drift region.
- The *p*-region contains two *n* regions and acts as a MOSFET source. An inversion layer can be formed by applying proper gate voltage.
- The cathode, i.e., the IGBT emitter is formed on the *n* source region.

Physical Operation

- The principle behind the operation of an IGBT is similar to that of a power MOSFET.
- **The IGBT operates in two modes:**
 - (i) The blocking or non-conducting mode
 - (ii) The ON or conducting mode.
- The **circuit symbol** for the IGBT is shown beside
- It is similar to the symbol for an *n-p-n bipolar-junction power transistor with the insulated-gate terminal replacing the base*.





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Thank You

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